

IELTS ACADEMIC READING

Passage 2 Collection

*A complete practice book of real and adapted IELTS Reading
Passage 2 texts, with full-text passages, original question types,
and a complete answer key*

IB

60

PASSAGES

804

QUESTIONS

Full

ANSWER KEY

Created by Teacher: Ibodullaev Bekhruz

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Introduction

Welcome to the IELTS Academic Reading — Passage 2 Collection, a practice book bringing together 60 real and adapted Passage 2 texts in one place, compiled and prepared by Teacher Ibodullaev Bekhruz.

Each passage is presented exactly as it would appear in the exam — the full text with paragraph labels, and every original question type (Matching Headings, Matching Information, Multiple Choice, Summary Completion, and more) — with blank spaces and answer circles left for you to write your own answers, just like on test day.

At the back of the book you will find a complete Answer Key for every passage, with the exact sentence from the passage that supports each correct answer, so you can check your work and understand exactly why an answer is correct.

How to use this book:

- Work through one passage at a time under timed conditions (about 20 minutes each).
- Write your answers directly in the circles and blanks provided.
- Check your answers against the key at the back, and read the supporting evidence for anything you got wrong.
- Revisit difficult passages after a few days to reinforce what you have learned.

Good luck with your preparation, and follow [**@bekhruzposts**](#) for more free IELTS materials.

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Passage 1

A New Look for Talbot Park

Talbot Park, a housing project in Auckland, New Zealand, was once described as a ghetto, troubled by high rates of crime and vandalism. However, it has just been rebuilt at a cost of \$48m and the project reflects some new thinking about urban design.

14 A The new Talbot Park is immediately eye-catching because the buildings look quite different to other state housing projects in Auckland. 'There is no reason why state housing should look cheap in my view,' says architect Neil Cotton, one of the design team. 'In fact, I was anticipating a backlash by those who objected to the quality of what is provided with government money.' The tidy brick and wood apartments and townhouses would not look out of place in some of the city's most affluent suburbs and this is a central theme of the Talbot Park philosophy.

15 B Talbot Park is a triangle of government-owned land, which in the early 1960s was developed for state housing built around a linear garden that ran through the middle. Initially, there was a strong sense of neighbourliness. Former residents recall the garden played a big part in their childhoods — a place where kids came together to play softball, cricket and bullrush. 'We had respect for our neighbours and they respected them by title — Mr and Mrs so-and-so,' recalls Georgie Thompson, who grew up there in the 1960s.

16 C Exactly what went wrong with Talbot Park is unclear. The community began to change in the late 1970s as more immigrants moved in. The new arrivals didn't always integrate with the community and a 'them and us' mentality developed. In the process, standards dropped and the neighbourhood began to look shabbier. The buildings themselves were also deteriorating and becoming run down, petty crime was on the rise and the garden was considered unsafe. In 2002, New Zealand decided the properties needed upgrading. The question was, how to avoid repeating the mistakes of the past?

17 D One controversial aspect of the upgrade is that the new development has actually made the density of housing in Talbot Park greater, putting 52 more homes on the same site. Doing this required a fresh approach that can be summed up as the 'mix and match'. The first priority was to mix up the housing mix by employing a variety of plans by different architects: some of the accommodation is free-standing houses, some semi-detached, some low-level, multi-apartment blocks. By doing this, the development avoids the uniform appearance of so many state housing projects, which residents complain denies them any sense of individual identity. The next goal was to prevent overspending by using efficient designs to maximise the sense of space from minimum room sizes. There was also a no-frills, industrial approach to kitchens, bathroom and flooring, to optimise durability and ensure the project did not go over budget.

18 E The population today is cosmopolitan: 50% Pacific Islanders, 20% Maori, 15% Asian, 10% New Zealand European and the rest composed of immigrants from Russia, Ukraine and Iran. 'It was important that the buildings were sufficiently flexible to cater for the needs of people from a wide variety of cultural backgrounds,' explains designer James Lundy.

19 F Despite the quality of the buildings, however, there should be no doubt that Talbot Park and its surrounding suburb of Tamaki are low socio-economic areas. Of the 5,000 houses there, 55% are state houses, 28% privately owned (compared to about 65% nationally) and 17% private rental. There are 85% high-density households with incomes in the \$500 to \$15,000 range and very few with an income over \$70,000. That's in sharp contrast to the more affluent suburbs in Auckland.

20 G Another important part of the new development is what Housing New Zealand calls 'intensive tenancy management'. Opponents of the project call it social control. 'The focus is on frequent inspections and setting clear guidelines and boundaries regarding the sort of behaviour we expect from tenants,' says Graham Bodman, Housing New Zealand's regional manager. The result is a code of sometimes strict rules: no loud parties after 10pm; no washing hung over balcony rails and a requirement to mow lawns and keep the property tidy. The Tenancy Manager walks the site every day, knows everyone by name and deals with problems quickly. 'It's all based on the intensification,' says project manager Stuart Bracey. 'We acknowledge that if you are going to ask people to live in these quite tightly-packed communities, you have to actually help them to get to know each other by organising morning teas and street barbecues.' So far it seems to be working and many involved in the project believe Talbot Park represents the way forward for state housing.

Questions 14–20

Reading Passage 2 has seven paragraphs, A–G.

Choose the correct heading for each paragraph from the list of headings below. Write the correct number, i–ix, in the circle before each paragraph.

- i Some of the problems that developed at Talbot Park
- ii Where the residents lived while the work was being completed
- iii The ethnic makeup of the new Talbot Park
- iv The unexpectedly high standard of the housing
- v Financial hardship in Talbot Park and a neighbouring community
- vi The experiences of one family living at Talbot Park today

- vii How to co-ordinate and assist the people who live at Talbot Park
- viii Raising the money to pay for the makeover
- ix A close community in the original Talbot Park development
- x Details of the style of buildings used in the makeover

Questions 21–23

Look at the following people (Questions 21–23) and the list of ideas below.

Match each person with the correct idea, A–F.

LIST OF IDEAS

- A Good tenant management involves supervision and regulation.
- B State housing must be built at minimum expense to the public.
- C Organizing social events helps tenants to live close together.
- D Mixed-race communities require adaptable and responsive designs.
- E Complaints were expected about the high standard of the development.
- F Too many rules and regulations will cause resentment from tenants.

Q	Statement	
21	James Lundy	
22	Graham Bodman	
23	Stuart Bracey	

Questions 24–26

Complete the summary below. Choose **ONE WORD ONLY** from the passage for each answer.

THE 'MIX AND MATCH' STRATEGY

One aspect of the Talbot Park project that some critics are concerned about is that the higher (24) _____ of accommodation would lead to the old social problems returning. To prevent this, a team of various (25) _____ worked on the project to ensure the buildings were not uniform. Further, they created pleasant, functional interiors that could still be built within their (26) _____. Finally, the absence of walls means Talbot Park is characterised by openness, making it easier to regulate behaviour within the community.

Passage 2

Achieving a Work–life Balance

Research suggests that working fewer hours may be beneficial

A Anthropologists calculate that in the Stone Age, approximately one to two million years ago, the ancestors of modern humans worked on average two or three hours a day hunting and gathering food. This amount of labour was all that was necessary for survival. Today, it's not uncommon to work 50–60 hours a week, and sometimes more. At the same time, there is a substantial body of international research showing that levels of work-related stress have never been higher. However, psychologist Dr Gina McDowell argues that the problem of stress caused by working too many hours still doesn't receive sufficient recognition and ought to be addressed with greater commitment by executives, employees and academics. Dr McDowell is one of many calling for a better work–life balance: spending an appropriate amount of time at work, but leaving sufficient hours in the day to enjoy family, friends and recreations.

B Perhaps leading the search for work–life balance are technology companies, who frequently allow employees greater flexibility in everything from working hours and dress code to office design. Global tech giant MEdia operates what it calls the '20% scheme': its engineers are free to use 20% of their working week (one full day) in any way they want. Ryan Schmidt, the director of human resources at MEdia, says that 'Productivity in the economy generally has increased significantly over the last 50 years but there's a common perception in the workforce that standards of living have not kept pace. Our company intends to turn that around.' Business analyst Carla Deng has made a study of MEdia's 20% scheme and says the results are overwhelmingly positive. Says Deng, 'Our research found that when employees have greater individual choice and responsibility for their work time, they still devote some of those "free" hours to fulfilling their normal working roles. And they also feel a lot happier.'

C An alternative approach taken by some other American companies is the four-day week. Ordinarily, Americans work eight hours a day, five days a week, making 40 hours in total. In a four-day week, they work four 10-hour days instead, still completing 40 hours in total but having three days off. In general, several studies have shown benefits, including reduced stress, increased productivity and more fulfilled employees. However, there may also be downsides, according to Dean Hazel, a professor of economics. For example, customers may expect a certain employee to be present five days a week and feel disappointed when they are not available. 'The truth is that such a four-day week has many attractions but it will not be appropriate in every instance,' says Professor Hazel. 'It depends very much on the company concerned, its size, internal structure and the nature of its relationships with customers, partners and suppliers.'

D Indeed, some commentators question the very idea of work–life balance. Faisal Hadi, a lecturer in sociology, argues that the term work–life balance implies that the two represent distinct aspects of life that are easily separable. But in fact, for many of us ‘work’ and ‘life’ intrude on each other all the time and trying to ‘balance’ them precisely represents a goal that can never be achieved. ‘This term work–life balance has become a cliché,’ says Hadi. ‘In reality, life is an intricate mix of work and leisure and the relationship between the two has so many strands that they cannot be neatly divided. The idea of work–life balance as it is often understood has limited value.’

E However, a recent experiment has certainly challenged this idea. Employees at the New Zealand company involved worked four eight-hour days a week, but were paid for five. The experiment was conducted at Certified Life, a firm which has sold insurance for more than six decades. At the end of the experiment, 78% of employees felt they were able to manage their work–life balance, up from 54%, when working a standard 40-hour week. The 240 employees at Certified Life completed surveys at the beginning and end of the experiment to provide information for the study. The employees themselves worked out how all the necessary tasks could be completed within deadlines while working fewer hours, and were then responsible for implementing those decisions. Lead researcher Dr Hannah Graham says that the experiment produced such positive results partly because employees were given a role in decision-making throughout it.

F The New Zealand experiment also showed a 7% drop in stress levels when working fewer hours, without any drop in productivity. The chief executive of Certified Life, Tane Robinson, says that the four-day week boosted the firm's reputation and made it easier to retain staff who might otherwise have moved on. Furthermore, cooperation between employees was seen to improve significantly when working shorter hours and team cohesion was significantly enhanced, according to the supervisors. Quite apart from the benefits at Certified Life, Dr Graham also suggests that such schemes might also address the issue of congestion in major cities by restricting the numbers going to work each day. Roland Gillespie, workplace relations minister in the New Zealand government, described the findings as ‘enormously encouraging’ and said that he was keen to see businesses experiment with similar new and improved work models. Meanwhile, Certified Life has plans to see the four-day week implemented long term. It may become an initiative that is replicated in other companies.

Questions 14–17

The text has six paragraphs, A–F.

Which paragraph contains the following information? Write the correct letter, A–F.

Q	Statement	
14	The claim that one workplace initiative may not be suitable for all businesses.	
15	An estimate of how much time ancient humans spent working.	
16	A reference to a politician who is interested in changing the way people's working lives are organized.	
17	The claim that today people are working harder but feel that their lives have not improved.	

Questions 18–22

Choose the correct person, A–F, for each statement.

LIST OF PEOPLE

- A Dr Gina McDowell
- B Carla Deng
- C Ryan Schmidt
- D Faisal Hadi
- E Dr Hannah Graham
- F Tane Robinson

Q	Statement	
18	One expression that is commonly used today may not be helpful because it oversimplifies a complicated situation.	
19	After the experiment people had a higher opinion of the business and staff were happy to continue working for us.	
20	Evidence shows that workers who are given more personal time actually spend some of that time on activities that benefit the company.	
21	Over-working is a more significant issue than is sometimes realised and should be taken more seriously by those involved.	
22	Involving staff at every stage of organising the trial was one of the reasons for its success.	

Questions 23–26

Answer the questions.

Write **ONE WORD ONLY** from the text in each gap.

- 23 What service was provided by the company in the New Zealand experiment?
- 24 What research method was used to gather data in the New Zealand experiment?
- 25 Which group of people reported that staff worked together more closely during the New Zealand experiment?
- 26 According to Dr Graham, what problem outside the workplace might be addressed by working shorter hours?

Passage 3

Australian Aboriginal Astronomy

Research is revealing a complex and functional astronomical knowledge used by Aboriginal people to navigate, find food and mark seasonal changes

A Australia's Aboriginal people, living under a dazzling canopy of constellations, absorbed the night skies into their cultural, social and spiritual life. The position of the stars, the motion of the planets, and astronomical events such as comets, meteorites and eclipses have informed their cosmology and traditions. 'Spirituality, sacred law, kinship, cultural rules about who you can marry, where you can go, what you can do, how society works...all of this social structure is written in the stars', says Dr Duane Hamacher, a lecturer at the Nura Gili Indigenous Programs Unit at the University of New South Wales, Australia.

B Hamacher, working closely with Aboriginal elders, is uncovering a wealth of astronomical knowledge. He takes the example of the Pleiades, one of the closest star clusters to Earth. He explains that the Pleiades rise early in the morning, just before the sun comes up, and are visible for about 15 minutes. This signals the time at which flowers appear on one type of tree, the start of winter, and the orca migrating north. These observations indicate that the early Aboriginal astronomers took an intellectual approach that sought meaning in, and application of, astronomical phenomena. And no more is this evidenced than in how they used this knowledge to navigate their vast, and sometimes featureless, island continent.

C Professor Ray Norris, an astrophysicist at the Australian Telescope National Facility and adjunct professor in Indigenous Astronomy at Macquarie University, Sydney, recounts an occasion while bushwalking with Bill Yidumduma Harney, an Aboriginal elder: 'Bill can name about 5,000 stars. Most Western astronomers can name only 20 or 30 on a good day...He looks up at the sky and knows how it changes with the seasons, with the time, in ways I don't actually quite understand. And for him it's completely intuitive....he looks at the sky and knows it reflects what's on the land.'

D William Stevens, an Aboriginal astronomy guide who conducts the Dreamtime Astronomy tour at Sydney Observatory, explains how some Aboriginal people use the constellation of Scorpius for navigation: 'We don't see a scorpion; it's actually a map for us,' says Stevens, adding that people use the stars to travel from one clan group to another. Norris considers the study of Aboriginal astronomy an opportunity for Aboriginal communities to gain access to information that may have been lost after European colonization of Australia. This giving back of knowledge, as Norris describes it, 'could promote community pride and provide educational material for young Aboriginal people'; this could also provide an opportunity to help foster a better understanding and appreciation of Aboriginal culture among the wider Australian society.

E Perhaps the most beguiling application of astronomical observation is associated with the behaviour of one of Australia's most iconic birds, the emu. 'The Emu in the Sky', as it is called, describes a carving that is clearly visible on a rock located in Ku-ring-gai Chase National Park, north of Sydney. It depicts an emu in a somewhat unnatural position for a real emu, with the legs folded behind it. That is, until it is observed that there is a huge and dark shape in one of the 'dark' areas of the Milky Way that resembles an emu. This 'emu' swings around the sky each night, its starting point at dusk changing as the year goes by and the seasons change. In April the body of the 'Emu in the Sky' reaches a certain angle, and people know it is the season to go and find fresh emu eggs, a rich, highly valued food source. When the angle of the representation of the emu on the rock face matches the 'Emu in the Sky', the harvest might begin. The folded legs signify that the emu is sitting on the nest. The Emu in the Sky exemplifies one of the key principles of Aboriginal cultures: what is in the sky is a reflection of what is on Earth. Aboriginal people also apply this concept to construct annual calendars. Often based on six seasons, Aboriginal calendars are relatively complex and are generally constructed from the heliacal rising of stars (i.e. when the star first becomes visible above the eastern horizon for a brief moment just before sunrise).

F Dr Philip Clarke of Federation University Australia has documented how the Kaurna Aboriginal people of South Australia use the rising of a star they call Parna, one of the brightest stars in the night sky. Its appearance just before sunrise indicates that the hot, dry summer is ending and the autumn rains will soon arrive. The lands of the Kaurna include the Adelaide Plains, which are prone to flooding. Therefore, knowledge of the imminent arrival of autumn allows them time to build their large, waterproof huts, known as wurlies. Not only were the positions and movements of individual stars used to predict seasonal changes, the scintillation of stars also informs Aboriginal astronomers of a change in the weather or season. 'They can tell by the degree to which a star twinkles or changes colour to gauge the amount of moisture in the atmosphere,' explains Hamacher. 'They then know whether a storm is approaching or the wet season is coming.'

G It is clear that Aboriginal cultures contain a wealth of astronomical knowledge. This knowledge incorporates a deep and sophisticated understanding of celestial and terrestrial events and should be viewed through the prism of an interconnected world-view: a paradigm in which the Aboriginal people saw themselves not as a separate external observer, but as an integral component of nature and the universe.

Questions 1–6

Reading Passage 2 has seven sections, A–G.

Which section contains the following information? Write the correct letter, A–G.

NB *You may use any letter more than once.*

Q	Statement	
1	an example of an Aboriginal person who can identify many stars in the sky	
2	a reference to an astronomical feature that predicts the movement of a particular animal	
3	a mention of an Aboriginal system of dividing the year into several periods	
4	an explanation of how the Aboriginal people viewed themselves as part of the environment	
5	an example of an Aboriginal art work that may seem strange at first	
6	a reference to a chance for greater public insight into Aboriginal ways	

Questions 7–12

Look at the following statements (Questions 7–12) and the list of people below.

Match each statement with the correct person, **A, B, C or D**.

NB You may use any letter more than once.

LIST OF PEOPLE

- A** Dr Duane Hamacher
- B** Professor Ray Norris
- C** William Stevens
- D** Dr Philip Clarke

Q	Statement	
7	The way a star shines gives information about the weather.	
8	An aboriginal person makes an instinctive connection between earth and sky.	
9	Astronomy provides a guide for Aboriginals about the suitability of partners.	
10	Astronomy helps aboriginal people make practical preparations for a change in the weather	
11	Current research may help aboriginal people today learn about what their ancestors knew	
12	A pattern in the stars helps people know which route to take when visiting people in other areas.	

Questions 13–16

Complete the summary below.

Choose **ONE WORD ONLY** from the passage for each answer.

THE EMU IN THE SKY

‘The Emu in the Sky’ is a (13) _____ on a rock in Ku-ring-gai Chase National Park. This portrays an emu that appears to be sitting in an odd way as its (14) _____ are in an unusual position. However, when (15) _____ comes, and emus lay their eggs, the rock art matches the shape of one of the dark areas in the Milky Way. Then it becomes clear that the rock art represents an emu on a (16) _____ and that the Aboriginal people see the sky as reflecting life on the land.

Passage 4

Biological Control of Pests

Why chemical pesticides are losing their war on agricultural pests, and what natural predators offer instead

The continuous and reckless use of synthetic chemicals for the control of pests which pose a threat to agricultural crops and human health is proving to be counter-productive. Apart from engendering widespread ecological disorders, pesticides have contributed to the emergence of a new breed of chemical-resistant, highly lethal superbugs.

According to a recent study by the Food and Agriculture Organisation (FAO), more than 300 species of agricultural pests have developed resistance to a wide range of potent chemicals. Not to be left behind are the disease-spreading pests, about 100 species of which have become immune to a variety of insecticides now in use.

One glaring disadvantage of pesticides' application is that, while destroying harmful pests, they also wipe out many useful non-targeted organisms, which keep the growth of the pest population in check. This results in what agroecologists call the 'treadmill syndrome'. Because of their tremendous breeding potential and genetic diversity, many pests are known to withstand synthetic chemicals and bear offspring with a built-in resistance to pesticides.

The havoc that the 'treadmill syndrome' can bring about is well illustrated by what happened to cotton farmers in Central America. In the early 1940s, basking in the glory of chemical-based intensive agriculture, the farmers avidly took to pesticides as a sure measure to boost crop yield. The insecticide was applied eight times a year in the mid-1940s, rising to 28 in a season in the mid-1950s, following the sudden proliferation of three new varieties of chemical-resistant pests.

By the mid-1960s, the situation took an alarming turn with the outbreak of four more new pests, necessitating pesticide spraying to such an extent that 50% of the financial outlay on cotton production was accounted for by pesticides. In the early 1970s, the spraying frequently reached 70 times a season as the farmers were pushed to the wall by the invasion of genetically stronger insect species.

Most of the pesticides in the market today remain inadequately tested for properties that cause cancer and mutations as well as for other adverse effects on health, says a study by United States environmental agencies. The United States National Resource Defense Council has found that DDT was the most popular of a long list of dangerous chemicals in use.

In the face of the escalating perils from indiscriminate applications of pesticides, a more effective and ecologically sound strategy of biological control, involving the selective use of natural enemies of the pest population, is fast gaining popularity – though, as yet, it is a new field with limited potential. The advantage of biological control in contrast to other methods is that it provides a relatively low-cost, perpetual control system with a minimum of detrimental side-effects. When handled by experts, bio-control is safe, non-polluting and self-dispersing.

The Commonwealth Institute of Biological Control (CIBC) in Bangalore, with its global network of research laboratories and field stations, is one of the most active, non-commercial research agencies engaged in pest control by setting natural predators against parasites. CIBC also serves as a clearing-house for the export and import of biological agents for pest control world-wide.

CIBC successfully used a seed-feeding weevil, native to Mexico, to control the obnoxious parthenium weed, known to exert devious influence on agriculture and human health in both India and Australia. Similarly the Hyderabad-based Regional Research Laboratory (RRL), supported by CIBC, is now trying out an Argentinian weevil for the eradication of water hyacinth, another dangerous weed, which has become a nuisance in many parts of the world. According to Mrs Kaiser Jamil of RRL, 'The Argentinian weevil does not attack any other plant and a pair of adult bugs could destroy the weed in 4–5 days.' CIBC is also perfecting the technique for breeding parasites that prey on 'disapene scale' insects – notorious defoliants of fruit trees in the US and India.

How effectively biological control can be pressed into service is proved by the following examples. In the late 1960s, when Sri Lanka's flourishing coconut groves were plagued by leaf-mining hispides, a larval parasite imported from Singapore brought the pest under control. A natural predator indigenous to India, *Neodumetia sangawani*, was found useful in controlling the Rhodes grass-scale insect that was devouring forage grass in many parts of the US. By using *Neochetina bruci*, a beetle native to Brazil, scientists at Kerala Agricultural University freed a 12-kilometre-long canal from the clutches of the weed *Salvinia molesta*, popularly called 'African Payal' in Kerala. About 30,000 hectares of rice fields in Kerala are infested by this weed.

Questions 14–17

Choose the correct letter, A, B, C or D.

- 14 The use of pesticides has contributed to
- A a change in the way ecologies are classified by agroecologists.
 - B an imbalance in many ecologies around the world.
 - C the prevention of ecological disasters in some parts of the world.
 - D an increase in the range of ecologies which can be usefully farmed.
- 15 The Food and Agriculture Organisation has counted more than 300 agricultural pests which
- A are no longer responding to most pesticides in use.
 - B can be easily controlled through the use of pesticides.
 - C continue to spread disease in a wide range of crops.
 - D may be used as part of bio-control's replacement of pesticides.
- 16 Cotton farmers in Central America began to use pesticides
- A because of an intensive government advertising campaign.
 - B in response to the appearance of new varieties of pest.
 - C as a result of changes in the seasons and the climate.
 - D to ensure more cotton was harvested from each crop.
- 17 By the mid-1960s, cotton farmers in Central America found that pesticides
- A were wiping out 50% of the pests plaguing the crops.
 - B were destroying 50% of the crops they were meant to protect.
 - C were causing a 50% increase in the number of new pests reported.
 - D were costing 50% of the total amount they spent on their crops.

Questions 18–21

Do the following statements agree with the claims of the writer in Reading Passage 2?

Write YES if the statement agrees with the claims of the writer, NO if it contradicts them, or NOT GIVEN if it is impossible to say what the writer thinks about this.

- 18 Disease-spreading pests respond more quickly to pesticides than agricultural pests do.
- 19 A number of pests are now born with an innate immunity to some pesticides.
- 20 Biological control entails using synthetic chemicals to try and change the genetic make-up of the pests' offspring.
- 21 Bio-control is free from danger under certain circumstances.

Questions 22–26

Complete each sentence with the correct ending, A–I, below. Write the correct letter in each box.

22 Disapene scale insects feed on

22 _____

23 Neodumetia sangawani ate

23 _____

24 Leaf-mining hispidus blighted

24 _____

25 An Argentinian weevil may be successful in wiping out

25 _____

26 *Salvinia molesta* plagues

26 _____

LIST OF ENDINGS

- A** forage grass
- B** rice fields
- C** coconut trees
- D** fruit trees
- E** water hyacinth
- F** parthenium weed
- G** Brazilian beetles
- H** grass-scale insects
- I** larval parasites

Passage 5

Biomimetics

How scientists borrow nature's designs to solve engineering problems

A What has fins like a whale, skin like a lizard, and eyes like a moth? The future of engineering. Almost all living organisms are uniquely adapted to the environment in which they live, some so well that scientists study them in the hope of replicating their natural designs in technology. This process, called biomimetics, is the crossroads where nature and engineering meet.

B Perhaps the best example of biomimetics is Velcro. In 1948 a Swiss scientist, George de Mestral, had trouble removing a plant's prickle which was stuck to his dog's fur, so he studied it under a microscope. Impressed by the stickiness of the prickle's hooks, he copied the design, engineering a fastener made of two pieces. One piece has stiff hooks like the prickly seedcase, while the other has soft loops that allow the hooks to stick. De Mestral named his invention Velcro – a combination of the words 'velour' and 'crochet.'

C Andrew Parker, a research fellow at the Natural History Museum in London and at the University of Sydney, is a leading proponent of biomimetics – applying designs from nature to problems in engineering, materials science, medicine and other fields. He has investigated iridescence in butterflies and beetles and antireflective coatings in moth eyes – studies that have led to brighter screens for cellular phones and an anticounterfeiting technique so secret he cannot say which company is behind it. He is working to make cosmetics that mimic the natural sheen of diatoms (a type of algae) and, with the British Ministry of Defence, to emulate the water-repellent properties of these same creatures. He even draws inspiration from nature's past: on the eye of a 45-million-year-old fly trapped in amber that he studied in a museum in Poland, Parker noticed microscopic corrugations that reduced light reflection. This feature is now being built into solar panels.

D To Parker, every species, even those that have become extinct, is a success story, optimized by millions of years of natural selection. He asks: why not learn from this? Parker explained how the metallic sheen and dazzling colors of certain birds derive not from pigments but from neatly spaced microstructures that reflect specific wavelengths of light. Such structural color, fade-proof and more brilliant than pigment, is of great interest to people who manufacture paint and holograms on credit cards. Glowworms produce a cool light with almost zero energy loss (a normal light bulb wastes 98 percent of its energy as heat), and bombardier beetles have a highly effective combustion chamber in their posterior that heats chemicals and fires them at would-be predators.

E For all nature's sophistication, many of its clever devices are made from simple substances like keratin, calcium carbonate, and silica, which are manipulated into structures of fantastic complexity and toughness. The abalone, for example, makes its shell out of calcium carbonate, the same stuff as soft chalk. Yet by coaxing this substance into walls of staggered, nanoscale bricks through a subtle play of proteins, it creates an armor 3,000 times harder than chalk. Understanding the microscale and nanoscale structures responsible for a living material's exceptional properties is critical to re-creating it synthetically.

F Though impressed by biological structures, Robert Cohen, an engineer at MIT in the United States, considers biology merely a starting point for innovation. 'You don't have to reproduce a lizard skin to make a water-collection device, or a moth eye to make an antireflective coating,' Cohen says. 'The biological structure provides a clue to what is useful. But maybe you can do it better.' Ultimately, he considers a biomimetics project a success only if it has the potential to make a useful tool for people. 'Looking at pretty structures in nature is not sufficient,' says Cohen. 'What I want to know is, can we actually transform these structures into something with true utility in the real world?'

G This, of course, is the tricky bit. Potentially one of the most useful embodiments of natural design is the bio-inspired robot, which could be deployed in places where people would be too conspicuous, bored to tears, or killed. But such robots are notoriously hard to build. Ronald Fearing, a professor of electrical engineering at the University of California, Berkeley, has taken on one of the biggest challenges of all: to create a miniature robotic fly that is swift, small, and maneuverable enough for use in surveillance or search-and-rescue operations.

H The key to making his micromechanical flying insect (MFI) work, Fearing says, is not to attempt to copy the fly, but to isolate the structures crucial to its feats of flying. 'The fly's wing is driven by 20 muscles, some of which only fire every fifth wing beat, and all you can do is wonder, "What on earth just happened there?"' says Fearing. 'Some things are just too mysterious and complex to be able to replicate.'

I For all the power of the biomimetics paradigm, and the brilliant people who practice it, bio-inspiration has led to surprisingly few mass-produced products, and arguably only one household word — Velcro. Some biomimeticists blame industry, whose short-term expectations about how soon a project should be completed and become profitable clash with the time-consuming nature of biomimetics research. Others lament the difficulty in coordinating joint work among diverse academic and industrial disciplines, which is required to understand natural structures and mimic what they do. But the main reason

biomimetics has not yet come of age is that from an engineering standpoint, nature is famously, fabulously complex. For the present, people cannot hope to reproduce such intricate nanopuzzles. Nonetheless, the gap with nature is gradually closing.

Questions 1–6

Reading Passage 2 has nine sections, **A–I**.

Which section contains the following information? Write the correct letter, **A–I**.

Q	Statement	
1	how an organism turns a basic material into something incredibly strong	
2	a claim that biomimetics has had limited commercial impact so far	
3	a difficulty that led to researcher's accidental discovery	
4	an example of nature being far more efficient than a common household object	
5	an ancient specimen that inspired a modern innovation	
6	situations where it is preferable to replace a human with a machine	

Questions 7–9

Look at the following statements (Questions 7–9) and the list of people below.

Match each statement with the correct person, **A–D**.

LIST OF PEOPLE

- A** George de Mestral
- B** Andrew Parker
- C** Robert Cohen
- D** Ronald Fearing

Q	Statement	
7	All living things provide something useful for humans to discover.	
8	Natural designs are sometimes impossible for people to copy.	
9	Biomimetics achieves nothing unless it has a practical application.	

Questions 10–13

Complete the sentences below.

Write **ONE WORD ONLY** from the passage for each answer.

- Andrew Parker is studying the shiny surface of diatoms in order to develop
⑩ _____.
- Paint companies are interested in the way some ⑪ _____ get their colour.
- Bombardier beetles protect themselves by shooting hot ⑫ _____ at their enemies.
- One scientist is studying the ⑬ _____ to build a tiny robot that can help people in danger.

Passage 6

Biophilic Design

How restoring the connection between people and nature is reshaping architecture, healthcare and education

A Biophilic design, a movement related to green architecture, has gained much momentum within the building community in recent years. The premise of biophilic design seeks both to avoid or minimize harmful impacts on the natural environment, and equally importantly, to provide and restore beneficial contacts between people and nature in the built environment.

B People have a psychologically developed need to commune with nature but this has frequently been neglected. All too often, architects have put creative originality before the needs of the people who have to live and work in their futuristic constructions, with scant regard for emotional, as well as practical, considerations. Biophilic design does not advocate tree houses or cave-dwelling, but it does provide the nature-based features that maximize human functioning and health. Though not technically a biophilic design, Fallingwater — the stunning house in rural Pennsylvania designed in the 1930s by Frank Lloyd Wright — arguably speaks to the human soul much more than the boxlike blocks of flats designed by his contemporary Le Corbusier.

C In the modern world, much of our built environment is obstructing the age-old connection between humans and nature. The rates of technological progress are far exceeding rates of psychological evolution, leaving us ill-equipped to cope with our lifestyle, which in turn leads to increased stress. Because biophilia attempts to integrate basic and current needs, it can alleviate the difficulties caused by the brain's constant attempts to function in a modern environment it has not yet evolved to handle. For example, one crucial element of the natural landscape to human health is sunlight. We are evolutionarily programmed to respond positively to sunny areas over dark or overcast ones, and sense that they will foster restoration, improve emotional well-being and promote health.

D Because of its tremendous impact on human psychology, biophilic design plays a vital role in healthcare and healthcare delivery. The current healthcare system contains many flaws, especially in its physical spaces. Hospitals, clinics, and offices are high-stress environments for patients, visitors and healthcare professionals alike. One well-known study looked at the impact of nature on patients after surgery. One group had a view of a tree; the others had a window looking onto a brick wall. The first group had shorter hospital stays, received fewer negative comments from the nurses, required fewer analgesics, and had slightly fewer post-operative complications.

E As a consequence of this and other studies, nature and nature-based design have been integrated into the physical design of many hospitals. For example, Dartmouth-Hitchcock Medical Center (DHMC) in New Hampshire boasts an atrium design, illuminating the entire facility. Natural elements also permeate the building, including wood, stone, and numerous live plants. While DHMC was built to incorporate these qualities, other hospitals have had biophilic features added to existing structures. The application of biophilia's concepts to interior design in hospitals has increased substantially as administrators have witnessed patients' positive responses to nature.

F Many existing buildings contain biophilic elements, but only a few have been built with the specific idea of biophilic design in mind. One such building is the Adam Joseph Lewis Center for Environmental Studies at Oberlin College, in Ohio. The director of Oberlin's Environmental Studies Program, David Orr, explained the building's goals were 'to create not just a place for classes but rather a building that would help to redefine the relationship between humankind and the environment — one that would expand our sense of ecological possibilities'. The Lewis Center is sustainable in a broader sense than in the typical application of the word. It harnesses solar power, utilizes both active and passive air systems, and monitors the weather to adapt to conditions. The Center's 'Living Machine' treats wastewater by combining traditional wastewater technology with wetland ecosystems' purification processes, producing water that can be used in the toilets and for irrigation. In their design, Orr and his team of architects engineered a healthy and comfortable space for students, while ensuring the surrounding environment was undamaged.

G Another example is the University of Guelph-Humber building in Ontario, Canada. It contains a centrally located bio-wall, vertically spanning the building. The wall is covered in dense foliage which not only connects people to nature on the inside of the building, but also functions as a new filtration system prototype. The wall purifies the air and has the potential to fulfil the building's fresh air intake requirements. This is another example of how biophilic design principles can be applied in a variety of contexts. These examples prove that the built environment need not interfere with biological human needs to commune with nature, nor with existing ecological systems.

H Ancient architects built for their cultures, which were almost always more in touch with the earth than Western society of the present. They mimicked nature's forms, producing magnificent structures with which we are still awed — though biophilic design is a novel concept, they certainly employed some of its recommendations. Today, we can add another layer to this tradition and ensure maximal benefit for our planet and ourselves.

Reading Passage 2 has eight paragraphs, **A–H**. Which paragraph contains the following information?

NB You may use any letter more than once.

Q	Statement	
14	a description of how rapid change has a negative effect on people	
15	a reference to an architect whose designs were uncharacteristic of biophilia	
16	a definition of the two main aims of biophilia	
17	a positive claim about early forms of architecture	
18	a reference to the fact that many architects are too focused on innovation	
19	a description of features which conserve energy in a biophilic design	

Questions 20–24

Complete the sentences below. Choose **ONE WORD ONLY** from the passage for each answer.

BIOPHILIC DESIGN NOTES

- In the modern world, rates of technological progress far exceed rates of (20) _____ evolution.
- Biophilia can help to reduce the (21) _____ caused by the difficulties of our current lifestyles.
- People instinctively feel more comfortable in places which have plenty of (22) _____, something which is necessary for health and well-being.
- An experiment found that patients who could see a (23) _____ made a faster recovery.
- A feature in the University of Guelph-Humber building improves the quality of the (24) _____.

Questions 25 and 26

Choose **TWO** letters, **A–E**. Which **TWO** of the following statements does the writer make about the Adam Joseph Lewis Center at Oberlin College?

25

- A** It was the first educational establishment designed with biophilic principles.
- B** It is beneficial for both students and the environment.
- C** It has established new concepts for classroom design.
- D** It aims to inspire new ways of thinking about the environment.
- E** It provides solar power to other buildings in the surrounding area.

26

- A** It was the first educational establishment designed with biophilic principles.
- B** It is beneficial for both students and the environment.
- C** It has established new concepts for classroom design.
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Passage 7

Children's Comprehension of Television Advertising

How well can young children distinguish television programmes from the commercials that interrupt them?

A In 1874, long before the advent of television, the English Parliament passed a law to protect children 'from their own lack of experience and from the wiles of pushing tradesmen and moneylenders' (James, 1965). The act is one of the earliest governmental policies to address children's vulnerability to commercial exploitation and was produced before major corporations earned huge profits by marketing products directly to children. Yet the issues underlying this 19th-century policy remain much the same in the 21st century.

B Television has long been the predominant medium that advertisers have chosen for marketing products to children. It is estimated that the average child sees more than 40,000 television commercials a year, most of which are 15 to 30 seconds in length (Kunkel, 2001). According to another estimate, children aged 14 years and under make \$424 billion in direct purchases and influence \$4190 billion in family purchases, underscoring the high stakes involved (McNeal, 1998). In addition, companies now recognize that brand loyalty built at an early age may reap economic rewards over a child's lifetime (McNeal, 1987).

C A number of factors have contributed to an unprecedented growth in both the amount and type of advertising directed at children. First and foremost, among these are changes in the media environment. In decades past, television programming targeted at children was limited and relegated to time slots unpopular with their parents, such as Saturday mornings (Turow, 1981). Today, the number of channels received in the average US home has escalated with the diffusion of cable television and satellite technologies. In this new multichannel era, there are numerous national program services devoted exclusively to children. Naturally, these channels deliver significant amounts of child-oriented marketing messages. Although parents may be pleased that their youngsters can now watch children's programming at any hour of the day, they may not recognize that such viewing opportunities entail much greater exposure to advertising than any previous generation of youth has experienced.

D Approximately 80% of all advertising targeted at children falls within four product categories: toys, cereals, candies, and fast-food restaurants (Kunkel et al, 1992). Commercials are highly effective at employing specific features designed to attract children's attention. For example, they use the strategy of introducing unique sound effects and rapidly moving images (Greer et al, 1982). The other most common persuasive strategy employed in advertising to children is to associate product with playfulness and

happiness, rather than to provide any actual product-related information (Kunkel et al, 1992). For example, a commercial featuring Ronald McDonald dancing, singing and smiling in McDonald's restaurants without any mention of the actual food products available reflects a playful or happy theme. This strategy is also found frequently with cereal ads, which often include cartoon characters to help children identify the product. In contrast, most commercials fail to mention even the major grain used in each cereal.

E Another common feature of advertising to children is the use of product disclosures such as 'batteries not included' or 'each part sold separately'. Studies make clear that young children do not comprehend the intended meaning of these disclaimers. For example, fewer than one in four kindergarten through second grade children could grasp the meaning of 'some assembly required' in a commercial. In contrast, the use of child-friendly language such as 'you have to put it together' more than doubled the proportion of children who understood the qualifying message (Liebert et al, 1977). The phrase 'part of a balanced breakfast' is also a frequent disclosure included in most cereal ads to combat the concern that sugared cereal holds little nutritional value for children. Research shows that most children below age 7 years have no idea what the term 'balanced breakfast' means (Palmer & McDowell, 1981). Rather than informing young viewers about the importance of a nutritious breakfast, this common disclaimer actually leaves many children with the misimpression that cereal alone is sufficient for a meal. This pattern of employing creative terminology in advertising content so as to obscure certain information that might be unhelpful to the sponsor is a long-standing practice that often misleads the consumer (Geis, 1982).

F Very young children do not recognize that there are two fundamentally different categories of television content: programs and commercials. Most children below the age of 4 or 5 exhibit low awareness of the concept of commercials, frequently explaining them as if they were a scene in the program itself. Once this confusion diminishes, children first recognize the difference between programs and commercials based on either affective ('commercials are funnier than TV programs') or perceptual ('commercials are short and programs are long') cues (Blatt et al, 1972).

G Although most children's programs indicate that a commercial break is coming (e.g. by saying 'We'll be right back after these messages'), research reveals that these 'separators' generally do not help children to recognize advertising content (Palmer & McDowell, 1979). This likely occurs because they are not perceptually distinct from the adjacent programming that surrounds them; in fact, many separators feature characters that appear in the same show that the commercial has just interrupted. When an ad includes one of the characters featured in a program, this is known as host-selling. This type of advertising

makes the task of discriminating between program and commercial content particularly difficult for young children (Kunkel, 1988) and is thus restricted in the US by the Federal Communications Commission during children's programs.

H In sum, because young children lack the cognitive skills and abilities of older children and adults, they do not comprehend commercial messages in the same way as more mature audiences, and are therefore uniquely susceptible to advertising influence.

Questions 14–18

Reading Passage 2 has eight paragraphs, **A–H**.

Which paragraph contains the following information? Write the correct letter, **A–H**. You may use any letter more than once.

Q	Statement	
14	a description of recent innovations in TV broadcasting	
15	a mention of the main goods and services advertised to children	
16	a reference to a current limitation on television advertising aimed at children	
17	two techniques used to encourage children to watch TV commercials	
18	a type of advertisement that may make children believe the opposite of what is true	

Questions 19–22

Look at the following statements (Questions 19–22) and the list of researchers below.

Match each statement with the correct researcher(s), **A–H**.

LIST OF RESEARCHERS

- A** Kunkel
- B** McNeal
- C** Turow
- D** Kunkel et al
- E** Greer et al
- F** Liebert et al
- G** Palmer & McDowell
- H** Geis

Q	Statement	
19	Advertisements often aim to teach children that a brand is fun rather than telling them about what is being sold.	
20	Originally children's programmes were only broadcast when adults rarely watched TV.	
21	Children have a significant impact on what adults buy.	
22	Tests showed that children can follow information if simple words are used.	

Questions 23–26

Complete the summary below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

How very young children perceive commercials

- Children below the age of 4 or 5 do not understand the difference between television programmes and commercials. In fact, these children often mistake an advertisement for a (23) _____ from the programme they are watching.
- This is despite the fact that children's TV programmes usually include announcements called (24) _____ to show that there is going to be a commercial break.
- The problem is made more difficult because of a technique called (25) _____ whereby a person or cartoon figure from the programme is used to sell a product during the commercial break.
- From the age of 4 or 5, children begin to realise that commercials are different from TV programmes: for example, they may then recognise that there is a difference in length, that advertisements are (26) _____ than actual TV.

Passage 8

Classical Music over the Centuries

How the idea of 'classical' music took shape, and how technology has reshaped the way we experience it

⑭ **A** The production of any great art form, and classical music is no exception, does not usually occur in a society dominated by the basic material demands of food and shelter. Art and music have flourished in those periods of history, and those parts of society, in which the luxury of free time and material wealth has allowed such a culture to take precedence over more material matters. In the medieval European world, it was thus primarily in the closed communities of the church and monastery, and royal courts that music, literature and learning were able to flourish.

⑮ **B** It was not until the 18th century that this situation changed to any great extent, and the rise of an economically independent middle class meant that concert going became a public activity for anyone who cared to buy a ticket. It is worth remembering that the idea of classical music widely accepted today did not exist until about 300 years ago. Performing music in concert halls to a paying audience, as something inherently pleasurable and significant, was pretty much unheard of until the 18th century, and not widely established until the 19th. The concert venue, the audience, and the idea of 'masterpieces' of classical music, were all effectively invented during the course of the 18th century – in London, Paris, Vienna, Berlin and other European cities where the arts in general were blossoming.

⑯ **C** Today, music that was originally written for a concert venue may appear, out of its original context, in an advert or film. Conversely, music written specifically for films is sometimes performed live. But nothing has changed music over the last century more radically than the invention and dissemination of recording technologies. However, although Thomas Edison originally developed the phonograph in 1877, and wax cylinders were used as early as the 1880s for recording music, commercial recordings of music were not generally available to the majority until the 1920s. From the mid-1980s onwards, the vinyl disc gradually gave way to the new technology of the CD, but just a decade later, the digital MP3 file was already displacing the CD as the favoured way to reproduce recorded music. Yet now, people have more music stored on their phones or computers – which they can call up with the touch of a finger – than would have been contained on all the metres of library shelves of a proud 'record collector' of the 20th century.

⑰ **D** Before recording, music was a social event – it involved one or more people coming together to make music. The music lasted for as long as the musicians sang or played and then it was over. Therefore, the only music that was heard tended to be composed by recent or living musicians, probably working in the locality; it was rare to hear music from

a past generation, distant place or culture. Even when music became professionalized, people who wanted to listen to music went to a specific venue, at a specific time, to hear musicians create a one-off event.

⑮ **E** These days, however, technology makes almost all the world's music instantly and constantly available to anyone with access to simple and cheap gadgets designed for playing it. Music thus floats free of any specific occasion or venue. It is no longer restricted to a particular audience or group of musicians. For the first time, music (any music) can be an entirely personal affair. This is one of the reasons that the 'classical' label becomes harder to pin down. One of its distinctive aspects – a performance defined by concert halls and opera houses – is dissolved by digital recording formats. As a consequence, all music, classical music included, can become any person's soundtrack for activities such as commuting, exercising or shopping.

⑯ **F** The ubiquity of music as recorded sound means that it's very easy to overlook perhaps the most definitive aspect of the classical music tradition: the fact that it is a written or notated music. Though classical music may lack a precise definition today and mean quite different things to different people, at its heart is the idea of a music that has remained viable over the years because it was written down in some form. The origin of what music historians think of as classical music dates from the ninth century, when a system of musical notation was first developed. Before this time, singers in religious services in cathedrals or monasteries had to learn by heart a huge repertory of chants. The first attempts to notate music were intended to help them remember these. Over the next thousand years, notation became more complex, incorporating such aspects as rhythm and pitch, allowing composers to rework and refine their musical ideas. Put very simply, the history of classical music, in all its varied forms, is the history of a tradition that grew out of the possibilities of musical notation.

Questions 14–19

Reading Passage 2 has six paragraphs, A–F.

Choose the correct heading for each paragraph from the list of headings below. Write the correct number, i–ix, in the circle before each paragraph.

- i Music comes to be enjoyed in a large variety of situations
- ii More people gain access to live music
- iii A focus on survival limits the practice of classical music
- iv A clash of musical styles takes place
- v A range of scientific advances brings music to a wider audience

- vi Listening to music being limited to live performances
- vii How classical music has managed to survive for centuries

Questions 20 and 21

Choose **TWO** letters, **A–E**. Which **TWO** of the following statements does the writer make about recording technologies?

20

- A The vinyl disc was relatively easy to damage.
- B The sound quality from wax cylinders was inferior to that of the phonograph.
- C Electronic storage allows people to keep a vast amount of music.
- D Recorded music sold well immediately after Edison invented the phonograph.
- E The CD was popular for a relatively brief period.

21

- A The vinyl disc was relatively easy to damage.
- B The sound quality from wax cylinders was inferior to that of the phonograph.
- C Electronic storage allows people to keep a vast amount of music.
- D Recorded music sold well immediately after Edison invented the phonograph.
- E The CD was popular for a relatively brief period.

Questions 22 and 23

Choose **TWO** letters, **A–E**. Which **TWO** of the following statements does the writer make about musical notation?

22

- A The way it is interpreted has changed over time.
- B It was originally designed as a memory aid.
- C It is often ignored by classical musicians today.
- D Classical music could not have survived without it.
- E Its importance diminished with the arrival of recording.

23

- A The way it is interpreted has changed over time.
- B It was originally designed as a memory aid.
- C It is often ignored by classical musicians today.
- D Classical music could not have survived without it.
- E Its importance diminished with the arrival of recording.

Questions 24–26

Complete the summary below. Choose **ONE WORD ONLY** from the passage for each answer.

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THE IMPACT OF TODAY'S TECHNOLOGY ON MUSIC

These days, the world's music is instantly and constantly available to almost everyone. Thus, music is no longer tied to a particular location or occasion, nor is it associated with a group of musicians or a specific (24) _____. It can become, uniquely in its history, completely personal to each and every individual. Thanks to digital recording, the need for venues such as opera houses or concert (25) _____ where concerts are performed has vanished. Digitization has also made it possible for people to treat music as a (26) _____ to their daily activities.

Passage 9

Coral Reefs

Why millions of the world's poorest coastal households depend on coral reefs for their livelihoods

A Coral reefs are found in shallow waters throughout the tropical world, and dominate the coastlines of many countries in the South Pacific, Southeast Asia and the Indian Ocean. It has been estimated that over half a billion people live within 100km of coral reef, with over 20 million living near reefs in South-east Asia and nearly 100 million living near reefs in the Indian Ocean (Bryant et al, 1998). While these figures are often quoted in the literature, it is not known exactly how many people depend on reefs, or what their level of dependence is. We do know, however, that there is a wide diversity of stakeholders.

B National and international interest in coral reefs are growing, for example as a focus for research studies by marine biologists, reefs are also increasingly seen as part of a national heritage, as a dumping ground for waste or a source of pleasure for tourists. Many of the people who live in adjacent coastal areas are, however, economically dependent on the reefs. Some of these local inhabitants are able to develop strategies that make full-time, regular use of the reef and its resources. Others can use the reef as a crucial safety net in difficult times. Still, others use the reef as a resource that they tap into at certain times of the year, when they cannot earn a living from other resources. For these groups, coral reefs provide a complex range of benefits.

C The reefs are formed from the skeletons of minute sea creatures, and provide cover for a wide variety of plants and animals. Raw materials include seasonally stable supplies of food, building materials and medicines. As well as an immediate resource, these can also act as a medium of exchange, or a source of income and status. In addition, reefs often give rise to islands that provide habitats for people, and lenses of fresh water for drinking and wave action, and protect lagoons and other productive areas, such as sea grasses and mangroves, which, in turn, provide a reserve of food in all climatic conditions. Furthermore, the physical structure of the reef dictates that many activities are done communally, and this means that work on the reefs can be socially unifying.

D Reefs offer multiple opportunities for direct exploitation by people with many different skills, as well as access to a wide range of different markets, including high-value export outlets. The structural and species diversity of the reef prohibits large-scale, industrial production, and favours small-scale production, thus preserving opportunities for those with few financial or physical resources. Furthermore, the fact that many reef resources are collectively owned means that people who are displaced from other sectors are able to share in them, especially in times of emergency, although the high degree of skill required to fully understand the reef limits the uptake of certain activities which require more complex harvesting strategies.

E For poor coastal households, particularly female headed households and vulnerable groups such as the elderly, shallow coral-reef resources are often their principal source of subsistence and income. Unlike many other types of fishery, where some are excluded from working, coral reefs offer opportunities for women and the elderly, because reef products can be collected on foot. This has significant benefits in empowering women in the household, and different reef-based occupations between men and women spread household risk. However, the ability of coral reefs to continue to provide benefits to the poor is now changing. Throughout the world the capacity of coral reefs to buffer risks and provide livelihood stability is eroding as a result of the diminishing availability of resources. This change is driven by a complex web of direct and indirect factors, over which the coastal poor have only a limited degree of control.

F One of the principal factors responsible for declining reef benefits is reef degradation: coral reef ecosystems are extremely sensitive to change, and easily suffer from disturbance. As coastal areas become ever more populated, increasing numbers of reef stakeholders compete for access to reef benefits. This situation has led to increasing pressure on reef resources, typically resulting in over harvesting and reef decline. In addition, lucrative markets for reef species often drive unsustainable and destructive extraction regimes. The reefs are also degraded by coastal and inland developments and the pollution these produce, as well as by the natural impacts of storms and predator outbreaks. However, perhaps the most significant large-scale cause of reef decline in the longer term is climate change, which threatens to damage large areas of reef worldwide.

G For the poor, these changes have resulted in an increasing reliance on reefs, as larger coastal populations, widespread development, and increasing global-market forces overwhelm and degrade alternative resources such as farmland. However, in many cases coastal tourism developments, external markets and well-meaning efforts to halt reef decline have excluded them from access to reef benefits too. In these cases, livelihoods have often become criminalized by legislation, increasing the burden of risk and transaction costs on the poor, who typically have few other alternatives.

Questions 1–6

Reading Passage 2 has seven paragraphs, A–G.

Which paragraph contains the following information? Write the correct letter, A–G. NB You may use any letter more than once.

Q	Statement	
1	how coral reefs offer shelter from the weather	
2	population statistics	

3	various causes of reef deterioration	
4	the location of coral reefs	
5	examples of coral-reef products	
6	the ease of physical access to reefs	

Questions 7–12

Do the following statements agree with the information given in Reading Passage 2?

Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- 7 Coral reefs are increasingly used for scientific purposes.
- 8 Some people only farm coral reefs to supplement other sources of income.
- 9 Foreign tourists are the main purchasers of coral-reef exports.
- 10 Coral reefs can support mass-production methods.
- 11 As with many fisheries, work on coral reefs is unsuitable for some sections of the population.
- 12 Coastal tourism developments have generally benefited the poor.

Question 13

Choose the correct letter, **A**, **B**, **C** or **D**.

- 13 Which is the best title for Reading Passage 2?
 - A What inhabits a coral reef
 - B How to repair a coral reef
 - C The economic importance of coral reefs
 - D The ecological sustainability of coral reefs

Passage 10

Early Approaches to Organisational Design

A review of the classical and neoclassical theories of organisational structures

Determining the 'best' type of organisational structure for a particular situation has long been an important task for managers in all types of organisations. All structures have advantages and disadvantages and managers face the challenge of developing the most appropriate design for changing circumstances.

The Classical Approach

Early management writers attempted to approach organisational design using a set of principles that would make an organisational structure perform efficiently in most situations, independent of external conditions and internal objectives. The sociologist Max Weber and management writers Frederick Taylor and Henri Fayol were major contributors to this so-called classical approach. They believed that the most efficient organisations had a legalised formal and hierarchical structure called a 'bureaucracy'. Members of the organisation were guided by a sense of duty to the organisation and by a set of rational rules and regulations. According to Weber, such organisations were characterised by the specialisation of tasks, appointment by merit, and an impersonal climate.

Today the word 'bureaucracy' sometimes has negative connotations. Early management writers, however, commended bureaucracy as an organisational design for its rationality, rules for decision-making, clear chain of command, and promotion of people based on ability and experience, rather than favouritism or whim. Weber also believed that clearly specified authority and responsibility made performance easier to evaluate and reward.

Criticism of the Classical Approach

Weber, Taylor and Fayol developed their theories when organisations that resembled this bureaucratic model were modern and efficient. It became evident, however, that some of the major advantages of the bureaucratic structure could become disadvantages if the theory were applied dogmatically. For example, the safeguards against favouritism could be rigidly imposed by adhering excessively to rules – resulting in both managers and subordinates becoming depersonalised.

The classical approach has been criticised from two major perspectives. First, the theory may not have a basis in reality. Have organisations like those described by Weber and the others ever existed? Second, it claims that organisations designed and managed according to bureaucratic principles will enjoy the predicted benefits. But critics argue that the world no longer fits the assumptions in Weber's model (if it ever did), and so a bureaucracy might not yield beneficial results.

Early human relations researchers and behavioural scientists attempted to deal with the major inadequacy of the classical bureaucratic model: neglect of the human element. They argued that an industrial organisation has two objectives: economic effectiveness and employee satisfaction. They also wrote that the bureaucratic structure could be improved by permitting more subordinate participation in decision-making. Because these researchers tried to improve, and not reject, the classical model, they are sometimes called neoclassical theorists and include Douglas McGregor, Chris Argyris and Rensis Likert.

The Neoclassical Approach

McGregor believed that the vertical division of labour that characterised bureaucratic organisations was derived from negative and false assumptions about workers which he called 'Theory X'. Managers assume lower-level employees lack ambition and need orders to work effectively. The rigid formal hierarchy is designed to maintain managers' power over subordinates. Decision-making takes place at upper levels of management while the decisions are carried out by people at lower levels. McGregor argued that organisations based instead on 'Theory Y' assumptions use their members' potential. Theory Y assumes that most people find work satisfying, commit themselves willingly to organisational goals and seek responsibility. Members have more independence than in bureaucratic organisations and lower-level participation in decision-making is encouraged.

Argyris was concerned that managers in bureaucratic organisations had near-total responsibility for controlling their subordinates' work. He argued that managerial domination makes subordinates passive and dependent, and decreases their sense of responsibility and self-control. Argyris argued for an alternative organisational design that acknowledged human needs and feelings, and increased workers' satisfaction. Like McGregor, he favoured giving subordinates more independence and decision-making power to create a more informal organisational culture.

Likert shared the perspectives of McGregor and Argyris. In his research, he found that managers who encouraged their subordinates could motivate them more than traditional authoritarian managers. Likert based a model of four possible systems on these findings. In System 1 power and authority are distributed strictly according to the classical management-subordinate relationship: a manager gives orders to lower-level members. In System 4 organisations, by contrast, there is extensive participation in decision-making and problem-solving groups. Some individuals in each group also belong to other work groups to ensure communication between them. System 4 represents Likert's view of an ideal organisation.

Criticisms of the Neoclassical Approach

The neoclassical approach to organisational design compensates for limitations in the traditional classical model, but it has also been criticised. First, the neoclassicists share the

classical assumption that there is one best way to design an organisation. They overlook environmental, technological, and other variables that might affect an organisation's design, and overemphasise psychological and behavioural variables. Second, Theories X and Y oversimplify human motivation and neglect individual differences. Not everyone is motivated by the non-monetary aspects of work, nor is all work satisfying. Finally, the coordination of work groups to achieve organisational goals may be more difficult than the neoclassicists suggest, particularly when the objectives of lower-level employees are not consistent with those of upper-level managers.

Questions 14 and 15

Choose **TWO** letters, **A–E**. According to the writer which **TWO** of the following are characteristics of the classical approach to organisational design?

14

- A a marked ranking order for employees
- B giving importance to everyone's work
- C the advancement of older workers
- D a neutral working environment
- E increased benefits for workers

15

- A a marked ranking order for employees
- B giving importance to everyone's work
- C the advancement of older workers
- D a neutral working environment
- E increased benefits for workers

Questions 16 and 17

Choose **TWO** letters, **A–E**. According to the writer, which **TWO** of the following are criticisms of the classical approach to organisational design?

16

- A Too many guidelines are proposed
- B Certain practices become negative if they are implemented too strictly
- C Managers and workers are unable to co-operate with each other
- D The administrative standards are unsuited to some work environments
- E Positive outcomes which were expected in the past would be unlikely today

17

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- B Certain practices become negative if they are implemented too strictly
- C Managers and workers are unable to co-operate with each other
- D The administrative standards are unsuited to some work environments
- E Positive outcomes which were expected in the past would be unlikely today

Questions 18 and 19

Choose **TWO** letters, **A–E**. According to the writer, which **TWO** of the following are aims of the neoclassical approach to organisational design?

18

- A to ensure workers are treated as individual people
- B to create a formal atmosphere in the workplace
- C to change the methods of production
- D to allow workers a greater say in what happens at the workplace
- E to standardise the procedures for promotion of workers

19

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- B to create a formal atmosphere in the workplace
- C to change the methods of production
- D to allow workers a greater say in what happens at the workplace
- E to standardise the procedures for promotion of workers

Questions 20 and 21

Choose **TWO** letters, **A–E**. According to the writer, which **TWO** of the following are criticisms of the neoclassical approach to organisational design?

20

- A It suggests that workers are involved in too many decisions
- B The effects of some psychological factors are given low importance
- C The effects of the workplace surroundings are ignored
- D It exaggerates the success of the organisations that use this approach
- E It assumes that all people work for enjoyment rather than financial gain

21

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- B The effects of some psychological factors are given low importance
- C The effects of the workplace surroundings are ignored
- D It exaggerates the success of the organisations that use this approach
- E It assumes that all people work for enjoyment rather than financial gain

Questions 22–26

Look at the following beliefs and the list of people below.

Match each belief with the correct person, **A–D**. You may use any letter more than once.

LIST OF PEOPLE

- A** Max Weber
- B** Douglas McGregor
- C** Chris Argyris
- D** Rensis Likert

Q	Statement	
22	The classical approach relied upon managers having incorrect views about workers.	
23	In the best organizational model, there is shared decision-making and interaction between teams at work.	
24	In an efficient organisation, workers develop expertise in particular areas.	
25	An organization must take into account the emotional demands of people.	
26	The classical approach allowed workers' skills to be assessed in a straightforward way.	

Passage 11

Farmers

An iconic chain store in New Zealand reaches a milestone

A The chain of stores known as Farmers began one hundred years ago as a mail order company. Today the Farmers chain employs 3,500 staff across 58 branches making it the most venerable institution of its type in New Zealand. 'Coming to Farmers once a week has become woven into the fabric of New Zealand culture,' says business historian Ian Hunter. By 1960, one in every ten people had an account with the company. It was not only a shop but also hosted the largest annual parade in the country, the Santa parade each Christmas time. Gary Blumenthal, whose family have been customers for three generations, understands well the role Farmers have played in New Zealand culture. 'We were on holiday in Auckland,' he explains, 'and I decided that the lookout tower on top of the Farmers building would be a unique place to fit the ring on my new fiancée's finger.' The couple has been married for 50 years.

B Farmers' longevity can be attributed to the philosophy fostered by Robert Laidlaw, the Scottish immigrant who established the business. Right from the outset, Laidlaw guaranteed customer satisfaction, and those who were not entirely happy with a purchase would receive a full refund with no questions asked. This was certainly the first time such a guarantee had been made by a New Zealand retailer and was probably one of the first worldwide. Laidlaw believed that the scheme could work if he ensured that every product sold was of the highest quality relative to its price.

C Another important aspect of the Farmers' philosophy was to internationalize the New Zealand market. By the time the Hobson Street store opened in 1920 in order to sell directly to customers for the first time, Laidlaw had established buying officers in New York and London so as to offer his customers the latest designs. This was a bold move in those days and won Farmers a devoted following. It's also true that whoever you speak to regarding Farmers in the early years, whether managers or workers, they all emphasize how the company represented a large family. Everyone employed by Laidlaw felt they were building something that would ultimately be to the benefit of the whole country. For example, to mark the centenary celebrations, 58 Farmers' stores around the country announced the neighborhood charities they would raise funds for, everything from guide dog services to volunteer fire brigades will benefit.

D Farmers has always had a heart, according to former board chairman Rawdon Busfield. 'One day I was in the Hobson Street store and a woman came in with two small children - they didn't have a lot of money, you could tell. That week we had a special on, a big bar of chocolate for a shilling. But the boy only had a penny. I took the penny and gave him the chocolate.' However, not everything has stayed the same. Farmers was once home to

genteel tea rooms, children's playgrounds, and an annual sale to celebrate the birthday of Hector the Parrot, who came to embody the store in the eyes of many shoppers and who died, aged 131, in the 1970s. You could buy houses from Farmers, and its saddle factory supplied the armed forces. In fact, although the company has grown considerably over the years, today it sells a narrower assortment of products than in the past.

E Walk through a Farmers store today and you'll see big changes. Its most recent branch in the suburb of Albany is filled with bright lights and big brands. The branch sells cosmetics from 16 different international companies and buys from approximately 500 suppliers, of which about 30 percent are locally owned. 'Eight or ten years ago,' says current chief executive Rod McDermott, 'lots of brands wouldn't partner with us. We weren't in trouble as such, but we could have been doing much better because we were focused too much on price and not on fashion. The reorientation of the last decade has seen a great improvement.'

F Farmers almost didn't pull through one economic crisis. In the mid-1980s, business was booming and the company had stores throughout the country. Then, with sales topping \$375 million, it was taken over by Chase Corporation. Lincoln Laidlaw, the son of the company's founder, remembers the dark days following the stock market crash of 1987 and the collapse of the Chase Corporation. 'The business was being divided up,' he says, 'and I believe that the original culture of the store was dispelled and it hasn't been recovered. That's a shame, but the business is still strong.' For a few turbulent years the stores were controlled first by a consortium of Australian banks, and later a local corporation. In 2003, it went back into 'family' ownership with the purchase by the James Pascoe Group, owned by an established New Zealand retail family. Brand power has made Farmers profitable again, according to McDermott. 'And now we're becoming the brand we used to be. It's like the rebirth of an icon.'

Questions 1–5

Reading Passage 2 has six paragraphs, A–F.

Which paragraph contains the following information? Write the correct letter, A–F.

Q	Statement	
1	a mention of the company being sold	
2	a reference to how the range of goods for sale has become more limited	
3	an example of an innovative offer made by the company's founder	
4	a description of a romantic event on the shop's roof	

5	a description of the latest store in the chain	
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Questions 6–9

Complete the sentences below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

- Farmers was set up as a/an (6) _____ business before it became a chain.
- Robert Laidlaw set up (7) _____ overseas so he could sell up-to-date fashions.
- According to employees, Farmers was like a/an (8) _____.
- Farmers held a/an (9) _____ once a year in honour of the company's famous pet.

Questions 10–13

Look at the following people (Questions 10–13) and the list of ideas below.

Match each person with the correct idea, **A–F**.

LIST OF IDEAS

- A** The character of the company has changed for the worse.
- B** The store is selling more local products than in the past.
- C** The company believes in offering lower prices than their competitors.
- D** The staff has always believed in generosity.
- E** The store is an integral part of the local way of life.
- F** The store needed to change its approach to business.

Q	Statement	
10	Ian Hunter	
11	Rawdon Busfield	
12	Rod McDermott	
13	Lincoln Laidlaw	

Passage 12

Giftedness and Intelligence

Why psychologists are questioning what IQ tests really measure

A There is a popular view that, by definition, gifted children have IQs of 140 upwards and that testing intelligence by psychologists using an individually administered intelligence test, such as the Wechsler Intelligence Scales for Children, the British Ability Scales or the Kaufmann Assessment Battery for Children, can help to pick out children who might otherwise not be identified as extraordinarily able. Used in this way and in conjunction with other data, educational or clinical psychologists have often advised parents or educationists on strategies to help children realize their abilities.

B However, today, many psychologists have doubts about the efficacy of these tests and about the ethics of using them. Unfortunately, the terms intelligence and IQ have become such a part of language that we tend to use them as if we knew what they meant. Many readers may have found it reassuring to read the first sentence of this article and think, 'Yes, now we know where we are', when, of course, we have said nothing more than, 'Lots of people think something which may or may not be true.'

C When Alfred Binet, as the director of the psychology laboratory at the Sorbonne in Paris, was invited in 1904 to construct a test to identify pupils who might need special assistance in their education, he reasoned that older children would perform better on mental tasks than younger children and that what was needed was a set of tasks arranged in chronological order. Children of 7 years of age who performed at the 5-year-old level, for instance, needed special help (his mental orthopaedics) in small classes to enable them to improve.

D It was all very simple, sensible and empirical. In Binet's words, 'It matters very little what the tests are, so long as they are numerous.' He avoided items like reading skills, which reflected schooling or rote learning and emphasised that he was not measuring intelligence, which was too complex to be measured, and that the results of his tests were purely diagnostic and did not represent innate or immutable qualities.

E Unfortunately, some psychologists in the UK and in the USA fastened upon Binet's tests and adopted his notions for purposes he never intended, and so the highly profitable testing industry was quickly spawned to serve the interests of armed forces, employers and educational administrators. On the one hand, they adopted the concept of intelligence as a single human attribute like height and, on the other, they maintained it was a faculty which was largely inherited. Some of the arguments about how much intelligence was inherited – 70 per cent, 80 per cent? – and how much environmentally determined, read like the discussions about how many angels could stand on the top of a needle.

F If one defines what is to be measured and always uses the same piece of knotted string to measure it, it is perfectly proper to compare the results, especially if one subjects them to stringent statistical analysis. But no matter how accurately one has measured, the value or significance of what has been measured may still be in doubt. Unfortunately, even measuring people's height accurately is not the simple matter it appears to be at first sight. How would we measure an abstract quality like strength and equate it to weight or height or age? The sensible advice we would give anyone who suggested embarking on such a project would be 'don't': 'strength' is an abstraction and, like 'intelligence', results from a number of different elements. Hence, if you want to investigate human performance, whether it is 'strength' or what is popularly called 'intelligence', it is better to do so by studying discrete elements in specific situations. Today, it is that direction in which studies of intellect have moved.

G There is another important reason why IQ tests have waned in popularity: even if they test abilities of one sort or another, which undoubtedly many of them do, they do not test life-skills such as personality, motivational and emotional factors, which may crucially affect future performance and achievement. We would suggest that taken alone, the result of an IQ test has little value and the view of a unitary, innate faculty such as intelligence diverts our attention from the diversity of abilities, skills and qualities of children.

H Again, if we are told, for instance, that children are highly intelligent but are grossly underperforming, what is significant is not their abilities but the factors which have caused their inability to perform, factors we should have been attending to all along. IQ tests are unlikely to have discovered why Churchill or Einstein was underperforming at school, if such was in fact the case, nor could they have predicted how they would develop. And we all know people who are intelligent but can do nothing. People who are all empty words and vain gestures.

I Instead of assessing children's intelligence as measured by intelligence tests, we would be better employed finding out what they can do and how well and quickly they can learn to do more. Tests of intelligence, sometimes masquerading as tests of mental ability, usually succeed in testing what they have been designed to test. But many have a strong verbal element which favours children with a highly verbal background, for example, children of professional people. They will not identify children with artistic or musical abilities and they are unlikely to reveal pupils of exceptional mathematical ability. Some sensitive or over-anxious children may underperform in test situations, and some exceptionally gifted children may find the questions so trivial that they will suspect a trick and spend time searching for hidden meanings.

Reading Passage 2 has nine paragraphs, **A–I**.

Which paragraph contains the following information? Write the correct letter, **A–I**.

NB You may use any letter more than once.

Q	Statement	
1	an explanation of the methodology behind Binet's tests	
2	a discussion of intelligence as a single measurable characteristic	
3	reasons for inadequacy of IQ tests	

Questions 4–7

Choose the correct letter, **A, B, C or D**.

- 4 The writer's purpose in the passage is to
- A describe the history of IQ tests.
 - B discuss the limitations of IQ tests.
 - C define the nature of intelligence.
 - D explain the design of IQ tests.
- 5 Alfred Binet's tests compared children of the same age in order to
- A identify children not performing satisfactorily.
 - B identify the most intelligent children.
 - C measure children's intelligence.
 - D decide the best age for children's education to begin.
- 6 The tests that Binet designed were
- A based on reading skills.
 - B based on mathematical skills.
 - C arranged according to age.
 - D different for different socio-economic groups.
- 7 Recent studies of intellect look at intelligence as
- A a single factor.
 - B a factor impossible to assess.
 - C a constantly changing factor.
 - D separate factors in particular circumstances.

Questions 8–13

Do the following statements agree with the claims of the writer in Reading Passage 2?

Write **YES** if the statement agrees with the claims of the writer, **NO** if it contradicts them, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 8 Alfred Binet believed that intelligence could not be measured.
- 9 Binet's tests were intended for use only in the schools of Paris.
- 10 Binet's tests were used by armies for purposes different to his original intention.
- 11 Intelligence can be measured like any other human characteristic.
- 12 We should assess children's separate abilities and learning capacity, not intelligence.
- 13 Intelligence tests accurately identify gifted and able children.

@bekhruzposts

Passage 13

From princes to paupers: how Goya's portraits tell the story of Spain

Across 160 surviving portraits, Francisco de Goya captured a nation's people, history and superstitions

A There are, according to current scholarship, 160 existing portraits by Francisco de Goya - about a third of his painted output. The real number, though, is much greater since there are no pure landscapes in Goya's work, in that everything he ever painted deals with people. In the work of no other great portraitist are a nation's people, history, traumas, folk traditions, and superstitions so comprehensively and relentlessly captured. His subject was a good one since Goya lived in interesting times: his lifetime was a period that saw Spain pass through the effects of the French Revolution, the Napoleonic wars, and the restoration of the monarchy; it was ruled by a succession of authoritarian and liberal governments; and was simultaneously marked by the Enlightenment. Put his work together - the public and private paintings, the personal drawings and sketches, and the unsettling and sinister etching series - and what you have is a portrait in the round of Spain itself.

B His work as a portraitist has its origins in the 45 tapestry cartoons he painted after he moved to Madrid in 1775. Although his designs don't show portraits, they do show types - the majos and majas who gave Madrid its street swagger, peasants and rich men, courting couples, singers, hunters, children, and young men and women playing blind man's buff or tossing a mannequin into the air. The pictures have a light, rococo palette and the faces are individualised without being those of individuals. But in them, Goya practised poses and groupings and a way of handling light on and around figures that was to be invaluable.

C The tapestry designs also show social roles rather than the people inhabiting them, but when it came to painting portraits proper, Goya would turn this on its head. His greatest strength as a portraitist is that regardless of the status of the sitter, be they a king and queen, the Duke of Wellington, or a doctor or writer, it was the person he showed first and their position second. It was this trait, most apparent in his royal portraiture, that has led him to be seen as satirising the Bourbon monarchy rather than as a painter who depicted what he saw without showing obeisance to the usual flattering conventions. In Robert Hughes's phrase, he did not 'pay reflexive homage to authority' but instead walked a fine line between respect and truth.

D Goya's success was rapid; in 1785 he was made deputy director of painting at the Royal Academy (his main message to his students was subversive - 'there are no rules in painting') and in 1786 became pintor de cámara, required to paint 'works required for

royal service'. He immediately spent some of his 15,000-reales salary on a two-wheeled gig (one of only three in Madrid) which he promptly crashed on his first outing. His amour propre untouched, he wrote to his childhood friend Martín Zapater: 'I have now established an enviable way of living: I do not wait on anyone in antechambers, and if anyone wants anything from me they must come to me; I have made myself more in demand and unless it is a person of rank or at the request of a friend I would do nothing for anyone...'

E Part of the reason for his success was that, unlike Gainsborough, for example, he did not resent portraiture as an economically necessary chore that ate away at the time he could devote to higher art. Portraiture fascinated him; it was, after all, part of that 'sacred science which requires so much study' and his portraits reveal a fully engaged artist. There is hardly any repetition in his poses (to help, he drew on prints of English portraits as well as classical statuary), he worked hard to give his sitters in repose a sense of latent movement, and he defined them by the careful depiction of the space around them. When it came to clothing he allowed himself extraordinary freedom, the frogging on uniforms or lace on a dress sketched in thick impasto rather than being laboriously defined. From close up such three-dimensional scumblings look out of focus but from a distance, they coalesce into silks, braiding, and gauze. Such varieties of technique within a painting keep the eye entertained and moving.

F 'My work is very simple. My art reveals idealism and truth,' Goya claimed. The truth though is always more apparent than the idealism. In the most important commission of his career, for example, the portrait of Charles IV and the royal family of 1800, he eschewed entirely the grandeur of majesty to portray the assembled Bourbons as they appeared - a motley grouping (a French visitor had likened Charles and his queen Maria Luisa to a butcher and his wife) in fancy clothes. Goya may have been a political liberal but he wouldn't have jeopardised his hard-won new position as first court painter – the first Spaniard since Velázquez to hold the role - in order to score political points at the expense of his patrons. That his sitters didn't regard the portrait as unflattering was evidenced by succeeding royal commissions, not least his 1815 portrait of Ferdinand VII. Ferdinand himself was physically unprepossessing, tyrannical, vicious, and small-minded and that is exactly how Goya shows him. The king, however, looking in Goya's particular mirror, saw no such characteristics staring back at him.

G The private side of Goya's portraiture was a world away from the public. The contrast between the Ferdinand portrait and the remarkable 1820 double portrait showing Goya with Dr Arrieta, represents a moral version of the three ages of man – the 73-year-old painter in the throes of a debilitating illness being tended by his doctor. If Ferdinand represents human malignity and Goya its frailty, then Arrieta stands for man's innate

kindness. While not all his portraits are successful (some have a doll-like stiffness), this is what Goya at his best could do: in the guise of representing individuals he showed, without judging, man as a universal being.

Questions 14–18

Reading Passage 2 has seven paragraphs, A–G.

Which paragraph contains the following information? Write the correct letter, A–G. You may use any letter more than once.

Q	Statement	
14	a weakness in some of Goya's paintings	
15	a reason that might explain why some artists did not enjoy painting portraits	
16	a genre Goya did not paint	
17	description of a reaction to a promotion	
18	a summary of the historical significance of Goya's works	

Questions 19–20

Choose **TWO** letters, A–E.

Which **TWO** statements are true about Goya's tapestry cartoons?

19

- A They featured lively civilians and labourers that he observed.
- B Individuals were portrayed without faces.
- C They were used to develop techniques for his future works.
- D People's social standings were a secondary focus.
- E He avoided painting political controversies.

20

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- B Individuals were portrayed without faces.
- C They were used to develop techniques for his future works.
- D People's social standings were a secondary focus.
- E He avoided painting political controversies.

Questions 21–22

Choose **TWO** letters, A–E.

Which **TWO** statements are true about Goya's portraits?

21

- A His depiction of royalty was typical of the time.
- B He was meticulous about the details when painting clothes.
- C He generally avoided using the same pose.
- D He paid very close attention to the background in his paintings.
- E He did not believe in using the works of other artists as inspiration.

22

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- C He generally avoided using the same pose.
- D He paid very close attention to the background in his paintings.
- E He did not believe in using the works of other artists as inspiration.

Questions 23–26

Complete the summary below.

Write **ONE WORD ONLY** from the passage for each answer.

Goya's Artistic Duality

Francisco Goya, in embracing the simplicity and truth in his art, masterfully navigated the delicate balance between idealism and reality. Among the many royal portraits of his career, his most significant (23) ————— was from Charles IV, in which he abandoned conventional majesty to faithfully reveal the monarch, whose depiction was later likened to a (24) —————. His subsequent royal paintings, including the 1815 unflattering portrait of Ferdinand VII, show that he was able to depict truth without jeopardizing his position.

However, Goya's private portraiture revealed a different narrative, as exemplified by his 1820 double portrait with Dr. Arrieta, in which an ailing Goya is seen in the care of the good (25) —————. It serves as a testament to Goya's pinnacle ability to portray humanity in both its finest and darkest moments, all while refraining from passing judgment; in the latter scenario, individuals may have to be dependent on the (26) ————— of their fellow men.

Passage 14

How Babies See Colour

Inside the University of Sussex Baby Lab, researchers track how infants learn to perceive the world in colour

① **A** Alice Skelton and fellow researchers at the Baby Lab at the University of Sussex in the UK are attempting to answer a basic question which has fascinated parents and scientists alike: when it comes to colour, what exactly can babies see? The Baby Lab study also hopes to develop ways to identify babies at risk of becoming colour blind and to discover ways to minimise the impact of this condition on their health and well-being. The technology used at the Baby Lab involves a sophisticated eye-tracking system which allows the researchers to follow the eye movements of the babies. While a camera locks its focus onto their faces, sensors register the corneal reflections in the babies' eyes and the position of the pupils in the eyes. With this data, the researchers can assess the babies' reactions to colours.

② **B** To illustrate one of their methods of studying babies, the Sussex researchers place an infant subject, four-month-old Teo Bosten-Lam, in a padded seat arranged so he gazes at a computer screen. The screen is grey, but in the top right-hand corner is a deep blue circle. When Teo shows he is aware of the circle by looking directly at it, it changes into a smiley face and a happy tune fills the room. At one point, Teo begins to look around the room. When this happens, suddenly a black and white spinning disc appears on the screen, making a 'bong' sound. A researcher explains, 'Babies can't resist the black and white swirl things.' These are played when they look away to get their attention back to the screen. And, when Teo shows he is getting tired of the whole process, the screen flashes a clip of an animated cartoon character. This results in Teo's eyes returning to the screen.

③ **C** To a baby, the world changes rapidly. At birth, everything is a blur, with visual acuity around 5% of that of a mature individual and faces initially only discernible at a distance of around 30 cm. But change is rapid. 'The early stages of learning to see colour and basic forms happen relatively quickly,' says Alex Wade, professor of psychology at Britain's University of York and an expert in visual processes. By the age of six months, he adds, a baby's visual acuity has developed to almost its full extent.

④ **D** Just how such changes occur, and their impact on a baby's understanding of the world, is the driving force behind baby labs around the world. The Sussex Baby Lab is attempting to discover how colour is seen and understood by infants; 'It is a myth that babies see in black and white,' says Anna Franklin, head of the Baby Lab, pointing out that studies have found that newborn babies can see large, bright patches of red on a grey

background. An expert on colour vision, Franklin is engaged in infancy research into why certain children have colour obsessions. Her research has aided the development of infant toys, as well as children's theatre and television shows.

⑤ **E** That we can see the world in glorious colour at all, Franklin points out, depends on specialist cells in the retinas of our eyes. Known as cones, these come in three types – those sensitive to long, medium, and short wavelengths of light. While babies are born with all three types of cones, it takes time for these to mature, and for the brain to make sense of the signals on which a baby's discrimination of colours depends. By two months, babies can tell red and green colours apart; a few weeks later, they can also recognise blues and yellows. But the intensity of the colours is crucial. Franklin notes that if you show a baby a kind of washed-out green, they won't be able to see it, even if they can see a strong green. While a gradual improvement in a baby's ability to see faint colours occurs as they mature, it isn't known whether all colours need to be just as strong for a baby to spot them.

⑥ **F** Skelton has found, through testing more than 40 babies, that even at four months they need blues and yellows to be stronger than reds and greens in order to be able to see them. Recently, Franklin and her team have been exploring the number of colour categories babies possess. More than 170 babies were recruited for the experiment, with each repeatedly shown two squares of the same colour, then two of different colours, and the babies' behaviour was monitored. 'The upshot is babies have got five colour categories, we think: red, green, blue, purple and yellow-brown,' says Skelton. Further categories, such as orange and pink, appear to emerge later with language. But, not all societies categorise colours in the same way. This may be linked to the words in different languages for particular colours. And the availability of words may depend on the needs of that society to be able to differentiate certain colours.

Questions 1–6

Reading Passage 2 has six paragraphs, A–F.

Choose the correct heading for each paragraph from the list of headings below. Write the correct number, i–ix, in the circle before each paragraph.

- i A comparison of the vision of babies with that of adults
- ii Devices for measuring colour intensity
- iii Ways in which bright colours are used to sell baby products
- iv Possible cultural influences on colour perception
- v A popular misconception about colours newly born babies can see
- vi Maintaining a baby's concentration during an experimental procedure

- vii The aims and potential medical benefits of the research
- viii Structures within the eye important for colour recognition

Questions 7 and 8

Choose **TWO** letters, **A–E**.

Which **TWO** of the following are mentioned with regard to the Sussex Baby Lab research?

OPTIONS

- A** A camera maintains direct attention on the baby's face
- B** When a baby notices a screen image, the image automatically changes.
- C** A baby smiles when he sees a new colour.
- D** A cartoon clip is used to show different bright colours
- E** A bright colour appears on a screen to signal the end of an experiment

Answer: (7) (8)

Questions 9 and 10

Choose **TWO** letters, **A–E**.

Which **TWO** of the following statements are made about a baby's ability to see?

OPTIONS

- A** Their clarity of vision will improve considerably in their first few months of life
- B** They can see black and white objects more clearly than coloured ones
- C** They are able to perceive a limited number of colours from birth
- D** They rapidly develop a preference for brightly coloured toys
- E** Their eyes contain fewer cones than those of an adult

Answer: (9) (10)

Questions 11–16

Complete the summary below.

Choose **ONE WORD ONLY** from the passage for each answer.

Cones are (11) _____ in the eye which serve a particular function in the recognition of colours. They all respond to light but vary in the different (12) _____ they respond to. Although babies have three types of cones from birth, their brains initially have difficulty understanding the (13) _____ coming from them. As these cones mature, the baby's power of (14) _____ develops between colours such as red and green. However, an important factor in the baby's ability to tell certain colours apart is the (15) _____, as weak colours are difficult for the baby to recognise. Over time, there is a slow (16) _____ in this.

Passage 15

How Plants Fight Back

New research reveals that plants sense danger and mount active chemical defences of their own

A Recent research has shown that plants are more aware of their environment and more active in their responses than was ever previously imagined. Simon Gilroy, a professor of botany at the University of Wisconsin-Madison in the USA, has spent much of his career trying to understand how plants work. Now, Gilroy and one of his post-doctoral researchers, Masatsugu Toyota, have produced a series of videos that shows how plants responded when they subjected them to wounds, including scissor cuts and caterpillar bites.

B Gilroy and Toyota discovered that when one part of a plant is attacked or damaged, a wave of calcium spreads throughout the rest of the plant. The calcium alerts the plant to danger and the need to deploy defence tactics. The team were able to see this by utilising a naturally occurring fluorescent-green protein which binds to the calcium, making its path visible. While scientists already knew that plants reacted to danger via an electrical charge that moves across the plant, they didn't know exactly how it happened. Gilroy and Toyota suspected it had something to do with calcium and were able to confirm their suspicions. They were also able to reveal how glutamate - an abundant neurotransmitter in animals - triggers this wave of calcium.

C The find was fortuitous given that Gilroy hadn't intended to study wounding at all. His real passion is understanding how plants sense gravity and seem to know which way is up - something that's proving extremely hard to work out. It was during the early stages of an experiment into gravity that Toyota came across the wounding response. 'We work very intensely on the calcium signal, because it's a ubiquitous signal. Biology uses it absolutely everywhere,' explains Gilroy. 'It makes your heart beat, it makes your muscles contract. Plants use it for a lot of their signalling machinery. We had some hints that the gravity-sensing system is based around the calcium signal, and so we were developing the technology to image calcium cells in real-time.' It was during this process that Gilroy and Toyota realised they'd captured something never usually visible to humans.

D The team found that the calcium travels at one millimetre per second, fast enough to spread to other leaves in just a couple of minutes. From the data collected up to now, it appears that how far the calcium travels depends on the extent of the wound, or, as Gilroy puts it, 'The more you hurt it, the louder it screams.' That 'scream' can result in a range of responses. 'Plants are masters of chemistry,' says Gilroy. 'We deal with the world by running away from it, plants deal with the world by growing in response to it, or by making a tonne of stuff.' That 'tonne of stuff' could be chemicals that poison a hungry insect, or

that make the plant unattractive, tough or unpalatable. Some plants even make proteins that block the ability of a caterpillar's gut to digest the plant material. This is dinner that fights back.

E The next step for Gilroy and his team is to delve deeper into the signalling response on a cellular level, dissecting the genes and proteins responsible. In contrast to our understanding of human nerve cells, Gilroy admits that the equivalent responses in plants are still barely understood. He is also going to widen the scope of the study and look at other signals that plants send out - signals regarding temperature and changes in light and touch. Gilroy explains that there may be a wider use for the research, albeit a long way in the future. Once scientists have managed to identify the specific genes that make the signalling process work and can understand what happens when you switch those genes on and off, it's not hard to think about the potential. 'You can imagine that we should be able to take a crop plant and switch on its defences on-call,' says Gilroy. 'We're nowhere near that point yet, but once we get there - say you're in a field and you predict there's going to be an outbreak of some pest, you could go in and pre-defend all of the plants in the field, but you do it on-call so the plants aren't wasting their resources defending themselves the whole time.'

F For now, though, Gilroy is happy to simply increase understanding of plants. He is energetic in his insistence that they are not the inactive and unreceptive organisms that people generally believe them to be. For that reason, he's as enthusiastic about the way the videos bring the response process to life as he is about the future potential of the research. 'When you look at a plant, just because it doesn't do what we do, and it doesn't move, that doesn't mean it isn't doing anything. They're hugely dynamic organisms,' he says.

Questions 14–18

Reading Passage 2 has six sections, A–F.

Which section contains the following information? Write the correct letter, A–F. NB You may use any letter more than once.

Q	Statement	
14	an explanation of how scientists were able to observe plants sending warning signals	
15	examples of the means used by scientists to provoke plants' signalling response	
16	reference to the idea that we could one day manipulate plants' signalling abilities	
17	the purpose of the research which led to Gilroy and Toyota's discovery	

18	a comparison regarding levels of scientific knowledge about responses in plants and humans	
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Questions 19–23

Complete the summary below.

Choose **ONE WORD ONLY** from the passage for each answer.

Gilroy and Toyota's research

- As part of their research at the University of Wisconsin-Madison, Simon Gilroy and Masatsugu Toyota made several (19) _____ revealing plants' reactions to cuts, bites and other (20) _____.
- These indicated how the spread of calcium warns plants about the presence of (21) _____. The use of a brightly coloured (22) _____ enabled the researchers to see the movement of calcium through the plant. The research also showed how (23) _____ activates the release of calcium in the plant.

Questions 24 and 25

Choose **TWO** letters, **A–E**. Which **TWO** aspects of the calcium signal in plants does the writer mention?

- (24)
- A the volume of calcium released following an attack
 - B the rate at which calcium flows through plants
 - C the different parts of the plant where calcium is produced
 - D the species of plants which produce calcium as a warning signal
 - E the link between the severity of an attack and the distance calcium moves
- (25)
- A the volume of calcium released following an attack
 - B the rate at which calcium flows through plants
 - C the different parts of the plant where calcium is produced
 - D the species of plants which produce calcium as a warning signal
 - E the link between the severity of an attack and the distance calcium moves

Questions 26 and 27

Choose **TWO** letters, **A–E**. Gilroy's team have made certain plans for the future. Which **TWO** plans are mentioned by the writer?

26

- A research plants' reactions to heat and cold
- B use their findings to help grow a greater variety of crops
- C investigate warning signals in organisms other than plants
- D analyse further the chemical processes involved in signalling responses
- E conduct studies to compare human cell and plant cell behavior

27

- A research plants' reactions to heat and cold
- B use their findings to help grow a greater variety of crops
- C investigate warning signals in organisms other than plants
- D analyse further the chemical processes involved in signalling responses
- E conduct studies to compare human cell and plant cell behavior

Passage 16

How do Plants Talk to Each Other?

Research shows that plants communicate and interact with each other in surprisingly subtle and sophisticated ways

14 A In 1983, plant scientists Jack Schultz and Ian Baldwin reported that the leaves of young maple trees increased their defense systems when exposed to maples that had been damaged by plant-eating (herbivorous) insects. The injured trees, they suggested, were alerting neighbors to the presence of a predator by releasing chemical signals into the air. But the plant research community did not accept this. The results were difficult to replicate, critics pointed out. Many also questioned how it could be evolutionarily stable if it benefited neighboring plants but not the plant releasing the signal. By the late 1980s, most ecologists felt that Schultz and Baldwin's idea had been discredited.

15 B A decade later, however, a number of more carefully designed experiments began to yield convincing indications to the contrary. In 2000, evolutionary ecologist Richard Karban showed that wild tobacco plants became resistant to herbivores when grown in close proximity to sagebrush plants whose leaves had been damaged by cutting. This change appeared to be in response to chemicals – known as volatile organic compounds or VOCs – released by the sagebrush plants. Other researchers soon reported similar VOC-induced defense responses in several other plants, including lima bean, broad bean, barley, and corn. And in 2006, Karban showed that VOCs released by damaged sagebrush induce herbivore resistance in plants growing at distances of up to 60 cm, well within the range of sagebrush neighbors in nature.

16 C But the question still remains: 'Why should a plant waste valuable resources on a function which has no obvious advantage for it?' One hypothesis is that external communication channels are merely an extension of within-plant signaling. In sagebrush, lima bean, and poplar, VOCs released from damaged parts of a plant induce resistance in intact sections of the same plant, suggesting that each individual plant uses the signals to coordinate its own physiological responses to protect itself. Karban agrees, saying: 'The interplant signaling we see may be a result of plants co-opting that process.' Alternatively, VOC-based signaling between plants may have been favored because it enhances the 'extended fitness' of the sender by aiding related plants of the same species: a strategy known as kin selection.

17 D Over the past few years, a team led by Ariel Novoplansky of Ben-Gurion University of the Negev in Israel seems to have found proof that distress signals can be passed through the plants' roots. They planted garden pea plants in rows and subjected the first in each row to conditions similar to those experienced in a drought. They then evaluated the response by measuring the microscopic holes on leaves, known as pores, which react when

there is a shortage of water. After fifteen minutes, the stressed plant was seen to be closing its pores, followed by all of its neighbors, one by one. Importantly, in a control setup where root contact between neighboring plants was blocked, pores stayed open. Meanwhile, David Johnson's team at the University of Aberdeen in Scotland have been studying the labyrinths of hair-like fungi that occur around the roots of most plants. These fungi are involved in an important two-way relationship: in exchange for sugars, they provide plants with much-needed phosphorus and nitrogen. Research in which broad bean plants were infested with aphids – small herbivorous insects – revealed that these networks also served as a channel for warning neighboring plants of the infestation.

18 E Monica Gagliano of the University of Western Australia believes that plants may even use sounds to alert their neighbors. In one study, she demonstrated that chili plants growing next to fennel plants developed more quickly than seedlings grown with other chili plants. Gagliano and her colleagues suspect the chili plants were compensating for the presence of the fennel, which is known to release chemicals that inhibit the growth of other plants. Remarkably, however, when plant communication pathways – airborne volatiles, root contact, and common fungal networks – were blocked, the results begged for an alternative explanation. 'We think this other channel of communication might be acoustic,' says Gagliano. But behavioral ecologist Carel ten Cate of the University of Leiden in the Netherlands points out that taking advantage of such benefits would require sensory mechanisms yet to be described in plants. Despite widespread skepticism, Gagliano has received some encouragement. Richard Karban, for example, is cautiously enthusiastic. 'Whether or not the explanation she favors is the right one,' he says, 'I think that she's gotten results that, as a field, we need to come to grips with.'

19 F Researchers believe that knowledge of this phenomenon could eventually be applied to agriculture, and lead to the cultivation of hardier crops. But all agree that there is still much work to be done: applying our limited knowledge of plant-communication mechanisms to agriculture is a big jump, says Johnson, 'but it is definitely on the horizon.' Nevertheless, one message has emerged loud and clear from those studying this new realm of botanical interaction: despite not possessing eyes, ears, or a nervous system, plants are anything but uncommunicative. 'When I started my PhD (in the late 1980s), all this stuff was considered very weird,' recalls Ariel Novoplansky. 'Today, there is no doubt. We now recognise that plants are capable of some very sophisticated exchanges of information with other plants. This idea is not strange anymore.'

Questions 14–19

Reading Passage 2 has six paragraphs, A–F.

Choose the correct heading for each paragraph from the list of headings below. Write the correct number, i–ix, in the circle before each paragraph.

- i Sending messages underground
- ii Potential advantages for farmers
- iii The widespread rejection of an idea
- iv One species accelerating growth in another
- v Some obstacles to plant communication
- vi Evidence leading to renewed belief in a theory
- vii New technologies which helped researchers
- viii A suggested benefit to the sender of plant communication

Questions 20–23

Complete the summary below. Choose **ONE WORD ONLY** from the passage for each answer.

PLANT COMMUNICATION RESEARCH IN ISRAEL AND SCOTLAND

Plants may send messages via their (20) ————— according to Ariel Novoplansky of Ben-Gurion University in Israel. In one study, Novoplansky's team planted rows of garden peas, and created conditions like a (21) ————— around the first plant in each row. The plants responded by closing the tiny holes on their leaf surfaces, known as pores. They then appeared to signal to other plants nearby, causing them to respond in a similar way.

Research at the University of Aberdeen in Scotland, meanwhile, has focused on the networks of (22) ————— which connect the plants beneath the soil. These structures, which extract sugars from the plants and supply them with phosphorus and nitrogen, also seem to play a key role in plant communication. Experiments involving broad bean plants and plant-eating insects called (23) ————— have also produced some interesting results.

Questions 24–26

Look at the following statements and the list of researchers below. Match each statement with the correct researcher, A–E.

LIST OF RESEARCHERS

- A** Richard Karban
- B** Ariel Novoplansky
- C** David Johnson
- D** Monica Gagliano
- E** Carel ten Cate

Q	Statement	
24	It is likely that research into plant communication will one day help to improve our food supply.	
25	Plant communication could occur as a consequence of internal mechanisms intended to help a single plant.	
26	Attitudes towards plant communication have changed greatly in recent decades.	

Passage 17

How the Petri Dish Supports Scientific Advances

A simple piece of scientific equipment is helping research in three institutions in Cambridge, UK

A Petri dishes, invented by German microbiologist Julius Richard Petri in 1887, rarely receive the appreciation or attention that their more complex lab companions like the microscope enjoy. They are simply, utilitarian little things and it's understandable that some people see them as just shallow dishes with lids. But Petri dishes deserve celebrating; they are still at the forefront of scientific discovery.

B The invention of the Petri dish, and the advances it has helped to create, are part of a bigger whole, of course – the development of glass scientific instruments, from microscope lenses to laboratory beakers. In *The Glass Bathyscaphe: How Glass Changed the World*, Alan Macfarlane argues that without glass, the Renaissance and the scientific revolution would never have happened. Around 70 per cent of what we know about the world comes in through our eyes, Macfarlane points out, and glass instruments enabled us to see better. Until about 1400, knowledge was based on what people had been told in the past. 'Glass allowed the growth of the experimental method. Don't trust what you are being told: see it for yourself. It was transformational,' he says.

C At the Wellcome-MRC Cambridge Stem Cell Institute, Professor Ludovic Vallier says that his first encounter with a Petri dish was a classic example of understanding the world in this way: students used the dishes to see which bacteria could grow in the presence of antibiotics. 'It's good to see things grow,' he says. 'It is a fascinating experience. Now, we grow cells in the Petri dish, and we don't use glass any more, but plastic.' Today, his team focuses on stem cells, which have the capacity to become any cell type in the human body: neurons, skin cells, liver cells, and so on. Vallier and his colleagues study them in order to understand how they do this, and how they can produce more cells. And to study them, they need to grow them. 'We put the stem cells on the dish and then we feed them and they grow,' he says. 'And then... we divide them and distribute them in new Petri dishes, and we grow them again. We feed them on a liquid medium that is basically food for cells: it tells them to grow and also what to do, as we want to produce new cells. So by feeding them this medium we can allow the cells to become neurons, cardiac cells, liver cells, and so on. We can then model disease in a dish, or produce cells for regenerative medicine applications.' This means that the Petri dish becomes a place where Vallier and his team can study what happens to those cells when disease strikes. 'We work a lot on fatty liver diseases, and in this case the liver cell in the dish becomes full of fat, which we can see ... We can't look inside a patient's liver to see what's happening. So we are reproducing those diseases in our dishes.'

D ‘Disease in a dish’ is also the focus of Dr Meritxell Huch’s team at the Gurdon Institute. They use between 50 and 150 Petri dishes every day to grow mouse liver and human liver cells, in order to study how the liver can regenerate itself. Huch’s team is examining the molecular mechanisms by which these cells decide to multiply. She says: ‘You can divide regeneration into different phases. The cells first have to realize that there is damage and activate the response. Once they activate the response, the cells will proliferate to compensate for the loss of cells owing to the damage. And once they have proliferated, they then have to become functional cells.’

E In the MRC Laboratory of Molecular Biology, Dr Madeline Lancaster and her team grow ‘mini-brains’ in hundreds of Petri dishes. Here, the dish has been specially treated to stop cells sticking to it and to encourage them to float freely. Dr Lancaster explains that they want the cells to develop in three, rather than two, dimensions as that’s the way our brains are. ‘If you can grow neurons on a dish in two dimensions, you can see individual neurons and see what they do, but you won’t be able to understand the architecture of those cells – their positioning relative to one another.’ She says that this new method gives you a structure that looks a lot more like that of an actual developing brain.

F The aim of this research is to look at exactly how neurons are made and how that differs in humans compared with other species. One day, says Lancaster, this work could translate into understanding far more about Alzheimer’s disease, Parkinson’s and schizophrenia. So in a world of cutting-edge and highly complex technology, Petri dishes, in their relative simplicity, remain a vital weapon in the fight against the world’s most serious diseases. And they also enable a hands-on approach that she finds satisfying. ‘It’s a bit like gardening,’ she says. ‘You’re taking care of this thing. You keep an eye on it and you check it every day. You change the media this day or that day to help it grow better. It’s rewarding to see something grow before your eyes. There’s something about the interplay between new, next-generation and classic technologies. They give you capabilities that were not possible before.’

Questions 17–22

Reading Passage 2 has six sections, A–F.

Which section contains the following information? Write the correct letter, A–F.

NB *You may use any letter more than once.*

Q	Statement	
17	a description of an experiment involving both human and non-human cells of a specific type	

18	possibilities for improved research into various medical conditions	
19	contrasting views of the importance of the Petri dish	
20	a change remarked on by one scientist in the material used for the Petri dish	
21	a claim that the Petri dish enables a scientist to monitor the progress of an experiment on a regular basis	
22	a reference to the importance of a material for different types of laboratory equipment	

Questions 23–28

Look at the following statements and the list of people below.

Match each statement with the correct person, **A, B, C or D**.

NB You may use any letter more than once.

LIST OF PEOPLE

- A** Alan Macfarlane
- B** Professor Ludovic Vallier
- C** Dr Meritxell Huch
- D** Dr Madeline Lancaster

Q	Statement	
23	To deal with the injury, cells must go through a series of activities in a particular order.	
24	One technological development formed the basis of all modern scientific research.	
25	A modification to the Petri dish allows experiments to provide more accurate information.	
26	Petri dishes allow observation of medical conditions that are normally impossible to observe.	
27	Visual evidence is a very important requirement for the provision of reliable information.	
28	Petri dishes can be used to help produce a range of new cells of many different kinds.	

Questions 29–32

Complete the summary below. Choose **ONE WORD ONLY** from the passage for each answer.

RESEARCH IN THE MRC LABORATORY OF MOLECULAR BIOLOGY

A team led by Dr Madeline Lancaster is using special Petri dishes which prevent brain cells from

29 _____ to them. The aim is to allow the neurons to grow in three

30 _____. This results in a 31 _____ that resembles a developing brain.

The technology could help scientists study how neuron production varies in different

32 _____, leading to possibilities for increased medical knowledge.

Passage 18

IQ Scores

How they are changing, and what the changes might reveal about human intelligence

A Studies from numerous countries suggest that IQ scores have been rising fast since at least the 1950s - enough to mean that someone with an IQ classed as average at that time could be labelled as having low intelligence today. However, psychologists disagree about what the upward trend in scores really means. Some researchers argue that people are not getting more intelligent overall, but just getting better at a particular sort of problem solving. Others say they have found echoes of rising IQ scores in 'real world' measures of intelligence. To confuse things still more, there are hints that the trend may be about to grind to a halt.

B James Flynn, a professor of political science, first showed that IQ scores were rising in the mid-1980s, and became famous after others dubbed the pattern he had observed the 'Flynn effect'. In a landmark paper he made detailed comparisons across fourteen countries of trends in the results of tests that measure reasoning power. Overall, these showed a real growth in scores of up to 25 points in a generation. To find out more, Flynn went on to look at scores for each type of reasoning that IQ tests set out to measure, such as verbal, numerical and visuo-spatial. This last test category usually includes 'Raven's matrices' - sequential patterns with one piece missing. The candidate has to choose, from a set of alternatives, which shape to put in the empty space. When Flynn took a closer look at the scores, the pattern was clear. The strongest gains in almost every case were on Raven's matrices and similar tests, while verbal and arithmetical scores showed more modest rises. Since presenting his original paper, Flynn has looked at data from further countries and found the same upward trend, with the same skills responsible for most of the increase.

C Flynn feels that his eponymous effect is meaningful, but does not interpret it as evidence that people are becoming cleverer. Our grandparents were not retarded and we are not geniuses, he says. More likely we have simply developed the skills and habits of mind that make us better at solving abstract problems, and - just as importantly - we take such problems seriously. Flynn believes that demands of visuo-spatial abilities have grown because of TV, computers and cars, and also that, in societies where intelligence has become something people focus on excessively, abstract problem-solving skills develop faster than other skills.

D Others have different views about what might account for the increase in IQ scores. John Rust of the University of London suggests that it is not just an increase in visuo-spatial activities that is pushing up IQ scores. He believes that there are several factors, including the fact that, in more complex societies, people have to think harder to find solutions to problems they face. He also points out that if just one mind grasps a new

concept, such as Einstein and the theory of relativity, it may then be easier for society in general to think in a more complex way. Others, such as experimental psychologist Robert Howard, argue that overall intelligence really has risen, helped by better nutrition, smaller families, wider access to education and other environmental changes.

E Howard's latest research has sent him on a different line of enquiry. Controversially, he now believes that the rise in IQ scores might have peaked in industrialised countries. To support this view, he cites data from Denmark. The Danish military still summons virtually all young men before a draft board to identify who is suitable to do military service. Since 1957, all 17-year-olds there have taken exactly the same qualifying IQ tests - including ones similar to Raven's matrices. Candidates must also be able to answer verbal questions such as 'Sun is to day as Moon is to ?', and number-series questions such as '2, 3, 5, 8, ?'. Results in these tests have been carefully analysed by Thomas Teasdale, a neuropsychologist. He found that during the 1960s and '70s, scores showed gains on a scale similar to those of other countries, but by the 1990s the rate of improvement had slowed right down to about one IQ point over the decade. Furthermore, the scores were improving only on the visuo-spatial questions; verbal and numerical performance was flattening out. Most intriguingly, however, scores from 1999 onwards show an overall downturn. One explanation might be that there is some absolute ceiling to people's abilities to perform these tasks, no matter how much the prevailing culture encourages them. However, as Teasdale points out, if such a ceiling had been reached, the test results should show scores pushing against it rather than actually falling.

F Flynn is keeping a close eye on the findings, but is not jumping to conclusions until data from other countries is available. However, he does wonder whether the conditions leading to dramatic improvements in mental ability disappear as societies become more prosperous. He points out that in human history, prosperity often led to decadence, and argues that the case of ancient Rome can be viewed as an illustration of this.

Questions 14–18

Look at the following statements (Questions 14–18) and the list of people below.

Match each statement with the correct person, **A**, **B**, **C** or **D**. You may use any letter more than once.

List of People

- A** James Flynn
- B** John Rust
- C** Robert Howard
- D** Thomas Teasdale

Q	Statement	
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14	Modern technology has influenced some aspects of human intelligence.	
15	IQ scores may not have attained their natural limit.	
16	Improved diet has a positive effect on mental powers.	
17	Exceptional individuals can have an impact on collective intelligence.	
18	As societies become more sophisticated the need for problem-solving skills increases.	

Questions 19–22

Reading Passage 2 has six paragraphs, A–F.

Which paragraph contains the following information? Write the correct letter, A–F.

Q	Statement	
19	an account of a selection procedure involving a particular group of people	
20	a description of one component of a visuospatial test	
21	an overview of the different interpretations of IQ score trends	
22	an example of the negative effect affluence may have on IQ trends	

Questions 23–26

Complete the summary below.

Write **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer.

James Flynn

During the (23) ———, James Flynn conducted research showing that IQ scores were increasing. Flynn then analysed specific kinds of reasoning ability separately. He found that tests called (24) ———, which measured visuo-spatial skills, accounted for the biggest score increases, whereas scores in numerical and (25) ——— components rose less steeply. Results from research in other (26) ———, which Flynn has examined more recently, are consistent with his earlier findings.

Passage 19

Is Graffiti Art or Crime?

Weighing the historical value of graffiti against the practical challenges of preventing and removing it

A The term graffiti derives from the Italian graffito meaning 'scratching' and can be defined as uninvited marking or writing scratched or applied to objects, built structures and natural features. It is not a new phenomenon: examples can be found on ancient structures around the world, in some cases predating the Greeks and Romans. In such circumstances it has acquired invaluable historical and archaeological significance, providing a social history of life and events at that time. Graffiti is now a problem that has become pervasive, as a result of the availability of cheap and quick means of mark-making.

B It is usually considered a priority to remove graffiti as quickly as possible after it appears. This is for several reasons. The first is to prevent 'copy-cat' emulation which can occur rapidly once a clean surface is defaced. It may also be of a racist or otherwise offensive nature and many companies and councils have a policy of removing this type of graffiti within an hour or two of it being reported. Also, as paints, glues and inks dry out over time they can become increasingly difficult to remove and are usually best dealt with as soon as possible after the incident. Graffiti can also lead to more serious forms of vandalism and, ultimately, the deterioration of an area, contributing to social decline.

C Although graffiti may be regarded as an eyesore, any proposal to remove it from sensitive historic surfaces should be carefully considered: techniques designed for more robust or utilitarian surfaces may result in considerable damage. In the event of graffiti incidents, it is important that the owners of buildings or other structures and their consultants are aware of the approach they should take in dealing with the problem. The police should be informed as there may be other related attacks occurring locally. An incidence pattern can identify possible culprits, as can stylised signatures or nicknames, known as 'tags', which may already be familiar to local police. Photographs are useful to record graffiti incident and may assist the police in bringing a prosecution. Such images are also required for insurance claims, and can be helpful to cleaning operatives, allowing them to see the problem area before arriving on site.

D There are a variety of methods that are used to remove graffiti. Broadly these divide between chemical and mechanical systems. Chemical preparations are based on dissolving the media; these solvents can range from water to potentially hazardous chemical 'cocktails'. Mechanical systems such as wire-brushing and grit-blasting attempt to abrade or chip the media from the surface. Care should be taken to comply with health and safety legislation with regard to the protection of both passers-by and any person carrying out the cleaning. Operatives should follow product guidelines in terms of application and removal,

and wear the appropriate protective equipment. Measures must be taken to ensure that run-off, aerial mists, drips and splashes do not threaten unprotected members of the public. When examining a graffiti incident it is important to assess the ability of the substrate to withstand the prescribed treatment. If there is any doubt regarding this, then small trial areas should be undertaken to assess the impact of more extensive treatment.

E A variety of preventive strategies can be adopted to combat a recurring problem of graffiti at a given site. As no two sites are the same, no one set of protection measures will be suitable for all situations. Each site must be looked at individually. Surveillance systems such as closed circuit television may also help. In cities and towns around the country, prominently placed cameras have been shown to reduce anti-social behavior of all types including graffiti. Security patrols will also act as a deterrent to prevent recurring attacks. However, the cost of this may be too high for most situations. Physical barrier such as a wall, railings, doors or gates can be introduced to discourage unauthorized access to a vulnerable site. However, consideration has to be given to the impact measures have on the structure being protected. In the worst cases, they can be almost as damaging to the quality of the environment as the graffiti they prevent. In others, they might simply provide a new surface for graffiti.

F One of the most significant problems associated with graffiti removal is the need to remove it from surfaces that are repeatedly attacked. Under these circumstances the repeated removal of graffiti using even the most gentle methods will ultimately cause damage to the surface material. There may be situations where the preventive strategies mentioned above do not work or are not a viable proposition at a given site. Anti-graffiti coatings are usually applied by brush or spray leaving a thin veneer that essentially serves to isolate the graffiti from the surface.

G Removal of graffiti from a surface that has been treated in this way is much easier, usually using low-pressure water which reduces the possibility of damage. Depending on the type of barrier selected it may be necessary to reapply the coating after each graffiti removal exercise.

Questions 14–19

Reading Passage 2 has seven paragraphs, A–G.

Which paragraph contains the following information? Write the correct letter, A–G. You may use any letter more than once.

Q	Statement	
14	why chemically cleaning graffiti may cause damage	

15	the benefit of a precautionary strategy on the gentle removal	
16	the damaging and accumulative impact of graffiti to the community	
17	the need for different preventive measures being taken to cope with graffiti	
18	a legal proposal made to the owner of building against graffiti	
19	the reasons of removing graffiti as soon as possible	

Questions 20–21

Choose **TWO** letters, **A–E**.

Which two statements are true concerning the removal of graffiti?

20

- A** Cocktail removal can be safer than water treatment.
- B** A small patch trial should be done before applying a large scale removal.
- C** Chemical treatments are the most expensive way of removing graffiti.
- D** There are risks for both chemical and mechanical methods.
- E** Mechanical removals are much more applicable than chemical treatments.

21

- A** Cocktail removal can be safer than water treatment.
- B** A small patch trial should be done before applying a large scale removal.
- C** Chemical treatments are the most expensive way of removing graffiti.
- D** There are risks for both chemical and mechanical methods.
- E** Mechanical removals are much more applicable than chemical treatments.

Questions 22–23

Choose **TWO** letters, **A–E**.

Which **TWO** of the following preventive measures against graffiti are mentioned as effective in the passage?

22

- A** Organise more anti-graffiti movements in the city communities.
- B** Increase the police patrols on the street.
- C** Build a new building with material repelling to water.
- D** Install more visible security cameras.
- E** Provide a whole new surface with a chemical coat.

23

- A Organise more anti-graffiti movements in the city communities.
- B Increase the police patrols on the street.
- C Build a new building with material repelling to water.
- D Install more visible security cameras.
- E Provide a whole new surface with a chemical coat.

Questions 24–27

Complete the summary of the paragraphs of Reading Passage 2 below.

*Use **NO MORE THAN TWO WORDS** from the passage for each answer.*

- Ancient graffiti is of significance and records the (24) _____ of details of life for that period.
- The police can recognise newly committed incidents of graffiti by the signature, which is called a (25) _____, that they are familiar with.
- Operatives ought to comply with relevant rules during the operation, and put on the suitable (26) _____.
- Removal of graffiti from a new type of coating surface can be made much more convenient by using (27) _____.

Passage 20

Keeping an Eye on Shoppers

How retailers use technology to track and analyse customer behaviour

A Several large retail chains utilise various technologies to monitor customer habits, increase profit margins, and enhance product delivery. Retailers, it seems, are buying the idea that by watching customers while they shop – what might be called retail surveillance – and redesigning their shops accordingly, they can get people to spend more.

B One system, called Smartlane, has been successfully trialled by several major retailers. It has two elements: sensors by the doors count the number of people entering and leaving the shop, and sensors by the tills work out how fast the queues are moving and how many ‘shopping units’ – groups of people, such as families or couples, that make one transaction between them – are standing in each queue. All this information can then be used to predict how many tills will be needed up to an hour in advance and monitor average waiting times and queue lengths. In some cases, a queue longer than one shopping unit triggers the opening of another till. One British retailer said last year that the monitoring system had reduced waiting time for customers and helped to boost the firm's revenue.

C This is not the only technology keeping tabs on shoppers. Another supplier of customer-monitoring systems uses dual-lens cameras, which are more accurate than single-lens cameras, in tracking and counting. Rather than monitoring queue lengths, however, the system, called BehaviorIQ, is used by retailers to gather data on where their customers go, how long they slow down to look at different products, and how they react to them. This information can then be compared with the store's transaction log to determine the effectiveness of store layout, product fixtures and other variables. In-store designs and marketing campaigns that work can be identified and worked on, and those that do not work can be replaced with something more profitable.

D One of the largest customer-monitoring projects undertaken so far is PRISM (or ‘Pioneering Research for an In-Store Metric’). The project involved the use of sensors at the entrances and exits and in some of the aisles of 160 stores across the US. These sensors recorded data on customer-traffic patterns, to which was added further information recorded by human observers. By comparing the resulting data with sales information, it was then possible to gain insight into shoppers' behaviour. For example, when observational data on the number of children present during a shopping trip was taken into account, the results were even more detailed: the PRISM results showed that although children took part in only 13 per cent of food-shopping trips, shoppers bought more when accompanied by them. Purchase of seasonal items was two and a half times more likely with children in tow but surprisingly there was no correlation found between the presence of children on a shopping trip and higher spending on sweets.

E A further adaptation of this technology is VideoMining, which uses a combination of video cameras and software to capture and analyse thousands of hours of footage of customer behaviour. As well as providing information about customer behaviour, the firm's software can also determine each shopper's sex, age range and ethnic group with an accuracy of around 80 per cent, says Jeff Hershey of VideoMining. The system can even be programmed to detect the characteristic behaviour of shoplifters as they steal goods from places in the store. Inevitably, all this raises doubts about privacy. Counting customers with motion sensors as they enter a supermarket is one thing, but sorting them into social or racial groups using video analysis is another. Retailers respond that none of the footage or information about individual shoppers is stored and that customers are generally willing to give up some personal information if they feel they are getting some benefit in return.

F Shoppers are not just being monitored by cameras and heat sensors, however; sometimes their speech is being recorded, too. Recordant, an in-store monitoring system operated by a company based in Atlanta, provides shop assistants with digital-audio recorders, to be hung around their necks, which record all their conversations with shoppers. At the end of each day the recording devices are plugged into special docking stations and the recordings are uploaded to Recordant's headquarters for analysis. Its software scans the recorded dialogue for particular words or phrases without the need for any human input, in order to provide accurate information on employee performance and customer behaviour. By providing precise measurements, the technology allows retailers to refine their employee training programmes. They can even programme the system to search for any mention of their competitors. Unlike systems based on video cameras or sensors, users of Recordant's technology are required to notify customers explicitly that their conversations are being recorded.

G As useful as customer-tracking systems may be for protecting harassed staff or minimising shoplifting, however, their primary objectives are clear. In an age when traditional 'bricks and mortar' retailers face stiff competition from online merchants, any means they can use to get customers into their shops and buying as much as possible is seen as a worthwhile investment.

Questions 14–17

Reading Passage 2 has seven paragraphs, A–G.

Which paragraph contains the following information?

Q	Statement	
14	a mention of the influence certain technology has on professional development of staff	

15	how technology can identify the profile of individual customers	
16	a mention of a significant current challenge to conventional shops	
17	how retailers use the data to evaluate the way their products are presented	

Questions 18–23

Match each statement with the correct technology, A–E.

NB You may use any letter more than once.

LIST OF TECHNOLOGIES

- A** Smartlane
- B** BehaviorIQ
- C** PRISM
- D** VideoMining
- E** Recordant

Q	Statement	
18	Computers search for specific language used in the store.	
19	Feedback is collected from people who watch customer activity.	
20	Staff wear the monitoring technology.	
21	It is possible to identify criminal behaviour in a store.	
22	Only customer arrivals, departures and lines are monitored.	
23	Cameras focus on the way customers move around the store and the places they stop.	

Questions 24–26

Complete the sentences below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

- In the Smartlane system, people who shop together are referred to as **24** _____.
- The PRISM project identified increased spending when shoppers had **25** _____ with them.

- The Recordant technology looks at customer behaviour and employee problems, and can even check whether shoppers talk about the shop's **26** _____.

@bekhruzposts

Passage 21

Learning from the Romans

How an ancient building material is influencing modern construction

A In a quest to make concrete more durable and sustainable, an international team of geologists and engineers has found inspiration in the concrete used by the ancient Romans. The chemical secrets of Roman concrete have been uncovered in samples taken from a concrete Roman breakwater. A breakwater is a barrier that is built out into the sea to protect coasts and harbors from the force of waves. This particular breakwater has spent the last 2,000 years submerged in seawater. The international team of researchers that collected the samples was led by Paulo Monteiro, a professor of civil and environmental engineering at the University of California, Berkeley. Analysis of samples from the ancient underwater site in Pozzuoli Bay near Naples in Italy pinpointed why the best Roman concrete was superior to most modern concrete.

B Concrete was the Roman Empire's construction material of choice. It was used in land monuments such as the Pantheon in Rome, as well as in underwater and partially underwater coastal and harbor structures. Monteiro and his team were particularly interested in how the coastal and harbor structures endured the unforgiving saltwater environment. Chemical analysis of Roman concrete showed that it differs from the modern kind in several ways. One is the content of the cement that binds the material in the concrete. The most commonly used cement in recent decades has been Portland cement. Portland cement is a compound of calcium, silicates, and hydrates (C-S-H). However, analysis of Roman concrete shows that it contains a significantly different cement. Roman cement contains aluminum, which is not found in Portland cement, and less silicon than is found in Portland cement. The resulting calcium-aluminum-silicate-hydrate (C-A-S-H) is an exceptionally stable cement.

C The recipe for the contents of Roman concrete was first described around 30 BC by Vitruvius, an engineer for the Roman Emperor Augustus. Volcanic ash was one of the ingredients, and it is now understood that this is crucial, as it is volcanic ash that contains the aluminum that ultimately gives Roman concrete its great durability. The Romans devised an efficient method of making concrete for coastal structures. They combined volcanic ash with lime, which added the calcium to the mix. This was then packed, together with stones and chunks of rock, into wooden molds, which were then immersed in seawater. The seawater instantly triggered a hot chemical reaction. The lime was hydrated by the seawater, which means that it incorporated water molecules into its structure, and reacted with the volcanic ash to cement the whole mixture together. This reaction formed the concrete that was used to build some of the most enduring structures in Western civilization.

D According to Marie Jackson, a member of the research team, Roman concrete is one of the most durable construction materials on the planet, and that was no accident. Shipping was the lifeline of political, economic, and military stability for the Roman Empire, so constructing harbors that would last was critical. ‘Over time, the Romans used less and less concrete. As the Roman Empire declined, and shipping declined, the need for the seawater concrete declined,’ said Jackson. ‘You could also argue that the original structures were built so well that, once they were in place, they didn’t need to be replaced,’ she added.

E Producing Roman concrete also left a smaller carbon footprint than its modern counterpart. ‘It’s not that modern concrete isn’t so good — it’s so good we use 19 billion tons of it a year,’ says Monteiro. The problem is that manufacturing Portland cement accounts for seven percent of the carbon dioxide that industry puts into the air. Portland cement holds most modern concrete together. However, making Portland cement releases carbon dioxide into the atmosphere from burning fuel in order to heat a mix of limestone and clays to 1,450 degrees Celsius. The production of Roman concrete, however, was much cleaner, as less heat was needed. A temperature that was a third lower than that required for making Portland cement was sufficient for making Roman concrete.

F Some modern concrete no longer contains Portland cement; instead, fly ash is used, which causes fewer greenhouse gas emissions. Fly ash is a waste product from burning coal, and the researchers are investigating whether volcanic ash would be a good, large-volume replacement in countries that cannot access fly ash easily. ‘There is not enough fly ash in this world to replace half of the Portland cement being used,’ said Monteiro. ‘Many countries don’t have fly ash, so the idea is to find alternative local materials that will work, including the kind of volcanic ash that Romans used. Using these alternatives could replace 40 percent of the world’s demand for Portland cement.’

G Cutting greenhouse gas emissions is one powerful incentive for finding a better way to provide the concrete the world needs; another is the need for stronger, longer-lasting buildings, bridges, and other structures. ‘In the middle 20th century, concrete structures were designed to last 50 years, and a lot of them are on borrowed time,’ Monteiro says. ‘Now we design buildings to last 100 to 120 years.’ Yet Roman harbor installations have survived 2,000 years of chemical attack from seawater and wave action. Stronger, longer-lasting modern concrete may be the legacy of a deeper understanding of how the Romans made their incomparable concrete.

Questions 1–7

Reading Passage 2 has seven paragraphs, A–G.

Which paragraph contains the following information? Write the correct letter, A–G.

Q	Statement	
1	an opinion on why the Romans reduced the amount of concrete they made for use in seawater	
2	a list of the contents of both Portland and Roman cement	
3	an argument for finding substitutes for a limited resource that is used in making cement	
4	information about the structure from which the scientists took their Roman concrete samples	
5	a comparison of the environmental impacts of making modern and Roman concrete	
6	a comparison of the durability of Roman concrete with concrete produced today	
7	details of how the Romans used seawater to make concrete	

Questions 8 and 9

Choose **TWO** letters, **A–E**.

According to the passage, which **TWO** of the following statements about fly ash are true?

OPTIONS

- A** Fly ash results in less damage to the environment than Portland cement.
- B** Fly ash was used by the Romans as an alternative to volcanic ash.
- C** Fly ash is already used in the production of some concrete.
- D** All countries have access to fly ash resources.
- E** Fly ash will soon replace volcanic ash.

Answer: (8) (9)

Questions 10–13

Complete the summary below.

Choose **ONE WORD AND/OR A NUMBER** from the passage for each answer.

THE ENVIRONMENTAL EFFECTS OF CONCRETE PRODUCTION

Roman concrete is better for the environment than modern concrete, which is made by using Portland cement. This is because a temperature of 1,450 degrees Celsius is needed to combine clays and ⑩ _____ to make Portland cement. This level of heat was reduced by a ⑪ _____ when making Roman concrete. Needing less heat meant less ⑫ _____ had to be burned and less carbon dioxide was produced. According to Monteiro, when considering the amount of carbon dioxide emitted by ⑬ _____, seven percent of it comes from the manufacture of Portland cement.

Passage 22

Lever Brothers' Sunlight Soap

A revolution in hygiene and industry

A Sunlight Soap, produced by the Lever Brothers in the late 19th century, was a groundbreaking innovation in the world of consumer goods. At a time when hygiene and cleanliness were not universally prioritised, Sunlight Soap not only introduced a revolutionary product but also played a crucial role in transforming industrial production, marketing strategies and public health. The success of Sunlight Soap laid the foundation for what would later become Unilever, one of the world's largest multinational corporations.

B The story of Sunlight Soap begins in 1884, when William Hesketh Lever and his brother James Darcy Lever established the Lever Brothers company in England. They sought to create a high-quality soap that was both affordable and effective in promoting hygiene. Traditional soaps at the time were often harsh on the skin and required extensive preparation. The Lever Brothers aimed to change this by producing a soap that was easy to use and gentle while maintaining strong cleansing properties.

The innovation behind Sunlight Soap lay in its production process. Unlike the traditional method of soap-making, which relied on animal fats and wood ash, Sunlight Soap was manufactured using a blend of vegetable oils and alkali, resulting in a more consistent and superior-quality product. This not only improved the soap's effectiveness but also made it a more ethical and sustainable option compared with its contemporaries.

C Beyond the product itself, the Lever Brothers were pioneers in marketing and branding. They understood the power of advertising and created memorable, persuasive campaigns for Sunlight Soap. These advertisements emphasised the soap's purity, its ability to fight germs, and its role in maintaining a happy and healthy home. The company used a variety of media, from newspapers and magazines to colourful posters and collectable cards, to reach a wide audience. This aggressive and innovative marketing strategy helped create a strong brand identity and fostered customer loyalty.

D The company also revolutionised industrial relations and worker welfare. In 1888, Lever Brothers built a purpose-built factory and village called Port Sunlight on the Wirral Peninsula. This model village was designed to provide high-quality housing, amenities, and social benefits for its employees. It featured gardens, schools, a hospital, and recreational facilities, which were considered revolutionary for the time. This approach was part of William Lever's philosophy of "prosperity sharing," where he believed that a

healthy and content workforce would lead to greater productivity and company success. Port Sunlight became a landmark in industrial planning and set a new standard for employee welfare.

E The impact of Sunlight Soap on public health was significant. By promoting the importance of cleanliness with an effective and accessible product, it contributed to improved hygiene practices among the general public. This was particularly important in an era before the widespread understanding of germ theory. The widespread use of soap like Sunlight played a role in reducing the incidence of infectious diseases, thereby improving overall public health standards in Britain and beyond.

F The phenomenal success of Sunlight Soap provided the capital and brand recognition for the Lever Brothers to expand their operations globally. They began acquiring other soap and food companies, steadily growing their portfolio. This expansion culminated in 1930 with a merger with the Dutch margarine company, Margarine Unie. This merger formed Unilever, a dual-headed company structure that remains unique to this day. The creation of Unilever marked the beginning of a new era, transforming a single soap product into a global consumer goods empire.

Questions 14–19

Reading Passage 2 has six paragraphs, A–F.

Which paragraph contains the following information?

Q	Statement	
14	a description of the advertising methods used to promote the soap	
15	the reason why a particular location was constructed for workers	
16	the way the success of one product led to the creation of a major corporation	
17	a comparison between the new product and the types of soap that existed previously	
18	the various health benefits for society resulting from the use of the soap	
19	the original business goal of the Lever brothers	

Questions 20–26

Complete the notes below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

SUNLIGHT SOAP: KEY FACTS

- **Founders:** William Hesketh Lever and (20) _____. Year company established: (21) _____.
- **Key Product Innovation:** Used (22) _____ and alkali instead of animal fats and wood ash. Resulted in a product that was more consistent, superior, and (23) _____.
- **Marketing & Branding:** Advertisements focused on purity and fighting (24) _____.
- **Worker Welfare:** Built a model village called (25) _____, based on the philosophy of “(26) _____”.

Passage 23

Mind Music

Scientists investigate "earworms", the music we can't get out of our head

A Ever had a song stuck in your head, playing on an endless loop? Scientists call them 'involuntary musical images', or 'earworms', and a wave of new research is shining light on why they occur and what can be learned from them. Some neuroscientists and cognitive psychologists are studying 'earworms' to explore the mysteries of memory and the part of the brain that is beyond our conscious control. "The idea that we have full control over our thought processes is an illusion," says psychologist Lauren Stewart, who founded the master's program in music, mind and brain at Goldsmiths, University of London, UK, where recent research has taken place. Researchers haven't been able to watch what happens in the brain when earworms occur, because they happen unpredictably. Much of what is known about them comes from surveys, questionnaires, diaries and lab experiments.

B A Goldsmiths study published in the journal *Memory and Cognition* this year showed that the singing we hear in our heads tends to be true to actual recordings. Researchers had 17 volunteers tap to the beat of any earworm they heard during a four-day period while a device attached to their wrist recorded their movements. The tapping tempos were within 10% of the tempos of the original recordings. Another Goldsmiths study, published this year in *Consciousness and Cognition*, found that people who report hearing earworms often, and find them most intrusive, have slightly different brain structures, with more gray matter in areas associated with processing emotions.

C Studies also show that the music in our heads often starts playing during times of 'low cognitive load', such as while showering, getting dressed, walking, or doing chores. Dr Stewart likens 'earworms' to 'sonic screen savers' that keep the mind entertained while it is otherwise unoccupied. She and her colleagues tested that theory by having volunteers listen to songs and giving them various tasks afterwards. The volunteers who sat idly for the next five minutes were the most likely to report hearing the music in their heads. Dr Stewart observed that the more challenging the activity, the less likely the volunteers were to hear the music. Diary studies also show songs tend to match people's moods and therefore they are not random. If you are energized and upbeat, an 'earworm' that occurs is likely to be uptempo too.

D Songs the brain fixates on are usually those it has been exposed to recently, surveys show, which is why tunes getting heavy radio play frequently top the earworm charts. Even tunes you may have heard but didn't pay attention to can worm their way into your subconscious, says Ira Hyman, a psychologist at Western Washington University in

Bellingham, USA. In an unpublished study there, participants who listened to music while doing other tasks were more likely to report that the songs returned as earworms later on, compared with participants who simply listened.

E Some earworms are just fragments of a song that repeat like a broken record. So, when the mind hits a part of a song it can't remember, it loops back rather than moving on. That could make an earworm even more entrenched, Dr Hyman says. According to a theory known as the Zeigarnik effect, named for a Soviet psychologist, Bluma Zeigarnik, unfinished thoughts and activities weigh on the mind more heavily than those that are completed, although experiments exposing students to interrupted songs have yielded mixed results.

F Researchers say they can't pinpoint a spot in the brain where earworms live. Imaging studies by Andrea Halpern at Bucknell University, in Lewisburg, USA, have shown that deliberately imagining music and actually listening to music activate many of the same neurological networks. Dr Halpern's earlier studies showed that when subjects listened to the first few notes of familiar music, areas in the right frontal and superior temporal portions of the brain became activated, along with the supplementary motor area at the top, which is typically involved in remembering sequences. When the same subjects were asked to voluntarily imagine music and were asked to recall it, there was activity in the left frontal portions of the brain instead.

G One factor that makes some songs stick might be repetition. 'Repetition leads to familiarity which leads to anticipation, which is satisfied by hearing the song,' says John Seabrook, author of *The Song Machine: Inside the Hit Factory*, about how producers pump pop songs full of 'hooks', the punchy melodic phrases designed to target the brain and leave it wanting more. The researchers are comparing the melodic structure of 100 often-mentioned songs with 100 similarly popular songs that weren't cited as earworms, to assess the difference. Songs with earworm potential appear to share certain features: a repeating pattern of ups and downs in pitch, and an irregular musical interval.

H The researchers plan next to test their results in reverse, and play ringtones from songs of both the earworm and non-earworm variety for volunteers several times a day to see which ones get stuck. Drs Stewart and Halpern are now working together to recruit survey participants for a study looking at whether people at different stages of life experience earworms differently. 'You can argue that older people might get them more often because they know more songs,' Dr Halpern says. 'But the few responses we have so far indicate that they have earworms less often. It could be that they don't play music as often as younger people do.'

Questions 14–17

The Reading Passage has eight paragraphs, **A–H**. Which paragraph contains the following information?

Q	Statement	
14	a description of the characteristics common to songs with earworms	
15	a justification for research into earworms	
16	a description of the brain's reaction to known and unknown songs	
17	details of proposed research into the frequency with which earworms occur in different age groups	

Questions 18–21

Complete the summary below. Choose **ONE WORD ONLY** from the passage for each answer.

GOLDSMITHS STUDY

Researchers from Goldsmiths concluded that the music we imagine in our minds is quite similar to recordings. They proved this by asking volunteers to record the rhythm of music using a monitor on their (18) _____. Further research has demonstrated that those who hear earworms more frequently have brains that may deal with (19) _____ differently from other people. Dr Stewart also believes that the brain is (20) _____ by earworms when it is not focused on a task. In fact, a reduction in the occurrence of earworms was found to be directly related to how (21) _____ the task was. Interestingly, volunteers' diaries revealed that the songs they heard inside their head reflected their moods, so the choice of music is not accidental.

Questions 22–26

Look at the following statements and the list of researchers below. Match each statement with the correct person, **A, B, C** or **D**.

NB You may use any letter more than once.

LIST OF RESEARCHERS

- A** Lauren Stewart
- B** Ira Hyman

- C Andrea Halpern
- D John Seabrook

Q	Statement	
22	Some musicians create music that is intentionally memorable.	
23	People are unable to completely regulate how they think.	
24	We can remember songs without knowing that we have heard them.	
25	Thinking about music has a similar effect on the brain to hearing music.	
26	Earworms are more persistent when only a short section of the song is constantly replayed.	

Passage 24

Nanotechnology: the science of the very small

How the manufacture of structures at the atomic scale is changing energy, water, food and healthcare

A Nanoscience and nanotechnology have become very fashionable in recent times. Their prominence has inspired science fiction and raised a lot of hope and hype about the possibilities. But what exactly is nanotechnology? Most definitions agree that it is the manufacture of structures that measure up to 100 nanometres (nm). The diameter of an atom is typically one or three tenths of a nanometre while that of the human hair is around 1,000,000 nm. Essentially, we are now able to build new structures from atoms and molecules as if we were using the bricks of a children's construction toy.

B Most of the electronic consumer goods that we use today contain components that measure less than 100 nm. The lasers in our CD and DVD players make use of the properties of semiconductor layers that are just a few atoms in thickness, so we clearly are in the nanotechnology age. There are many other everyday items that we take for granted that have nanoparticles in them. One of these is sunscreen. In the 1990s, there were safety issues with many of the sunscreen products available, so scientists at Oxford University set about finding a solution. The product they developed uses nanoparticles of titanium dioxide that are doped with the element manganese. This reduces the hazard of potentially skin-damaging agents known as free radicals. The product is almost transparent to the eye when spread on the skin, and it is effective for much longer than many of the traditional preparations. In this case, we see an example of nanotechnology providing a solution to an existing problem rather than opening up new product areas.

C The technique of using nanotechnology to adapt existing products can also be applied to garments, upholstery, paint and plastics, to improve their longevity. Bearing this in mind, it is interesting to ask whether nanotechnology can help with some of the world's major current issues. The recent growth in climate change awareness calls a range of things into question. Can we use energy more effectively? Can we reduce the carbon dioxide in the atmosphere? Can we use and purify water more effectively? Can healthcare be improved? Nanotechnology may very well provide the solutions. There are few times in history when the application of science to improving the human condition has been so apparent.

D We are already making new types of solar cells and batteries by using our new-found ability to manipulate matter at the atomic and molecular scale. If we can capture some of the huge quantities of energy that arrive at the Earth's surface from the Sun, and store it effectively, we can reduce the need for fossil fuel-based power stations. We are already using nanoparticles and nanoporous materials to capture harmful greenhouse gases, and

we may possibly convert some of them back into fuel. There are, however, many technical problems involved, and it will be another decade or so before some of these ideas are deployed.

E Water purification is increasingly challenging and many believe that water, which we take for granted, is going to become more valuable than oil. Currently we do not use water efficiently, and many industrial and food processes leave contamination in water that is potentially toxic and difficult to remove. Nanofiltration may provide the solution to these problems, and may also offer new routes to the desalination of brackish water and groundwater in some parts of the world.

F Food production and processing are becoming more important as the world's population increases. There are concerns that potentially harmful nano-products might enter the body via the gut. Such concerns are being addressed, but it is even more important to ask questions about how nanotechnology might increase the nutritious yield of food and be adapted to help preserve and package food. There is already evidence to show that it might also be helpful in the primary production of raw food materials, by way of better delivery of pesticides in the field and better extraction of protein and carbohydrates.

G Healthcare is an area where the application of nanotechnology is already exceeding the expectations of ten years ago. It is happening across several fronts: better appreciation of what the 'natural' nanoparticles circulating in body fluid do; better methods of delivering drugs to the body via nanoparticles that target the diseased body part; improvements to the diagnostic imaging techniques of ultrasound, magnetic resonance imaging and X-ray computerised tomography. We are now at the threshold of a new subject, 'theranostics', whereby nanoparticles will first deliver the diagnostic information to the clinician and then, using the same particle, deliver a form of therapy.

H Nanotechnology is a life-changing technology, making the quality of life better, enabling us to make use of our resources more effectively, thereby protecting our environment, and creating new business opportunities. There are many other examples in the world of 'new materials' that also make use of nanotechnology. The possibilities may well be limitless.

Questions 14–16

Complete the summary below.

*Write **ONE WORD ONLY** from the text for each answer.*

Nanotechnology

In the same way that a child may use (14) _____, scientists are now able to create new structures by combining the smallest particles known to man. We can fully appreciate the microscopic scale of this work by comparing the exact (15) _____ of one atom with that of a single human hair.

A lot of today's electronic consumer goods, such as lasers, have components that are microscopic in size. They are not the only products relying on nanotechnology. For example, a long-lasting (16) _____ widely available today, relies on minute nanoparticles for its effectiveness. This was developed in the 1990s as a result of concerns about the safety of existing products.

Questions 17–22

Reading Passage 2 has eight sections, A–H.

Which paragraph contains the following information? Write the correct letter, A–H. You may use any letter more than once.

Q	Statement	
17	a comparison between two of the world's vital natural resources	
18	a mention of how nanotechnology could be dangerous for humans	
19	a claim that nanotechnology has influenced modern authors	
20	a reference to the fact that nanotechnology has achieved more than was anticipated a decade ago	
21	a suggestion that nanotechnology could provide financial as well as ecological benefits	
22	an acknowledgement that more time is required to realise some of nanotechnology's potential	

Questions 23–24

Choose **TWO** letters, A–E.

Which **TWO** of the following statements does the writer make about nanotechnology and energy resources?

23

- A Nanotechnology means that solar energy will replace fossil fuel energy.
- B We have made solar batteries for the first time by applying nanotechnology.
- C Further research is needed to turn nanotechnology into practical energy solutions.
- D Nanotechnology is being used to remove dangerous gases from the atmosphere.
- E Nanotechnology has helped create cheaper sources of energy.

24

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Questions 25–26

Choose **TWO** letters, **A–E**.

In which **TWO** ways does the writer think nanotechnology could improve food production?

25

- A by enabling crops to grow faster
- B by keeping food fresh for longer
- C by making food tastier
- D by protecting food from insects better
- E by increasing the variety of food

26

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Passage 25

Next to Nature, But What Is Art?

Julian Barnes explores the questions posed by Life-Casts, an exhibition of plaster moulds of living people and objects originally used for scientific purposes

A Art changes over time and our idea of what art is changes too. For example, objects originally intended for devotional, ritualistic or recreational purposes may be recategorised as art by members of other later civilisations, such as our own, which no longer respond to these purposes.

B What also happens is that techniques and crafts which would have been judged inartistic at the time they were used are reassessed. Life-casting is an interesting example of this. It involved making a plaster mould of a living person or thing. This was complex, technical work, as Benjamin Robert Haydon discovered when he poured 250 litres of plaster over his human model and nearly killed him. At the time, the casts were used for medical research and, consequently, in the nineteenth century life-casting was considered inferior to sculpture in the same way that, more recently, photography was thought to be a lesser art than painting. Both were viewed as unacceptable shortcuts by the 'senior' arts. Their virtues of speed and unwavering realism also implied their limitations, they left little or no room for the imagination.

C For many, life-casting was an insult to the sculptor's creative genius. In an infamous lawsuit of 1834, a moulder whose mask of the dying French emperor Napoleon had been reproduced and sold without his permission was judged to have no rights to the image. In other words, he was specifically held not to be an artist. This judgement reflected the view of established members of the nineteenth-century art world such as Rodin, who commented that life-casting 'happens fast but it doesn't make Art'. Some even feared that 'if too much nature was allowed in, it would lead Art away from its proper course of the Ideal'.

D The painter Gauguin, at the end of the nineteenth century, worried about future developments in photography. If ever the process went into colour, what painter would labour away at a likeness with a brush made from squirrel-tail? But painting has proved robust. Photography has changed it, of course, just as the novel had to reassess narrative after the arrival of the cinema. But the gap between the senior and junior arts was always narrower than the traditionalists implied. Painters have always used technical back-up such as studio assistants to do the boring bits, while apparently lesser crafts involve great skill, thought, preparation and, depending on how we define it, imagination.

E Time changes our view in another way, too. Each new art movement implies a reassessment of what has gone before. What is done now alters what was done before. In some cases, this is merely self-serving, with the new art using the old to justify itself. It seems to be saying, ‘Look at how all of that points to this! Aren’t we clever to be the culmination of all that has gone before?’ But usually it is a matter of re-alerting the sensibility, reminding us not to take things for granted. Take, for example, the cast of the hand of a giant from a circus, made by an anonymous artist around 1889, an item that would now sit happily in any commercial or public gallery. The most significant impact of this piece is on the eye, in the contradiction between unexpected size and verisimilitude. Next, the human element kicks in. You note that the nails are dirt-encrusted, unless this is the caster’s decorative addition, and the fingertips extend far beyond them. Then you take in the element of choice, arrangement, art if you like, in the neat, pleated, buttoned sleeve-end that gives the item balance and variation of texture. This is just a moulded hand, yet the part stands utterly for the whole. It reminds us slyly, poignantly, of the full-size original.

F But is it art? And, if so, why? These are old, tediously repeated questions to which artists have often responded, ‘It is art because I am an artist and therefore what I do is art’. However, what doesn’t work for literature works much better for art – works of art do float free of their creators’ intentions. Over time the ‘reader’ does become more powerful. Few of us can look at a medieval altarpiece as its painter intended. We believe too little and aesthetically know too much, so we recreate and find new fields of pleasure in the work. Equally, the lack of artistic intention of Paul Richer and other forgotten craftsmen who brushed oil onto flesh, who moulded, cast and decorated in the nineteenth century is now irrelevant. What counts is the surviving object and our response to it. The tests are simple: does it interest the eye, excite the brain, move the mind to reflection and involve the heart? Further, is an apparent level of skill involved? Much currently fashionable art bothers only the eye and briefly the brain but it fails to engage the mind or the heart. It may, to use the old dichotomy, be beautiful but it is rarely true to any significant depth. One of the constant pleasures of art is its ability to come at us from an unexpected angle and stop us short in wonder.

Questions 1–5

Reading Passage 2 has six paragraphs, A–F.

Which paragraph contains the following information? Write the correct letter, A–F.

Q	Statement	
1	an example of a craftsman’s unsuccessful claim to ownership of his work	

2	an example of how trends in art can change attitudes to an earlier work	
3	the original function of a particular type of art	
4	ways of assessing whether or not an object is art	
5	how artists deal with the less interesting aspects of their work	

Questions 6–11

Do the following statements agree with the claims of the writer in Reading Passage 2?

*Write **YES** if the statement agrees with the claims of the writer, **NO** if it contradicts them, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.*

- 6 Nineteenth century sculptors admired the speed and realism of life-casting.
- 7 Rodin believed the quality of life-casting would improve if a shower process were used.
- 8 The importance of painting has decreased with the development of colour photography.
- 9 Life-casting requires more skills than sculpture does.
- 10 New art encourages us to look at earlier work in a fresh way.
- 11 The intended meaning of a work of art can get lost over time.

Questions 12–13

*Choose the correct letter, **A**, **B**, **C** or **D**.*

- 12 The most noticeable contrast in the cast of the giant's hand is between the
 - A dirt and decoration.
 - B size and realism.
 - C choice and arrangement.
 - D balance and texture.
- 13 According to the writer, the importance of any artistic object lies in
 - A the artist's intentions.
 - B the artist's beliefs.
 - C the relevance it has to modern life.
 - D the way we respond to it.

Passage 26

Orientation of Birds

How migrating birds find their way home across vast, often unfamiliar, distances

A For many of us, the way birds are able to orientate is both astounding and difficult to appreciate fully. For instance, the annual migration of the golden plover of the Pacific takes it from Alaska to Hawaii on a flight of well over 3000 kilometres, and if it were to deviate by only one degree, it would miss the island on which it nests.

B The first systematic studies on orientation in birds were made possible by the 'homing instinct' exhibited by so many species. Birds are caught at a time when they show an attachment to their territory, especially during the nesting season. They are taken to some spot, released, and the percentage of returns is recorded. The distance can be varied, and the direction, as well as the method of transporting them, and then the influence of climatic and other factors on their ability to find their way home can be studied. These experiments have shown a wide variation in ability to home, and three types of homing behaviour have been identified.

C In the first type, birds methodically explore the area in which they are released until they pick up some familiar feature, and then they quickly find their way back to the nest. Such birds possess a highly developed visual memory, as experiments with pigeons have shown. Domestic pigeons have been trained to peck at a certain point on an aerial photograph, with a system of rewards, and four years later the birds were still able to respond to this training when placed on the aerial photograph. Birds' eyes have a power of resolution two to three times greater than ours, enabling them to pick up very fine details. If a bird uses only this type of homing behaviour, however, it can only succeed if the point of release is not too far away. If the birds are transported 800 kilometres from their nest, it is only by good fortune that they find their way back as a result of long exploratory flights. Usually, the area known to a bird is its feeding territory. Released within this area, the birds soon make their return; release them outside it and far fewer return. However, if a bird is released for a second time in the same place, its visual memory comes into play, and the bird, no longer requiring tedious exploratory flights, will return much more quickly.

D The second type of homing behaviour is shown by birds that are capable of choosing their flight direction and holding to it for the rest of their journey. How do they decide what direction to take? They appear to choose their normal migration direction even if they are released in a different place from their usual starting point. If, for example, birds which normally fly to the north-east to reach latitude 45 degrees north are released at that latitude, they will immediately start flying north-east anyway. So if they're released further to the west, they maintain the correct direction, but fly west of their destination, and so fail to arrive.

E The third type of homing behaviour shows the highest degree of orientation. Released at one point, the birds immediately take stock of it, compare its position with that of the nest, decide on the direction and fly off. This happens even if the birds are in a country right off their migration routes, where they have never been before. In one example, a Laysan albatross returned to its nesting area on Midway Island in the middle of the Pacific, having flown over 5000 kilometres from the west coast of the USA in just over ten days. This is a perfect example of the third type of homing, for the albatross clearly couldn't rely on any landmarks over the vast expanse of the Pacific Ocean.

F The percentage of successful birds varies greatly, being highest in those species with a strong migratory behaviour. Thus the lesser black-backed gull is more migratory than the herring gull and more often reaches 'home'. Great migrants such as the swift have the highest percentage of returns. In one case, seven out of nine alpine swifts were recaptured at their nests after being displaced some 1400 kilometres; one made the journey in three days.

G What part does heredity play in all this? Two research studies suggest that instinctive, i.e. genetically inherited, behaviour patterns play a part in navigation. The first was carried out by Ernst Schüz and it is highly significant. Schüz caught first-year European storks and released them later, after the departure of the adult storks at a time when they normally make their south-west autumn migration to Africa. The recaptures showed that, in spite of the fact that there were no adults to guide them, the birds unanimously headed southwest. This was a striking finding, for it showed that the birds had an innate and unlearned attraction for the African wintering area that they have occupied for thousands of years.

H The case of starlings is a little different. These birds have a great aptitude for homing, but this behaviour differs in the different age groups. Birds that were shifted to the south-east of their normal migration route split into two lots. The adults, in full possession of their gift for orientation, found their wintering area by modifying their direction by 90 degrees, whereas the juveniles sought their winter quarters to the southeast of their real position.

Questions 14–18

Complete the summary below.

*Choose **NO MORE THAN TWO WORDS** from the passage for each answer.*

Types of homing behaviour

First type

- Birds rely on their sophisticated (14) _____. However, they are generally most successful if they are released within their feeding territory.

Second type

- Birds select their accustomed (15) _____, no matter where they are released. As a result, they may miss their (16) _____.

Third type

- Birds orientate correctly, even when they are released in an unfamiliar place and have no (17) _____ to make use of. One bird with this type of skill is the (18) _____.

Questions 19–22

Reading Passage 2 has eight paragraphs, A–H.

Which paragraph contains the following information? Write the correct letter, A–H. You may use any letter more than once.

Q	Statement	
19	the effects of distance on some birds' ability to find their nests	
20	a methodology for testing the general ability of birds to find their nests	
21	one aspect of physical ability in humans and birds	
22	how some birds' migration was delayed for experimental purposes	

Questions 23–26

Look at the following types of birds (Questions 23–26) and the list of points which the author wishes to illustrate below.

Match each bird with the point which it illustrates, A–G.

LIST OF POINTS

- A an ability to orientate without previous training
- B the speed at which birds fly
- C the ability to remember things seen previously
- D the effect of age on homing ability

- E** the strength required to fly a great distance
- F** a high success rate in finding nests
- G** the importance of seasonal cues for migrating birds

Q	Statement	
23	domestic pigeon	
24	alpine swift	
25	European stork	
26	starling	

Passage 27

Robotic Approach to Crop Breeding

Jennifer Manyweathers takes a look at a robot that is being used to identify drought-tolerant crop varieties

A The Australian sunflower industry is the major source of polyunsaturated fatty acids found in margarines and spreads. Recognised as the type of fatty acid most able to protect against heart disease, it is in everybody's best interest that Australia has a competitive and healthy sunflower industry, but in Australia, there is a constant struggle with the harsh climate. However, thanks to one special robot, farmers may be able to win the battle against drought.

B Dr Chris Lambrides, a research fellow at the University of Queensland, is nearing the end of a project that aims to develop more drought tolerant sunflowers by selecting flowers that use water more efficiently. He's done this with the help of a robot developed by the Australian National University's Research School of Biological Sciences.

C Plants undergo photosynthesis to produce energy in the form of sugar. This involves allowing carbon dioxide to enter the leaves through pores called stomata. Transpiration is the mechanism by which plants lose water through their leaves. This system is thought to facilitate the passage of minerals through the plant and is vital for healthy plants.

D However, in conditions of drought, the plants that can use the available water efficiently and lose less to the environment will be more likely to thrive and, in a commercial sense become more profitable. These plants are classified as having a high transpiration efficiency. When plants transpire, the leaves become cooler due to evaporation. Therefore, by measuring the temperature of the leaves, scientists can determine how much water is being lost through transpiration.

E When the project first began, the researchers used hand-held infrared thermometers to measure the temperature difference between leaves of different varieties of sunflowers in an experimental plot. Wind can affect leaf temperature, and the research team discovered that its initial approach did not cater for changes in wind speed, which could not be controlled as an experimental variable. The team, therefore, needed a technique to measure temperature continuously that would allow it to examine the effects of other variables such as humidity. They needed a robot.

F They designed a robot with two infrared thermometers set at 180° to each other. The robot runs on an oblong track around the experimental plot and the thermometers operate on each side of the track. In order to minimize any variables from the two thermometers, they are rotated 180° at the beginning of each run and the results are averaged. The infrared thermometers can be rotated on an angle to examine different parts of the foliage.

G The robot is also able to detect light intensity. It has a garage on the track, where it waits until the light intensity is high enough to give useful results. If the skies darken due to rain, heavy cloud cover or sunset, the robot makes its way back to the garage to wait.

H The main difficulty faced by the research group was to find an agronomist who could grow the perfect crop of sunflowers. The sunflower canopy had to be complete, with no visible soil, so that the thermometers would only measure the temperature of the plants and not the surrounding environment. Eight varieties of sunflowers were examined. The data collected by the robot has been used by the research team to determine which variety has the highest transpiration efficiency.

I This is not the first time such methods have been used to determine drought-resistance in plants. The team and their robot have already made a major breakthrough in the Australian wheat industry with Drysdale Wheat, which signalled the arrival of a new technique for selecting drought-resistant species.

Questions 1–4

Complete the sentences below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

- In terms of our health, sunflowers are important in defending humans against
① _____.
- The research team wanted to find a sunflower that could cope well in ② _____ conditions.
- The name of the process which is believed to help keep plants in good condition is
③ _____.
- The research team had to rethink their initial approach when they realized they needed to measure the impact of external conditions such as ④ _____.

Questions 5–12

The reading passage has nine paragraphs labelled **A–I**.

Which paragraph contains the following information? Write the correct letter, **A–I**. **NB** You may use any letter more than once.

Q	Statement	
---	-----------	--

5	the precise growing conditions are required to allow the experiment to work	
6	a description of the how the robot operates	
7	an explanation of two important processes used by plants	
8	a reference to a previous study using a different crop	
9	details of what the robot does when conditions are poor	
10	the name of the group responsible for making the robot	
11	the number of different types of sunflowers tested	
12	the purpose of taking the temperature of the plants	

Passage 28

Roller Coaster: The Great Fairground Attraction

How gravity, chains, and 400 years of tinkering built the modern thrill ride

A Like a passenger train, a roller coaster consists of a series of connected cars that move on a track. But unlike a passenger train, it has no engine or power source of its own. For most of the ride, it is moved only by the forces of inertia and gravity. The only exertion of energy occurs at the very beginning of the ride when the coaster train is pulled up the lift hill.

B The traditional lifting mechanism is a long length of chain running up the hill under the track. The chain is fastened in a loop, which is wound around a gear at the top of the hill and another one at the bottom of the hill. The gear at the bottom of the hill is turned by a motor. This turns the chain so that it continually moves up the hill like a long conveyor belt. The coaster cars grip onto the chain, which simply pulls them to the top of the hill. At the summit, the train is released and starts its descent.

C The purpose of this initial ascent is to build up a sort of reservoir of potential energy, which simply means that as the coaster gets higher in the air, there is a greater distance gravity can pull it down.

D As the train starts coasting down the hill, this potential energy is converted into kinetic energy (energy of motion), and the train speeds up. At the bottom of the hill, this has reached its maximum, and this propels the train up the second hill, again building up the potential energy level.

E In this way, the course of the track is constantly converting energy from kinetic to potential and back again. This fluctuation in acceleration is what makes roller coasters so much fun. At its most basic level, this is all a roller coaster is – a machine that uses gravity and inertia to send a train along a winding track.

F Roller coasters have a long, fascinating history. Their direct ancestors were ice slides, popular in Russia in the 16th and 17th centuries. They consisted of a long, steep, wooden slide covered in ice. Riders walked up a ladder or set of stairs to the top of the slide, as high as 21 metres up. At the top, they climbed into sleds made out of wood or blocks of ice and shot down the slope. At the base of the slide, the sleds would crash-land in a sand pile.

G It seems that the idea was then imported into France. For most of the year, the warmer climate would melt the ice, so the French started building waxed slides instead. To help the sleds move down these slides, they added wheels, and in 1817, for the first time, a train was attached to the track. The French continued to expand on this idea, coming up with more complex track layouts, with multiple cars and all sorts of twists and turns.

H The first American roller coaster was built in the mountains of Pennsylvania in the mid-1800s, originally to provide an easy way to send coal to the railway 29km down to mountain. When the track was first built, a crew at the bottom of the mountain would attach the cart to a team of mules after emptying the load, and the mules would drag it back up to the top. They were eventually replaced with steam engines, to make the system more efficient.

I Soon after these improvements were made, the railway company built a new tunnel that brought the freight trains much closer to the coal mine. Now no longer required for its original purpose, the roller coaster was configured as a 'scenic tour'. For one dollar, tourists got a leisurely ride up to the top of the mountain, followed by a wild, bumpy ride straight down. This was soon a resounding success, attracting thousands of tourists every year.

J Scenic rides like this continued to thrive and were joined by wooden roller coasters similar to the ones we know today. These coasters were the main attraction at popular amusement parks throughout the United States, such as the many parks of Coney Island in New York. By the 1920s, roller coasters were in full swing, with some 2,000 rides in operation around the country.

K Following the Great Depression, a decline in roller coaster production began in the early 1930s, but a second roller-coaster boom in the 1970s and the beginning of the 1980s revitalised the amusement park industry, and introduced a slew of innovative tubular steel coasters.

L This was followed by a decline in interest for the rest of the decade, but since the early 1990s, the amusement-park industry has experienced another coaster boom of sorts. New launching techniques and other technological developments have opened up a world of options for designers, so in some rides you feel as if you are flying. In the next few years we can expect to see many faster, taller and more twisted rides popping up in amusement parks around the world.

Questions 14–16 — Label the diagram

Choose **ONE WORD ONLY** from the passage for each answer.

Lifting mechanism of a roller coaster

- Chain running up the hill (loop around gears) → 14 _____
- Toothed wheel at bottom of hill → 15 _____
- Machine that turns the gear → 16 _____

Questions 17–21 — Summary Completion

Complete the summary below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

History of roller coasters

- Modern roller coasters are descended from 16th-century Russian slides with a surface of (17) _____
- People climbed to the top, and travelled down in sleds. In France, because of the higher temperatures, the wooden surface on the slides was (18) _____
- and (19) _____ were attached to the cars to ease the descent.
- The first US roller coaster was used for transporting (20) _____ down a mountainside in carts.
- Initially, these were pulled by mules but in time power was produced by (21) _____.

Questions 22–26

Do the following statements agree with the information given in the passage?

Write **TRUE** if the statement agrees with the information, **FALSE** if the statement contradicts the information, or **NOT GIVEN** if there is no information on this.

- (22) The earliest modifications to the basic slide were made in France.
- (23) Roller coasters continued to increase in popularity throughout the 1920s and 1930s.
- (24) New roller coaster technology was introduced in the 1970s in response to public demand.
- (25) Roller coasters were less popular for most of the 1980s than in the 1970s and 1990s.
- (26) The design of roller-coaster rides became more varied in the 1990s.

Passage 29

Saturn Spectacular

Solving the puzzles of the ringed planet

① **A** In 1610 the Italian astronomer Galileo pointed his crude telescope at the planet Saturn and was dumbfounded by what he saw: ‘The planet Saturn is not alone, but is composed of three, which almost touch each other, and never move or change with respect to one another.’ Even more surprising, the two bulging planets on either side of the main planet had disappeared when he looked again a few months later. Eventually, the frustrated Galileo decided never to look at Saturn again. Now, of course, we have much better telescopes, and we know that Galileo was looking at the planet’s unique set of wide, thin rings. Seen broadside, they resembled companion planets through Galileo’s crude telescope. Even with much better telescopes today, Saturn still teases astronomers. Saturn’s magnificent rings, for example, consist of trillions of pieces of ice, some no bigger than a speck of dust. The ring particles are so small that they could be expected to scatter and fall into the planet, yet they are still there. Scientists are unsure about the origin of the rings, their exact chemical composition, and why they behave in the way that they do.

② **B** However, as spectacular as Saturn’s rings are, the fuzzy orange ball of Saturn’s giant moon Titan is even more interesting. Although it is larger than the planet Mercury, Titan probably resembles other large moons in the Solar System. It is a rock covered with ice and dirt. However, it has something no other moon has: a substantial atmosphere whose pressure is 50% greater than Earth’s. Even more intriguing, Titan’s atmosphere consists mainly of nitrogen (the primary component of our own air), mixed with carbon-based compounds. Cloaked in this atmosphere, Titan’s surface may resemble the surface chemistry of the early Earth, but astronomers cannot see through the moon’s maddeningly opaque orange fog.

③ **C** All sorts of Earth-like chemical interactions could happen on Titan. Scientists hesitate to speak of the possibility of life when they speak of Titan, but it is a distant possibility. Living on Titan would be no walk in the park. From the vantage point of the Saturn system, the sun is a rather dim bulb. Titan is therefore a very cold place. Surface temperatures average about –300 degrees Fahrenheit. At those temperatures, water is nowhere to get a drink. In addition, there is no oxygen in Titan’s atmosphere – it is all locked up in water ice. The only hope for life as we know it, and it is an exceedingly slim one, is that water mixed with ammonia may get warm enough deep below the surface to liquefy. If so, life could possibly eke out an underground living much like the hardy microbes that surround Earth’s hydrothermal vents (deep cracks on the seabed through which boiling water and hot gases escape).

④ **D** Titan could contain information on the prebiotic chemistry that led to life on Earth and perhaps elsewhere in the Universe. The bitterly cold temperatures that make Titan so forbidding for life in some ways make it more intriguing. Titan's chilly climate keeps things in a state of preservation, so that it can hang on to most of the substances that it has acquired during its 4.5 billion years in the Solar System. The organic reactions that may have established the starting conditions for life on the Earth are long gone, erased by our planet's high-speed chemical and geologic evolution. On Titan, similar reactions may still be sitting in deep storage.

⑤ **E** Measured against its showy rings and haze-obscured moon, Saturn itself seems downright ordinary. Yet the 75,000-mile-wide planet – the second largest in the Solar System – holds some serious interest of its own. Like the planet Jupiter, Saturn is a gas giant: a relatively small ball of rock surrounded by a vast envelope of helium, hydrogen and various hydrogen compounds. Saturn is only half as dense as Jupiter, even less dense than water. It releases less heat than Jupiter, but, given its smaller size, scientists are not sure why it radiates any heat at all. Saturn's storms are, surprisingly, more powerful than Jupiter's, and its jet streams are much faster. The planet looks blander, however, because a thick haze of ammonia crystals obscures the colourful banding seen so easily on Jupiter.

⑥ **F** Scientists hope that studying these differences will reveal how giant planets form, how weather systems work under different conditions and what planets around other stars might be like. A recent spacecraft that investigated Saturn and Titan (the Cassini-Huygens probe) has produced results that Galileo would have dearly appreciated back in 1610: photographs that finally show Saturn with crystalline clarity. After all the data from Cassini-Huygens has been interpreted, even though that might take 40 years, all questions on these topics may have been answered once and for all.

Questions 1–6

Reading Passage 2 has six paragraphs, A–F.

Choose the correct heading for each paragraph from the list of headings below. Write the correct number, i–ix, in the circle before each paragraph.

- i Comparing two planets
- ii A surprisingly small moon
- iii A longstanding puzzle
- iv Saturn's fascinating moon
- v The importance of Titan's gravity
- vi Expectations of possible solutions

- vii The problems for life on Titan
- viii An advantage of the low temperatures

Questions 7–10

Choose the correct letter, **A, B, C or D**.

- 7 Galileo's observations of Saturn were influenced by
 - A the frequency of his attempts.
 - B the quality of his experiment.
 - C what he expected to see.
 - D observations made by others.
- 8 The writer suggests that scientists are interested in Titan because
 - A it is larger than Earth, although less dense.
 - B its atmosphere may contain oxygen.
 - C its surface is invisible because of a thick, coloured fog.
 - D its atmosphere has similar elements to Earth's atmosphere.
- 9 The writer uses the phrase 'no walk in the park' because life forms in Titan would
 - A probably exist in water.
 - B be less developed than life forms on Earth.
 - C face considerable difficulties.
 - D be unable to move around.
- 10 How does the writer compare Saturn and Jupiter?
 - A Saturn gives off more heat than Jupiter.
 - B Jupiter is less dense than Saturn.
 - C Saturn is less colourful than Jupiter.
 - D Jupiter has more violent storms than Saturn.

Questions 11–13

Complete the summary below.

Choose **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer.

THE CASSINI-HUYGENS PROBE

The Cassini-Huygens probe has produced clearer ⑪ _____ of Saturn. However, it may take up to ⑫ _____ before all the data from the probe has been analysed. It is expected that this data will assist in the investigation of the way ⑬ _____ function, and the nature of other planets and stars.

Passage 30

Skyscraper Farming

With a global food crisis predicted, a group of scientists is advocating an innovative alternative to conventional farming

⑭ **A** Today's environment scientists are in no doubt that world's resources of fertile soil are rapidly deteriorating, and new land for agriculture is becoming ever more sparse. Intensive farming, urbanization, desertification and sea-level rises are all putting growing pressure on the planet's agricultural land and therefore on food supplies. Currently 24 per cent of the world's 11.5 million hectares of cultivated land has already undergone human-induced soil degradation, particularly through erosion, according to a recent study by the UK Government Office for Science.

⑮ **B** The global population is expected to exceed nine billion by 2050 - up a third from today's level - and studies suggest that food production will have to go up by 70 per cent if we are to feed all of those new mouths. This means that scientists will have to develop new ways of growing crops if we are to avoid a humanitarian crisis. Indeed, UK Food and Agriculture Organization figures suggest that the number of undernourished people is already growing. And with escalating climate change, crop yields in many areas have been affected.

⑯ **C** With this in mind, some scientists and agricultural experts are advocating an innovative alternative to traditional farming whereby skyscrapers packed with shelf-based systems for growing vegetables on each storey - known as 'vertical farms' - could hold the key to revolutionizing agriculture. Columbia University professor Dickson Despommier claims that vertical farming could boost crop yields many times over. A single 20-storey vertical farm could theoretically feed 50,000 people, according to Despommier. And if the theory translates into reality as proposed, 160 skyscraper-sized vertical farms could feed the entire population of New York City, while 180 would be needed to feed London, 289 to feed Cairo and 302 to feed Kolkata.

⑰ **D** It's a compelling vision, and one that has already been put into practice in Asia, albeit on a smaller scale. 'But there are problems - such as initial investment and operating costs are too great,' says a spokesman for Japan's Ministry of Agriculture, Forestry and Fisheries. Nevertheless, Tokyo-based mushroom producer Hokuto Corporation is a model example of how a vertical farm can be profitable. With 28 vertical mushroom farms operating across the country, it produces some 68,000 tons of mushrooms annually. 'Vertical mushroom farms have more advantages than ground-level farms,' says Hokuto's Ted Yamanoko. Yamanoko goes on to highlight the relative cost-effectiveness of his organization's farming practices together with reduced emissions of greenhouse gases.

18 E And the impact vertical farms could extend beyond feeding established urban populations. Despommier sees them as being capable of helping centers of displaced persons - such as refugee camps - in much the same way that Mobile Army Surgical Hospital (MASH) units are deployed in emergency situations. 'Developing an emergency-response system for crop production inside specially constructed modular and highly transportable greenhouses would allow for humanitarian interventions, at least for refugees that are forced out of their countries by political turmoil,' he says. 'If you have three or four storeys of food already growing some place, they could become mobile units that could be picked up by helicopters and dropped into the middle of a crisis zone. The food would be ready to pick and eat. It could be designed to supply people with all the nutrition they need to make it through the crisis.'

19 F But it isn't only about increasing food production, Despommier is concerned about the harm which farming has done to the world's landscape over a relatively short time span, particularly the elimination of hardwood forests. 'Farming is only 12,000 years old,' he points out. 'We have been a species for over 200,000 years. Producing food in tall buildings will allow us for the first time to feed everyone on earth and still return land to its original ecological function.' Natalie Jeremijenko, associate professor at New York University, agrees. 'The challenge that we have now is how we can design urban agriculture systems that not only reduce food miles, but also improve the world's ecosystems,' she says. By significantly reducing the amount of land required for food production, vertical farms could help to enrich biodiversity. And according to Jeremijenko, this can, in turn, help to improve the productivity of conventional farms, as the health of agricultural land is often tied to the health of the surrounding ecosystems. Furthermore, vertical farming could dramatically cut the utilization of fossil fuels, and also reduce geopolitical tensions in countries where poor farming conditions cause conflict and malnutrition.

Questions 14–19

The Reading Passage has six paragraphs, A–F.

Choose the correct heading for each paragraph from the list of headings below. Write the correct number, i–ix, in the circle before each paragraph.

- i Potential production capacities of vertical farms
- ii Opposition to new ideas about food production
- iii A successful application of vertical farming technology
- iv The potential to provide urgent relief
- v The original inspiration for vertical farming

- vi Various environmental benefits of vertical farming
- vii An increasing problem for farmers worldwide
- viii A return to traditional farming methods
- ix A rising demand of food

Questions 20–22

Complete the sentences below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

- A UK government study found that (20) _____ is a significant factor contributing to worldwide levels of soil degradation.
- Disadvantages of vertical farming projects include the expense of setting them up, as well as their high (21) _____.
- (22) _____ could potentially be used to take vertical farming facilities to areas where there is a critical food shortage.

Questions 23–26

Look at the following statements (Questions 23–26) and the list of people below.

Match each statement with the correct person, **A**, **B** or **C**. **NB** You may use any letter more than once.

LIST OF PEOPLE

- A** Dickson Despommier
- B** Ted Yamanoko
- C** Natalie Jeremijenko

Q	Statement	
23	Vertical farming can have financial benefits.	
24	Traditional farming has had a negative effect on the natural world.	
25	Vertical farming could dramatically increase world food production.	
26	Traditional farms may benefit from wider use of vertical farming.	

Passage 31

Sports Science

When the first Olympics took place in Greece 3,000 years ago, athletes could get by with little more than raw strength. These days, however, talent and guts just aren't enough to make it on the international circuit

A Olympic athletes today train with a dedicated team of sports scientists, each applying the latest research and technology to their quest for success. Everything from the fibres in their muscles to the cells in their brains is put through a rigorous workout programme to ensure that, on the big day, they walk out of their changing room with a perfectly designed body and a focused mind. It's not difficult to find examples of this, but what's behind this never-ending increase in performance? Most experts agree that part of it is down to huge advances in sports science, bringing not only a better understanding of the body and mind, but massive improvements in equipment design.

B Sports science can be split into four areas: biomechanics, physiology, psychology and technology. Biomechanics is the science that applies engineering principles to the motion of the body. Biomechanists analyse an athlete's movements using video, motion tracking, force transducers and instruments to measure electrical muscle activity and gauge internal and external forces on the body. 'We need to know which muscles are working when, and how hard, to understand technique and co-ordination,' says Dr Neil Fowler, a biomechanist at Manchester Metropolitan University and biomechanics chair for the British Association of Sport and Exercise Sciences.

C Over the years, Fowler has worked with his fair share of elite athletes, including Olympic javelin throwers and long jumpers, and has plenty of examples of when biomechanics has made a difference to performance. 'We found that in the long jump, it's best if the foot is moving backwards when it hits the board, like a kind of pawing movement. One of our elite jumpers made a radical jumping strategy change as a result of this advice and that season there was a substantial increase in their personal best.'

D But to get the best from biomechanics, an athlete has to be physically capable of making the changes — and that's where the physiology comes in. Physiologists often work closely with biomechanists to fill the physiological gaps that could make the difference between success and failure. What physiologists measure varies from sport to sport and even between events. For an endurance athlete, for example, a priority is to get enough oxygen to the muscles so they can work aerobically for as long as possible. Once your body reaches the maximum rate at which it can process oxygen, your muscles begin to work anaerobically and produce lactic acid, which leads to muscle fatigue. With this in mind, physiologists try to establish what is the maximum sustainable speed where lactic acid levels no longer rise.

E ‘We know in general that if you want to get stronger, you lift a lot (bones become bigger and there’s an increase in density leading to more strength); if you want to be a good endurance runner, you run a lot — but it’s really about targeting each of these determinants of performance and training at the correct intensity, for the correct duration, at the correct frequency,’ says Dr Greg Whyte, physiologist and sports science co-ordinator for the English Institute of Sport.

F But it does seem there is a limit to what the body can do, and in some parts there may now be little room for improvement. So this is where equipment can play an important role. The Sports Engineering Group at Sheffield University is just one group which is designing high-tech sporting equipment that can make changes. However, sometimes not everyone wants the same from technology. ‘We get it from all angles,’ says the group’s Dr Matt Carre. ‘Within industry, a company might want to make tennis rackets that can hit balls faster, but we also get governing bodies who want to know what’s happening. Obviously they want new technology, but if it starts to spoil the game then they need to bring in some rule changes to stop that happening.’

G Even with a perfect body and the best equipment, the athlete’s mind could let them down on the day. Professor Ian Maynard from Sheffield Hallam University is psychology advisor to the British Olympic Association, and works with the sailing and diving teams. As he explains, mental preparation can begin up to two months before the event, with competitors striving to maintain a positive frame of mind. ‘The whole idea is that consistent preparation leads to consistent performance,’ says Maynard. ‘They might have videos, music, arrange to meet friends and family, anything that would be a positive distraction.’ They are also trained to refocus quickly and put themselves back on track in case something goes wrong mid-event.

H Visualisation can add an extra dimension to training. ‘Reliving your best performances is one of the best ways to build confidence, so we go through a performance in the mind’s eye, reliving the emotions and the technical aspects of it,’ says Maynard. Research also suggests visualisation is almost as good as practice. ‘The neurophysiological explanation is that if you imagine a movement, you go through the same synaptic pathways in the brain as if you were actually executing it,’ he says.

Questions 1–3

Reading Passage 2 has eight paragraphs, A–H.

Which paragraph contains the following information? Write the correct letter, A–H.

Q	Statement	
---	-----------	--

1	a reference to a particular sports event which has benefited from close analysis of performance	
2	a reference to the importance for athletes of recalling past successes	
3	examples of devices used to gather data	

Questions 4–8

Look at the following statements (Questions 4–8) and the list of people below.

Match each statement with the correct person, **A, B, C or D**.

NB You may use any letter more than once.

LIST OF PEOPLE

- A** Dr Neil Fowler
- B** Dr Greg Whyte
- C** Dr Matt Carre
- D** Professor Ian Maynard

Q	Statement	
4	He mentions the difficulty in satisfying conflicting demands.	
5	He aims to prevent athletes from being deterred by unforeseen problems.	
6	He describes an occasion when a small adjustment in technique led to improved performance.	
7	He explains the need to observe athletes in action.	
8	He mentions the importance of research in helping to decide upon the right amount of physical preparation.	

Questions 9–13 — Label the diagram

The diagram shows how sports science applies to different body systems.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

Sports science and the body

- **Nervous System** – imagining a movement uses the same brain pathways as actually performing it, so the use of → **9** —————

- **Mind** – athletes are encouraged to participate in any activities which promote a →

⑩ _____

- Training is also given in how to → ⑪ _____

- **Respiratory System** – for endurance athletes there should be enough oxygen for the muscles to function → ⑫ _____

- Any more oxygen than this will not be processed and will result in the production of →

⑬ _____

Passage 32

Stress Less

How senior executives manage – and sometimes thrive on – the pressures of a demanding workload

How busy is too busy? For some it means sometimes having to have a short lunch; for others it may mean missing lunch altogether. For a few, it is not being able to take a day off once a month. Then there is a group of people for whom working every evening and weekend is normal, and feeling stressed is taken for granted. For most senior executives, workloads swing between extremely busy and frenzied. Neil Plumridge, vice-president of a management consultancy company, says that his weeks vary from a 'manageable' 45 hours to 80 hours, but average 60 hours.

Three signs warn Plumridge about his workload: sleep, scheduling and family. He knows he is doing too much when he gets less than six hours' sleep for three consecutive nights, when he is constantly having to reschedule appointments, and when he misses a family birthday or anniversary. 'Then,' he says, 'I know things are out of control.' Plumridge states that stress is often caused by his having unrealistic expectations of himself. 'I'll promise a client that I'll do something tomorrow, and then I'll promise another client that I'll do the same thing, when I know it's not going to happen. I could have said: "Why don't I give that to you in 48 hours?" The client wouldn't care.'

Over-committing is something people experience as an individual problem. However, new research indicates that people may be designed to over-commit. A study in the Journal of Experimental Psychology shows that people always believe that they will be less busy in the future. This is a misapprehension according to the authors of the report, Gal Zauberman of the University of North Carolina and John Lynch of Duke University. 'On average, an individual will be just as busy two weeks or a month from now as he or she is today. But that is not how it appears in everyday life,' they say. 'People make commitments long in advance that they would never make if the same commitments required immediate action. They discount future time investments relatively steeply.'

Being 'too busy' is highly subjective, but for any individual there are some concrete signs of stress: disturbed sleep and declining mental and physical health are the most common examples. Figures for National Workers' Compensation (insurance against injury caused by work) show that stress causes the most lost time of any workplace injury: employees suffering stress are absent from work for an average of 16.6 weeks. The effects of stress are also expensive. The Australian Government insurer reports that in 2003–2004, claims for stress-related psychological injury accounted for 7% of insurance claims, but almost 27% of the amount paid out in claim payments.

Experts say the key to dealing with stress is not to focus on relief – a game of golf or a massage – but to reassess workloads. Neil Plumridge says he makes it a priority to work out what has to change: that might mean allocating extra staff to a job, allowing more time, or changing expectations. He also relies on the advice of colleagues, saying that his peers coach each other on business problems: ‘Just a fresh pair of eyes over an issue can help,’ he states.

Executive stress is not confined to big organisations. Vanessa Stoykov has been running her own advertising agency and public relations business for seven years. Her company has grown so fast that it appeared on the Business Review Weekly ‘Fast 100’ list of fastest-growing small enterprises in 2004, just after Stoykov had her first child. In 2005, revenue was projected to double to \$2.4 million. She had just had her second child. With a fast-growing business and two small children, Stoykov says she has mastered the art of caring for children, typing, and talking on the phone at the same time. But, unlike many others, she appears to thrive on the mental stimulation of running her own business, despite the stress.

Jan Elsner, a psychologist who specialises in executive coaching, says that doing well on a demanding workload is typical of senior executives. Some people work best with high-adrenaline periods followed by quieter times, while others thrive under sustained pressure. Elsner’s practice is based on a movement known as positive psychology, a school of thought that argues that positive experiences – feeling engaged and challenged, for example – do not balance out negative experiences such as stress; instead they help people increase their resilience over time. Elsner says that many of the senior business people she coaches are increasingly relying more on regulating stress through methods such as meditation. She points to research showing that meditation can alter the biochemistry of the brain and the way that brains and bodies react to stress.

Some experts believe there is too much emphasis on treating job stress as an individual problem. Tony LaMontagne, of the University of Melbourne, says that while personality traits do have some effect on stress, it is systemic stressors – qualities of job roles and organisations themselves – that have a far greater effect. His recent research shows that the major predictor of stress is the level of job control a person has. The best type of job combines challenging work with high autonomy. The worst jobs combine challenging work and low control. People with demanding jobs but little autonomy have up to four times the probability of depression and more than double the risk of heart attack.

Questions 1–5

Match each statement with the correct person or people, A, B, C or D.

NB *You may use any letter more than once.*

LIST OF PEOPLE

- A Neil Plumridge
- B Gal Zauberman and John Lynch
- C Jan Elsner
- D Tony LaMontagne

Q	Statement	
1	High-level workers tend to react positively to stress	
2	Stress levels are increased by trying to please customers	
3	Support from other workers may relieve stress	
4	Lack of independence at work is often responsible for stress	
5	Workers commonly expect their workloads to lessen over time	

Questions 6–8

Choose the correct letter, **A**, **B**, **C** or **D**.

- 6 Which of the following is **NOT** mentioned by Neil Plumridge as an indication that his workload is too heavy?
- A an inability to keep to his schedule
 - B inattention to family celebrations
 - C a lack of concentration on a task
 - D a period of insufficient sleep
- 7 Which method of lessening work stress is **NOT** suggested by Neil Plumridge?
- A rethinking ideas of what can be achieved
 - B extending the deadline for completing the task
 - C using more workers on a project
 - D taking more time off for sport or other recreation
- 8 According to Jan Elsner, meditation offers a method of
- A taking a worker's mind off his troubles
 - B changing physical response to stress
 - C resting more effectively
 - D encouraging executives to take breaks

Questions 9–13

Complete the summary below.

Choose **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer.

STRESS: ITS EFFECTS AND HOW TO REDUCE IT

Statistics on workers' compensation show that people take more time off work due to stress than for any other (9) ____ at work. On average, workers who take time off because of stress stay away for (10) _____. This absence comes at a high price – while the number of insurance claims due to stress amount to only (11) _____ of the total, they account for a much higher proportion of the cost of claims payments. Experts believe that seeking to relieve stress through physical therapies such as sport or (12) _____ may be less effective than simply reviewing your (13) _____.

Passage 33

Tackling Hunger in Msekeni

A free school-lunch programme in Malawi shows how nutrition and education reinforce each other

① **A** There are not enough classrooms at the Msekeni primary school, so half the lessons take place in the shade of yellow-blossomed acacia trees. Given this shortage, it might seem odd that one of the school's purpose-built classrooms has been emptied of pupils and turned into a storeroom for sacks of grain. But it makes sense. Food matters more than shelter.

② **B** Msekeni is in one of the poorer parts of Malawi, a landlocked southern African country of exceptional beauty and great poverty. No war lays waste Malawi, nor is the land unusually crowded or infertile, but Malawians still have trouble finding enough to eat. Half of the children under five are underfed to the point of stunting. Hunger blights most aspects of Malawian life, so the country is as good a place as any to investigate how nutrition affects development, and vice versa.

③ **C** The headmaster at Msekeni, Bernard Kumanda, has strong views on the subject. He thinks food is a priceless teaching aid. Since 1999, his pupils have received free school lunches. Donors such as the World Food Programme (WFP) provide the food: those sacks of grain (mostly mixed maize and soya bean flour, enriched with vitamin A) in that converted classroom. Local volunteers do the cooking – turning the dry ingredients into a bland but nutritious slop and spooning it out on to plastic plates. The children line up in large crowds, cheerfully singing a song called “We are getting porridge”.

④ **D** When the school's feeding programme was introduced, enrolment at Msekeni doubled. Some of the new pupils had switched from nearby schools that did not give out free porridge, but most were children whose families had previously kept them at home to work. These families were so poor that the long-term benefits of education seemed unattractive when set against the short-term gain of sending children out to gather firewood or help in the fields. One plate of porridge a day completely altered the calculation. A child fed at school will not howl so plaintively for food at home. Girls, who are more likely than boys to be kept out of school, are given extra snacks to take home.

⑤ **E** When a school takes in a horde of extra students from the poorest homes, you would expect standards to drop. Anywhere in the world, poor kids tend to perform worse than their better-off classmates. When the influx of new pupils is not accompanied by an increase in the number of teachers, as was the case at Msekeni, you would expect standards to fall even further. But they have not. Pass rates at Msekeni improved dramatically, from 30% to 85%. Although this was an exceptional example, the nationwide

results of school feeding programmes were still pretty good. On average, after a Malawian school started handing out free food it attracted 38% more girls and 24% more boys. The pass rate for boys stayed about the same, while for girls it improved by 9.5%.

⑥ **F** Better nutrition makes for brighter children. Most immediately, well-fed children find it easier to concentrate. It is hard to focus the mind on long division when your stomach is screaming for food. Mr Kumanda says that it used to be easy to spot the kids who were really undernourished. “They were the ones who stared into space and didn’t respond when you asked the question,” he says. More crucially, though, more and better food helps brains grow and develop. Like any other organ in the body, the brain needs nutrition and exercise. But if it is starved of the necessary calories, proteins and micronutrients, it is stunted, perhaps not as severely as a muscle would be, but stunted nonetheless. That is why feeding children at schools works so well. And the fact that the effect of feeding was more pronounced in girls than in boys gives a clue to who eats first in rural Malawian households. It isn’t the girls.

⑦ **G** On a global scale, the good news is that people are eating better than ever before. Homo sapiens has grown 50% bigger since the industrial revolution. Three centuries ago, chronic malnutrition was more or less universal. Now, it is extremely rare in rich countries. In developing countries, where most people live, plates and rice bowls are also fuller than ever before. The proportion of children under five in the developing world who are malnourished to the point of stunting fell from 39% in 1990 to 30% in 2000, says the World Health Organisation (WHO). In other places, the battle against hunger is steadily being won. Better nutrition is making people cleverer and more energetic, which will help them grow more prosperous. And when they eventually join the ranks of the well off, they can start fretting about growing too fast.

Questions 1–7

The reading passage has seven paragraphs, A–G.

Choose the correct heading for paragraphs A–G from the list below. Write the correct number, i–ix, in the circle before each paragraph.

- i Why better food helps students’ learning
- ii Becoming the headmaster of Msekeni
- iii Surprising use of school premises
- iv Global perspective
- v Why students were undernourished
- vi Surprising academic outcome

- vii An innovative program to help girls
- viii How food program is operated
- ix How food program affects school attendance
- x None of the usual reasons
- xi How to maintain academic standard

Questions 8–11

Complete the sentences below.

Use **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer.

- 8 In Kumanda's school _____ are given to girls after the end of the school day.
- 9 Many children from poor families were sent to collect _____ from the field.
- 10 Thanks to the free food program, _____ of students passed the test.
- 11 The modern human is _____ bigger than before after the industrial revolution.

Questions 12 and 13

Choose **TWO** letters, **A–F**.

Which **TWO** of the following statements are true?

OPTIONS

- A Some children are taught in the open air.
- B Bernard Kumanda became the headmaster in 1991.
- C No new staffs were recruited when attendance rose.
- D Girls are often treated equally with boys in Malawi.
- E Scientists have devised ways to detect the most underfed students in school.
- F WHO is worried about malnutrition among kids in developing countries.

Answer: 12 13

Passage 34

The City of Great Zimbabwe

The ruins of a unique stone city testify to the sophistication and reach of the Shona civilisation

A The ruins of the city of Great Zimbabwe in southern Africa are a unique testimony to the civilisation of the Shona people. The Shona word 'zimbabwe' means a large stone house, but the ruins consist of the largest stone complex in Africa built before the modern era, aside from the monumental architecture of ancient Egypt. Their method of construction is unique to African architecture and, although there are examples of similar work elsewhere, none are as distinguished and imposing as Great Zimbabwe. The city was an important trading centre and has been renowned for centuries. The modern country of Zimbabwe, where the ruins are situated, was named after it, and in 1986 Great Zimbabwe was designated both a national monument and a World Heritage Site.

B Great Zimbabwe is located on a plain at an altitude of some 1,100 metres in a sparsely populated region. The city, built between 1100 and 1450 AD, extends over seven square kilometres, and is divided into three areas: the Hill Complex, the Great Enclosure and the Valley Complex. The Hill Complex, forming a huge granite mass on top of a north-east/south-west-facing ridge, was continuously occupied from the 12th to the 15th century, and traces of human settlements are found in numerous layers. Rough granite rubble-stone blocks form distinct enclosures, accessed by narrow, partly covered passageways. This complex is generally considered a 'royal city'; the west enclosure is thought to have been the residence of successive chiefs and the east enclosure, where six soapstone posts topped with figurines of birds were found, is considered to have served a ritual purpose.

C The Great Enclosure, which is oval shaped, is located to the south of the hills, and dates to the 13th century. The external walls were built of cut granite blocks, laid in regular courses, and within these walls there are community and living areas, and a narrow passage leading to a high conical tower, which may have functioned as a granary warehouse. Interior walls mark off each family's area, generally comprising a kitchen, two living huts and a yard. Huts were made of mud bricks, composed of a mixture of granitic sand and clay.

D The Valley Complex, which dates back to the 15th century, is a series of living quarters scattered throughout the valley. Each one has similar characteristics: many of the structures are in brick (huts, indoor flooring, benches, basins, etc.) and dry stone walls provide insulation. Construction work was carried out to a high standard of craftsmanship, incorporating an impressive display of V-shaped and checked wall decorations.

E Scientific research has shown that Great Zimbabwe was founded around 1100 on a site which had been sparsely occupied in the prehistoric period by the Shona. By the 14th century, it was the principal city of a major state extending over the gold-rich area; its population exceeded 10,000. However, in about 1450 the capital was abandoned. The reasons for this are debated, but suggestions include a shortage of food for the overpopulated city, and deforestation. The resulting migration benefited Khami, which then became the most influential city in the region, and signalled Great Zimbabwe's waning political power.

F Archaeological excavations at Great Zimbabwe have revealed glass beads and porcelain from China and Persia, and gold and Arab coins from the Indian Ocean island of Kilwa, which testify to long-standing trade with the outer world. Other evidence, including fragments of pottery and ironware, gives further insight into the city's socio-economic complexity and into farming and pastoral activities in the surrounding area. A monumental granite cross, located at a traditionally revered and sacred site, also illustrates contact with Christian missionaries. In the late 19th century, numerous soapstone figurines in the form of birds were found mounted on columns, buried among the ruins. These are testimony to the use of the site as a place of worship from the 12th century to the present day. The birds, which had some human features, were subsequently described as the Zimbabwe Birds, and a design based on them became a national symbol, incorporated into the Zimbabwe flag and shown in other prestigious contexts.

G Despite the importance of the site, it has suffered some deterioration due to variations in temperature and soil moisture content, as well as pressure from tourism, encroaching invasive vegetation, and incorrect preservation methods. The site would benefit if attention was given to conservation techniques and the materials employed, as well as to the need for conservation standards that meet the international benchmark, but which are balanced with the traditional activities still taking place there. The site has been legally protected since 1893, and today is managed by the Zimbabwean Ministry of Home Affairs. Funding for the management and conservation of the site comes mainly from central government, with limited revenue generated by the sale of publications and charging an entrance fee for the site.

Questions 1–8

Reading Passage 2 has seven sections, A–G.

Which section contains the following information? Write the correct letter, A–G.

NB *You may use any letter more than once.*

Q	Statement	
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1	a list of factors which have resulted in some damage to Great Zimbabwe	
2	reference to the high status of some of the original inhabitants	
3	proof of the former commercial significance of Great Zimbabwe	
4	an indication of the global importance attached to Great Zimbabwe nowadays	
5	mention of the authority currently responsible for the area	
6	an overview of the layout of the archaeological site	
7	a reference to the quality of comparable buildings in other places	
8	theories about the eventual desertion of Great Zimbabwe	

Questions 9–12

Look at the following aspects of Great Zimbabwe (Questions 9–12) and the list of areas below.
Match each aspect with the correct area, **A, B or C**.

NB You may use any letter more than once.

LIST OF AREAS

- A** Hill Complex
- B** Great Enclosure
- C** Valley Complex

Q	Statement	
9	Its outer walls are curved.	
10	Part of it may have been used for ceremonies.	
11	Some of its buildings feature geometric patterns.	
12	It includes a structure that could have been used for storage.	

Questions 13–16

Complete the sentences below.

Choose **ONE WORD ONLY** from the passage for each answer.

- **13** An image based on the bird sculptures found at Great Zimbabwe now appears on the national _____.

- 14 The impact of nature on Great Zimbabwe, as well as that of _____, needs to be carefully managed.
- 15 When choosing methods of conservation for Great Zimbabwe, it is important to adopt global _____.
- 16 A small part of Great Zimbabwe's income comes from selling tickets and _____.

@bekhruzposts

Passage 35

The Dingo Debate

Graziers see them as pests, and poisoning is common, but some biologists think Australia's dingoes are the best weapon in a war against imported cats and foxes

A A plane flies a slow pattern over Carlton Hill station, a 3,600 square kilometre ranch in the Kimberley region in northwest Australia. As the plane circles, those aboard drop 1,000 small pieces of meat, one by one, onto the scrubland below, each piece laced with poison; this practice is known as baiting. Besides 50,000 head of cattle, Carlton Hill is home to the dingo, Australia's largest mammalian predator and the bane of a grazier's (cattle farmer's) life. Stuart McKechnie, manager of Carlton Hill, complains that graziers' livelihoods are threatened when dingoes prey on cattle. But one man wants the baiting to end, and for dingoes to once again roam Australia's wide-open spaces. According to Chris Johnson of James Cook University, 'Australia needs more dingoes to protect our biodiversity.'

B About 4,000 years ago, Asian sailors introduced dingoes to Australia. Throughout the ensuing millennia, these descendants of the wolf spread across the continent and, as the Tasmanian tiger disappeared completely from Australia, dingoes became Australia's top predators. As agricultural development took place, the European settlers found that they could not safely keep their livestock where dingoes roamed. So began one of the most sustained efforts at pest control in Australia's history. Over the last 150 years, dingoes have been shot and poisoned, and fences have been used in an attempt to keep them away from livestock. But at the same time, as the European settlers tried to eliminate one native pest from Australia, they introduced more of their own.

C In 1860, the rabbit was unleashed on Australia by a wealthy landowner and by 1980 rabbits had covered most of the mainland. Rabbits provide a huge prey base for two other introduced species: the feral (wild) cat and the red fox. The interaction between foxes, cats and rabbits is a huge problem for native mammals. In good years, rabbit numbers increase dramatically, and fox and cat populations grow quickly in response to the abundance of this prey. When bad seasons follow, rabbit numbers are significantly reduced – and the dwindling but still large fox and cat populations are left with little to eat besides native mammals.

D Australian mammals generally reproduce much more slowly than rabbits, cats and foxes – an adaptation to prevent overpopulation in the arid environment, where food can be scarce and unreliable – and populations decline because they can't grow fast enough to replace animals killed by the predators. Johnson says dingoes are the solution to this problem because they keep cat and fox populations under control. Besides regularly eating the smaller predators, dingoes will kill them simply to lessen competition. Dingo packs live

in large, stable territories and generally have only one fertile female, which limits their rate of increase. In the 4,000 years that dingoes have been in Australia, they have contributed to few, if any, extinctions, Johnson says.

E Reaching out from a desolate spot where three states meet, for 2,500 km in either direction, is the world's longest fence, two metres high and stretching from the coast in Queensland to the Great Australian Bight in South Australia; it is there to keep dingoes out of the southeast, and the fence separates the main types of livestock found in Australia. To the northwest of the fence, cattle predominate; to the southwest, sheep fill the landscape. In fact, Australia is a land dominated by these animals – 25 million cattle, 100 million sheep and just over 20 million people.

F While there is no argument that dingoes will prey on sheep if given the chance, they don't hunt cattle once the calves are much past two or three weeks old, according to McKechnie. And a study in Queensland suggests that dingoes don't even prey heavily on the newborn calves unless their staple prey disappears due to deteriorating conditions like drought. This study, co-authored by Lee Allen of the Robert Wicks Research Centre in Queensland, suggests that the aggressive baiting programs used against dingoes may actually be counter-productive for graziers. When dingoes are removed from an area by baiting, the area is recolonized by younger, more solitary dingoes. These animals aren't capable of going after large prey like kangaroos, so they turn to calves. In their study, some of the highest rates of calf predation occurred in areas that had been baited.

G Mark Clifford, general manager of a firm that manages over 200,000 head of cattle, is not convinced by Allen's assertion. Clifford says, 'It's obvious if we drop or loosen control on dingoes, we are going to lose more calves.' He doesn't believe that dingoes will go after kangaroos when calves are around. Nor is he persuaded of dingoes' supposed ecological benefits, saying he is not convinced that they manage to catch cats that often, believing they are more likely to catch small native animals instead.

H McKechnie agrees that dingoes kill the wallabies (small native animals) that compete with his cattle for food, but points out that in parts of Western Australia, there are no foxes, and not very many cats. He doesn't see how relaxing controls on dingoes in his area will improve the ecological balance. Johnson sees a need for a change in philosophy on the part of graziers. 'There might be a number of different ways of thinking through dingo management in cattle country,' he says. 'At the moment, though, that hasn't got through to graziers. There's still just one prescription, and that is to bait as widely as possible.'

Questions 1–7

Reading Passage 2 has eight sections, A–H.

Which section contains the following information? Write the correct letter, **A–H**.

NB You may use any letter more than once.

Q	Statement	
1	a description of a barrier designed to stop dingoes, which also divides two kinds of non-native animals	
2	how dingoes ensure that rival species do not dominate	
3	a reference to a widespread non-native species that other animals feed on	
4	a mention of the dingo's arrival in Australia	
5	research which has proved that dingoes have resorted to eating young livestock	
6	a description of a method used to kill dingoes	
7	the way that the structure of dingo groups affects how quickly their numbers grow	

Questions 8–10

Look at the following statements (Questions 8–10) and the list of people below.

Match each statement with the correct person, **A, B, C or D**.

LIST OF PEOPLE

- A** Stuart McKechnie
- B** Chris Johnson
- C** Lee Allen
- D** Mark Clifford

Q	Statement	
8	Dingoes tend to hunt native animals rather than hunting other non-native predators.	
9	The presence of dingoes puts the income of some people at risk.	
10	Dingoes have had little impact on the dying out of animal species in Australia.	

Questions 11–13

Complete the sentences below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

- 11 The dingo replaced the _____ as the main predatory animal in Australia.
- 12 Foxes and cats are more likely to hunt native animals when there are fewer _____.
- 13 Australian animals reproduce at a slow rate as a natural way of avoiding _____.

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Passage 36

The Fascinating World of Attine Ants

How leaf-cutter ants farm a fungus, and the parasite that keeps their whole system in check

A Leaf-cutting ants and their fungus (fungus — an organism such as a mushroom which obtains its food from decaying material) ‘farm’ are a marvel of nature and perhaps the best-known example of symbiosis — the mutual dependence of two species. Ants cultivate a mushroom-like fungus in ‘farms’. Both the ants and their so-called ‘agriculture’ have been extensively studied over the years, but recent research has uncovered intriguing new findings. Ants invented agriculture 50 million years before people did, and the leaf-cutters, members of the large attine ant family, practice the most sophisticated example of it. They grow their fungus in underground chambers that can reach the size of a football. A single leaf-cutter nest may contain a thousand such chambers, embedded in an underground metropolis up to 18 feet deep, and support a society of more than a million ants.

B These ant communities are the dominant plant-eaters of the Neotropics, the region comprising South and Central America, Mexico and the Caribbean. Biologists believe 15 percent of the leaf production of tropical forests disappears down the nests of leafcutter ants. In the nest, the leaves are shredded and added to the fungus, which digests the leaves and is in turn eaten by the ants. The attine ants’ achievement is remarkable because it allows them to consume, courtesy of their mushroom’s digestive powers, the harvest of tropical forests whose leaves are laden with poisonous chemicals.

C There are more than 200 known species of the attine ant tribe, divided into 12 groups, or genera. The leaf-cutters use fresh vegetation; the other groups, known as the lower attines because their nests are smaller and their techniques more primitive, feed their gardens with similar leaves which have fallen on the ground and insects that lie on the forest floor. Lower attine ants are all a similar size. However, leaf-cutter worker ants come in made-to-fit sizes — large ants to saw off leaves, medium ones to shred them, and miniature workers to seed them with fungus and clean off alien growths.

D In 1994, biologists from the United States Department of Agriculture analyzed the DNA of ant funguses. They found the leaf-cutters’ fungus was descended from a single pure strain, propagated for at least 23 million years. However, the funguses grown by lower attine ants fell into four different groups, as if the ants had domesticated wild funguses at least four times in evolutionary history. What could be driving these two patterns of fungus gardening, the pure clone cultivation of the leaf-cutters and multiple varieties of the lower attines?

E The answer has been suggested by Cameron Currie of University of Toronto. The pure strain of fungus grown by the leaf-cutters, it seemed to him, resembled the single crops grown by humans to the exclusion of all others, such as potato growing. These ‘monocultures’, which lack the genetic diversity to respond to changing environmental threats, are particularly vulnerable to parasites — organisms which live and feed on their host, often causing harm. Currie felt there had to be a parasite in the ant-fungus system. But a century of ant research did not provide any evidence for his idea. Textbooks describe how leaf-cutter ants scrupulously weed their gardens of all foreign organisms. ‘People kept telling me, the ants keep their gardens free of parasites,’ said Currie. Nevertheless, after three years of sifting through attine ant gardens, Currie discovered several alien organisms, particularly a family of parasitic molds called ‘Escovopsis’.

F Escovopsis is a deadly disease that can devastate a fungus garden in a couple of days. It blooms like a white cloud which envelops the whole garden. Other ants won’t go near it and the ants associated with the garden just starve to death. Evidently, the ants usually manage to keep Escovopsis and other parasites under control. Nevertheless, with any lapse in control Escovopsis will quickly burst forth. Although new leaf-cutter gardens start off free of Escovopsis, within two years some 60 percent become infected.

G The discovery of Escovopsis’s role brings a new level of understanding to the evolution of the attine ants. In the last decade, evolutionary biologists have been increasingly aware of the role of parasites as driving forces in evolution. With Currie’s work, there is now a possible reason for the different varieties of fungus in the lower attine mushroom gardens — to stay one step ahead of the relentless Escovopsis. Interestingly, the leaf-cutters had in general fewer alien molds in their gardens than the lower attines, yet more Escovopsis infections. Clearly, the price they pay for cultivating a pure variety of fungus is a higher risk from Escovopsis.

H So how do attine ants keep this parasite under control? People have known for a hundred years that ants have a whitish growth on their body surface. It was thought to be wax but, after examining it under a microscope, Currie discovered a specialized patch on the ants’ bodies that harbors a particular kind of bacterium, one well known to the pharmaceutical industry and the source of half the antibiotics used in medicine. This bacterium is potent poisoner of Escovopsis inhibiting its growth and suppressing spore formation. Astoundingly, the leaf-cutter ants are accomplishing feats beyond the power of humans: they are growing a monocultural crop year after year without disaster, and they are using an antibiotic apparently so wisely that, unlike people, they are not provoking antibiotic resistance in the target disease-producing organism.

Reading Passage 2 has eight paragraphs, **A–H**.

Which section contains the following information?

NB You may use any letter more than once.

Q	Statement	
14	two things at which leaf-cutter ants have succeeded but humans have failed	
15	a comparison between the nests of leaf-cutter and lower attine ants	
16	an assessment of the impact leaf-cutter ants have on their environment	
17	the effect Escovopsis has on ant communities	
18	the advantage for lower attine ants of growing a range of funguses	
19	the discovery of the age of the attine ant funguses	

Questions 20–24

Classify the following features as belonging to

A Leaf-cutting ants **B** Lower attines **C** Both

- 20 the use of dead vegetation to cultivate their fungus
A Leaf-cutting ants
B Lower attines
C Both
- 21 very small ants that keep the fungus free of foreign organisms
A Leaf-cutting ants
B Lower attines
C Both
- 22 the ability to safely eat harmful plants
A Leaf-cutting ants
B Lower attines
C Both
- 23 the cultivation of a single fungus
A Leaf-cutting ants
B Lower attines
C Both
- 24 a nest with a very large number of rooms for growing fungus
A Leaf-cutting ants
B Lower attines
C Both

Questions 25–26

Choose the correct letter, A, B, C or D.

- 25 What does the writer say about Cameron Currie's research?
- A No previous work had been done in this area.
 - B Earlier studies did not support his theory.
 - C Textbooks on this subject lacked specific detail.
 - D Currie's initial theory had proven to be incorrect.
- 26 Using a microscope Currie was the first to discover that the body of attine ants
- A has a white covering.
 - B is covered in wax.
 - C is poisonous to humans.
 - D has a substance useful to humans.

Passage 37

The Gender Gap in New Zealand's High School Results

Results from New Zealand's new national examination for secondary schools are giving that country some cause for concern

A The issue is the difference in pass rates between the sexes: at each level of the examination and across all school types, the difference is about 10 percentage points. Girls are doing better in every subject, and those in girls-only schools are taking top honours. The results are not a surprise: high school girls have been outperforming boys academically for more than a decade. It is an international phenomenon, and within Australia was the subject of much debate and controversy. Within New Zealand back in the 1980s, there was a concerted campaign, called 'Girls Can Do Anything', which was aimed at lifting girls participation rates, achievement levels and aspirations. This was so successful that the pendulum has now swung to the other extreme. Views differ on how worried people should be. After all, for much of history, girls were excluded from any form of education, and this new phenomenon could be seen as a temporary over-correction before the balance is righted.

B However the New Zealand State Ministry of Education says it is taking the issue seriously. It is working with a reference group on boys' education which has been set up, and it has commissioned an Australian academic to report on interventions that have been found to work for boys, drawing particularly on Australia's experience. But some, such as former prison manager Celia Lashlie, the author of a book for parents of teenage boys, believe there is still resistance within the Education Ministry towards doing anything about the problem.

C Education Ministry learning policy manager Steve Benson says that the 'National Certificate in Educational Achievement', or NCEA, as New Zealand's high school exams are called, is useful to employers and to universities because it provides a fine-grained picture of pupils performance in every aspect of a subject, rather than just a pass or fail in an overall area. 'In most parts of the curriculum, for example in maths, there isn't really a gender gap. But literacy is a different matter. Even boys who are good at writing tend not to write so much. There's actually a quantity issue.'

D The discrepancy in reading and writing skills between males and females shows up as early as preschool, and the difference is clear by the time these children enter high school. Not being good at literacy was not such a problem in the old days when many students left school for manual jobs after Year 11. But nowadays many more stay on to higher education, and almost all jobs require literacy skills. Roger Moses, the headmaster of Wellington

College, says that the written content of NCEA papers is more demanding than the previous system of secondary school qualifications in New Zealand, even in subjects such as statistics and accounting.

E New Zealand 15-year-olds score close to average in international reading tests, but beneath this average lies a wide variance, with New Zealand European girls most represented at the top and New Zealand Pacific Island boys at the bottom. Yet some European girls drop out, and some Pacific Island boys excel. In other words, the range in performance within each gender group is much greater than the gender differences. Ethnic differences, and differences in socio-economic status, may be more significant than the simple boy/girl explanation.

F This makes the Education Ministry nervous about pushing solutions that emphasise stereotyped gender differences, rather than looking at underachievement as a whole. Rob Burroughs, principal of Linwood High School in Christchurch, agrees. For three years his school ran separate boys' classes to try to address the disparity in performance, before abandoning them. The research showed that the boys did better in their own class than in the co-educational environment. But when he looked at which teachers they had, and how well those teachers' other classes did, it became clear that the difference was, instead, to do with the quality of instruction.

G At Onslow College, Dr Stuart Martin would do away with the NCEA Level 1 exam if he could. He says that in Year 11, aged 15, boys are simply not mature enough to cope. 'They tend to think that just passing is enough, and that it's not necessary to work hard for a Merit or an Excellence grade. Often they are busy with other activities and part-time jobs. Boys' competitive instinct tends to come out later in their schooling years, especially if there is money attached or other tangible rewards. By 17, boys are catching up academically with the girls, and by the end of Year 13, boys are again winning the top prizes.'

H Boys in single-sex schools do better in NCEA across all levels, something economist Brian Easton reported after analysing data from the first year of NCEA's implementation. He said the results were valid, even when socioeconomic status was taken into account. Dr Paul Baker, head of Waitaki Boys' High School in Oamaru, agrees. He thinks that although it is possible for all schools to do more to boost boys' performance, it is easier in a boys' school, where activities cannot be 'captured by girls'.

Questions 1–3

Complete the summary below.

*Choose **NO MORE THAN TWO WORDS** from the passage for each answer.*

High school assessment in New Zealand

- New Zealanders are worried at the outcomes of their high school assessment system, because the (1) _____ of girls are higher than those of boys by 10%.
- A gender gap has been apparent for over a (2) _____.
- This situation is not unique to New Zealand and has been noticed in (3) _____ also.

Questions 4–7

Reading Passage 2 has eight paragraphs, A–H.

Which paragraph contains the following information? Write the correct letter, A–H.

Q	Statement	
4	an advantage of New Zealand's secondary school tests	
5	a mention of current government initiatives to boost male achievement	
6	when gender difference in literacy skills first becomes evident	
7	findings that relate academic achievement to race	

Questions 8–13

Look at the following people (Questions 8–13) and the list of statements below.

Match each person with the correct statement, A–H.

LIST OF STATEMENTS

- A Boys gain lower marks on NCEA if they attend an all-boys' school.
- B Boys are disadvantaged by girls tending to take over at school.
- C Good teaching is more important than whether classrooms are single-sex or mixed.
- D Mathematical skills were not so important in the past.
- E The difference in achievement between school boys and girls is only evident in some subjects.
- F Older boys are more motivated to study than younger boys.
- G The NCEA exams have higher literacy standards than past exams did.
- H The New Zealand government is reluctant to take action on behalf of boys.

Q	Statement	
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8	Celia Lashlie	
9	Steve Benson	
10	Roger Moses	
11	Rob Burroughs	
12	Stuart Martin	
13	Paul Baker	

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Passage 38

The Global Effect of Food Systems

How the way we grow, transport and waste food is reshaping the climate

A Did you know that what's on your plate plays a larger role in contributing to climate change than the car you drive? Many people, especially in wealthier countries, are becoming worried about how much their individual carbon footprint is contributing to climate change. However, when they consider these issues, they'll usually think about what vehicle they drive and how much electricity they use in their home, but not so much about farming machinery, processed meats or food waste. Few consider the impacts of the food they eat, despite the fact that worldwide food systems account for roughly one quarter of all manmade emission of greenhouse gases. That's more than the entire global transportation sector.

B The most immediate threat from climate change for most of the global population will be at the dinner table, as our ability to grow critical staple crops is being affected by the global warming we've already experienced. Between 1980 and 2008, for instance, wheat yields fell by 5.5% and maize yields by 3.8% due to rising temperatures. Climate change threatens the food security of millions of poor people around the world. And yet what we see is that while food and agriculture are massively impacted by climate change, they are also, simultaneously, major contributors to it. What all of this tells us is that our food systems, as currently structured, are facing major challenges.

C So what are food systems? Everything from seed and soil, to the supermarket, to the plate, to the landfill site. Food systems include the growing, harvesting, processing, packaging, transporting, marketing, consumption and disposal of food and food-related items. While farming alone accounts for 10–12% of global greenhouse gas emissions, when we look at entire food systems the contributions to climate change more than double. A recent report published by the non-for-profit organisation Meridian Institute lays out the many factors throughout food systems that spell trouble for the climate, and also explains why a broad, systems-wide perspective is needed for implementing effective changes.

D Consider the impact of deforestation as forests are cleared for the purpose of making land available for other uses. Worldwide, 80% of deforestation is carried out to create farmland, with potentially serious consequences for climate change. The world's forests are massive carbon sinks, vital natural 'reservoirs' which remove carbon dioxide from the atmosphere and store it. So is soil, which locks away – or 'sequesters' – two to three times as much carbon as there is present in the atmosphere. However, there are ways to produce food without adding to climate change. Environmentally responsible farming can help restore ecosystem function by producing crops and livestock in productive ways that sequester carbon and preserve forests.

E Or consider food waste. Not just the scraps that we throw away, but throughout the entire food system. Every year, a staggering 30–40% of the food produced in the world is never eaten. Some never gets harvested, some spoils before it reaches consumers, and a lot is tossed away by supermarkets, restaurants and at home. For the sake of comparing emissions, if food waste were a country in its own right, it would be the world's third largest contributor to global warming, after only China and the United States. This says nothing of the gross injustice of wasting so much food while so many in the world go hungry. In the developing world, improving infrastructure along the food chain – including cold storage – would prevent much good food being lost. In the developed world, retailers can prevent large amounts of waste by finding outlets for slightly flawed or blemished goods, and consumers can limit waste by buying food in amounts they actually want and need.

F The complex, dynamic and widely diverse forms of the world's many food systems yield some wildly divergent outcomes in terms of nutrition, health, and environmental and climate impacts. Just as there's no universal crop that grows everywhere, there's no 'one size fits all' model food system to implement across the world. It is critical we start to better examine what works in some systems and what must be improved in others, in order to produce more just and sustainable outcomes around the world. It's time to look beyond farming and agriculture and to see the whole picture, to create systems that cause less harm to the climate and are more resilient to the impacts we're already suffering from global warming. Food is a fundamental human need and to eat is a basic human right. Our food systems must deliver that need, without worsening the impacts of climate change.

Questions 1–6

Reading Passage 2 has six paragraphs, **A–F**.

Which paragraph contains the following information? Write the correct letter, **A–F**.

NB You may use any letter more than once.

Q	Statement	
1	a figure indicating the environmental impact of agricultural practices	
2	an outline of two basic aims which food systems need to achieve	
3	an illustration of a drop in the amount of basic foodstuffs being produced	
4	a mention of a lack of public awareness of how food systems affect climate change	
5	a mention of the two-way relationship between farming and climate change	
6	a reference to how food systems need to vary according to region	

Questions 7–10

Complete the summary below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

THE PUBLIC'S PERCEPTION OF THEIR CONTRIBUTION TO CLIMATE CHANGE

People, especially in richer countries, are increasingly concerned about the impact of their personal (7) _____ on the environment. When thinking about how their lifestyles affect climate change, people tend to focus on the car they use or on the amount of (8) _____ that they consume. Not many people consider the environmental effects of what they eat, even though food systems are responsible for a large proportion of the (9) _____ being released. In fact, the impact of food systems exceeds the contribution to climate change of all (10) _____ throughout the world.

Questions 11 and 12

Choose **TWO** letters, **A–E**.

Which **TWO** facts about the forests does the writer mention?

OPTIONS

- A** how the presence of forests benefits the environment
- B** the proportion of the world's forests which have been destroyed
- C** the principal reason for which forests are cut down
- D** the species of crops which have the potential to protect forests
- E** how long it will take to reverse the damage caused by the forest clearances

Answer: (11) (12)

Questions 13 and 14

Choose **TWO** letters, **A–E**.

Which **TWO** facts about food waste does the writer mention?

OPTIONS

- A** the countries in the world which waste most food
- B** the percentage of food which is wasted by retailers
- C** the total annual proportion of food wasted worldwide
- D** the impact of food waste on global warming
- E** the rate at which global food waste is increasing each year

Answer: (13) (14)

Questions 15 and 16

Choose **TWO** letters, **A–E**.

Which **TWO** solutions to the problem of food waste does the writer mention?

OPTIONS

- A** selling products which have minor imperfections
- B** limiting the range of perishable goods on offer in retail outlets
- C** encouraging consumers to keep food for longer rather than discarding it
- D** shifting food production to countries whose inhabitants are undernourished
- E** increasing refrigeration facilities in poorer regions of the world

Answer: (15) (16)

Passage 39

The Impact of Invasive Species

How non-native organisms disrupt ecosystems, and what can be done about it

A Invasive species are among the leading threats to the native wildlife of most countries, with approximately 42 percent of endangered species at risk from them. Invasive species can be any kind of organism — for example, a mammal, amphibian, fish, insect or plant — that is not native to an ecosystem. Often they reproduce and spread with great speed. Contrary to popular belief, a plant or animal classified as an invasive species does not necessarily originate in another country. For example, lake trout are native to the Great Lakes of North America, but are considered to be an invasive species in Yellowstone Lake in Wyoming because they compete with native cutthroat trout for habitat.

B When a new species is introduced into an ecosystem, native wildlife may struggle to compete with it for food and other resources. Invasive species can change the food web in an ecosystem by destroying or replacing native food sources, while providing little or no food for local wildlife. In extreme cases, the invader may prey on the native species. Above all, invasive species threaten biodiversity in many habitats. For instance, the climbing plant species kudzu (which is native to East Asia) can easily replace a habitat that had a wide range of plants with a monoculture consisting solely of kudzu, as has started to happen in the south east of the USA.

C The phenomenon is not an exclusively modern one. Humans have always transferred a variety of species from one region to another, but the development of rapid means of transportation has increased the frequency of such introductions. Aquatic organisms can be shipped across the oceans, while insects can easily get into the wooden crates that are transported around the world in this way. In addition, climate change has enabled some invasive plant species to colonise new areas. Other invasive species include pets such as snakes or turtles which are intentionally set free into the wild.

D Invasive species do not all present the same level of threat to native ecosystems and can be classified into three types. The first of these may be introduced species which can maintain themselves in a limited range of habitats without upsetting the ecological equilibrium of the area. Some scientists have even argued that in these cases the introduction enhances the diversity of animal and plant life in that specific area. A second group of invaders present a greater threat because their spread is at the expense of one native species. The North American grey squirrel, for example, was introduced to the UK in 1876 largely because wealthy landowners thought it would be a fashionable and attractive addition to the local wildlife on their estates. It spread widely, leading directly to the dramatic decline in the population of the native red squirrel. However, it would appear that this is the only definite negative impact of the grey squirrel.

E There is a third level of threat in which the dominance of the introduced species has an extremely destructive effect on the entire ecosystem. One of the most damaging examples involved a species of comb jellyfish. Native to estuaries along the western Atlantic coast from the northern United States to the Valdés peninsula in Argentina, this species was released from a ship into the Black Sea in Eastern Europe in 1982, almost certainly by accident. The Black Sea has levels of industrial waste which are by international standards exceptionally high. Despite this, fishing boats were still able to catch healthy numbers of fish. But when the invading jellyfish underwent a population explosion in the space of just six years, the entire marine ecosystem was transformed, and fish numbers declined dramatically because they were in competition with the jellyfish, preying on exactly the same microscopic creatures. The jellyfish had a more serious impact on the ecosystem than the heavily polluted water.

F More than a century after its introduction outside its native range on the Amazon river in South America, a plant known as a water hyacinth can be found in tropical lakes, streams, and rivers around the world. Its beauty attracted botanists seeking exotic plants for botanical gardens and they imported it to a horticultural exhibition in New Orleans in 1884. Visitors were so impressed they planted it in many locations during the 1880s and 1890s, resulting in the aquatic ecosystems of the south eastern United States being progressively colonised by vast, floating, dense carpets of water hyacinth. Today it is present around the globe, damaging boat engines and even blocking cooling pipes for power plants, occasionally leading to massive blackouts. The impact of the water hyacinth on native water plants is largely unstudied, as is unfortunately the case for most ecosystems invaded by new plant species.

G In the United States, more than 7,000 introduced species have established themselves, of which at least 15 percent cause ecological damage. As the number of invasive species expands, legislation to deal with this problem is rare or non-existent in the majority of countries. Unfortunately, ordinary people outside the scientific community have a very limited understanding of the threat posed by invasive species, which means that other environmental threats receive considerably more media attention. The introduction of new species can initially seem highly desirable, but the full extent of their impact is consistently underestimated.

H Although ultimately measures need to be taken at an international level, limited action is possible by individuals. One way is for people to plant native plants in their gardens rather than species from abroad. It is also useful to learn to identify invasive species, and report any sightings to wildlife organisations. Regularly cleaning clothing, boots, boats, tires, and any other equipment regularly used outdoors can remove insects and plant parts that may introduce invasive species into new locations.

Questions 14–19

Reading Passage 2 has eight paragraphs, **A–H**. Which paragraph contains the following information? Write the correct letter, **A–H**.

Q	Statement	
14	a suggestion that people have moved numerous species across the globe throughout history	
15	an example of how an ecosystem can be damaged very rapidly	
16	a description of what can be done to restrict the spread of invasive species	
17	a reference to the lack of research on the effects of some invasive species	
18	a mention of a current lack of public awareness of the problem of invasive species	
19	that an introduced species may benefit a specific ecosystem	

Questions 20–23

Choose **TWO** letters for each question.

20–21. Which **TWO** reasons for the spread of invasive species are mentioned in the text?

OPTIONS

- A** the wish to eliminate undesirable native species
- B** the recent expansion of international trade in agricultural products
- C** a lack of checks on some of the cargo on board ships
- D** the deliberate release of non-native animals
- E** an extension of their geographical range as a result of global warming

Answer: (20) (21)

22–23. Which **TWO** statements does the writer make about the water hyacinth?

OPTIONS

- A** It is native to almost every region of the world.
- B** It was brought to North America in the late nineteenth century.
- C** Its beauty has led people to ignore the negative effects it has.
- D** Its spread has caused some practical problems in recent years.
- E** Scientists recommended its introduction to the USA.

Answer: (22) (23)

Questions 24–26

Complete the sentences below. Choose **ONE WORD ONLY** from the passage for each answer.

Kudzu has reduced the (24) _____ of certain areas in the south east of the USA.

Some introduced species present a low level of threat if they remain within a small area and do not disturb the (25) _____ of the surrounding ecosystem.

The effect of invasive jellyfish in the Black Sea was greater than that of (26) _____ from factories.

Passage 40

The Importance of Law

Why legal understanding shapes almost every part of modern life — and why lawyers deserve a fairer hearing

⑭ **A** The law influences all of us virtually all the time, it governs almost all aspects of our behaviour, and even what happens to us when we are no longer alive. It affects us from the embryo onwards. It governs the air we breathe, the food and drink we consume, our travel, family relationships, and our property. It applies at the bottom of the ocean and in space. Each time we examine a label on a food product, engage in work as an employee or employer, travel on the roads, go to school to learn or to teach, stay in a hotel, borrow a library book, create or dissolve a commercial company, play sports, or engage the services of someone for anything from plumbing a sink to planning a city, we are in the world of law.

⑮ **B** Law has also become much more widely recognised as the standard by which behaviour needs to be judged. A very telling development in recent history is the way in which the idea of law has permeated all parts of social life. The universal standard of whether something is socially tolerated is progressively becoming whether it is legal, rather than something that has always been considered acceptable. In earlier times, most people were illiterate. Today, by contrast, a vast number of people can read, and it is becoming easier for people to take an interest in law, and for the general population to help actually shape the law in many countries. However, law is a versatile instrument that can be used equally well for the improvement or the degradation of humanity.

⑯ **C** This, of course, puts law in a very significant position. In our rapidly developing world, all sorts of skills and knowledge are valuable. Those people, for example, with knowledge of computers, the internet, and communications technology are relied upon by the rest of us. There is now someone with IT skills or an IT help desk in every UK school, every company, every hospital, every local and central government office. Without their knowledge, many parts of commercial and social life today would seize up in minutes. But legal understanding is just as vital and as universally needed. The American comedian Jerry Seinfeld put it like this, 'We are all throwing the dice, playing the game, moving our pieces around the board, but if there is a problem, the lawyer is the only person who has read the inside of the top of the box.' In other words, the lawyer is the only person who has read and made sense of the rules.

⑰ **D** The number of laws has never been greater. In the UK alone, about 35 new Acts of Parliament are produced every year, thereby delivering thousands of new rules. The legislative output of the British Parliament has more than doubled in recent times from 1,100 pages a year in the early 1970s, to over 2,500 pages a year today. Between 1997 and

2006, the legislature passed 365 Acts of Parliament and more than 32,000 legally binding statutory instruments. In a system with so much law, lawyers do a great deal not just to vindicate the rights of citizens and organizations but also to help develop the law through legal arguments, some of which are adapted by judges to become laws. Law courts can and do produce new law and revise old law, but they do so having heard the arguments of lawyers.

18 E However, despite their important role in developing the rules, lawyers are not universally admired. Anti-lawyer jokes have a long history going back to the ancient Greeks. More recently the son of a famous Hollywood actor was asked at his junior school what his father did for a living, to which he replied, 'My daddy is a movie actor, and sometimes he plays the good guy, and sometimes he plays the lawyer.' For balance, though, it is worth remembering that there are and have been many heroic and revered lawyers such as the Roman philosopher and politician Cicero and Mahatma Gandhi, the Indian campaigner for independence.

19 F People sometimes make comments that characterise lawyers as professionals whose concerns put personal reward above truth, or who gain financially from misfortune. There are undoubtedly lawyers that would fit that bill, just as there are some scientists, journalists and others in that category. But, in general, it is no more just to say that lawyers are bad because they make a living from people's problems than it is to make the same accusation in respect of nurses or IT consultants. A great many lawyers are involved in public law work, such as that involving civil liberties, housing and other issues. Such work is not lavishly remunerated and the quality of the service provided by these lawyers relies on considerable professional dedication. Moreover, much legal work has nothing to do with conflict or misfortune, but is primarily concerned with drafting documents. Another source of social disaffection for lawyers, and disaffection for the law, is a limited public understanding of how law works and how it could be changed. Greater clarity about these issues, maybe as a result of better public relations, would reduce many aspects of public dissatisfaction with the law.

Questions 14–19

The Reading Passage has six paragraphs, A–F.

Choose the correct heading for each paragraph from the list of headings below. Write the correct number, i–ix, in the circle before each paragraph.

- i Different areas of professional expertise
- ii A negative regard for lawyers
- iii The disadvantages of the legal system

- iv The law applies throughout our lives
- v The law has affected historical events
- vi Growth in laws
- vii Public's increasing ability to influence the law
- viii Reasons why it is unfair to criticise lawyers

Questions 20 and 21

Choose **TWO** letters, **A–E**. Which **TWO** of the following statements does the writer make about legal skills in today's world?

20

- A There should be a person with legal training in every hospital.
- B Lawyers with experience in commercial law are the most in demand.
- C Knowledge of the law is as important as having computer skills.
- D Society could not function effectively without legal experts.
- E Schools should teach students about the law.

21

- A There should be a person with legal training in every hospital.
- B Lawyers with experience in commercial law are the most in demand.
- C Knowledge of the law is as important as having computer skills.
- D Society could not function effectively without legal experts.
- E Schools should teach students about the law.

Questions 22–26

Complete the summary below. Choose **ONE WORD ONLY** from the passage for each answer.

LAWYERS AS PROFESSIONALS

People sometimes say that (22) _____ is of little interest to lawyers, who are more concerned with making money. This may well be the case with some individuals, in the same way that some (23) _____ or scientific experts may also be driven purely by financial greed. However, criticising lawyers because their work is concerned with people's problems would be similar to attacking IT staff or (24) _____ for the same reason. In fact, many lawyers focus on questions relating, for example, to housing or civil liberties, which requires them to have (25) _____ to their work. What's more, a lot of lawyers' time is spent writing (26) _____ rather than dealing with people's misfortunes.

Passage 41

The International Style

How a stripped-down, functional architectural movement came to dominate – and then fall out of favour with – the modern city skyline

A In the early decades of the 20th century, many Western cities experienced a steep rise in demand for commercial and civic premises, due to population growth and expansion of the white-collar professions. At the same time, architects were growing discontented with the ornamental spirals and decorative features in the prevailing design ethos of art deco or art moderne. Once considered the height of sophistication, these styles were quickly becoming seen as pretentious and old-fashioned. In this confluence of movements, a new style of architecture emerged. It was simple, practical and strong; a new look for the modern city and the modern man. It was named ‘the international style’.

B Although the international style first emerged in Western Europe in the 1920s, it found its fullest expression in American architecture and was given its name in a 1932 book of the same title. The first hints of it in America can be seen on the Empire State Building in New York City, which was completed in 1931. The top of the building, with its tapered crown, is decidedly art deco, yet the uniform shaft of the lower two thirds represents a pronounced step in a new direction. Later efforts, such as the United Nations Secretariat building (1952) and the Seagram Building (1954) came to exemplify the ‘true’ international style.

C The architects of the international style broke with the past by rejecting virtually all nonessential ornamentation. They created blockish, flat-roofed skyscrapers using steel, stone and glass. A typical building facade in this style has an instantly recognisable ribbon design, characterised by strips of floor-to-ceiling windows separated by strips of metal panelling. Interiors showcased open spaces and fluid movements between separate areas of the building.

D Fans of the international style of modern buildings celebrated their sleek and economical contribution to modern cityscapes. While pre-modern architecture was typically designed to display the wealth and prestige of its landlords or occupants, the international style in some ways exhibited a more egalitarian tendency. As every building and every floor looked much the same, there was little attempt to use these designs to make a statement. This focus on function and practicality reflected a desire in mid-century Western cities to ‘get on with business’ and ‘give everyone a chance’, rather than lauding the dominant and influential institutions of the day through features such as Romanesque columns.

E Detractors, however, condemned these buildings for showing little in the way of human spirit or creativity. For them, the international style represented not an ethos of equality and progress, but an obsession with profit and ‘the bottom line’ that removed spiritual and creative elements from public life and public buildings. Under the dominance of the international style, cities became places to work and do business, but not to express one’s desires or show individuality. It is perhaps telling that while banks and government departments favoured the international style, arts organisations rarely opted for its austerity.

F By the mid-1970s, the international style was ubiquitous across key urban centres, dominating skylines to such an extent that many travellers complained they could get off a plane and not know where they were. By their nature, buildings in this style demanded very little of architects in the way of imagination, and a younger generation of designers was yearning to express their ideas and experiment in novel and unexpected ways. The outcome was a shift toward postmodernism, which celebrated much of what the international style had dismissed: decoration, style without function, and an overall sense of levity. By the turn of the 1980s, the international style was considered outdated and was falling rapidly out of favour.

Questions 1–6

Reading Passage 2 has six paragraphs, **A–F**.

Which paragraph contains the following information? Write the correct letter, **A–F**.

Q	Statement	
1	a description of how international style buildings look on the inside	
2	a reference to institutions that didn't like to use international style buildings	
3	a reason why architects didn't like the international style	
4	a building which combined art deco and international features	
5	types of materials commonly used in international style buildings	
6	an architectural feature previously associated with prominent organisations	

Questions 7–11

Complete the sentences below.

Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

- 7 The development of the international style was prompted by an increased need for _____ buildings.
- 8 Designers used hardly any _____ on international style buildings.
- 9 International style buildings are easily identified from the outside because of the _____.
- 10 Demonstration of _____ was often an important factor in the design of old buildings.
- 11 The similarity of international style constructions reflects the concern of architects with _____.

Questions 12 and 13

Choose the correct letter, A, B, C or D.

- 12 Some people didn't like the international style because they felt it focused too much on
 - A the public sector
 - B differences between people
 - C new ideas
 - D making money
- 13 In the mid-1970s
 - A the best architects were no longer using the international style
 - B there was a lot of international style architecture in major cities
 - C young architects were becoming interested in the international style
 - D people visited specifically to see international style buildings

Passage 42

The Lost City

Sally Palmer reports on how remote-sensing techniques are revealing the secrets of a ruined city in Turkey

A The low granite mountain, known as Kerkenes Dag, juts from the northern edge of the Cappadocian plain in Turkey. Sprawled over the mountainside are the ruins of an enormous city, contained by crumbling defensive walls seven kilometers long. Many respected archaeologists believe these are the remains of the fabled city of Pteria, the sixth-century BC stronghold of the Medes that the Greek historian Herodotus described in his famous work *The Histories*. The short-lived city came under Median control and only fifty years later was sacked, burned and its strong stone walls destroyed.

B British archaeologist Dr Geoffrey Summers has spent ten years studying the site. Excavating the ruins is a challenge because of the vast area they cover. The 7 km perimeter walls run around a site covering 271 hectares. Dr Summers quickly realized it would take far too long to excavate the site using traditional techniques alone. So he decided to use modern technology as well to map the entire site, both above and beneath the surface, to locate the most interesting areas and priorities to start digging.

C In 1993, Dr Summers hired a special hand-held balloon with a remote-controlled camera attached. He walked over the entire site holding the balloon and taking photos. Then one afternoon, he rented a hot-air balloon and floated over the site, taking yet more pictures. By the end of the 1994 season, Dr Summers and his team had a jigsaw of aerial photographs of the whole site. The next stage was to use remote sensing, which would let them work out what lay below the intriguing outlines and ruined walls. "Archaeology is a discipline that lends itself very well to remote sensing because it revolves around space," says Scott Branting, an associated director of the project. He started working with Dr Summers in 1995.

D The project used two main remote-sensing techniques. The first is magnetometry, which works on the principle that magnetic fields at the surface of the Earth are influenced by what is buried beneath. It measures localised variations in the direction and intensity of this magnetic field. "The Earth's magnetic field can vary from place to place, depending on what happened there in the past," says Branting. "If something containing iron oxide was heavily burnt, by natural or human actions, the iron particles in it can be permanently reoriented, like a compass needle, to align with the Earth's magnetic field present at that point in time and space." The magnetometer detects differences in the orientations and intensities of these iron particles from the present-day magnetic field and uses them to produce an image of what lies below ground.

E Kerkenes Dag lends itself particularly well to magnetometry because it was all burnt once in a savage fire. In places the heat was sufficient to turn sandstone to glass and to melt granite. The fire was so hot that there were strong magnetic signatures set to the Earth's magnetic field from the time – around 547 BC – resulting in extremely clear pictures. Furthermore, the city was never rebuilt. “If you have multiple layers, it can confuse pictures, because you have different walls from different periods giving signatures that all go in different directions,” says Branting. “We only have one going down about 1.5 meters, so we can get a good picture of this fairly short-lived city.”

F The other main sub-surface mapping technique, which is still being used at the site, is resistivity. This technique measures the way electrical pulses are conducted through sub-surface soil. It's done by shooting pulses into the ground through a thin metal probe. Different materials have different electrical conductivity. For example, stone and mudbrick are poor conductors, but looser, damp soil conducts very well. By walking around the site and taking about four readings per metre, it is possible to get a detailed idea of what is where beneath the surface. The teams then build up pictures of walls, hearths and other remains. “It helps a lot if it has rained, because the electrical pulse can get through more easily,” says Branting. “Then if something is more resistant, it really shows up.” This is one of the reasons that the project has a spring season, when most of the resistivity work is done. Unfortunately, testing resistivity is a lot slower than magnetometry. “If we did resistivity over the whole site it would take about 100 years,” says Branting. Consequently, the team is concentrating on areas where they want to clarify pictures from the magnetometry.

G Remote sensing does not reveal everything about Kerkenes Dag, but it shows the most interesting sub-surface areas of the site. The archaeologists can then excavate these using traditional techniques. One surprise came when they dug out one of the gates in the defensive walls. “Our observations in early seasons led us to assume that we were looking at a stone base from a mudbrick city wall, such as would be found at most other cities in the Ancient Near East,” says Dr Summers. “When we started to excavate, we were staggered to discover that the walls were made entirely from stone and that the gate would have stood at least ten metres high.” After ten years of study, Pteria is gradually giving up its secrets.

Questions 1–4

Which paragraph contains the following information?

Write the correct letter, A–G.

Q	Statement	
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1	the reason for the deployment of a variety of investigative methods	
2	an example of an unexpected find	
3	how the surface of the site was surveyed from above	
4	the reason why experts are interested in the site	

Questions 5–12

Complete the summary below.

Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

Exploring the ancient city of Pteria. Archaeologists began working ten years ago. They started by taking photographs of the site from the ground and then from a distance in a (5) _____. They focused on what lay below the surface using a magnetometer, which identifies variations in the magnetic field. These variations occur when the (6) _____ in buried structures have changed direction as a result of great heat. They line up with the surrounding magnetic field just as a (7) _____ would do. The other remote-sensing technique employed was resistivity. This uses a (8) _____ to fire electrical pulses into the earth. The principle is that building materials like (9) _____ and stone do not conduct electricity well, while (10) _____ does this much more effectively. This technique is mainly employed during the (11) _____, when conditions are more favourable. Resistivity is mainly being used to (12) _____ some images generated by the magnetometer.

Question 13

Choose the correct letter, **A, B, C or D**.

- (13) How do modern remote-sensing techniques help at the Pteria site?
- A They detect minute buried objects for the archaeologists to dig up.
 - B They pinpoint key areas, which would be worth investigating closely.
 - C They remove the need for archaeologists to excavate any part of the site.
 - D They extend the research periods as they can be used at any time of year.

The Mbuti of the Ituri Forest

How hunter-gatherers and villagers in Zaire depend on each other for survival

① **A** For perhaps two millennia, the Ituri Forest of northern Zaire in Africa has been the home for both Mbuti hunter-gatherers and Bantu village-dwelling cultivators. The cultivators are settled in isolated villages using only small areas of the forest. The population density in the forest including both Mbuti and cultivators is low, averaging less than three persons per km². More than 70% of the forest has probably never been cut and large areas remain uninhabited. Although never totally out of touch with villagers, Mbuti spend a great deal of time in the uncultivated parts of the forest. They are considered by all, including themselves, as the guardians of the forest, being the most familiar with the forest and the most efficient at extracting its resources. This alone, however, is not enough to assure Mbuti security. All Mbuti live as forest specialists in association with cultivators. Such interaction is key to their welfare and their future.

② **B** Economic aspects of the reciprocity between Mbuti and villagers have been described in some detail. Starch foods from villagers' gardens make up a significant part of Mbuti diet year-round. In turn, Mbuti provide villagers on an irregular basis with 'prestige foods', for example honey from the forest. They help clear and harvest the gardens and participate in seasonal fishing expeditions. Mbuti also provide diverse forest supplies, such as thatching and construction materials, firewood and medicinal plants. Trade relations between Mbuti and the villagers are not rigidly defined; there are no fixed terms. In most cases no cash is involved, and donations may even take the form of gifts. Mbuti, as nomads, are highly unpredictable in what they will provide and when they will provide it. Villagers, on the other hand, have a fairly predictable yearly cycle of activities centered on the preparation, planting and harvesting of their gardens.

③ **C** Important elements in Mbuti-villager reciprocity extend beyond its economic base. For lack of a better term, these elements might be categorized as 'cultural reciprocity'. Under various circumstances, Mbuti use the more hierarchical and formalized social structure of villager societies. For example, in the southern and central Ituri Forest, villagers play a prominent role in Mbuti marriages and deaths, and in mediating outbreaks of conflict among Mbuti. Although Mbuti themselves will decide who will marry whom, and although they may not abide by the recommended terms in cases of arbitration, they nevertheless turn to villagers when wealth or formality is needed. Villagers often organize the resources needed for a wedding feast and delegate responsibilities at a funeral. Even if a death happens in a remote forest camp, the burial takes place in the village. Likewise, individuals involved in any crime occurring in the forest are brought by Mbuti to a village tribunal.

4 D At other times, the villagers use Mbuti to act as intermediaries with the forest. Just as Mbuti use the more elaborate political structures of villagers to regulate certain aspects of their own social system, so villagers require the spiritual unity Mbuti have with the forest to assure their own relationship with the land. In the harvest festivals of cultivators of the southern Ituri, Mbuti even take precedence over village ritual authorities. At the start of the harvest, the first crop of bananas is for the ancestors and is left to rot on the plant. The second goes to Mbuti, and only the third is for the mwami (king). Mbuti ease and independence in the forest is acknowledged to be a manifestation of their special relationship with the forest, a unity which the villagers cannot achieve in their more hierarchical, agriculturally-orientated lives. Mbuti participate at major ceremonies, including coronation ceremonies of the mwami, and the opening of new barazza (men's communal house).

5 E The exchange relationship between Mbuti and villagers has been flexible enough to survive a number of migrations and political upheavals in the past, and the Bantu are not the only villagers to benefit from this relationship. Immigrants from north Africa entered the forest more than a century ago, and their descendants, the Baungwana, remain there today. The Baungwana are a Swahili-speaking agricultural group, who have not only established themselves in the forest, but have had Mbuti associated with their villages for at least three or four generations. These relatively recent relationships resemble the established Mbuti-Bantu interactions in many ways. The Baungwana come from a different cultural background, but in the Ituri Forest they are also small-scale, subsistence cultivators. Not only do they need forest products, but also, even more so than the indigenous forest villagers, they see the forest as powerful and alien. Therefore, they too value Mbuti for their forest skills.

Questions 1–5

Reading Passage 2 has five paragraphs, A–E.

Choose the correct heading for paragraphs A–E from the list of headings below. Write the correct number, i–ix, in the circle before each paragraph.

- i the symbolic status of the Mbuti
- ii new methods of agriculture
- iii the forest and its inhabitants
- iv the exchange of produce
- v a potential conflict of interests
- vi traditions which survive despite changing circumstances

vii assistance given to Mbuti by villagers on important occasions

Questions 6–9

Classify the following statements as referring to

A Mbuti only B Villagers only C Both Mbuti and villagers

- 6 they depend on another group to fulfill certain needs.
A Mbuti only
B Villagers only
C Both Mbuti and villagers
- 7 their lives generally follow a pattern.
A Mbuti only
B Villagers only
C Both Mbuti and villagers
- 8 they have probably inhabited the forest for two thousand years
A Mbuti only
B Villagers only
C Both Mbuti and villagers
- 9 their superior forest-management abilities are generally acknowledged.
A Mbuti only
B Villagers only
C Both Mbuti and villagers

Questions 10–13

Complete the sentences below.

*Write **ONE WORD ONLY** from the passage for each answer.*

- 10 For every square kilometre of forest there are under _____ inhabitants.
- 11 Mbuti occasionally provide villagers with luxuries such as _____.
- 12 When trading goods, Mbuti and the villagers do not often use _____.
- 13 Mbuti have the priority over the villagers' king at celebrations taking place at the time of the _____.

Passage 44

The Museum Dilemma

Can museums and art galleries make works of art both accessible to the public and protected at the same time?

On any day of the week, tourists flock to museums and galleries such as the Louvre in Paris and The Met in New York, willingly paying their steep entrance fees. This is in stark contrast to the 55,000 local museums around the globe, who often struggle when it comes to attracting visitors and the resulting much-needed funds. These institutions may be a source of pride to locals, but are too often perceived as dusty cabinets – useful when it comes to storing ancient things, but not very interesting to look at. The constant dilemma for museum curators is that increasing visitor numbers also brings a far greater chance of damage, which can be deliberate, incidental or accidental.

Deliberate damage cannot be controlled for, as the perpetrators act in a determined and destructive way. This was the case with the Leonardo da Vinci cartoon damaged by gunshot in the National Gallery in London in 1987, despite the painting being protected behind a glass screen. Incidental damage is easier to anticipate and often results from a visitor's innate curiosity and instinctive urge to touch. This is mainly managed through the use of signage, gallery attendants, or a physical barrier such as a rope. Gallery attendants are the more expensive but preferred option as written warnings tend to be ignored. However, cutbacks in funding mean fewer and fewer attendants.

Rope barriers are commonplace but their very nature renders them ineffective – visitors can still get quite close to a painting. It was in recognition of this that staff at Huashan 1914 Creative Park in Taipei decided to place a raised platform between the barrier and a valuable 17th century oil painting as a reminder to visitors to not get too close.

Unfortunately, the platform inadvertently contributed to extensive damage when a young boy tripped on it and put his hand through the painting while trying to break his fall.

Many galleries would rather not use barriers at all because they tend to spoil the overall look, and some have resorted to technology to get around this issue. In the past, museums such as the Stedelijk Museum in Amsterdam used alarms triggered by lasers to alert visitors to their proximity to a painting. Perhaps unsurprisingly, the trade-off between aesthetics and noise means this idea has not been taken up universally.

Nevertheless, technology does appear to solve the issues of attracting visitors, and there are more interactive displays popping up. Even so, incorporating the occasional exhibit that encourages visitor engagement can mean that visitors then assume it is acceptable to get up close and personal with all works of art. The ensuing surface damage is often not immediately apparent, but when thousands of subsequent visitors reach out to touch a

beautiful sculpture, the effect builds up cumulatively causing irreversible damage to the patina of materials such as bronze. While the surface of a painting does not offer quite the same tactile allure, they too can suffer similar consequences.

Despite these challenges, technology use in museums seems likely to grow, especially as they are uniquely placed to take advantage of augmented reality, where real people and ancient artefacts can be juxtaposed in a virtual world. Tools such as these are being used to construct exciting experiences that can bring a dull museum visit to life while also keeping visitors away from precious and fragile objects.

One such example is an augmented reality project that was initially trialled at White Sands National Monument in New Mexico with great success. Staff there showed visitors how, by scanning a code with a smart phone camera, a troop of mammoths would appear to walk over the horizon. Curiously though, when the same idea was later deployed with The Etches Collection, an exhibit in Dorset, on Britain's Jurassic Coast, no one engaged with it. Rather than a lack of interest in the technology, the failure appears to have been due to the reluctance of visitors to download the museum's app onto their own phone. The best technology in the world can't fix that.

A key challenge is the lack of insight into what visitors actually want and expect from a museum visit, and a recent study at several cultural sites in Scotland has tried to provide this. Through questionnaires and interviews, researchers made some surprising discoveries. While it had been assumed that more visual experiences would need less narrative, the study shows the opposite is true: visitors still see information about the place as important, whether the experience is virtual or not. Interestingly, although audiences do enjoy immersive visitor attractions, if an exhibition is purely a simulation, they like to be able to handle objects at the same time for extra realism, such as at Culloden Battlefield, whose visitor centre has artefacts such as 18th-century weapons.

The clear message is that, although technology has much to offer the museums, and is arguably essential to their survival, there is clearly some way still to go. It can bring museums to life to the benefit rather than the detriment of the precious artefacts and artworks they are home to, but only if it gives visitors the experience they want.

Questions 1–6

Look at the following museums and galleries (Questions 1–6) and the list of comments below.

Match each place with the correct comment, A–J.

LIST OF COMMENTS

- A** This museum shows the benefits of using a physical barrier
- B** An attempt to prevent damage had the opposite effect
- C** This is a good example of the importance of alarm systems
- D** This experience shows the potential benefits of technology for museums
- E** Holding something real helps improve a virtual experience
- F** An occurrence here shows that certain damage cannot be prevented
- G** This shows that technology will only benefit museums if it is used
- H** Some people prefer to have a real experience rather than a virtual one
- I** A system used to protect artwork proved to be unpopular elsewhere
- J** Visitors are not at all interested in using technology

Q	Statement	
1	National Gallery, London	
2	Huashan 1914 Creative Park, Taipei	
3	Stedelijk Museum, Amsterdam	
4	White Sands National Monument, New Mexico	
5	The Etches Collection, Dorset	
6	Culloden Battlefield, Scotland	

Questions 7–13

Do the following statements agree with the claims of the writer in Reading Passage 2?

Write **YES** if the statement agrees with the claims of the writer, **NO** if it contradicts them, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.

- 7 Even the most famous museums in the world generally struggle for funds.
- 8 The majority of people admit they have never visited their local museum.
- 9 Putting up written signs is the best way to avoid incidental damage.
- 10 Rope barriers have been shown to cause visitors to trip.
- 11 Some interactive exhibits can lead to more incidental damage.
- 12 Surface damage to paintings is very different to that of sculptures.
- 13 Even with simulated experiences, visitors still want to be told the history of a place.

Passage 45

The Plan to Bring an Asteroid to Earth

Moving in orbit around our sun are millions of rocks known as asteroids. Now scientists have plans to capture one.

14 **A** Send a robot into space, catch an asteroid and bring it back to Earth's orbit. This may sound like science fiction, but it is being taken seriously by scientists and engineers at the California Institute of Technology (Caltech). A four-day workshop was dedicated to investigating the feasibility of capturing a near-Earth asteroid, bringing it closer to our planet and using it as a base for future manned spaceflight missions. This is not something the scientists are imagining could be done some day off in the future. It is possible with the technology we have today and could be accomplished within a decade. A robotic probe could anchor to an asteroid with simple magnets, or with a specialized spacecraft with claws. Alternatively, a large spacecraft could use its gravitational field to shift the orbit of a larger asteroid, sending it towards Earth. 'Once you get over the initial reaction – You want to go what?! – it actually starts to seem like a reasonable idea,' said project leader John Brophy, who helped organize the workshop. In fact, many of these ideas have been on the drawing board for years as part of the NASA planetary defense program, which is aimed at large space-based objects that might threaten Earth. There's no shortage of potential targets. NASA estimates there are 19,500 asteroids at least 100 meters wide within 45 million kilometers of Earth.

15 **B** Though rearranging the heavens may seem an excessive undertaking, the US mission has its merits. The US already has plans to send astronauts to a near-Earth asteroid, a mission that would mean confining them in a tiny capsule for three to six months and involve all the risks of deep-space voyage. Instead, robots could bring an asteroid close enough for them to get there in just a month. An asteroid would provide a stationary base from which to launch missions farther into space. There are several advantages to this. For one, launching missions carrying materials from Earth requires a lot of power, fuel, and consequently money, because of our planet's strong force of gravity. Since this would be far weaker on an asteroid, materials mined there could be more easily taken off the asteroid and shuttled around the solar system. And many asteroids have a lot to offer. Some are full of metals, which can be mined and used to build space-based habitats, or brought back to Earth. Others are up to one-quarter water, which would either be used for life-support or broken down into hydrogen and oxygen to make fuel. And astronomers would have the chance to get a close-up look at one of the solar system's earliest relics, generating important scientific data. 'Executing the asteroid retrieval plan would help demonstrate and greatly expand mankind's space-based engineering capabilities,' says engineer Louis Friedman, another co-organizer of the Caltech workshop.

'For instance, the mission would teach engineers how to capture an uncooperative target, which could be useful practice for planetary defense missions in the event of a threat from a meteoroid or comet from space approaching our planet,' he adds.

16 C Former astronaut Rusty Schweickart, cofounder of the B612 Foundation, an organization dedicated to protecting Earth from asteroid strikes, points out that though it would be technically feasible, shifting such a hefty and substantial target would not be easy: 'You're moving the largest mother lode imaginable,' he says. Engineers would need to be absolutely certain they could control such a potentially dangerous object. 'It's the opposite of planetary defense, if you do something wrong you have a Tunguska event,' says engineer Marco Tantardini from the Planetary Society, referring to the powerful 1908 explosion above a remote Russian region thought to have been caused by a meteoroid or comet.

17 D Still, these obstacles only add to the appeal of the project for engineers, who love to go over every potential difficulty in order to solve it. And if the challenges for a large asteroid seem too daunting, researchers could always start with a smaller asteroid perhaps 2 to 10 meters across. Last year, Brophy helped conduct a study to look at the feasibility of bringing a two-meter, 1,000 kilogram asteroid – of which there might conceivably be millions – to the International Space Station. The study suggested the asteroid could be captured robotically in something as simple as a large bag. Of course, such a small object might not have the same emotional impact as a larger target. 'NASA isn't going to want to go to something that is smaller than our spaceships,' says engineer Dan Mazanek from NASA Langley Research Center.

18 E No matter the size of the asteroid, these plans would require hefty investments. Even capturing a small asteroid would consume at least a billion dollars. Convincing taxpayers to foot such a bill could be difficult. However, private industry might be interested in getting involved. One possibility would be to push the asteroid to near Earth orbit and then invite anyone who wants to develop the capabilities to reach and mine the object. However, though the undertaking might be scientifically exciting, and provide great insight into the solar system formation, this is not enough on its own to justify the expense of bringing an asteroid to Earth. Investigations of asteroids can be done much more cheaply with an unmanned spacecraft, says chemist Joseph A Nuth from NASA's Goddard Space Flight Center. According to Brophy, ultimately, we would be working towards bringing an asteroid closer to Earth in order to help move out into the solar system.

Questions 14–18

Reading Passage 2 has five paragraphs, A–E.

Choose the correct heading for each paragraph from the list of headings below. Write the correct number, i–ix, in the circle before each paragraph.

- i The need for skill and care
- ii Choosing the richest asteroid
- iii The safest way to protect an asteroid
- iv Obtaining support for the project
- v An achievable goal, not an impossible dream
- vi The need for global cooperation
- vii Beginning with a less challenging task
- viii Practical, economic and research justifications

Questions 19–22

Look at the following experts (Questions 19–22) and the list of statements below.

Match each expert with the correct statement, A–G.

LIST OF STATEMENTS

- A A mistake could have serious consequences for Earth.
- B It might be difficult to arouse interest in an asteroid of limited size.
- C The project could be an early step in space exploration.
- D An asteroid's weight makes the project extremely challenging.
- E The skill gained could save Earth from future danger.
- F An asteroid could be a useful landing place for a spaceship.
- G Capturing an asteroid would not be an efficient method of research.

Q	Statement	
19	Louis Friedman	
20	Rusty Schweickart	
21	Dan Mazanek	
22	Joseph A Nuth	

Questions 23–26

Complete the summary below. Choose **ONE WORD ONLY** from the passage for each answer.

THE MERITS OF THE US MISSION

Capture of an asteroid would reduce the time that astronauts travelling to it needed to spend in space. An asteroid could also act as a (23) _____ for further space travel and exploration. The fact that an asteroid would have weaker (24) _____ would allow easier movement of resources.

These resources include (25) _____ which could be used in space or on Earth and (26) _____ which could be consumed or used as a source of power.

Passage 46

The Power of Music

Robert Matthews looks at research into the effects of music

A Music is becoming ever more popular electronically. To meet our craving for music, internet sites are using increasingly sophisticated ways of putting us in touch with artists we may not even know we like. Most work by trawling our existing files or online listening habits and looking for patterns so they can recommend new artists for their subscribers to listen to. The search often turns up surprises. But is it possible to tease apart our likes and dislikes to identify precisely what it is about some music that thrills us or leaves us cold?

B For centuries composers have sought to create unforgettable music using accepted notions about the emotional appeal of certain combinations of sounds, yet only now are scientists starting to uncover what it is about these combinations that can have such a dramatic effect on our minds. Given that archaeologists have found musical instruments played by Neanderthals at least 50,000 years ago, why have scientists taken so long to investigate such a source of pleasure?

C 'For psychologists, who are always desperate to show that their work is rigorous, there's an image problem in tackling the emotionality of music,' says Professor Norman Cook of Kansai University in Osaka, Japan, one of the pioneers of the new science of music. 'Emotion is such a slippery topic.' The other problem, says Cook, is the long-standing principle among psychologists that our response to music is an acquired one, rather than something that is stimulated by the effect of sound on our brain cells. Yet one of the first insights to emerge from this new branch of psychology is that music affects our brains at a very basic level.

D Together with his colleague, Professor Takefumi Hayashi, Cook has been investigating one of the best-known examples of the emotional impact of music: the difference between major and minor chords. For centuries, composers have known that notes arranged to form major chords sound happy and upbeat, while those in minor chords sound mournful. In tests, even three-year-olds have been shown to link music in a major mode to happy faces and minor modes to sad faces.

E According to Cook, analysis of how people respond to notes suggests a link with how our brains interpret certain sounds in everyday life. He points out that sad-sounding minor chords can be formed by raising the pitch of any of a set of notes, while dropping the pitch produces a major chord. The same change in pitch works as an emotional telltale in communication between some mammals, where rising pitch is used to communicate

weakness or defeat, while falling pitch signals social dominance. It's also present in our speech. 'A rising inflection is used to denote questions, politeness or deference, whereas a falling inflection signals dominance,' says Cook.

F This suggests that music in major and minor modes taps into some very basic features of how we relate to the world and each other — perhaps dating back millions of years. Could music in general be doing something similar? Quite possibly, according to research into how music triggers certain types of brain activity. At McGill University in Canada, Professor Robert Zatorre and his colleagues have carried out studies in which volunteers listen to different types of music while their brain activity is monitored. 'The biggest surprise was the evidence that pleasurable music activates brain circuitry which has been in existence in the human brain for thousands of years,' says Zatorre. 'We share it with rats and other distant relatives on the evolutionary tree — and it's typically associated with biological rewards, like food, for example.'

G At the University of Oxford, Dr Joyce Chen has been looking into another celebrated feature of music — the irresistibility of rhythm. Her interest was sparked by studies involving patients with movement difficulties. 'If music that had a strong rhythm — say, a marching band — was played to these patients, they were able to improve their walking ability,' says Chen. In an attempt to find out why the simple act of listening to music might help disabled patients, Dr Chen and colleagues from the International Laboratory for Brain, Music and Sound Research in Montreal carried out brain scans on volunteers who were listening to rhythmic sounds. The criteria for selecting these volunteers were that they should be in first-rate physical health but musically untrained. The results have been another revelation. Chen and her colleagues found the rhythms triggered activity in parts of the brain linked to hearing, but something even more surprising was that the rhythms also triggered activity in the motor regions of the brain, linked to active movement.

H 'Somehow, the mere act of just listening triggers motor-neural activity. Maybe this is one reason why we often tap our feet, move or dance when hearing music,' says Chen. She believes the discovery of this deep connection between music and movement may cast light on why disabled patients can benefit from listening to music — and could also prove useful with other impairments such as those involved in sound production. 'It's been shown that people who talk with a stutter might have problems in this auditory-motor loop.'

I For researchers working in this new area of science, these early discoveries hold the promise of much more to come. Zatorre and his colleagues are investigating whether some people have more musical brains than others. 'We can see certain subtle brain features that can tell us how well somebody can do things like identify a slight change in a melody,' explains Zatorre. 'This ability could be enhanced by training — just like someone born with a predisposition to building strong muscles can enhance them by taking up weightlifting.'

Questions 14–18

Reading Passage 2 has nine paragraphs, A–I.

Which paragraph contains the following information? Write the correct letter, A–I.

Q	Statement	
14	a reference to studies involving children	
15	a mention of the discovery of significant artefacts	
16	reasons why a particular aspect of music has not been researched	
17	a mention of an unexpected discovery involving two different areas of the brain	
18	a comparison of tone variations produced by certain animals and humans	

Questions 19–22

Complete the summary below. Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

A STUDY INVOLVING COLLABORATION BETWEEN RESEARCHERS IN OXFORD AND MONTREAL

The participants in this study led by Dr Chen were chosen because they were not musicians, and they demonstrated a good state of (19) _____. The participants were given (20) _____, while music with a very noticeable rhythm was being played. Previous research had indicated that listening to this type of music seemed to be of assistance to some (21) _____ people. By listening to it, their (22) _____ ability had definitely got better. The findings of Dr Chen's study proved most informative.

Questions 23–26

Look at the following statements and the list of researchers below.

Match each statement with the correct researcher, A, B or C.

NB You may use any letter more than once.

LIST OF RESEARCHERS

- A Professor Norman Cook
- B Professor Robert Zatorre
- C Dr Joyce Chen

Q	Statement	
23	Research into the brain activity set off by music may help people with speech defects.	
24	It may be possible in time to improve a person's ability to recognise certain musical characteristics.	
25	The way listeners react to certain musical combinations may be similar to the way they react to other noises.	
26	When a person reacts positively to music, the same parts of the brain are stimulated as when certain animals react to a positive outcome.	

Passage 47

The Purpose of Facial Expressions

Do facial expressions reflect inner feelings? Or are they social devices for influencing others?

A The use of facial expressions for measuring people's emotions has dominated psychology since the late 1960s. This was when Paul Ekman of the University of California pioneered the idea that, by carefully measuring facial expression, he could evaluate people's true emotions. Ekman carried out experiments in which people were asked to label photos of basic facial expressions — such as a smile, frown or scowl — with words for emotions. He found, for example, that a smile represented joy and a scowl represented anger. In fact, since the 1970s Ekman has dominated the field of emotion research with his theory that when an emotion occurs, a large number of electrical impulses also occur, creating specific facial expressions and other physiological changes, such as increased or decreased heart rate or heightened blood pressure.

B Many took Ekman's work to imply that facial expressions precisely indicated people's feelings. However, this theory has been attacked by a number of psychologists, such as Alan Fridlund of the University of California, who claim that there is no one-to-one correspondence between facial expressions and emotions. Expressions evolved to set off certain behaviors in other people, says Fridlund. So a smile may prompt people to approach, a scowl may warn them to stay away, and a look of sadness may elicit words of sympathy and reassurance. In other words, in Fridlund's view, facial expressions are inherently social — they involve not one, but two or more people. Even when people are alone they are holding a dialog with another person internally, or imagining themselves in a social situation.

C Thus Fridlund thinks of facial expressions as tools for influencing social interactions, a view which, he claims, enables us to begin predicting when certain facial expressions will occur. This in turn will allow more precise theories about social interactions. His studies find that expressions occur most often during pivotal points, the turning points in social interactions — during greetings, social crises, or times of appeasement, for example. At these pivotal points there is an approach, or closeness, or more intimacy, and facial expressions, as well as gestures, open up the possibilities of various social interactions.

D Although much work on the emotions relies on a link between facial expression and emotions, psychologist James Russell, of the University of British Columbia, says there is very little evidence supporting such a connection. 'There's some sense in which faces express emotion, but only in the sense that everything expresses emotion,' says Russell, a long-time critic of the expression-emotion link. 'Music does, posture does, words do, tone of voice does, your behavior does. The real question is, "Is there anything special about

faces?" And there we really do not know much. What is more likely, argues Russell, is that facial expressions tell others something about a person's overall mood and context, rather than provide details about specific emotions.

E Others, including Ekman, argue that the face can display information about emotions, but they admit that it is not reliable one hundred percent of the time. And those who only examine faces when trying to study emotion will jump to false conclusions. But according to Ekman, to say, as Fridlund does, that there is the connection at all between facial expressions and emotions is simply wrong. 'There is a link between facial expression and emotion,' agrees developmental psychologist Linda Camras of DePaul University, 'but it's not a one-to-one kind of relationship as many once thought.' She believes there are many situations where emotion is experienced, yet no basic facial expression is displayed. And there are times when a facial expression appears with no corresponding emotion.

F Ekman's theory states that if the emotion comes on slowly or is rather weak, the feeling might not be strong enough to trigger the expression. This would explain why there can sometimes be emotion without expression, he argues. In addition, cultural rules — which determine when and whether people of certain cultures display emotional expressions — can prevent this otherwise automatic process from being completed. Facial expressions evolved in humans as signalers to others about how they feel, says Ekman. At times, though, it may be uncomfortable or inconvenient for us to let others know our emotions. But in the long run, over the course of evolution, it was useful to us as signalers: an angry look on someone's face may be a warning that they are preparing to behave in an angry fashion.

G Although Fridlund disagrees with Ekman on certain matters, the two basically share the opinion that facial expressions indicate people's future actions. The area of dispute between Fridlund and Ekman draws attention away from their major areas of agreement, says Joseph Campos of the University of California. Indeed, he says, 'there is profound agreement that the face, along with the voice, body posture and hand gestures, forecasts to outside observers what people will do next.' He goes on to say, 'The face is a component [of emotion], but to make it the center of study of the human being experiencing an emotion is like saying the only thing you need to study in a car is the transmission. Not that the transmission is unimportant, but it's only part of an entire system.'

Questions 14–18

Reading Passage 2 has seven paragraphs, A–G.

Which paragraph contains the following information? Write the correct letter, A–G.

Q	Statement	
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14	a recognition that facial expressions do not always provide a true guide to feelings	
15	examples of ways that a person's facial expression can affect someone else's actions	
16	a reference to reasons for hiding emotions	
17	examples of times when facial expressions are used especially frequently	
18	examples of changes inside the body when an emotion is felt	

Questions 19–23

Look at the following statements (Questions 19–23) and the list of researchers below.

Match each statement with the correct researcher, **A–E**.

NB You may use any letter more than once.

LIST OF RESEARCHERS

- A** Paul Ekman
- B** Alan Fridlund
- C** James Russell
- D** Linda Camras
- E** Joseph Campos

Q	Statement	
19	Focusing on the face in researching emotions is similar to researching a large structure by looking at just one small unit of it.	
20	Facial expressions developed in order to encourage other people to react in particular ways.	
21	Although certain researchers have different opinions about various points, they share some important ideas.	
22	Both emotion and expression can exist independently of each other in certain circumstances.	
23	It cannot be proved that there is a connection between facial expression and real emotion.	

Questions 24–26

Complete the sentences below. Choose **ONE WORD ONLY** from the passage for each answer.

In Fridlund's view a (24) _____ on someone's face may be intended as an invitation to come nearer. According to Fridlund, when we are by ourselves we still use facial expressions because we are having a (25) _____ with someone in our minds. Fridlund considers facial expressions to be (26) _____ that can affect our contact with other people.

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Passage 48

The Return of Monkey Life

Rain forest trees growing anew on Central American farmland are helping scientists find ways for monkeys and agriculture to benefit one another

A Hacienda La Pacifica, a remote working cattle ranch in Guanacaste province of northern Costa Rica, has for decades been home to a community of mantled howler monkeys. Other native primates — white-faced capuchin monkeys and spider monkeys — were once common in this area, too, but vanished after the Pan-American Highway was built nearby in the 1950s and most of the surrounding land was cleared for cattle-raising. At Hacienda La Pacifica, however, an enlightened ranch owner chose to leave some strips of native trees growing. He used these as windbreaks to protect both cattle and their food crops from dry-season winds. In the process, the farmer unwittingly founded a unique laboratory for the study of monkeys.

B Ken Glander, a primatologist from Duke University in the USA, is studying La Pacifica's monkeys in an effort to understand the relationship between howlers and regenerating forests at the edges of grazing lands. Studying such disturbed woodlands is increasingly important because throughout much of the New World Tropics, these are the only forests left. In the 18th century, tropical dry forests once covered most of Central America, but by the 1980s less than two percent remained undisturbed, and less than one percent was protected.

C Howlers persist at La Pacifica, Glander explains, because they are leaf-eaters. They eat fruit when it is available but, unlike capuchin and spider monkeys, do not depend on large areas of fruiting trees. Glander is particularly interested in howlers' ability to thrive on leaves loaded with toxins — poisonous substances designed to protect the plants. For leaf-eaters, long-term exposure to a specific plant toxin can increase their ability to neutralize the poisonous substances and absorb the leaf nutrients. Watching generations of howlers at La Pacifica has shown Glander that the monkeys keep their systems primed by sampling a variety of plants and then focusing on a small number of the most nutritious food items. The leaves that grow in regenerating forests, like those at La Pacifica, are actually more howler-friendly than those produced by the centuries-old trees that survive farther south. In younger forests, trees put most of their limited energy into growing wood, leaves, and fruit, so they produce much lower levels of toxin than do well-established, old-growth trees.

D The value of maturing forests to primates is also a subject of study at Santa Rosa National Park, about 35 miles northwest of La Pacifica. Large areas of Santa Rosa's forests had at one time been burnt to make space for cattle ranching and coffee farming, thereby devastating local monkey habitat. But in 1971 the government protected the area by

designating it a National Park, and species of Indigenous trees which had been absent for decades began to invade the abandoned pastures. Capuchins were the first to begin using the reborn forests, followed by howlers. Eventually, even spider monkeys, fruit-eaters that need large areas of continuous forest, returned. In the first 28 years following protection of the area, the capuchin population doubled, while the number of howlers increased sevenfold.

E Some of the same traits that allow howlers to survive at La Pacifica also explain their population boom in Santa Rosa. Howler reproduction is faster than that of other native monkey species. They give birth for the first time at about 3.5 years of age, compared with seven years for capuchins, and eight or more for spider monkeys. Also, while a female spider monkey will have a baby about once every four years, well-fed howlers can produce an infant every two years. Another factor is diet. Howlers are very adaptable feeders, and only need a comparatively small home range. Spider monkeys, on the other hand, need to occupy a huge home range. Also crucial is the fact that the leaves howlers eat hold plenty of water, so the monkeys can survive away from open streams and water holes. This ability gives them a real advantage over capuchin and spider monkeys, which have suffered during the long, ongoing drought in the area.

F Alejandro Estrada, an ecologist at Estación de Biología Los Tuxtlas in Veracruz, Mexico, has been studying the ecology of a group of howler monkeys that thrive in a habitat totally altered by humans: a cacao plantation in Tabasco state, Mexico. Cacao plants need shade to grow, so 40 years ago the owners of Cholula Cacao Farm planted figs, monkeypod and other tall trees to form a protective canopy over their crop. The howlers moved in about 25 years ago after nearby forests were cut. This strange habitat seems to support about as many monkeys as would a same-sized patch of wild forest. The howlers eat the leaves and fruit of the shade trees, leaving the valuable cacao pods alone.

G Estrada believes the monkeys bring underappreciated benefits to such plantations, dispersing the seeds of fruits such as fig and other shade trees, and fertilizing the soil. Spider monkeys also forage for fruit here, though they need nearby areas of forest to survive in the long term. He hopes that farmers will begin to see the advantages of associating with wild monkeys, which could include potential ecotourism projects. 'Conservation is usually viewed as a conflict between farming practices and the need to preserve nature,' Estrada says. 'We're moving away from that vision and beginning to consider ways in which commercial activities may become a tool for the conservation of primates in human-modified landscapes.'

Reading Passage 2 has seven paragraphs, A–G. Which paragraph contains the following information? **NB** You may use any letter more than once.

Q	Statement	
14	a reason why newer forests provide howlers with better feeding opportunities than older forests	
15	a reference to a change in farmers' attitudes towards wildlife	
16	a description of the means by which howlers select the best available diet for themselves	
17	figures relating to the reduction of natural wildlife habitat over a period of time	

Questions 18–21

Complete the summary below. Choose **ONE WORD ONLY** from the passage for each answer.

WHY DO HOWLERS HAVE AN ADVANTAGE OVER OTHER CENTRAL AMERICAN MONKEYS?

Howler monkeys have a more rapid rate of (18) _____ than either capuchin or spider monkeys. Unlike the other local monkey species, howlers can survive without eating (19) _____, and so can live inside a relatively small habitat area. Their diet is more flexible, and they are able to tolerate leaves with high levels of (20) _____. Howlers can also survive periods of (21) _____ better than the other monkey species can.

Questions 22–26

Look at the following features and the list of locations below. Match each feature with the correct location, A, B, or C. **NB** You may use any letter more than once.

LIST OF LOCATIONS

- A Hacienda La Pacifica
- B Santa Rosa National Park
- C Cholula Cacao Farm

Q	Statement	
22	It has seen the return of native tree species.	
23	It supports only one species of native monkey.	

24	Its monkey population helps the agriculture of the area.	
25	It is home to populations of all three local monkey species.	
26	Its landscape was altered by the construction of a transport link.	

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Passage 49

The Role of Mothers in the Origins of Music

A leading researcher has proposed that the key to understanding the origin of music is in the playful language used between mother and child

A In a recent lecture Richard Parncutt, a professor of systematic musicology at the University of Graz, Austria, discussed the idea that music originated from 'motherese' — the playful voices mothers adopt when speaking to infants and toddlers. As the theory goes, evolutionary changes occurring between one and two million years ago resulted in earlier births, more fragile infants and a critical need for stronger relationships between mothers and their newborns. According to Parncutt, 'motherese' arose as a way to strengthen this maternal bond and to help infants survive. 'If babies were born earlier, it is clear they would need better care in order to survive ... this would involve the baby communicating its state and needs more clearly to the mother,' he says. Parncutt adds that this makes it very likely that motherese evolved as an evolutionary adaptation, and that there is a large amount of modern empirical evidence for the musical nature of motherese.

B Although it might appear amusing or even nonsensical on the surface, 'motherese' is actually a sophisticated form of communication. It contains structural musical elements such as rhythm and melody, and codes that babies and mothers understand. It also contains cross cultural similarities with regard to the physical gestures and movements it incorporates, an important consideration when examining the origin of music. 'The sonic-gestural vocabulary tells both mother and infant about the current physical and emotional state of the other, as well as the current state of play between them,' says Parncutt. 'Here, emotions such as surprise and disappointment are learned for the first time in a social and musical context. It is about survival in that it motivates the mother to care for the infant and gives her information about the infant's needs.' For example, mothers can understand when their babies are tired or hungry. Motherese also helps infants to acquire language.

C Parncutt explains that a fetus begins hearing nearly four months before birth, as it regularly hears its mother's voice, heartbeat, and digestive sounds, all of which provide information about its mother's emotional state. After birth, the baby recognises these sound patterns and its mother's correlating mood. Gary McPherson, head of the University of Melbourne's School of Music, reiterates the importance of this prenatal association, saying that music is the very first form of intelligence to reveal itself even before birth.

D However, there are several competing theories about the origins of music. Parncutt developed a grading system to assess these theories, the objective being to measure their validity against a set of criteria which he called 'universal aspects of music'. Some key aspects are that music is meaningful and can be used to communicate information; it involves movement and gesture; it has structural components like rhythm, melody,

repetition and form; it performs certain social functions; music is emotional and can influence behaviour; it is intentional, requiring mental sophistication; and it is intrinsically spiritual.

E According to Parncutt, a strong theory about the origin of music should be able to explain or align with all the universal features he outlined. Using these standards, he reviewed several existing theories. One theory proposes that musical ability is innate in animal behavior, such as birds singing during courtship rituals or wolves howling to establish their territory. Others argue that music exists to bring us pleasure, similarly to the effect of a drug; some believe it evolved as a method for men to attract sexual partners, with the most talented musicians earning the best mates; and another idea is that music developed as a tool for play, helping children learn and gain other non-musical skills.

F Although Parncutt's research has not invalidated these theories, it proposes that 'motherese' offers the most comprehensive explanation for the origins of music. Parncutt states, 'I believe it can explain nearly every universal characteristic of music, including both its similarities to language and its unique aspects,' describing it as the 'least speculative' theory among those considered. McPherson adds, 'In conducting this research, he confronts existing theories directly. He does not simply accept a popular idea; instead, he critically examines the entire field. By evaluating all these theories, he constructs a compelling argument. It's an excellent contribution'.

G Despite the importance of Parncutt's work, the evidence supporting the 'motherese' theory remains inconclusive, and some specialists question whether the question of music's origin can ever be definitively answered. Sandra Trehub, a psychology professor at the University of Toronto Mississauga who has extensively studied 'motherese,' warns, 'Many ideas about music's beginnings are speculative — quite wild speculations at that. Vocal music, like speech, leaves no physical traces that could help us determine its ancient origins... None of these theories are provable'.

H Another point of critique is that the 'motherese' theory does not explain the origins of non-vocal music. Neil McLachlan, a music psychologist at the University of Melbourne, highlights the role of music in work and warfare traditions. He states, 'Music has served many social functions. It's very likely that some musical behaviors evolved from needs unrelated to learning. Music was used to motivate marching armies in war and to distract them from fear of danger. In work, music could help distract from monotonous tasks.' According to McLachlan, the origins of music are probably multiple and span a historical timeline. 'Motherese might represent the earliest source, but it does not cover all music — especially non-vocal forms like rhythmic entertainment.' Parncutt himself concedes the

evidence is not definitive and that multiple factors may have influenced music's origin. However, he maintains that 'motherese' currently provides the most thorough explanation, particularly due to its connection with spirituality.

Questions 14–18

Reading Passage 2 has eight paragraphs, **A–H**. Which paragraph contains the following information?

Write the correct letter, **A–H**.

Q	Statement	
14	an observation that the nature of vocal sound makes it impossible to find evidence of how music began	
15	an example of how one person's memory was improved over time	
16	examples of characteristics that one researcher believes all music must possess	
17	a claim that motherese is not as simple as it might seem when first heard	
18	an explanation of the origins of motherese	

Questions 19–23

Look at the following statements (Questions 19–23) and the list of researchers below.

Match each statement with the correct researcher, **A–D**. **NB** You may use any letter more than once.

List of Researchers

A Richard Parncutt

B Gary McPherson

C Sandra Trehub

D Neil McLachlan

Q	Statement	
19	Theories about how music first began are just guesswork.	
20	Music can help people cope with boredom in their jobs.	
21	An analysis of the various accounts for the origin of music has greatly enriched understanding.	
22	Motherese reinforces the connection between a mother and her baby.	

23	The motherese theory fails to explain the beginnings of music that is not produced by the voice.	
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Questions 24–26

Complete the summary below.

Choose **ONE WORD ONLY** from the passage for each answer.

Theories of the origin of music

Parncutt assessed a number of theories of the origins of music. One of these suggested that animals have natural musical ability. An example of this is the sounds birds make as part of their

- 24 ————. Another theory claimed that the purpose of music was to bring humans
 25 ————. Others believed it was a way to entice a mate, or that music was a
 26 ———— that could teach children things that had nothing to do with music.

Passage 50

The Science of Sleeping on It

Why a good night's sleep may be the key to solving the problems that stump us during the day

A Each one of us has experienced a wide array of dreams, from the funny and romantic to the terrifying and bizarre. In dreams, bits of reality, imagination, and memory intermingle in nonlinear narratives that often make little sense upon awakening. Dreams seem to be related to a reshuffling of memories, and indeed, it is often said that studying before sleep is a good way to retain information. When people say they will "sleep on" a decision or problem, it is likely because they know they will awaken the next day with a clearer solution in mind, and scientific evidence has shown that this is in fact the case.

B To test whether or not sleep really does play a role in solving difficult problems, researchers at Northwestern University conducted a study in which they presented participants with a series of tricky, carefully designed puzzles, each of which participants had two minutes to figure out. During this time, a unique sound clip would play in the background, and a new sound clip would be played with each new puzzle. After this initial stage of the experiment, which always took place in the early evening, members of the study group were sent home with some specialized equipment that monitored their sleep and played some of the sound clips they heard while attempting the puzzles in the lab. The ensuing morning, they all returned to the lab to try to solve the puzzles that stumped them the previous night. Their success rate improved across the board, but the most marked improvement was in puzzles that were associated with sound clips they had heard while sleeping, solving 32% of the sleep-cued puzzles, compared to only 21% of the un-cued puzzles.

C A study on the benefits of napping, headed by Sara Mednick, assistant professor of psychiatry at University of California San Diego, and Denise Cai, a graduate student, revealed that rapid eye movement (REM) sleep enhances creative problem solving. The findings of this study may have implications for how sleep, and REM sleep in particular, promotes the formation of associative creative problems that you've already been working on - the passage of time is enough to find solutions. However, for new problems, only REM sleep enhances creativity. Mednick believes that REM sleep does this because it stimulates associative networks, which allow the brain to make connections between unrelated concepts or ideas.

D According to Robert Stickgold, professor of psychiatry at Harvard Medical School, "when you get into bed, the brain does a triage and rifles through the events of the day and sees what's left unfinished." The brain picks out emotions that occurred during or shortly after something happened, and uses these memory cues to indicate that the event was important and needs to be analyzed further. Stickgold believes that there are two key

elements at play here, the first being that the prefrontal cortex shuts down. This region of the brain is in charge of executive decision-making, along with rational thinking and impulse control. With the prefrontal cortex shut down, there are no categories into which ideas can be sorted, allowing the brain to freely associate and process information in the background. The other factor at play is that when we enter REM sleep, the neuromodulators norepinephrine and serotonin are turned off. Norepinephrine enhances focus on immediate issues, and it is believed that the brain is biased into placing higher value on loose connections when serotonin is turned off. With both neurochemicals dormant, parts of ideas come together as one.

E Mark Beeman, a professor of psychology at Northwestern University, stated: "There was this idea that during sleep the brain is resting, but now we know there's a lot of important work being done" following research by his team. While we sleep, our brains sort, consolidate, and store new information, and we "walk through" memories of recent events. This rehearsal appears to distinguish meaningful patterns from unhelpful "background noise." Beeman believes that this can be useful in problem-solving and provides an explanation as to why playing sound clips associated with certain puzzles facilitate this type of rehearsal. "This research adds to a growing list of literature suggesting that sleep can reorganise information to facilitate problem-solving," says Kristin Sanders, first author of the study and a doctoral student at Northwestern University. "It also suggests that a replay of the problem memory during sleep is critical for this reorganisation." Sanders also goes on to explain that "problem solving is part of everyone's daily life; while we use tricky puzzles in our study, the underlying cognitive processes could relate to solving any problem on which someone is stuck or blocked by an incorrect approach." However, she warns that this can only be applied to situations in which one already has some background knowledge or information. For example, this effect would not help someone find a cure for a rare disease if they are not already working on this problem.

F Some researchers also believe that dreams have an evolutionary basis, and that our ability to dream may have indeed evolved from the need to solve problems. "Dreams are thinking or problem-solving in a different biochemical state from that of waking," writes the Harvard psychologist Deirdre Barrett in *The New Science of Dreaming*. According to Barrett, all functions of the brain must have some practical usage, and are rooted in evolution and natural selection, which may explain why the brain utilizes dreams to help break down, analyze, and solve problems.

G Before we fall asleep, the brain wants to acknowledge a thought and move on to the next one, but in reality, we tend to latch onto one thought and ruminate on it, and then feelings of regret start to set in, triggering an adrenaline rush, according to Stickgold. It is helpful to acknowledge a thought and let it go by without passing judgment, similar to the calming methods used in meditation. However, if a particular thought won't go away, there are

calming steps that can be taken. Stickgold states that "there is nothing wrong with checking," and for issues that cannot be resolved right away, it is acceptable to keep a notepad and pencil nearby to write down a reminder for the following day. Getting up in the middle of the night to work on a problem, however, is not particularly helpful, as it will just leave you exhausted in the morning without having gotten any closer to an answer. The best solution is in fact to just let the ideas pass through your mind unimpeded.

Questions 1–4

Reading Passage 2 has seven paragraphs, A–G.

Write the correct letter, A–G, in boxes 1–4 on your answer sheet.

Q	Statement	
1	a mention of the limitations of using sleep to help solve problems	
2	a reference to how an activity before sleep can help with information recall	
3	a contrast between what the brain wants to do and what we actually do	
4	an explanation of why we have dreams	

Questions 5–9

Complete the summary below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

Sleep and Memory

- A study at Northwestern University was conducted in which participants had to solve puzzles that were accompanied by a (5) _____.
- Following this first phase, they went home with equipment that monitored their sleep and played sounds that they had heard in the lab, and the following morning, their (6) _____ improved, especially for puzzles cued with sounds that they had heard in their sleep.
- A study by Sara Mednick showed that REM sleep was beneficial for creative (7) _____, and that REM sleep allows the brain to create associations between previously (8) _____ ideas.

- Professor Robert Stickgold mentions that the brain selects **9** ————— that are connected to an event and uses these memory signals to decide if something was important.

Questions 10–13

Look at the following statements (Questions 10–13) and the list of researchers below.

Match each statement with the correct researcher, **A, B, C, D** or **E**.

LIST OF RESEARCHERS

- A** Sara Mednick
- B** Robert Stickgold
- C** Mark Beeman
- D** Kristin Sanders
- E** Deirdre Barrett

Q	Statement	
10	going over memories helps differentiate between useful and trivial information	
11	staying up late is not effective for trying to solve problems	
12	information related to a problem can be restructured during sleep	
13	all the functions of the brain must have evolved to serve a particular purpose	

Passage 51

The Shape of Bird Eggs

A Princeton-led team uncovers a surprising link between egg shape and a bird's flying ability

A A sandpiper's egg is shaped like a teardrop, an owl's looks a bit like a golf ball, and a hummingbird's resembles a tiny bean. Now, for the first time, scientists in the US have come up with a convincing explanation for this variation. Princeton University evolutionary biologist Mary Stoddard has long been fascinated by the fact that eggs are so diverse in shape even though they all basically serve one function: nourish and protect the developing chick inside. She recently headed an interdisciplinary team of evolutionary biologists, computer scientists, mathematicians and physicists, with the expectation of bringing together different ways of looking at bird egg shapes and achieving a better understanding of them.

B Fortunately, over the past century, the Museum of Vertebrate Zoology at the University of California, Berkeley has amassed thousands of egg shells from 1,400 species, representing about 14% of all birds, and put digital photos of them online. Using this database, Stoddard and her team at Princeton University wrote a computer program, which they called Eggxtractor, that can select the image of any egg and calculate its length, width and shape. The team used these calculations to determine how far from perfectly spherical each of nearly 50,000 eggs in their sample was – that is, how pointed or elongated each was.

C Next, the research team attempted to answer how and why eggs might have acquired these varying shapes. Rather than looking at the outer hard shell, as one might expect, the researchers concentrated on the egg's soft thin inner membrane, which is, in fact, essential in fixing the egg's shape. Stoddard worked with Harvard University physicist L. Mahadevan and Ee Hou Yong of China's Nanyang Technological University to devise a mathematical representation based on the membrane's properties and how much pressure it received from the unhatched chick within the egg. They then used their model to create many different egg shapes by altering the membrane's stiffness and changing the pressure. 'Adjusting these [features] allows us to generate the entire diversity of egg shapes that we observe in nature,' Stoddard says.

D When Stoddard and her colleagues made a diagram showing the relationship between some 1,000 bird species, they realized that members of each closely associated species tended to lay eggs with the same characteristic shape. The researchers then investigated why egg shapes might be so spectacularly diverse. Some scientists had previously believed that the shape might depend on nest location: cliff-nesting birds, it was thought, lay pointed eggs so that if the eggs are knocked, they spin in a circle rather than rolling off the

cliff. Other scientists suggested that birds lay eggs in shapes that pack together most economically in a nest. Stoddard and her researchers found neither of these hypotheses to be persuasive.

E In fact, Stoddard's team were surprised by their findings: that egg shape is strongly correlated with a measure of wing shape, which in turn reflects how well the birds can fly and therefore their frequency of flight. 'There was an obscure hypothesis that egg shape could be related to flight ability that no one had paid any attention to,' Stoddard says. To her team's surprise, they found that egg shape does depend on how much the species flies. Good fliers such as sandpipers tend to lay eggs that are more elongated and more asymmetrical. This is probably because a bird which spends lots of time in the air requires a compact, long, streamlined body, and this best accommodates an elongated egg. Meanwhile, birds that spend little or no time in the air, like tropical pittas and trogons, do not need elongated bodies and therefore have more spherical eggs. For such round eggs to be laid, the bird requires a wide pelvis. By contrast, birds needing aerodynamically shaped bodies have smaller, less heavy skeletons and their eggs have evolved to fit through their relatively narrow pelvises.

F Martin Sander, a paleontologist at Bonn University in Germany, says that scientists can quite accurately predict how good a flier a bird species is just by looking at the shape of its eggs. 'What's cool is you have the [overall] formula for egg shape,' Sander says. 'You can take this study and look at the egg and immediately get some general information.' Surprisingly, though, penguins lay pointed, asymmetrical eggs too, even though they are flightless. Stoddard says that penguins' bodies may be an adaptation to allow them to swim, and so perhaps the same processes that influence egg shape in flying birds are at work in swimming birds.

G The work is significant on two levels, Stoddard says. For one, the results of the research 'could be of value to the egg industry', she says, perhaps by helping in the production of more durable eggs. But for her, just solving the puzzle is reward in itself. A specialized egg like that of modern birds made it possible for young to survive on land, she notes, and thus allowed our land vertebrate ancestors to leave the seas about 36 million years ago. Stoddard adds that she is eager to explore how eggs changed shape at the point when birds evolved from dinosaurs. She believes that both egg symmetry and flight evolved roughly at this time, raising the intriguing possibility that the emergence of flight is associated with egg shape variation.

Questions 1–7

Reading Passage 2 has seven paragraphs, A–G.

Which paragraph contains the following information? Write the correct letter, A–G.

Compiled by @bekhruzposts

Q	Statement	
1	a reference to a part of the egg focused on by the Stoddard team	
2	an explanation of why egg shape is related to flying	
3	an exception to the theory connecting egg shape and flight	
4	a mention of possible commercial uses of the Stoddard research findings	
5	a reference to the kinds of egg measurements the Stoddard researchers made	
6	examples of incorrect explanations of why egg shapes vary	
7	a reference to the essential purpose of eggs	

Questions 8 and 9

Choose **TWO** letters, **A–E**.

Which **TWO** things do we learn about the Princeton University team?

OPTIONS

- A** They collected egg shells of a majority of all bird species.
- B** They were specialists in the nourishment of unhatched chicks.
- C** They had a variety of backgrounds offering different perspectives.
- D** They attempted to identify all known birds with spherical-shaped eggs.
- E** They developed a tool for precisely examining digital photos of eggs.

Answer: (8) (9)

Questions 10 and 11

Choose **TWO** letters, **A–E**.

Which **TWO** things do we learn about how and why eggs have varying shapes?

OPTIONS

- A** The inner membrane of a bird egg is important in establishing the egg shape.
- B** An unhatched chick's movements can change the shape of its egg.
- C** Birds in related species usually have similarly shaped eggs.
- D** The shape of bird eggs is determined by where the nests are built.

E Bird eggs are shaped so they can fit efficiently in their nest.

Answer: 10 11

Questions 12–16

Complete the summary below.

Choose **ONE WORD ONLY** from the passage for each answer.

THE STODDARD RESEARCH FINDINGS

Stoddard and her team found that the egg shape of a bird species is determined at least in part by the species' ability to fly and the 12 _____ with which it flies. They found that birds with aerodynamic bodies and lightweight 13 _____ require long eggs so that they are able to pass through the birds' pelvises. Likewise, the shape of the bodies and eggs of penguins is probably an evolutionary 14 _____ to their need to swim. The findings of Stoddard's research could prove beneficial to the sale of eggs if this leads to the 15 _____ of tougher eggs that last longer. Stoddard says she hopes to study changes in egg shape which took place during the evolutionary transition from 16 _____ to birds.

Passage 52

The Study of Laughter

Humans don't have a monopoly on laughter, says behavioural biologist Silvia Cardoso – and too little of it could harm our mental, physical and social well-being

Laughter is a universal phenomenon, and one of the most common things we do. We laugh many times a day, for many different reasons, but rarely think about it, and seldom consciously control it. We know so little about the different kinds and functions of laughter, and my interest really starts there. Why do we do it? What can laughter teach us about our positive emotions and social behaviour? There's so much we don't know about how the brain contributes to emotion and I think we can get at understanding this by studying laughter.

Only 10 or 20 per cent of laughing is a response to humour. Most of the time it's a message we send to other people – communicating joyful disposition, a willingness to bond and so on. It occupies a special place in social interaction and is a fascinating feature of our biology, with motor, emotional and cognitive components. Scientists study all kinds of emotions and behaviour, but few focus on this most basic ingredient. Laughter gives us a clue that we have powerful systems in our brain which respond to pleasure, happiness and joy. It's also involved in events such as release of fear.

Many professionals have always focused on emotional behaviour. I spent many years investigating the neural basis of fear in rats, and came to laughter via that route. When I was working with rats, I noticed that when they were alone, in an exposed environment, they were scared and uncomfortable. Back in a cage with others, they seemed much happier. It looked as if they played with one another real rough-and-tumble and I wondered whether they were also laughing. The neurobiologist Jaak Panksepp has shown that juvenile rats make short vocalisations, pitched too high for humans to hear, during rough-and-tumble play. He thinks these are similar to laughter. This made me wonder about the roots of laughter.

You only have to look at the primates closest to humans to see that laughter is clearly not unique to us. I don't find this too surprising, because we're only one among many social species and there's no reason why we should have a monopoly on laughter as a social tool. The great apes, such as chimpanzees, do something similar to humans. They open their mouths wide, expose their teeth, retract the corners of their lips, and make loud and repetitive vocalisations in situations that tend to evoke human laughter, like when playing with one another or with humans, or when tickled. Laughter may even have evolved long before primates. We know that dogs at play have strange patterns of exhalation that differ from other sounds made during passive or aggressive confrontation. But I think we need to

be careful about over-interpreting panting behaviour in animals at play. It's nice to think of it as homologous to human laughter, but it could just be something similar but with entirely different purposes and evolutionary advantages.

Everything humans do has a function, and laughing is no exception. Its function is surely communication. We need to build social structures in order to live well in our society and evolution has selected laughter as a useful device for promoting social communication. In other words, it must have a survival advantage for the species.

The brain scans are usually done while people are responding to humorous material. You see brainwave activity spread from the sensory processing area of the occipital lobe, the bit at the back of the brain that processes visual signals, to the brain's frontal lobe, which analyses the words and structure of jokes, while the right side does the intellectual analyses required to 'get' jokes. Finally, activity spreads to the motor areas of the brain controlling the physical task of laughing. We also know about these complex pathways involved in laughter from neurological illness and injury. Sometimes after brain damage, tumours, stroke, or brain disorders such as Parkinson's disease, people get 'stone-faced' syndrome and can't laugh.

We are sure that laughter differs between the sexes, particularly the uses to which the sexes put laughter as a social tool; for instance, women smile more than laugh, and are particularly adept at smiling and laughing with men as a kind of 'social lubricant'. It might even be possible that this has a biological origin because women don't or can't use their physical size as a threat, which men do, even if unconsciously.

It's undoubtedly the best medicine. For one thing, it's exercise. It activates the cardiovascular system, so heart rate and blood pressure increase, the arteries dilate, causing blood pressure to fall again. Repeated short, strong contractions of the chest muscles, diaphragm, and abdomen increase blood flow into our internal organs, and forced respiration – the ha! ha! – makes sure that this blood is well oxygenated. Muscle tension decreases, and indeed we may temporarily lose control of our limbs, as in the expression 'weak with laughter'. It may also release brain endorphins, reducing sensitivity to pain and boosting endurance and pleasurable sensations. Some studies suggest that laughter affects the immune system by reducing the production of hormones associated with stress and that when you laugh the immune system produces more T-cells. But no rigorously controlled studies have confirmed these effects. Laughter's social role is definitely important.

Today's children may be heading for a whole lot of social ills because their play and leisure time is so isolated and they lose out on lots of chances for laughter. When children stare at computer screens, rather than laughing with each other, this is at odds with what's natural for them. Natural social behaviour in children is playful behaviour, and in such situations,

laughter indicates that make-believe aggression is just fun, not for real, and this is an important way in which children form positive emotional bonds, gain new social skills and generally start to move from childhood to adulthood. I think parents need to be very careful to ensure that their children play in groups, with both peers and adults, and laugh more.

Questions 1 and 2

Which of the following claims and arguments are presented in the passage above? Choose **TWO** letters, **A–E**.

OPTIONS

- A** All animals share the phenomenon of laughter
- B** Laughter can influence both adult and child health
- C** Laughter is not unique to humans
- D** Human mental, physical and social well-being are closely related
- E** Laughter teaches us how to behave

Answer: ① ②

Questions 3–7

Do the following statements agree with the writer in Reading Passage 2?

Write **YES** if the statement agrees with the writer, **NO** if it contradicts the writer, or **NOT GIVEN** if there is no information about this in the passage.

- ③ Laughter is one of the most common expressions shared by humans.
- ④ There are complicated systems in human brain which take responsibility of our emotions as happiness and fear.
- ⑤ Communication is the only purpose of laughter.
- ⑥ Reduced blood pressure would lead to a stimulated cardiovascular system.
- ⑦ With the mass production of T-cells from laughter, stress hormones would be deducted from immune system.

Questions 8–13

Complete the notes below.

Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

Emotional behaviour takes academic concerns. For years scientists have been examining the origin of (8) _____ and laughter that comes from the same route as rats. Within an open environment, they have been noticed to be (9) _____. When they are alone, happier when they are back with the others. Jack Panksepp even found that rats make (10) _____ when they are in a chaotic state. It is well understood that humans are not only living species that laughs and laughter may have developed long before (11) _____. Despite such facts, we need to pay attention when we explain various animal behaviour, as they may express with differed (12) _____ and (13) _____.

Passage 53

The Tasmanian Tiger

A look at the extinction — or possible survival — of Australia's most famous vanished predator

The Tasmanian tiger, or thylacine, was a carnivorous marsupial (a meat-eating mammal which carries its young in a pouch). It was given the name 'tiger' because it had striped fur, and because it was ferocious. Between 24 million and 15 million years ago, many types of thylacine roamed across Australia, their powerful jaws playing a role in maintaining a balance in the ecosystems of their day. Some species were fox-sized, while others were barely the size of kittens.

But when a period of climate change cooled Australia about 12 million years ago, the numbers of these ancient thylacines began to decline. By about 3 million years ago, only one species was left. About 4,000 years ago, these vanished completely from the Australian mainland, so that Tasmania, a large island to the south of Australia, was then the last remaining place where thylacines existed. They ruled the animal life of that island unchallenged until Europeans with sheep, dogs and a great indifference to native flora and fauna, seem to have brought about their extinction. In 1936, the last captive Tasmanian tiger died in Hobart Zoo. Since then, many expeditions have searched for tigers in the Tasmanian bush, but no definitive evidence has been found. Despite this, there are many who keep searching.

In 1981, Dutch-born zoologist Hans Naarding was in Tasmania conducting a survey of Latham's snipe, a species of endangered bird. One night he saw an animal in the light from the searchlight mounted on his vehicle. He described it as about the size of a large dog, but with slightly sloping hindquarters and a fairly thick tail continuing straight on from its backbone. He said that it had 12 distinct stripes on its back, running down to the point where the tail began. He reported the sighting to the Director of Tasmania's National Parks. When the news broke, said Naarding, 'I was besieged by television crews, including four or five from Japan, and others from the United Kingdom, Germany, New Zealand and South America.'

Government and private search parties combed the region, but no further sightings were made. The tiger, as always, had escaped to its lair – a place that many insist exists only in the imagination. Others disagree, there have been more than 4,000 claimed sightings of the animal since it supposedly died out, and the average number of claims reported to the authorities each year is now 150. So is it out there? Even experts differ in opinion.

Randolph Rose, Associate Professor of Zoology at the University of Tasmania, says that he dreamed of seeing a thylacine, but is now convinced that his dream will go unfulfilled. The consensus among conservationists is that any animal with a population base of less than

1,000 is headed for extinction within 60 years. 'Sixty years ago,' he says, 'there was only one thylacine that we know of, and that was in Hobart Zoo. Take it from me, the tiger is gone.'

But Dr David Pemberton, curator of zoology at the Tasmanian Museum, states that, despite scientific thinking that a relatively large number of animals is required to sustain a population, 'the Florida panther is down to a dozen or so animals, and, while it does have some inbreeding problems, is still ticking along!' After all, animals can be notoriously elusive. The strange fish known as the coelacanth, with its 'proto legs', was thought to have died out with the dinosaurs until a specimen was dragged to the surface in a shark net off the coast of South Africa in 1938.

Wildlife biologist Nick Mooney has the unenviable task of investigating all so-called sightings of the tiger. It was Mooney who was first consulted in late February 2005 about the authenticity of new digital photographic images of a thylacine allegedly taken by a tourist. On face value, Mooney says, this particular account of a sighting and the photographs submitted as proof amount to one of the most convincing cases for the species' survival that he has seen. Many other 'sightings' have been hoaxes, and many sincere seekers are victims of obsession. 'It is a blind optimism that something is, rather than cynicism that something isn't,' Mooney says. 'If something crosses the road, it's not a case of "I wonder what that was?" Rather, it is a case of "That's a thylacine!"'

However, Mooney treats all sightings on face value. 'I never try to embarrass people,' he says, '...but the fact that I don't pack the car immediately after they telephone can often be taken as ridicule. Obsessive characters get angry that someone in my position is not out there when they think the thylacine is there.'

Hans Naarding, whose sighting of a striped animal two decades ago was the highlight of a lifetime of animal spotting, remains puzzled by the time and money people waste on tiger searches. He says resources would be better applied to saving another endangered animal, the Tasmanian devil, and helping declining migratory bird populations. Could the thylacine still be out there? 'Sure,' Naarding says. 'I know the vast south-west wilderness of Tasmania well. They could survive ... [But] if this is the case, it will not be long before they do disappear completely.' Naarding believes that any discovery of surviving thylacines would be 'rather pointless'. 'How do you bring a species back from extinction?' he asks. 'What could you do with it? If there are thylacines out there, they are better off right where they are.'

Questions 14–18

Complete the summary below.

Choose **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer.

The Tasmanian Tiger

- The thylacine was a dog-like animal which had a (14) _____ coat and was carnivorous. It was originally spread widely throughout the mainland of (15) _____, before a period of climate change reduced that area around (16) _____ ago.
- In the end, thylacines were found only on the island of (17) _____, until the arrival of (18) _____ with their farming practices brought about a drastic reduction in thylacine numbers. The last one is thought to have died in Hobart Zoo in 1936.

Questions 19–24

Look at the following statements (Questions 19–24) and the list of people below.

Match each statement with the correct person, **A–D**.

LIST OF PEOPLE

- A** Hans Naarding
- B** Randolph Rose
- C** David Pemberton
- D** Nick Mooney

Q	Statement	
19	There is no longer any hope of finding a surviving Tasmanian tiger.	
20	It would be preferable not to disturb any surviving Tasmanian tigers.	
21	Many who claim to have seen Tasmanian tigers are not objective witnesses.	
22	Expert estimates of numbers needed to ensure species survival may be inaccurate.	
23	There is a great deal of international interest in Tasmanian tiger stories.	
24	Some fresh evidence provided by a visitor to Tasmania seems credible.	

Questions 25–26

Choose the correct letter, **A, B, C or D**.

- 25 Has Naarding's sighting of a Tasmanian tiger resulted in
- A the capture of the tiger.
 - B an extensive follow-up.
 - C many other sightings.
 - D the death of the tiger.
- 26 The example of the coelacanth is used to show that
- A new animal species are still evolving.
 - B animals can possess surprising physical characteristics.
 - C species of sea animals can be saved from extinction.
 - D opinions regarding the extinction of animal species can be mistaken.

Undoing Our Emotions

New research suggests positive emotions can reverse the lingering effects of stress — and may even help us live longer

A Three generations ago, 180 young women wrote essays describing why they wanted to join a convent (a religious community of nuns). Years later, a team of psychological researchers came across these autobiographies in the convent's archives. The researchers were seeking material to confirm earlier studies hinting at a link between having a good vocabulary in youth and a low risk of Alzheimer's disease in old age. What they found was even more amazing. The researchers found that, although the young women were in their early twenties when they wrote their essays, the emotions expressed in these writings were predictive of how long they would live: those with upbeat autobiographies lived more than ten years longer than those whose language was more neutral. Deborah Danner, a psychologist at the University of Kentucky who spearheaded the study, noted that the results were particularly striking because all members of the convent lived similar lifestyles, eliminating many variables that normally make it difficult to interpret longevity studies. 'It was a phenomenal finding,' she says. 'A researcher gets a finding like that maybe once in a lifetime.' However, she points out that no one has been able to determine why positive emotions might have such life-extending effects.

B Barbara Fredrickson, Professor of Psychology at the University of Michigan, believes that part of the answer is the 'undo effect'. According to this theory, positive emotions help you live longer by shutting down the effects of negative ones. Fredrickson's theory begins with the observation that negative emotions, like fear and stress, enhance our fight-or-flight response to very real threats. However, even when the emergency is gone, negative emotions produce lingering effects. Brooks Gump, a stress researcher at the State University of New York, explains that one of these effects is excessive cardiovascular reactivity. Behaviourally, Gump says, this reactivity is related to excessive vigilance: the state of being constantly on guard for potential dangers. Not only is it physically draining to live in a perpetual state of high vigilance, but high cardiovascular reactivity could be linked to increased chances of a heart attack.

C Fredrickson believes positive emotions work their magic by producing a rapid unwinding of pent-up tension, restoring the system to normal. People who quickly bounce back from stress often speed the process by harnessing such emotions as amusement, interest, excitement, and happiness, she says. To test her theory, Fredrickson told a group of student volunteers that they had only a few minutes to prepare a speech that would be critiqued by experts. After letting the students get nervous about that, Fredrickson then told them they wouldn't actually have to deliver their speeches. She monitored heart rates

and blood pressure. Not surprisingly, all students got nervous about their speeches, but those who viewed the experiment with good-humoured excitement saw their heart rates return to normal much more quickly than those who were angry about being fooled. In a second experiment, Fredrickson reported that even those who normally were slow to bounce back could be coached to recover more quickly by being told to view the experiment as a challenge, rather than a threat.

D Fredrickson believes that positive emotions make people more flexible and creative. Negative emotions, she says, give a heightened sense of detail that makes us hypersensitive to minute clues related to the source of a threat. But that also produces 'tunnel vision' in which we ignore anything unrelated to the danger. Fredrickson speculated that just as positive emotions can undo the cardiovascular effects of negative ones, they may also reverse the attention-narrowing effects of negative feelings: broadening our perspectives.

E To verify her theory, Fredrickson showed a group of students some film clips – some saw frightening clips, some saw humorous ones or peaceful ones. They then did a matching test in which they were shown a simple drawing and asked which of two other drawings it most resembled. The drawings were designed so that people would tend to give one answer if they focused on details, and another answer if they focused on the big picture. The results confirmed Fredrickson's suspicion that positive emotions affect our perceptions. Students who had seen the humorous or peaceful clips were more likely to match objects according to broad impressions.

F This fits with the role that positive emotions might have played in early human tribes, Fredrickson says. Negative emotions provided focus, which was important for surviving in life-or-death situations, but the ability to feel positive emotions was of long-term value because it opened the mind to new ideas. Humour is a good example of this. She says: 'The emotions are transient, but the resources are durable. If you are building a friendship through being playful, that friendship is a lasting resource.' So while the good feelings may pass, the friendship remains. On an individual level, Fredrickson's theory also says that taking time to do things that make you feel happy isn't simply self-indulgent. Not only are these emotions good for the individual, but they are also good for society.

G Other researchers are intrigued by Fredrickson's findings. Susan Folkman, of the University of California, has spent two decades studying how people cope with long-term stresses such as bereavement, or caring for a chronically ill child. Contrary to what one might expect, she says, these people frequently experience positive emotions. 'These emotions aren't there by accident,' she adds. 'Mother Nature doesn't work that way. I think that they give a person time out from the intense stress to restore their resources and keep going. This is very consistent with Fredrickson's work.'

Questions 14–19

Reading Passage 2 has seven paragraphs, **A–G**.

Which section contains the following information? Write the correct letter, **A–G**. You may use any letter more than once.

Q	Statement	
14	a conclusion that it is possible to train people to deal with anxiety	
15	conclusive evidence that lifespan can be influenced by emotions	
16	an explanation of the way negative emotions affect what people concentrate on	
17	an experiment that showed how a positive outlook can help people adjust to a stressful situation faster than others	
18	a discovery beyond what researchers were investigating	
19	an experiment where the nature of a material seen by participants affected the way they performed a task	

Questions 20–23

Look at the following statements and the list of researchers below.

Match each statement with the correct researcher, **A–D**. You may use any letter more than once.

LIST OF RESEARCHERS

- A** Deborah Danner
- B** Barbara Fredrickson
- C** Brooks Gump
- D** Susan Folkman

Q	Statement	
20	People whose daily lives are stressful often have surprisingly positive emotions.	
21	The body's reaction to a crisis may trigger a life-threatening event.	
22	It is unusual to have a study group whose circumstances were very alike.	
23	The reasons for a link between positive emotions and a longer life have not been established.	

Questions 24–26

Complete the sentences below.

*Choose **ONE WORD ONLY** from the passage for each answer.*

- In early tribes, negative emotions gave humans (24) _____ the emergencies that they needed to deal with.
- Fredrickson believes that a passing positive emotion can lead to an enduring asset such as a (25) _____, which is useful in times to come.
- Fredrickson also believes that both individuals and (26) _____ benefit from positive emotions.

Passage 55

Urban Wildlife and Citizen Science

How ordinary people with garden cameras are helping scientists track and protect Britain's wild mammals

A About half the world's human population currently live in urban areas, which cover about 3% of the Earth's land surface. Both figures are increasing rapidly and, by 2050, it is estimated that two thirds of the world's population will live in an urban area. This growing trend of urbanisation represents the most extreme form of habitat loss for most plants and animals. As towns and cities grow, the natural habitats are removed and replaced with hard, impermeable structures such as roads and buildings. In a recent global study, researchers estimated that cities accommodate only 8% of the bird species and 25% of the plants that would have lived in those areas prior to urban development.

B Until recently, we knew relatively little about how many of the species that do live in towns and cities were coping. With a growing human population, it is now more important than ever for scientists and the public to work together to monitor wildlife and biodiversity effectively. When data is limited, it is difficult to understand the bigger picture: we can't know if animal populations are becoming more or less abundant and why, or whether conservation is needed.

C One way that hundreds of ordinary people in the UK are helping to assess biodiversity is by setting up cameras in their gardens to record and then report any animal activity they capture on film. They are taking part in a project known as the MammalWeb database. Anyone with access to a camera can register to take part and become a 'spotter'. Using the general public in this way gives the ability to have far more cameras out in the field than any single researcher could manage, resulting in a much more comprehensive data set to analyse. The database has now amassed over 500,000 photographs of local wildlife, and recorded 34 species, ranging from the largest UK land mammal – the red deer – right down to some of the smallest, such as hedgehogs and bank voles.

D Many of the participants have been surprised by what the animals were doing in their own back yard. At times the cameras have revealed an animal coexisting happily with one of its known predators. Another remarkable discovery was a North American raccoon, living wild in the north-east of England. It is not known how long the raccoon was roaming free and, without the aid of the public, it may never have been spotted, which highlights just how easy it is for urban wildlife to go unnoticed. Once discovered, the authorities were able to locate the animal and transfer it to a wildlife park, where it was given a more suitable home. The raccoon is not the only American visitor to have made itself at home in the UK. In fact, another – the American grey squirrel – is the most frequent sighting on MammalWeb, far outnumbering the native red squirrel.

E In many European cities, the red squirrel appears well adapted to modern urban living, and they are abundant in countries such as Finland, France and Poland. They once thrived in the UK, too. However, since the grey squirrel was introduced in the 1800s, the population has declined drastically, and they are now classed as endangered. Several studies have shown that the introduction of the grey squirrel is the main factor in the red squirrel's decline, due to competition for food and shelter and the spread of the squirrelpox virus (which grey squirrels transmit to red squirrels).

F However, again thanks in large part to the efforts of ordinary citizens, one area where the reds haven't disappeared is a small coastal town in the north-west of England called Formby, one of few red squirrel strongholds in England. Red squirrels can easily be spotted in gardens throughout the town, and the local residents are passionate about protecting them, with many volunteering with a local conservation group. This voluntary organisation manages the extensive woodland nearby, supplying additional food, and employing dedicated "squirrel officers" who help maintain "grey squirrel-free" habitats.

G Elsewhere in the UK, most research and conservation is carried out in more rural areas. However, given the predicted future increases in urbanisation, managing urban sites like the one in Formby may be a better alternative, particularly as it makes the most of the benefits to animals of living alongside people, such as easy access to food and shelter. Of course, there are downsides too: road traffic poses an ever-present threat, as do pets. Even supplemental feeding can have unintended consequences, drawing animals from the safety of their nests and lairs and encouraging the spread of disease. Still, the benefits appear to outweigh the risks, and it is also worth noting that many native plant and bird species continue to exist in cities that were never designed with biodiversity protection in mind.

H Humans rely on biological diversity, either directly for food, or indirectly, through nutrient cycling and pollination. As these community-based conservation management programmes show, with cameras offering fascinating insights into the secret lives of mammals, and local volunteers safeguarding endangered species, there are many courses of action we can take to help to counteract the damage brought by urbanisation and ensure that animals not only survive, but thrive in our towns and cities.

Questions 1–7

Which paragraph contains the following information?

Write the correct letter, A–H.

NB *You may use any letter more than once.*

Q	Statement	
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1	the pros and cons for animals living in cities	
2	an example illustrating the benefit to research of working with non-scientists	
3	an explanation for the drop in numbers of one type of animal	
4	the likely proportion of local wildlife remaining, once a location has been urbanized	
5	the activities of a programme designed to help a particular at-risk species	
6	the consequences of having too little information about wildlife numbers	
7	an argument for more conservation programmes in cities rather than country areas	

Questions 8–12

Complete the notes below.

Choose **NO MORE THAN ONE WORD AND/OR A NUMBER** from the passage for each answer.

The MammalWeb Database

- 8 It is a UK wildlife programme aiming to measure
- 9 Members of the public can apply to be something called a

Findings

- 10 A total of
different types of animal have been recorded
- 11 The most common animal recorded is a type of
- 12 One unusual report was of a
(it was later taken to a wildlife park).

Question 13

Choose the correct letter, **A, B, C or D**.

13 Which of the following is the most suitable title for this reading passage?

- A The hidden world of garden animals
- B It's time to limit urban development
- C How local residents aid conservation
- D Why the future looks bad for urban wildlife

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Passage 56

Viking Ireland

A recent excavation in Dublin challenges long-held ideas about when the Scandinavian raiders known as Vikings arrived in Ireland

A When Irish archaeologists working under Dublin's South Great George's Street unearthed the remains of four young men buried with fragments of Viking shields, daggers and personal ornaments, the excavation appeared to be simply more evidence of the Viking presence in Ireland. At least 77 Viking burial sites have been found across the basis of artefacts that accompanied them, and the South Great George's Street burials seemed to be further examples. Yet when archaeologist Linzi Simpson sent the remains for analysis, the tests showed that men had been buried in Irish soil years, or even decades, before the accepted date for the establishment of the first year-round Viking settlement in Dublin.

① **B** Simpson's findings are now adding new weight to an idea gaining growing acceptance – that instead of a sudden, calamitous invasion, the arrival of the Vikings in Ireland started with small-scale settlements and trade links connected Ireland with northern Europe. And, further, that those trading contacts may have occurred generations before the violent raids described in contemporary texts, works written by monks living in isolated monasteries. These were often the only places where literate people lived and were especially targeted by Viking raiders for their food supplies and treasures. Scholars are continuing to examine the texts written by monks, but are also considering the limitations of using them. 'Most researchers accept now that the raids were not the first contact, as the old texts suggest,' says Viking expert Gareth Williams. 'How did the Vikings know where all those monasteries were? It's because there was already contact. They were already trading before those raids happened.'

② **C** Although the earlier dates for a Viking presence in Dublin that have been identified by Simpson and independent archaeologists differ from the later dates by only a few decades, when combined with other evidence, they are challenging the chronology of Viking settlement in Ireland. Since the 1960s, archaeologists have been gathering information about the mid-ninth-century settlement that lay under the sidewalks of Fishamble Street in Dublin. According to archaeologist Ruth Johnson, the Vikings started with sporadic summer raids, but after some years of profitable plunder, they decided to stay, and built settlements for the winter.

③ **D** Carbon dating, which measures the age of organic materials based on the amount of radioactive carbon 14 remaining in a specimen, usually gives a range of likely dates for the time of death. The older the material, the wider the range. In the case of the four individuals excavated at the South Great George's Street site, Simpson found that two of them had a 95 per cent probability of having died between 670 and 680, with a 68 per cent

probability of death occurring between 690 and 790. Thus, the entire most likely range was before the first documented arrival of Vikings in 795. A third individual lived slightly later, with a 95 per cent probability of having died between 680 and 882. The dates were not what Simpson had thought they would be. ‘These dates seem impossibly early and difficult to reconcile with the available historical and archaeological sources,’ she says.

④ **E** The fourth individual excavated at South Great George’s Street was the most intact of the group, and revealed the most about the lives and hardships of Vikings at this time. A powerfully built man in his late teens or early 20s, he was approximately 1.70m tall by the day’s standards, with the muscular torso and upper limbs that would have come from hard, ocean-going rowing. His bones showed stresses associated with heavy lifting beginning in childhood. Unlike the three other men, he was not buried with weapons. Like one of the other men found at the site, he had a congenital deformity at the base of his spine, perhaps indicating they were relatives. Carbon dating gave a wider range for his lifetime, showing a 95 per cent probability he died between 786 and 955.

⑤ **F** Tests were also carried out on the four South Great George’s Street men’s isotopic oxygen levels. Such tests indicate where a person spent their childhood based on a chemical signature left by groundwater in developing teeth. The results showed that the two men with the spinal deformity had spent their childhood in Scandinavia. However, the other two had spent their childhoods in Ireland or Scotland, another sign of permanent settlement by Viking families and not just summertime raids by warriors.

⑥ **G** The evidence of an earlier-than-expected Viking presence in Ireland, based as it is on forensic tests conducted on a handful of burials, may seem slight. But seemingly small pieces of evidence can overturn well-established conventions in archaeology. Both Simpson and Johnson stress more excavations and tests will be needed before anyone can rewrite the history of Viking settlement, and such work is years away. Williams adds, ‘There are two possibilities raised by (Simpson’s) work. Either there was Viking activity earlier than we’ve realised in Ireland, or there is something in the water or soil in Dublin skewing the data, and both possibilities need further research.’ Nevertheless, Williams agrees with Simpson and others that the chronology of the Viking presence in Ireland is uncertain, and that they were possibly trading or raiding in Ireland before 795. ‘It’s a poorly documented part of history,’ says Williams. But before there was Viking settlement, there was this big trading zone in the North Sea. Did it extend to the Irish Sea? We don’t have any evidence to say that, but it could be just a question of time.’

Questions 1–6

Reading Passage 2 has seven sections, A–G.

Choose the correct heading for each section from the list of headings below. Write the correct number, i–ix, in the circle before each paragraph.

NB Section A has been done for you as an example (heading **ii**).

- i** A possible genetic link between the Vikings and the Irish
- ii** An assumed similarity with previous discoveries (Example)
- iii** The need for additional data
- iv** An insight into the lifestyle of a particular Viking
- v** Doubts about the truth of historical documents
- vi** A research technique providing unexpected information
- vii** The locations particular Vikings grew up
- viii** A decision to remain in Ireland for longer periods

Questions 7–10

Look at the following statements and the list of experts below.

Match each statement with the correct expert, **A, B or C**. **NB** You may use any letter more than once.

- 7** The Vikings were aware of the financial benefits of staying in Ireland.
 - A** Linzi Simpson
 - B** Gareth Williams
 - C** Ruth Johnson
- 8** Written accounts reporting when the Vikings arrived in Ireland may not be accurate.
 - A** Linzi Simpson
 - B** Gareth Williams
 - C** Ruth Johnson
- 9** The inconsistency in sets of data came as a surprise.
 - A** Linzi Simpson
 - B** Gareth Williams
 - C** Ruth Johnson
- 10** It may be the case that the archaeological evidence gathered so far is being affected by geological factors.
 - A** Linzi Simpson
 - B** Gareth Williams
 - C** Ruth Johnson

Questions 11–15

Complete the summary below.

Choose **ONE WORD ONLY** from the passage for each answer.

THE FOURTH VIKING OF SOUTH GREAT GEORGE'S STREET

When the remains of the fourth Viking were excavated at South Great George's Street, it became clear that they were of a male who was tall by (11) _____ of that time. Due to strenuous physical activity, his (12) _____ as well as his arms were well developed, but several of his bones indicated stresses that would have dated from his (13) _____. He also had a genetic abnormality in his lower (14) _____. His burial differed from that of the other three Vikings discovered at the site, as no (15) _____ were included.

Passage 57

Walking and shoes in eighteenth-century London

A look at changing fashions in footwear and leisure

A It was difficult to get about in eighteenth-century London. During the first half of that century, the physical mobility of the upper classes in the city had been considerably restricted: the poor condition of the streets did not allow walking for pleasure outside private parks or pleasure gardens. Streets were dirty, rubbish was everywhere and the British climate made matters worse. Walking could be a risky business. The terrible state of roads, always flooded and muddy, led to the employment of cleaners and street sweepers.

B References to walking are not abundant in early eighteenth-century literature, except for writing about the countryside, where conditions were far better. Walking in London was regarded as inconvenient and even dangerous. For Londoners who wanted to travel, choices were limited to the use of boats and coaches. Walking in urban areas was generally associated with the poor, especially those who could not afford a coach.

C This situation began to change in the mid-eighteenth century, with measures implementing the provision of public paving. The first legislation concerning paving in London was passed in 1762, and during the following decades, similar measures were enacted in many cities. By the end of the century, areas for walking were commonplace, and far more people were enjoying the pastime than before. Sophie von la Roche writes about metropolitan street life in the diary of her visit to London of 1786. She comments on how many more pedestrians there were on the broad, clean pavements, alongside the houses where, she says, "many thousands of neatly clad people, eminent men and dressy women, pursue their way safe from the carriages, horses and dirt". Although the pavements were not yet a continuous feature of London's streets, their introduction marked a profound shift in how the city was experienced. Walking was no longer solely a necessity for the poor but had become a respectable activity for the upper classes.

D The transformation of London's streets had significant implications for footwear. In the early part of the century, shoes were designed primarily for protection rather than comfort or style. They were typically made of stout leather with thick soles, intended to keep the wearer's feet dry and clean when negotiating muddy, refuse-strewn thoroughfares. Pattens—overshoes with raised iron rings—were commonly worn by women to lift their delicate footwear above the filth. These contraptions were noisy and awkward, clattering against the stones and requiring a mincing gait to avoid tripping. For men, sturdy riding boots were the norm, even when walking, as they provided the best defence against the mire.

E As paving spread and streets became cleaner, shoe design evolved to reflect new possibilities. Lighter materials, including softer leathers and even silk and satin for evening wear, became fashionable. Heels grew more delicate, and soles thinner. This was particularly evident in women's footwear, where shoes became objects of display rather than mere utility. The French fashion for ornate buckles, often set with paste jewels, was widely adopted by London's elite. Men's shoes also became more refined, with the heavy riding boot giving way to the buckled court shoe for those not obliged to travel on horseback. This was not merely a matter of aesthetics; it represented a fundamental change in how Londoners inhabited their city. To walk in comfort and style was to announce one's participation in the new urban order.

F The rise of promenading as a social ritual further accelerated these trends. By the 1780s, certain London spaces had become established venues for the display of fashionable clothing, including footwear. Parks such as St James's and Kensington Gardens attracted crowds of strollers during the season, their circuits providing opportunities for social encounter and the mutual inspection of attire. Shoemakers responded to this demand by producing ever more varied and decorative styles. The craft of shoemaking itself was transformed, with specialist lasts—the forms around which shoes were shaped—being developed for left and right feet, a refinement not universally applied in earlier decades. This improved fit and comfort, encouraging yet more walking.

G The new culture of walking also had its critics. Moralists complained that the freedom to stroll encouraged idleness and frivolous display. Young people of both sexes, it was feared, might use the promenade as an opportunity for unsuitable liaisons. The very anonymity of the crowd, so different from the close-knit community of the village, was seen by some as a threat to social order. Yet such complaints were themselves a testament to how thoroughly walking had been transformed. What had once been a dangerous necessity for the poor had become a leisure activity pursued by all ranks of society, accompanied by its own etiquette, its own spaces, and its own distinctive footwear. The pedestrian had become a familiar figure in the London scene, and the shoe had become far more than a mere protection for the foot.

Questions 14–19

Reading Passage 2 has seven sections, A–G.

Which section contains the following information? Write the correct letter, A–G. You may use any letter more than once.

Q	Statement	
14	A reference to the employment of people to keep streets clean.	

15	Examples of the specific locations where fashionable people gathered to walk.	
16	An account of the awkward devices women used to protect their shoes from dirt.	
17	Mention of the first laws introduced to improve the condition of streets.	
18	Criticism of the new culture of walking from those concerned about morality.	
19	A comparison between walking in the city and walking in the countryside.	

Questions 20–23

Complete the sentences below.

Choose **ONE WORD ONLY** from the passage for each answer.

Walking and Shoes

- In the early eighteenth century, walking in London was associated mainly with people who were (20) _____.
- Sophie von la Roche observed that pedestrians could walk safely because pavements kept them away from carriages, horses, and (21) _____.
- In the early part of the century, shoes were made from thick (22) _____ to protect the feet.
- The development of specialist lasts for left and right feet improved the (23) _____ of shoes.

Questions 24–26

Choose the correct letter, **A**, **B**, **C**, or **D**.

- (24) What was the main reason for the change in shoe design in the late eighteenth century?
- A** French fashion became more influential in London.
 - B** The improved condition of streets allowed for lighter footwear.
 - C** Shoemakers developed new techniques for mass production.
 - D** The upper classes wanted shoes that were suitable for horse riding.

- 25 According to the passage, the introduction of paving in London
- A was completed throughout the city by 1762.
 - B was opposed by the poor who could not afford new shoes.
 - C made walking a respectable activity for all social classes.
 - D led to an immediate decrease in the use of coaches.
- 26 What is the main purpose of section G?
- A to describe the etiquette of promenading in London parks
 - B to explain how shoemakers responded to new fashions
 - C to show that not everyone approved of the changes in walking culture
 - D to compare village communities with city life in the eighteenth century

Passage 58

Waste Production

How the USA and other nations are tackling the growing problem of garbage and recycling

The United States of America leads the world in waste production, according to the Environmental Protection Agency (EPA), generating some 200 million tons a year, enough to fill a convoy of garbage trucks stretching eight times around the globe. Each US citizen discards 3.6 pounds a day, almost twice as much as the average German. Others feed on Americans' high-quality leavings. Japan takes their twisted metal and sells it back to them as cars. Other tree-poor nations buy their used paper to make more paper. Scrap iron and waste paper top the exports leaving New York harbor.

Landfills get two-thirds of what's left behind. But much of the heavily populated east coast is expected to run out of landfill space in this decade, and the EPA estimates that within the same time span, 80 per cent of landfills nationwide will close and new sites must be found. Western states have room for more, although dumps and their toxic by-products are seldom welcome neighbors. Where will our waste go?

We burn 16 per cent now and could burn more. But that raises concerns about air pollution from heavy metals such as lead, cadmium, copper and mercury, which vaporize in intense heat. We could reduce waste by cutting down on the packaging that surrounds our products: it accounts for one-third of our trash. And we could re-use more of it. 'Recycling works, and it makes economic sense even when a material is plentiful,' says Phil Bailey of the National Recycling Coalition in Washington, DC. 'Recycling glass requires less energy than making it from sand. Recycling steel is less expensive than mining ore. This is where the environment and the economy meet. It's becoming an integral part of business and industry.' Recycling has grown beyond a feel-good trend.

Here's how recycling works, generally. A community decides to divert waste from a landfill. In addition to the regular garbage trucks, new haulers with separate compartments must now go out to pick up newspapers, beer cans, pickle jars, detergent bottles and other reclaimables.

At a recycling center the materials are further sorted and compacted for sale to a manufacturer who makes them into new products. Yard trimmings and wooden construction debris – two of the largest components of landfills after paper – may be composted at the center and offered directly to consumers.

Most collection is operated or subsidised by local and state governments. The pacesetter is Seattle. It recovers 45 per cent of its waste compared with a 17 per cent average for the rest of the country. Success is pinned to a 'pay as you throw' program: residents who recycle best pay the least for removal of their trash. 'Our recyclables are picked up for nothing,'

explained Sally Kentch, who lives in a north Seattle neighborhood. 'But we pay \$15 a month for a 30-gallon can for garbage that can't be recycled. If we don't recycle well, we may need two garbage cans, which costs twice as much.' With the fees charged for the garbage cans, Seattle covers its weekly garbage pickup and pays two companies to pick up the recyclables. The companies sort the materials and sell them for profit.

But not everyone praises recycling; there are economic arguments against it as well. 'Implementing programs is expensive, and it's difficult to sell the materials,' says Lynn Scarlett of the Reason Foundation, a Los Angeles think tank. 'It works sometimes, but it doesn't make sense to ship recyclables to markets halfway across the country from areas where there is still landfill space.' Programs that look promising on paper sometimes flounder in practice. In Germany, for example, law-makers were praised in 1991 for requiring manufacturers to recycle their packaging – not literally to take it back, but to pay a second party to recycle the volume of packaging they were producing. Many businesses signalled their compliance by marking their products with a green dot. But without any system of enforcement, some dots appeared that were not backed by recycling contracts. And the packaging piled up beyond the recyclers' ability to process it. Green dots began showing up in landfills.

In the United States the first surge in recycling was also marked by mountains of unclaimed bottles and paper. The gluts have shrunk but not vanished. Supporters say it just takes time for regional markets to evolve and close the loop from manufacturer to consumer and back to manufacturer again. 'We underestimated people's response to recycling,' said David Dougherty, director of the Clean Washington Center, a state agency in Seattle that develops markets for recyclables. 'They did it so well, an over-supply resulted. But seeing those materials go begging has made people realize the need to create markets for them.'

'We've got to do it right,' says Dougherty. 'We've got to create local markets, make recycling a natural part of the economy so it becomes a part of our lifestyle. Of all the environmental concerns that have come up through the years, this is the most personal. People are uncertain what they can do about saving whales or the rainforest. But they can recycle the waste of their everyday lives.'

Questions 1–6

Answer the questions below.

Write **NO MORE THAN THREE WORDS AND/OR A NUMBER** from the passage for each answer.

- ① Which country buys metal waste from the USA?

- ② Which part of the USA has most space left for landfills?
- ③ What is the main objection to burning waste?
- ④ Which product is cheaper to recycle than to produce from raw materials?
- ⑤ What is the largest component of the landfills?
- ⑥ What percentage of waste do most US towns recycle?

Questions 7–13

Do the following statements agree with the claims of the writer in Reading Passage 2?

*Write **YES** if the statement agrees with the claims of the writer, **NO** if it contradicts them, or **NOT GIVEN** if it is impossible to say what the writer thinks about this.*

- ⑦ The USA should burn more of its waste.
- ⑧ People in Seattle recycle because they have to pay for unrecycled waste.
- ⑨ There is no system reinforcing the recycling laws in Los Angeles.
- ⑩ Germany is the country with the most advanced recycling laws.
- ⑪ Some German manufactures made false claims about recycling their packaging.
- ⑫ Given time, the recycling systems in the USA will be able to cope with the amounts of waste produced.
- ⑬ People in Seattle reacted to recycling more enthusiastically than expected.

Passage 59

What's Happening to Our Food?

Is the way we produce and process food making it less nourishing than in our grandparents' day?

A We now know a lot about what food does to the body and the importance of a healthy diet. But what if modern intensive farming methods have affected the mineral and vitamin content of what we eat? Donald Davis, at the University of Texas, has found notable declines in nutrients in crops including tomatoes, eggplants and squash. Davis blames agricultural practices that prioritize quantity over quality. High-yielding crops produce more food, more rapidly, but they can't make or absorb nutrients at the same pace, so the nutrition is diluted. 'It's like taking a glass of orange juice and adding water to it. If you do that, the concentration of nutrients that was in the original juice drops,' he says. But the idea that modern farming produces less nourishing crops remains controversial, since nutrient levels can vary widely according to the variety of plant, the year of harvest and the time of harvest.

B But intensive farming has also led to a huge increase in food supply, which has undoubtedly had a positive effect on our diet and health. 'Evidence suggests that some nutrients have fallen, particularly trace elements such as copper in vegetables,' says Paul Finglas, at the Institute of Food Research in Norwich. 'Foods are now bred for yield, and not necessarily nutritional composition. But I don't think that is a problem, because we eat a wider range of foods today than we did 10 years ago, let alone 40 years ago.' Eric Decker, professor of food science at the University of Massachusetts in Amherst, agrees. If nutrients are declining, the losses are insignificant, he says. 'Over the last century, lifespans have got longer, people are bigger and stronger, and a lot of that has to do with the food supply being better.' Even Davis agrees that any differences in nutrient levels are relatively small. 'Despite their decline, fruit and vegetables are still our richest source of many nutrients, and you can make up for it by eating more,' he says. 'But we know that many people don't get the recommended amounts of nutrients such as iron, magnesium, and calcium. They aren't overt deficiencies in the usual sense, but they increase susceptibility to lots of different problems.'

C In addition to changes in its production, food is being transported for ever-increasing distances. Fruit and vegetables in supermarkets might look shiny and fresh, but often they were picked several days earlier. Some nutrients, particularly vitamin C and folic acid, begin to deteriorate as soon as picking happens, but manufacturers can take action to minimise this. 'Lots of these reactions are driven by enzymes,' says Carol Wagstaff, from the University of Reading, UK, who points out that chilling is the best way of slowing down such reactions. That said, if you are choosing between organic fruit or vegetables from a distant country or locally grown, non-organic ones, always opt for home-grown, she says.

'From a nutritional standpoint, go for the shortest possible supply chain rather than the production method.' Many kinds of mass-produced fruit and veg, most famously tomatoes, are harvested unripe before being transported to supermarkets. Wagstaff agrees that this may cause some loss of flavor, compared with a tomato that you have grown at home, and left on the plant until it's absolutely ready to eat. 'But there's no way you could do that at a commercial level, because of the bruising and other types of damage that would occur if ripe fruits were transported through a typical supply chain,' she says. In addition, each method of shipping and storing foods has different effects on the compounds they contain. Vitamin C, for example, breaks down in the dark, whereas glucosinolates — found in vegetables like cabbage — deplete in the light.

D Surprisingly, frozen fruit and vegetables are often nutritionally better than fresh. 'Frozen peas are much more nutritious than those you buy ready to shell,' says Catherine Collins, principal dietician at St George's Hospital in London. What's more, frozen foods often have fewer add-ons. 'Freezing is a preservative,' she says. Similarly, processing has become a maligned word in the context of food, but there are some cases where it enhances a food's health benefits. For example, lycopene — a compound tomatoes are rich in, and which has been shown to protect against certain diseases — is much more readily absorbed by humans from tomato paste than fresh tomatoes. A recent trend is the sale of 'fresh-cut' fruit and vegetables — peeled potatoes, ready chopped carrots and bagged salads. One in five adults in the UK regularly buys fruit and vegetables in this form every week, according to market research firm Mintel. Surely this cutting and peeling speeds up the degradation of nutrients? 'There is a chance that ready prepared vegetables may have a lower content of some vitamins,' says Judy Buttriss, of the British Nutrition Foundation in London. 'But if their availability means that such vegetables are consumed in greater quantities, then the net effect is beneficial.'

E The bottom line is that although aspects of today's food production, processing and storage might make what we eat a bit less nutritious, they are also making foods more available — and this is far more important. 'The most important thing you can do is eat more fruits, vegetables and wholegrains, and cut down on highly refined, processed foods, vegetable oils and added sugars,' says Davis. He believes this will make a far greater difference to us than worrying about growing methods and transportation.

Questions 14–18

Reading passage 2 has 5 sections, A–E.

Which section contains the following information?

NB *You may use any letter more than once.*

Q	Statement	
14	an admission that if one type of food is harvested too soon, its taste may be affected	
15	the view that there is not a great difference between the quantity of nutrients in our food now and in the past	
16	a comparison which illustrates why developments in agriculture may reduce the amounts of nutrients in foods	
17	a warning that customers could be deceived by the attractive appearance of a fruit or vegetable	
18	evidence of the popularity of a new development in food processing which aims to save customers trouble	

Questions 19–22

Look at the following statements and the list of people below.

Match each statement with the correct person, **A–F**.

LIST OF PEOPLE

- A** Donald Davis
- B** Paul Finglas
- C** Eric Decker
- D** Carol Wagstaff
- E** Catherine Collins
- F** Judy Buttriss

Q	Statement	
19	People are more likely to become ill if they have low levels of some nutrients.	
20	Making healthy foods easy to cook may be more important than their vitamin content.	
21	An improved diet has made people nowadays live longer and be healthier than in the past.	
22	People's diets are more varied now than they were in the past.	

Questions 23–26

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

FOOD TRANSPORTATION

In order to prevent loss of nutrients when transporting fruit and vegetables, chilling is used to slow down the effect that (23) _____ have on them.

Some foods, such as tomatoes, must be picked before they are ripe to avoid problems such as (24) _____ during transportation.

Other foods, such as cabbage, lose nutrients when kept in the (25) _____.

Vegetables such as (26) _____, which are picked fresh and transported to the supermarket, may be less nutritious than those which are frozen.

Passage 60

Why Do We Need Sleep?

From car crashes to obesity, the surprising cost of not getting enough rest – and what our brains are actually doing while we sleep

A According to the United States (U.S.) Centers for Disease Control and Prevention (CDC), more than 80 million American adults are chronically sleep deprived, meaning they sleep less than the recommended minimum of seven hours a night. The resulting fatigue contributes to more than a million auto accidents each year, as well as to a significant number of medical errors. Even small adjustments in sleep can be problematic. The Monday after a daylight-savings time change in the U.S. – when the clocks are moved ahead one hour – there is a 24 percent increase in heart attacks and a jump in fatal car crashes, compared with other Mondays.

B During their lifetimes, about a third of Americans will suffer from at least one of the recognized sleep disorders. They range from common sleep disorders to much rarer and stranger conditions such as sleep apnea and restless leg syndrome. For example, in exploding head syndrome, booming noises seem to reverberate in your brain as you try to sleep. Also, people with Kleine-Levin syndrome will, every few years, sleep nearly nonstop for a week or two. However, insomnia is by far the most prevalent of these problems and the main reason 4 percent of U.S. adults take sleeping pills in any given month.

C Evolution endowed us with sleep that is flexible in its timing and easily disturbed, so that attention can be directed to higher priorities. The brain has an automatic override system that can wake us in all stages of sleep when it perceives an emergency like the cry of a child. But the problem is that, in the modern world, our ancient innate wake-up call is constantly triggered by something that is non-life threatening, like worrying before an exam or the unexpected sound of a car alarm. Before the industrial revolution, which brought us alarm clocks and fixed work schedules, we could often counteract a sleep deficit by simply sleeping in. That is no longer possible.

D The first segment of the brain that begins to fail when we don't get enough sleep is the prefrontal cortex, the center of decision-making and problem-solving. Dr Chira Cirelli, a neuroscientist in the U.S., suggests that, 'Every cognitive function to some extent seems to be affected by sleep loss,' and she comments that people are often more irritable, moody, and irrational. For instance, sleep-deprived suspects held by the police will often confess to anything in exchange for rest. Other research has also found that people who regularly sleep less than six hours a night have an elevated risk of developing depression, as well as other mental and physical illnesses. In fact, lack of sleep has been directly tied to obesity:

without enough sleep, the stomach and other organs overproduce ghrelin, the hunger hormone, causing us to eat more. However, proving a cause-and-effect relationship in this case is challenging because subjecting humans to the necessary experiments is unethical.

E NREM (Non-Rapid Eye Movement) makes up the first of two distinct and repetitive phases of a night's sleep. As we fall into NREM sleep, our brain stays active and begins an editing process – deciding which memories to keep and which ones to ignore. The first of NREM's four stages is called the shallow end of sleep and is characterized by a distinctive regular pattern of brain activity, as shown on an EEG (electroencephalogram) device – a machine that measures electrical impulses. In stage two, the EEG measures electric sparks that strike the cerebral cortex (grey matter covering the outer layer of the brain) in regular half-second intervals. Researchers believe that this electrical activity helps the cortex preserve and store recently acquired information. Stages three and four have been described as a deep, coma-like sleep that is as essential to our brain as food is to our body. In stage three, big, rolling waves of brain activity as measured on an EEG are present less than half the time; in stage four, more than half the time. It's in this deep sleep that our cells produce most growth hormones, which are needed throughout life to service bones and muscles.

F REM (Rapid Eye Movement) sleep was at first believed to be only a variation of a stage in NREM sleep, and not particularly significant. However, once scientists documented the distinctive eye movements in REM, and found that virtually all dreaming takes place during this period, REM sleep was recognized as the second major phase of sleep. Today, scientists believe that the content of our approximately two hours of dreams each night in REM sleep is important in the processing of new memories. Some sleep theorists argue that REM sleep is when we are our most intelligent, insightful, creative, and free. It's when we truly come alive. 'REM sleep may be the thing that makes us the most human, both for what it does for the brain and body, and for the sheer experience of it,' says Professor Michael Perlis.

G The problem of sleep loss is not easily solved – whether by power naps or pharmaceuticals. Dr Jeffrey Ellenbogen, a sleep scientist at Johns Hopkins University in the U.S., argues, 'It's tempting to manipulate sleep with drugs or devices, but we don't yet understand sleep enough to risk artificially manipulating the parts.' Dr Ellenbogen and other experts argue against shortcuts, especially the notion that we can mostly do without sleep. In fact, Dr Steven Lockley, of Brigham Women's Hospital in Boston, suggests that sleep may be more essential to us than food.

Questions 1–7

Reading Passage 2 has seven paragraphs, A–G.

Which paragraph contains the following information? Write the correct letter, **A–G**.

NB You may use any letter more than once.

Q	Statement	
1	an example of how sleeplessness can influence a person accused of a crime	
2	description of recordings of different types of brain activity during sleep	
3	a description of how present-day society interrupts sleep	
4	information about the relationship between being overweight and sleep	
5	a suggestion that medication is an ineffective solution for sleeping problems	
6	a discovery that changed ideas about how sleep is understood	
7	examples of the life-threatening consequences of inadequate sleep	

Questions 8 and 9

Which **TWO** of these statements describe characteristics of *NREM* sleep? Choose **TWO** letters, **A–E**.

OPTIONS

- A** NREM sleep is important for maintaining brain function.
- B** NREM sleep consists of two distinct parts.
- C** The brain remains passive throughout NREM sleep.
- D** Each stage of NREM sleep has a unique EEG pattern.
- E** Hormones are released continuously throughout NREM sleep.

Answer: (8) (9)

Questions 10 and 11

Which **TWO** of these statements describe the characteristics of *REM* sleep? Choose **TWO** letters, **A–E**.

OPTIONS

- A** The REM stage is the longest phase of sleep.
- B** REM sleep can be identified by an unusual physical feature.
- C** REM sleep is important for removing unpleasant memories.
- D** REM sleep is another part of NREM sleep.

E Some scientists consider REM sleep to be a source of human innovation.

Answer: 10 11

Questions 12–16

Complete the sentences below.

Choose **ONE WORD ONLY** from the passage for each answer.

- 12 According to the CDC, lack of sleep leads to _____, which is often a factor in car crashes.
- 13 The most common sleep disorder in the U.S. is _____.
- 14 The brain responds to any type of _____ by activating a built-in alarm.
- 15 Researchers claim that routinely sleep-deprived people are more likely to be obese and suffer from illnesses like _____.
- 16 The processing of new memories is crucially linked to our _____ during sleep.

Answer Key & Explanations

All 60 Passages

Passage 1 — A New Look for Talbot Park

14. **iv**

would not look out of place in some of the city's most affluent suburbs. Answer: iv.

15. **ix**

there was a strong sense of neighbourliness. Answer: ix.

16. **i**

petty crime was on the rise and the garden was considered unsafe. Answer: i.

17. **x**

some of the accommodation is free-standing houses, some semi-detached, some low-level, multi-apartment blocks. Answer: x.

18. **iii**

50% Pacific Islanders, 20% Maori, 15% Asian, 10% New Zealand European. Answer: iii.

19. **v**

Talbot Park and its surrounding suburb of Tamaki are low socio-economic areas. Answer: v.

20. **vii**

The focus is on frequent inspections and setting clear guidelines and boundaries. Answer: vii.

21. **D**

Lundy: buildings were sufficiently flexible to cater for the needs of people from a wide variety of cultural backgrounds. Answer: D.

22. **A**

Bodman: the focus is on frequent inspections and setting clear guidelines and boundaries. Answer: A.

23. **C**

Bracey: help them to get to know each other by organising morning teas and street barbecues. Answer: C.

24. density

24: the new development has actually made the density of housing in Talbot Park greater.

25: employing a variety of plans by different architects. **26:** ensure the project did not go over budget.

Answers: density / architects / budget.

25. architects

24: the new development has actually made the density of housing in Talbot Park greater.

25: employing a variety of plans by different architects. **26:** ensure the project did not go over budget.

Answers: density / architects / budget.

26. budget

24: the new development has actually made the density of housing in Talbot Park greater.

25: employing a variety of plans by different architects. **26:** ensure the project did not go over budget.

Answers: density / architects / budget.

Passage 2 — Achieving a Work-life Balance

14. C

Para C: the truth is that such a four-day week has many attractions but it will not be appropriate in every instance. Answer: C.

15. A

Para A: in the stone age... the ancestors of modern humans worked on average two or three hours a day hunting and gathering food. Answer: A.

16. F

Para F: roland gillespie, workplace relations minister in the new zealand government... said that he was keen to see businesses experiment with similar new and improved work models. Answer: F.

17. B

Para B: productivity in the economy generally has increased significantly over the last 50 years but there's a common perception in the workforce that standards of living have not kept pace. Answer: B.

18. D

Faisal Hadi: life is an intricate mix of work and leisure and the relationship between the two has so many strands that they cannot be neatly divided. Answer: D.

19. F

Tane Robinson: the four-day week boosted the firm's reputation and made it easier to retain staff who might otherwise have moved on. Answer: F.

20. **B**

Carla Deng: **when employees have greater individual choice and responsibility for their work time, they still devote some of those free hours to fulfilling their normal working roles.**

Answer: **B.**

21. **A**

Dr Gina McDowell: **the problem of stress caused by working too many hours still doesn't receive sufficient recognition and ought to be addressed with greater commitment.** Answer: **A.**

22. **E**

Dr Hannah Graham: **the experiment produced such positive results partly because employees were given a role in decision-making throughout it.** Answer: **E.**

23. **insurance**

certified life, a firm which has sold insurance for more than six decades. Answer: **insurance.**

24. **surveys**

the 240 employees at certified life completed surveys at the beginning and end of the experiment. Answer: **surveys.**

25. **supervisors**

team cohesion was significantly enhanced, according to the supervisors. Answer: **supervisors.**

26. **congestion**

Dr Graham suggests such schemes might address **the issue of congestion in major cities.** Answer: **congestion.**

Passage 3 — Australian Aboriginal Astronomy

1. **C**

Para C: **Bill can name about 5,000 stars.** Most Western astronomers can name only 20 or 30. Answer: **C.**

2. **B**

Para B: the rising of the Pleiades signals **the orca migrating north.** Answer: **B.**

3. **E**

Para E: Aboriginal calendars are **often based on six seasons.** Answer: **E.**

4. **G**

Para G: Aboriginal people saw themselves not as a separate external observer, but as **an integral component of nature and the universe**. Answer: **G**.

5. **E**

Para E: the carving depicts an emu **in a somewhat unnatural position for a real emu, with the legs folded behind it**. Answer: **E**.

6. **D**

Para D: this could help **foster a better understanding and appreciation of Aboriginal culture among the wider Australian society**. Answer: **D**.

7. **A**

Hamacher: **they can tell by the degree to which a star twinkles or changes colour to gauge the amount of moisture in the atmosphere**. Answer: **A**.

8. **B**

Norris recounts how Bill **looks at the sky and knows it reflects what's on the land**, completely intuitively. Answer: **B**.

9. **A**

Hamacher: **cultural rules about who you can marry... all of this social structure is written in the stars**. Answer: **A**.

10. **D**

Clarke: knowing autumn is imminent allows the Kaurna time to **build their large, waterproof huts**. Answer: **D**.

11. **B**

Norris: the study is an opportunity **to gain access to information that may have been lost after European colonization**. Answer: **B**.

12. **C**

Stevens: Scorpius is actually **a map for us**; people use the stars to travel from one clan group to another. Answer: **C**.

13. **carving**

it is a **carving** clearly visible on a rock. Answer: **carving**.

14. **legs**

the emu is depicted with its **legs** folded behind it. Answer: **legs**.

15. **April**

in **April** the body of the Emu in the Sky reaches a certain angle, and it is the season to find fresh emu eggs.
Answer: **April**.

16. **nest**

the folded legs signify that the emu is sitting on the **nest**. Answer: **nest**.

Passage 4 — Biological Control of Pests

14. **B**

pesticides have caused **widespread ecological disorders** and created chemical-resistant superbugs.
Answer: **B**.

15. **A**

more than 300 species of agricultural pests have developed resistance to a wide range of potent chemicals. Answer: **A**.

16. **D**

farmers **took to pesticides as a sure measure to boost crop yield**. Answer: **D**.

17. **D**

50% of the financial outlay on cotton production was accounted for by pesticides. Answer: **D**.

18. **NOT GIVEN**

the passage gives counts for both groups – **about 100 species of which have become immune to a variety of insecticides now in use** – but never compares how quickly each group responds. Answer: **NOT GIVEN**.

19. **YES**

pests **bear offspring with a built-in resistance to pesticides**. Answer: **YES**.

20. **NO**

biological control **involving the selective use of natural enemies of the pest population** – not synthetic chemicals. Answer: **NO**.

21. **YES**

when handled by experts, bio-control is safe. Answer: **YES**.

22. **D**

disapene scale insects – **notorious defoliants of fruit trees**. Answer: **D**.

23. **H**

Neodumetia sangawani, was found useful in controlling the Rhodes grass-scale insect.
Answer: **H**.

24. **C**

Sri Lanka's flourishing coconut groves were plagued by leaf-mining hispid. Answer: **C**.

25. **E**

an Argentinian weevil for the eradication of water hyacinth. Answer: **E**.

26. **B**

About 30,000 hectares of rice fields in Kerala are infested by this weed. Answer: **B**.

Passage 5 — Biomimetics

1. **E**

Para E: **it creates an armor 3,000 times harder than chalk**, made from the same substance as soft chalk. Answer: **E**.

2. **I**

Para I: **bio-inspiration has led to surprisingly few mass-produced products, and arguably only one household word** — Velcro. Answer: **I**.

3. **B**

Para B: de Mestral **had trouble removing a plant's prickle which was stuck to his dog's fur, so he studied it under a microscope**. Answer: **B**.

4. **D**

Para D: glowworms waste almost no energy, while **a normal light bulb wastes 98 percent of its energy as heat**. Answer: **D**.

5. **C**

Para C: **a 45-million-year-old fly trapped in amber** inspired a design now used in solar panels — **this feature is now being built into solar panels**. Answer: **C**.

6. **G**

Para G: bio-inspired robots are useful in places **where people would be too conspicuous, bored to tears, or killed**. Answer: **G**.

7. **B**

Andrew Parker: **every species, even those that have become extinct, is a success story, optimized by millions of years of natural selection.** Answer: **B**.

8. **D**

Ronald Fearing: **some things are just too mysterious and complex to be able to replicate.** Answer: **D**.

9. **C**

Robert Cohen: **he considers a biomimetics project a success only if it has the potential to make a useful tool for people.** Answer: **C**.

10. **cosmetics**

he is working to make cosmetics that mimic the natural sheen of diatoms. Answer: **cosmetics**.

11. **birds**

the metallic sheen and dazzling colors of certain birds derive not from pigments but from neatly spaced microstructures, of interest to paint manufacturers. Answer: **birds**.

12. **chemicals**

bombardier beetles have a highly effective combustion chamber in their posterior that heats chemicals and fires them at would-be predators. Answer: **chemicals**.

13. **fly**

Ronald Fearing aims **to create a miniature robotic fly that is swift, small, and maneuverable enough for use in surveillance or search-and-rescue operations.** Answer: **fly**.

Passage 6 — Biophilic Design

14. **C**

The rates of technological progress are far exceeding rates of psychological evolution, leaving us ill-equipped to cope with our lifestyle, which in turn leads to increased stress. Answer: **C**.

15. **B**

arguably speaks to the human soul much more than the boxlike blocks of flats designed by his contemporary Le Corbusier. Answer: **B**.

16. **A**

seeks both to avoid or minimize harmful impacts on the natural environment, and equally importantly, to provide and restore beneficial contacts between people and nature. Answer: **A**.

17. **H**

They mimicked nature's forms, producing magnificent structures with which we are still awed.
Answer: **H**.

18. **B**

architects have put creative originality before the needs of the people who have to live and work in their futuristic constructions. Answer: **B**.

19. **F**

It harnesses solar power, utilizes both active and passive air systems, and monitors the weather to adapt to conditions. Answer: **F**.

20. **psychological**

The rates of technological progress are far exceeding rates of psychological evolution.
Answer: **psychological**.

21. **stress**

the brain's constant attempts to function in a modern environment... which in turn **leads to increased stress.** Answer: **stress**.

22. **sunlight**

one crucial element of the natural landscape to human health is **sunlight**. We are evolutionarily programmed to respond positively to sunny areas. Answer: **sunlight**.

23. **tree**

One group had a view of a tree; the others had a window looking onto a brick wall. The first group had shorter hospital stays. Answer: **tree**.

24. **air**

The wall purifies the air and has the potential to fulfil the building's fresh air intake requirements.
Answer: **air**.

25. **B or D**

(B) Orr and his team of architects **engineered a healthy and comfortable space for students, while ensuring the surrounding environment was undamaged.** (D) a building that would help to **redefine the relationship between humankind and the environment — one that would expand our sense of ecological possibilities.** Answers: **B, D**.

26. **B or D**

Same evidence as Q25 — this pair shares one answer set. Answers: **B, D**.

Passage 7 — Children's Comprehension of Television Advertising

14. **C**

Para C: **the number of channels received in the average US home has escalated with the diffusion of cable television and satellite technologies.** Answer: C.

15. **D**

Para D: **approximately 80% of all advertising targeted at children falls within four product categories: toys, cereals, candies, and fast-food restaurants.** Answer: D.

16. **G**

Para G: **host-selling is thus restricted in the US by the Federal Communications Commission during children's programs.** Answer: G.

17. **D**

Para D: **they use the strategy of introducing unique sound effects and rapidly moving images... to associate product with playfulness and happiness.** Answer: D.

18. **E**

Para E: 'part of a balanced breakfast' leaves children with **the misimpression that cereal alone is sufficient for a meal.** Answer: E.

19. **D**

to associate product with playfulness and happiness, rather than to provide any actual product-related information (Kunkel et al, 1992). Answer: D.

20. **C**

television programming targeted at children was limited and relegated to time slots unpopular with their parents, such as Saturday mornings (Turow, 1981). Answer: C.

21. **B**

children influence \$4190 billion in family purchases, underscoring the high stakes involved (McNeal, 1998). Answer: B.

22. **F**

the use of child-friendly language such as 'you have to put it together' more than doubled the proportion of children who understood the qualifying message (Liebert et al, 1977). Answer: F.

23. **scene**

frequently explaining them as if they were a scene in the program itself. Answer: **scene.**

24. separators

these 'separators' generally do not help children to recognize advertising content. Answer: separators.

25. host-selling

when an ad includes one of the characters featured in a program, this is known as host-selling. Answer: host-selling.

26. shorter

'commercials are short and programs are long'. Answer: shorter.

Passage 8 — Classical Music over the Centuries

14. iii

the production of any great art form... does not usually occur in a society dominated by the basic material demands of food and shelter. Answer: iii.

15. ii

the rise of an economically independent middle class meant that concert going became a public activity for anyone who cared to buy a ticket. Answer: ii.

16. v

nothing has changed music over the last century more radically than the invention and dissemination of recording technologies. Answer: v.

17. vi

people who wanted to listen to music went to a specific venue, at a specific time, to hear musicians create a one-off event. Answer: vi.

18. i

all music, classical music included, can become any person's soundtrack for activities such as commuting, exercising or shopping. Answer: i.

19. vii

the history of classical music, in all its varied forms, is the history of a tradition that grew out of the possibilities of musical notation. Answer: vii.

20. **C or E**

(C) **people have more music stored on their phones or computers... than would have been contained on all the metres of library shelves of a proud record collector of the 20th century.**
(E) **from the mid-1980s onwards, the vinyl disc gradually gave way to the new technology of the cd, but just a decade later, the digital mp3 file was already displacing the cd.** Answers: C, E.

21. **C or E**

Same evidence as Q20 — this pair shares one answer set. Answers: C, E.

22. **B or D**

(B) **the first attempts to notate music were intended to help them remember these [chants].**
(D) **the history of classical music, in all its varied forms, is the history of a tradition that grew out of the possibilities of musical notation.** Answers: B, D.

23. **B or D**

Same evidence as Q22 — this pair shares one answer set. Answers: B, D.

24. **venue**

24: music is no longer tied to **any specific occasion or venue.** **25:** "one of its distinctive aspects — a performance defined by **concert halls** and opera houses — is dissolved by digital recording formats." **26:** music can become any person's **soundtrack** for activities such as commuting, exercising or shopping.
Answers: **venue / halls / soundtrack.**

25. **halls**

24: music is no longer tied to **any specific occasion or venue.** **25:** "one of its distinctive aspects — a performance defined by **concert halls** and opera houses — is dissolved by digital recording formats." **26:** music can become any person's **soundtrack** for activities such as commuting, exercising or shopping.
Answers: **venue / halls / soundtrack.**

26. **soundtrack**

24: music is no longer tied to **any specific occasion or venue.** **25:** "one of its distinctive aspects — a performance defined by **concert halls** and opera houses — is dissolved by digital recording formats." **26:** music can become any person's **soundtrack** for activities such as commuting, exercising or shopping.
Answers: **venue / halls / soundtrack.**

Passage 9 — Coral Reefs

1. **C**

Para C: **reefs often give rise to islands that provide habitats for people, and lenses of fresh water for drinking and wave action, and protect lagoons and other productive areas.** Answer: C.

2. **A**

Para A: **over half a billion people live within 100km of coral reef, with over 20 million living near reefs in south-east asia and nearly 100 million living near reefs in the indian ocean.**

Answer: **A.**

3. **F**

Para F: **one of the principal factors responsible for declining reef benefits is reef degradation.**

Answer: **F.**

4. **A**

Para A: **coral reefs are found in shallow waters throughout the tropical world, and dominate the coastlines of many countries in the south pacific, southeast asia and the indian ocean.**

Answer: **A.**

5. **C**

Para C: **raw materials include seasonally stable supplies of food, building materials and medicines.** Answer: **C.**

6. **E**

Para E: **coral reefs offer opportunities for women and the elderly, because reef products can be collected on foot.** Answer: **E.**

7. **TRUE**

national and international interest in coral reefs are growing, for example as a focus for research studies by marine biologists. Answer: **TRUE.**

8. **TRUE**

still, others use the reef as a resource that they tap into at certain times of the year, when they cannot earn a living from other resources. Answer: **TRUE.**

9. **NOT GIVEN**

the passage mentions high-value export outlets and tourism separately, but never states that tourists are the main buyers of exported reef products. Answer: **NOT GIVEN.**

10. **FALSE**

the structural and species diversity of the reef prohibits large-scale, industrial production, and favours small-scale production. Answer: **FALSE.**

11. **FALSE**

unlike many other types of fishery, where some are excluded from working, coral reefs offer opportunities for women and the elderly. Answer: **FALSE.**

12. FALSE

coastal tourism developments, external markets and well-meaning efforts to halt reef decline have excluded them from access to reef benefits too. Answer: **FALSE**.

13. C

the whole passage focuses on the economic benefits, livelihoods, income and resources coral reefs provide to coastal communities, not on marine biology, repair techniques, or ecological sustainability specifically. Answer: **C**.

Passage 10 — Early Approaches to Organisational Design

14. A or D

(A) organisations were **characterised by the specialisation of tasks... clear chain of command** — a marked ranking order. (D) an **impersonal climate** — a neutral working environment. Answers: **A, D**.

15. A or D

Same evidence as Q14 — this pair shares one answer set. Answers: **A, D**.

16. B or E

(B) **some of the major advantages of the bureaucratic structure could become disadvantages if the theory were applied dogmatically**. (E) critics argue **the world no longer fits the assumptions in weber's model... so a bureaucracy might not yield beneficial results**. Answers: **B, E**.

17. B or E

Same evidence as Q16 — this pair shares one answer set. Answers: **B, E**.

18. A or D

(A) an alternative design that **acknowledged human needs and feelings, and increased workers' satisfaction**. (D) **permitting more subordinate participation in decision-making**. Answers: **A, D**.

19. A or D

Same evidence as Q18 — this pair shares one answer set. Answers: **A, D**.

20. C or E

(C) neoclassicists **overlook environmental, technological, and other variables that might affect an organisation's design**. (E) **not everyone is motivated by the non-monetary aspects of work, nor is all work satisfying**. Answers: **C, E**.

21. C or E

Same evidence as Q20 — this pair shares one answer set. Answers: **C, E**.

22. **B**

McGregor: the bureaucratic hierarchy **was derived from negative and false assumptions about workers which he called 'theory x'**. Answer: B.

23. **D**

Likert: **in system 4 organisations... there is extensive participation in decision-making and problem-solving groups. some individuals in each group also belong to other work groups to ensure communication between them.** Answer: D.

24. **A**

Weber: organisations were **characterised by the specialisation of tasks.** Answer: A.

25. **C**

Argyris **argued for an alternative organisational design that acknowledged human needs and feelings.** Answer: C.

26. **A**

Weber believed **clearly specified authority and responsibility made performance easier to evaluate and reward.** Answer: A.

Passage 11 — Farmers

1. **F**

Para F: **it was taken over by chase corporation... in 2003, it went back into 'family' ownership with the purchase by the james pascoe group.** Answer: F.

2. **D**

Para D: **today it sells a narrower assortment of products than in the past.** Answer: D.

3. **B**

Para B: **laidlaw guaranteed customer satisfaction, and those who were not entirely happy with a purchase would receive a full refund with no questions asked. this was certainly the first time such a guarantee had been made by a new zealand retailer.** Answer: B.

4. **A**

Para A: **the lookout tower on top of the farmers building would be a unique place to fit the ring on my new fiancée's finger.** Answer: A.

5. **E**

Para E: **its most recent branch in the suburb of albany is filled with bright lights and big brands.** Answer: E.

6. **mail order**

the chain of stores known as farmers began one hundred years ago as a mail order company.
Answer: **mail order**.

7. **buying officers**

laidlaw had established buying officers in new york and london so as to offer his customers the latest designs. Answer: **buying officers**.

8. **large family**

they all emphasize how the company represented a large family. Answer: **large family**.

9. **annual sale**

an annual sale to celebrate the birthday of hector the parrot. Answer: **annual sale**.

10. **E**

coming to farmers once a week has become woven into the fabric of new zealand culture.
Answer: **E**.

11. **D**

farmers has always had a heart. Answer: **D**.

12. **F**

we were focused too much on price and not on fashion. the reorientation of the last decade has seen a great improvement. Answer: **F**.

13. **A**

i believe that the original culture of the store was dispelled and it hasn't been recovered.
Answer: **A**.

Passage 12 — Giftedness and Intelligence

1. **C**

Para C: Binet **reasoned that older children would perform better on mental tasks than younger children and that what was needed was a set of tasks arranged in chronological order**. Answer: **C**.

2. **E**

Para E: some psychologists **adopted the concept of intelligence as a single human attribute** like height. Answer: **E**.

3. **G**

Para G: **even if they test abilities of one sort or another, which undoubtedly many of them do, they do not test life-skills such as personality, motivational and emotional factors, which may crucially affect future performance and achievement.** Answer: **G**.

4. **B**

Throughout, the writer raises doubts about validity, ethics, and coverage of IQ tests – e.g. **many psychologists have doubts about the efficacy of these tests and about the ethics of using them.** The overall aim is to discuss their limitations, not simply narrate history or define intelligence. Answer: **B**.

5. **A**

children of 7 years of age who performed at the 5-year-old level, for instance, needed special help in small classes to enable them to improve. Answer: **A**.

6. **C**

what was needed was a set of tasks arranged in chronological order. Answer: **C**.

7. **D**

the writer argues **it is better to do so by studying discrete elements in specific situations.** **Today, it is that direction in which studies of intellect have moved.** Answer: **D**.

8. **YES**

Binet emphasised **he was not measuring intelligence, which was too complex to be measured.** Answer: **YES**.

9. **NOT GIVEN**

The passage says Binet worked in Paris, but never states his tests were restricted to Paris schools only. Answer: **NOT GIVEN**.

10. **YES**

psychologists **adopted his notions for purposes he never intended,** serving the interests of armed forces, employers and educational administrators. Answer: **YES**.

11. **NO**

no matter how accurately one has measured, the value or significance of what has been measured may still be in doubt – unlike simple physical measures such as height. Answer: **NO**.

12. **YES**

the writer suggests **we would be better employed finding out what they can do and how well and quickly they can learn to do more.** Answer: **YES**.

13. NO

tests **will not identify children with artistic or musical abilities and they are unlikely to reveal pupils of exceptional mathematical ability**. Answer: NO.

Passage 13 — From princes to paupers: how Goya's portraits tell the story of Spain

14. G

a weakness in some of Goya's paintings ≈ para G: **"some have a doll-like stiffness"**. Answer: G.

15. E

why some artists did not enjoy painting portraits ≈ para E: unlike Gainsborough, Goya "did not resent portraiture as **an economically necessary chore that ate away at the time he could devote to higher art**" — implying other artists like Gainsborough did resent it for that reason. Answer: E.

16. A

a genre Goya did not paint ≈ para A: **"there are no pure landscapes in Goya's work"**. Answer: A.

17. D

reaction to a promotion ≈ para D: after becoming pintor de cámara, he bought and promptly crashed a gig, then **"he wrote to his childhood friend Martín Zapater"** boasting about his new enviable way of living. Answer: D.

18. A

summary of the historical significance of Goya's works ≈ para A: after listing the historical upheavals Goya lived through, "what you have is **a portrait in the round of Spain itself**". Answer: A.

19. A or C

Correct pair is A & C ≈ para B: **"the majos and majas who gave Madrid its street swagger, peasants and rich men, courting couples, singers, hunters, children"** (A), and **"Goya practised poses and groupings and a way of handling light on and around figures that was to be invaluable"** (C).

20. A or C

Same pair as Q19: **A & C** — see explanation above (either letter counts for either blank).

21. C or D

Correct pair is C & D ≈ para E: **"There is hardly any repetition in his poses"** (C), and **"he defined them by the careful depiction of the space around them"** (D).

22. **C or D**

Same pair as Q21: **C & D** — see explanation above (either letter counts for either blank).

23. **commission**

his most significant ____ was from Charles IV ≈ para F: "In the most important commission of his career, for example, the portrait of Charles IV...". Answer: **commission**.

24. **butcher**

whose depiction was later likened to a ____ ≈ para F: "a French visitor had likened Charles and his queen Maria Luisa to a butcher and his wife". Answer: **butcher**.

25. **doctor**

in the care of the good ____ ≈ para G: the ailing painter is shown "being tended by his doctor". Answer: **doctor**.

26. **kindness**

dependent on the ____ of their fellow men ≈ para G: "Arrieta stands for man's innate kindness". Answer: **kindness**.

Passage 14 — How Babies See Colour

1. **vii**

Para A: the Baby Lab **hopes to develop ways to identify babies at risk of becoming colour blind and to discover ways to minimise the impact of this condition**. Answer: **vii**.

2. **vi**

Para B: the smiley face, spinning disc and cartoon clip are all used to **keep Teo's attention on the screen during the experiment**. Answer: **vi**.

3. **i**

Para C: **visual acuity around 5% of that of a mature individual** at birth, developing to almost full extent by six months — a direct baby-vs-adult comparison. Answer: **i**.

4. **v**

Para D: **it is a myth that babies see in black and white**, says Franklin — correcting a popular misconception. Answer: **v**.

5. **viii**

Para E: colour vision depends on **specialist cells in the retinas of our eyes... known as cones**. Answer: **viii**.

6. **iv**

Para F: **not all societies categorise colours in the same way. This may be linked to the words in different languages.** Answer: **iv.**

7. **A / B**

A: a camera locks its focus onto their faces. B: when Teo looks at the circle, it changes into a smiley face. Answers: **A, B.**

8. **A / B**

A: a camera locks its focus onto their faces. B: when Teo looks at the circle, it changes into a smiley face. Answers: **A, B.**

9. **A / C**

A: by the age of six months... a baby's visual acuity has developed to almost its full extent. C: newborn babies can see large, bright patches of red on a grey background. Answers: **A, C.**

10. **A / C**

A: by the age of six months... a baby's visual acuity has developed to almost its full extent. C: newborn babies can see large, bright patches of red on a grey background. Answers: **A, C.**

11. **cells**

cones are **specialist cells in the retinas of our eyes.** Answer: **cells.**

12. **wavelengths**

cones are sensitive to long, medium, and short **wavelengths** of light. Answer: **wavelengths.**

13. **signals**

the brain must make sense of the **signals** from the cones. Answer: **signals.**

14. **discrimination**

a baby's **discrimination** of colours develops, e.g. telling red and green apart. Answer: **discrimination.**

15. **intensity**

the **intensity** of the colours is crucial. Answer: **intensity.**

16. **improvement**

a gradual **improvement** in a baby's ability to see faint colours occurs as they mature. Answer: **improvement.**

Passage 15 — How Plants Fight Back

14. **B**

Para B: **the team were able to see this by utilising a naturally occurring fluorescent-green protein which binds to the calcium, making its path visible.** Answer: B.

15. **A**

Para A: **how plants responded when they subjected them to wounds, including scissor cuts and caterpillar bites.** Answer: A.

16. **E**

Para E: **you can imagine that we should be able to take a crop plant and switch on its defences on-call.** Answer: E.

17. **C**

Para C: **his real passion is understanding how plants sense gravity and seem to know which way is up.** Answer: C.

18. **E**

Para E: **in contrast to our understanding of human nerve cells, gilroy admits that the equivalent responses in plants are still barely understood.** Answer: E.

19. **videos**

Gilroy and Toyota have produced **a series of videos that shows how plants responded when they subjected them to wounds.** Answer: **videos.**

20. **wounds**

how plants responded when they subjected them to wounds, including scissor cuts and caterpillar bites. Answer: **wounds.**

21. **danger**

the calcium alerts the plant to danger and the need to deploy defence tactics. Answer: **danger.**

22. **protein**

a naturally occurring fluorescent-green protein which binds to the calcium, making its path visible. Answer: **protein.**

23. **glutamate**

how glutamate - an abundant neurotransmitter in animals - triggers this wave of calcium. Answer: **glutamate.**

24. **"B**

(B) **the calcium travels at one millimetre per second** — its rate of flow. (E) **how far the calcium travels depends on the extent of the wound** — severity linked to distance. Answers: **B, E**.

25. **"B**

Same evidence as Q24 — this pair shares one answer set. Answers: **B, E**.

26. **"A**

(A) they will **look at other signals that plants send out - signals regarding temperature and changes in light and touch** — heat/cold. (D) they will **delve deeper into the signalling response on a cellular level, dissecting the genes and proteins responsible**. Answers: **A, D**.

27. **"A**

Same evidence as Q26 — this pair shares one answer set. Answers: **A, D**.

Passage 16 — How do Plants Talk to Each Other?

14. **iii**

By the late 1980s, most ecologists felt that Schultz and Baldwin's idea had been discredited. Answer: **iii**.

15. **vi**

a number of more carefully designed experiments began to yield convincing indications to the contrary. Answer: **vi**.

16. **viii**

VOC-based signaling between plants may have been favored because it enhances the 'extended fitness' of the sender by aiding related plants. Answer: **viii**.

17. **i**

distress signals can be passed through the plants' roots. Answer: **i**.

18. **iv**

chili plants growing next to fennel plants developed more quickly than seedlings grown with other chili plants. Answer: **iv**.

19. **ii**

knowledge of this phenomenon could eventually be applied to agriculture, and lead to the cultivation of hardier crops. Answer: **ii**.

20. roots

20: distress signals can be passed through the plants' roots. 21: conditions similar to those experienced in a drought. 22: the labyrinths of hair-like fungi that occur around the roots. 23: broad bean plants were infested with aphids. Answers: roots / drought / fungi / aphids.

21. drought

20: distress signals can be passed through the plants' roots. 21: conditions similar to those experienced in a drought. 22: the labyrinths of hair-like fungi that occur around the roots. 23: broad bean plants were infested with aphids. Answers: roots / drought / fungi / aphids.

22. fungi

20: distress signals can be passed through the plants' roots. 21: conditions similar to those experienced in a drought. 22: the labyrinths of hair-like fungi that occur around the roots. 23: broad bean plants were infested with aphids. Answers: roots / drought / fungi / aphids.

23. aphids

20: distress signals can be passed through the plants' roots. 21: conditions similar to those experienced in a drought. 22: the labyrinths of hair-like fungi that occur around the roots. 23: broad bean plants were infested with aphids. Answers: roots / drought / fungi / aphids.

24. C

applying our limited knowledge of plant-communication mechanisms to agriculture is a big jump, says Johnson, 'but it is definitely on the horizon'. Answer: C.

25. A

each individual plant uses the signals to coordinate its own physiological responses to protect itself. Karban agrees, saying. Answer: A.

26. B

When I started my PhD (in the late 1980s), all this stuff was considered very weird,' recalls Ariel Novoplansky. 'Today, there is no doubt. Answer: B.

Passage 17 — How the Petri Dish Supports Scientific Advances

17. D

they use between 50 and 150 petri dishes every day to grow mouse liver and human liver cells. Answer: D.

18. F

one day, says lancaster, this work could translate into understanding far more about Alzheimer's, Parkinson's and schizophrenia. Answer: F.

19. **A**

some people see them as just shallow dishes with lids. but petri dishes deserve celebrating.
Answer: A.

20. **C**

use glass any more, but plastic. Answer: C.

21. **F**

you keep an eye on it and you check it every day. Answer: F.

22. **B**

the development of glass scientific instruments, from microscope lenses to laboratory beakers. Answer: B.

23. **C**

Huch: the cells first have to realize that there is damage and activate the response — then proliferate, then become functional. Answer: C.

24. **A**

Macfarlane: without glass, the renaissance and the scientific revolution would never have happened. Answer: A.

25. **D**

Lancaster: without the modification you won't be able to understand the architecture of those cells.
Answer: D.

26. **B**

Vallier: can't look inside a patient's liver, "so we are reproducing those diseases in our dishes".
Answer: B.

27. **A**

Macfarlane: around 70 per cent of what we know about the world comes in through our eyes.
Answer: A.

28. **B**

Vallier: feeding them this medium we can allow the cells to become neurons, cardiac cells, liver cells. Answer: B.

29. sticking

29: the dish has been specially treated to stop cells sticking to it and to encourage them to float freely. 30: they want the cells to develop in three, rather than two, dimensions. 31: this new method gives you a structure that looks a lot more like that of an actual developing brain. 32: how neurons are made and how that differs in humans compared with other species. Answers: sticking / dimensions / structure / species.

30. dimensions

29: the dish has been specially treated to stop cells sticking to it and to encourage them to float freely. 30: they want the cells to develop in three, rather than two, dimensions. 31: this new method gives you a structure that looks a lot more like that of an actual developing brain. 32: how neurons are made and how that differs in humans compared with other species. Answers: sticking / dimensions / structure / species.

31. structure

29: the dish has been specially treated to stop cells sticking to it and to encourage them to float freely. 30: they want the cells to develop in three, rather than two, dimensions. 31: this new method gives you a structure that looks a lot more like that of an actual developing brain. 32: how neurons are made and how that differs in humans compared with other species. Answers: sticking / dimensions / structure / species.

32. species

29: the dish has been specially treated to stop cells sticking to it and to encourage them to float freely. 30: they want the cells to develop in three, rather than two, dimensions. 31: this new method gives you a structure that looks a lot more like that of an actual developing brain. 32: how neurons are made and how that differs in humans compared with other species. Answers: sticking / dimensions / structure / species.

Passage 18 — IQ Scores

14. A

modern technology... influenced human intelligence ≈ para C: Flynn believes "demands of visuo-spatial abilities have grown because of TV, computers and cars". Answer: A (James Flynn).

15. D

may not have attained their natural limit ≈ para E: "as Teasdale points out, if such a ceiling had been reached, the test results should show scores pushing against it rather than actually falling" — the falling scores argue against a ceiling having been reached. Answer: D (Thomas Teasdale).

16. C

improved diet... positive effect on mental powers ≈ para D: Howard argues intelligence has risen, "helped by better nutrition". Answer: C (Robert Howard).

17. **B**

exceptional individuals... impact on collective intelligence ≈ para D: Rust points out that "if just one mind grasps a new concept, such as Einstein and the theory of relativity, it may then be easier for society in general to think in a more complex way". Answer: B (John Rust).

18. **B**

more sophisticated... need for problem-solving skills increases ≈ para D: "in more complex societies, people have to think harder to find solutions to problems they face". Answer: B (John Rust).

19. **E**

selection procedure involving a particular group of people ≈ para E: "The Danish military still summons virtually all young men before a draft board to identify who is suitable to do military service". Answer: E.

20. **B**

description of one component of a visuospatial test ≈ para B: Raven's matrices are "sequential patterns with one piece missing. The candidate has to choose, from a set of alternatives, which shape to put in the empty space". Answer: B.

21. **A**

overview of the different interpretations of IQ score trends ≈ para A: "psychologists disagree about what the upward trend in scores really means", with several competing views summarised. Answer: A.

22. **F**

negative effect affluence may have on IQ trends ≈ para F: "prosperity often led to decadence", illustrated by "the case of ancient Rome". Answer: F.

23. **the mid-1980s**

During the ____, Flynn conducted research showing IQ scores were increasing ≈ para B: Flynn "first showed that IQ scores were rising in the mid-1980s". Answer: the mid-1980s.

24. **Raven's matrices**

tests called ____, which measured visuo-spatial skills ≈ para B: "This last test category usually includes 'Raven's matrices'". Answer: Raven's matrices.

25. **verbal**

numerical and ____ components rose less steeply ≈ para B: "verbal and arithmetical scores showed more modest rises" — "arithmetical" = the summary's "numerical", leaving verbal as the answer.

26. **countries**

research in other ____, examined more recently ≈ para B: "Flynn has looked at data from further countries and found the same upward trend". Answer: **countries**.

Passage 19 — Is Graffiti Art or Crime?

14. **D**

chemically cleaning... cause damage ≈ para D: it's important to "assess the ability of the substrate to withstand the prescribed treatment" before applying chemical solvents. Answer: **D**.

15. **G**

benefit... on the gentle removal ≈ para G: because of the anti-graffiti coating (precautionary strategy from para F), "removal of graffiti from a surface that has been treated in this way is much easier, usually using low-pressure water". Answer: **G**.

16. **B**

accumulative impact... to the community ≈ para B: graffiti can lead to vandalism and, ultimately, "the deterioration of an area, contributing to social decline". Answer: **B**.

17. **E**

different preventive measures ≈ para E: "no one set of protection measures will be suitable for all situations. Each site must be looked at individually". Answer: **E**.

18. **C**

proposal made to the owner ≈ para C: "it is important that the owners of buildings or other structures and their consultants are aware of the approach they should take in dealing with the problem". Answer: **C**.

19. **B**

reasons of removing graffiti as soon as possible ≈ para B: "It is usually considered a priority to remove graffiti as quickly as possible after it appears. This is for several reasons." Answer: **B**.

20. **B or D**

Correct pair is **B & D** ≈ para D: "small trial areas should be undertaken to assess the impact of more extensive treatment" (B), and both chemical solvents ("potentially hazardous") and mechanical abrasion require care around bystanders and operators (D) — risk exists for both methods.

21. **B or D**

Same pair as Q20: **B & D** — see explanation above (either letter counts for either blank).

22. **B or D**

Correct pair is **B & D** ≈ para E: "**Security patrols will also act as a deterrent to prevent recurring attacks**" (B), and "**prominently placed cameras have been shown to reduce anti-social behavior**" (D).

23. **B or D**

Same pair as Q22: **B & D** — see explanation above (either letter counts for either blank).

24. **social history**

records the ____ of details of life ≈ para A: ancient graffiti provides "**a social history of life and events at that time**". Answer: **social history**.

25. **tag**

signature, called a ____ ≈ para C: "**stylised signatures or nicknames, known as 'tags'**". Answer: **tag**.

26. **protective equipment**

put on the suitable ____ ≈ para D: operatives should "**wear the appropriate protective equipment**". Answer: **protective equipment**.

27. **low-pressure water**

made much more convenient by using ____ ≈ para G: "**usually using low-pressure water which reduces the possibility of damage**". Answer: **low-pressure water**.

Passage 20 — Keeping an Eye on Shoppers

14. **F**

Para F: **the technology allows retailers to refine their employee training programmes**. Answer: **F**.

15. **E**

Para E: the software can **determine each shopper's sex, age range and ethnic group**. Answer: **E**.

16. **G**

Para G: **traditional 'bricks and mortar' retailers face stiff competition from online merchants**. Answer: **G**.

17. **C**

Para C: data is **compared with the store's transaction log to determine the effectiveness of store layout, product fixtures and other variables**. Answer: **C**.

18. **E**

Para F (Recordant): **its software scans the recorded dialogue for particular words or phrases without the need for any human input.** Answer: E.

19. **C**

Para D (PRISM): **to which was added further information recorded by human observers.** Answer: C.

20. **E**

Para F (Recordant): **provides shop assistants with digital-audio recorders, to be hung around their necks.** Answer: E.

21. **D**

Para E (VideoMining): **the system can even be programmed to detect the characteristic behaviour of shoplifters.** Answer: D.

22. **A**

Para B (Smartlane): **sensors by the doors count the number of people entering and leaving the shop, and sensors by the tills work out how fast the queues are moving.** Answer: A.

23. **B**

Para C (BehaviorIQ): **gather data on where their customers go, how long they slow down to look at different products.** Answer: B.

24. **shopping units**

Para B: **groups of people, such as families or couples, that make one transaction between them** are called shopping units. Answer: **shopping units.**

25. **children**

Para D: **showed that although children took part in only 13 per cent of food-shopping trips, shoppers bought more when accompanied by them.** Answer: **children.**

26. **competitors**

Para F: **they can even programme the system to search for any mention of their competitors.** Answer: **competitors.**

Passage 21 — Learning from the Romans

1. **D**

Jackson says, 'as the Roman Empire declined, and shipping declined, **the need for the seawater concrete declined**'. Answer: D.

2. **B**

Roman cement contains aluminum, which is not found in Portland cement, and less silicon than is found in Portland cement. Answer: **B**.

3. **F**

there is not enough fly ash in this world to replace half of the Portland cement being used, so alternatives are needed. Answer: **F**.

4. **A**

samples came from the ancient underwater site in Pozzuoli Bay near Naples in Italy. Answer: **A**.

5. **E**

Producing Roman concrete also left a smaller carbon footprint than its modern counterpart. Answer: **E**.

6. **G**

modern structures are designed to last 100–120 years, yet Roman harbor installations have survived 2,000 years of chemical attack. Answer: **G**.

7. **C**

packed into wooden molds which were immersed in seawater: the seawater instantly triggered a hot chemical reaction. Answer: **C**.

8. **A / C**

A: fly ash is used, which causes fewer greenhouse gas emissions than Portland cement. C: some modern concrete no longer contains Portland cement; instead, fly ash is used. Answers: **A, C**.

9. **A / C**

A: fly ash is used, which causes fewer greenhouse gas emissions than Portland cement. C: some modern concrete no longer contains Portland cement; instead, fly ash is used. Answers: **A, C**.

10. **limestone**

heat a mix of limestone and clays to 1,450 degrees Celsius. Answer: **limestone**.

11. **third**

a temperature that was a third lower than that required for Portland cement. Answer: **third**.

12. **fuel**

making Portland cement releases carbon dioxide from burning fuel. Answer: **fuel**.

13. industry

manufacturing Portland cement accounts for seven percent of the carbon dioxide that **industry** puts into the air. Answer: **industry**.

Passage 22 — Lever Brothers' Sunlight Soap

14. C

Para C: **the company used a variety of media, from newspapers and magazines to colourful posters and collectable cards.** Answer: C.

15. D

Para D: **this model village was designed to provide high-quality housing, amenities, and social benefits.** Answer: D.

16. F

Para F: the success of Sunlight Soap **provided the capital and brand recognition for the Lever Brothers to expand their operations globally,** leading to Unilever. Answer: F.

17. B

Para B: **traditional soaps at the time were often harsh on the skin and required extensive preparation.** Answer: B.

18. E

Para E: **the widespread use of soap like Sunlight played a role in reducing the incidence of infectious diseases.** Answer: E.

19. B

Para B: **they sought to create a high-quality soap that was both affordable and effective in promoting hygiene.** Answer: B.

20. James Darcy Lever

Para B: William Hesketh Lever and **his brother James Darcy Lever.** Answer: **James Darcy Lever.**

21. 1884

Para B: the story **begins in 1884.** Answer: **1884.**

22. vegetable oils

Para B: manufactured **using a blend of vegetable oils and alkali.** Answer: **vegetable oils.**

23. ethical / sustainable

Para B: **made it a more ethical and sustainable option.** Answer: **ethical / sustainable.**

24. **germs**

Para C: emphasised its purity and **its ability to fight germs**. Answer: **germs**.

25. **Port Sunlight**

Para D: built a purpose-built factory and **village called Port Sunlight**. Answer: **Port Sunlight**.

26. **prosperity sharing**

Para D: William Lever's **philosophy of "prosperity sharing,"** Answer: **prosperity sharing**.

Passage 23 — Mind Music

14. **G**

Songs with earworm potential appear to share certain features: **a repeating pattern of ups and downs in pitch, and an irregular musical interval**. Answer: **G**.

15. **A**

Some neuroscientists and cognitive psychologists are studying 'earworms' to explore the mysteries of memory and the part of the brain that is beyond our conscious control. Answer: **A**.

16. **F**

when subjects listened to the first few notes of familiar music, areas in the right frontal and superior temporal portions of the brain became activated — a different pattern from when the same subjects were asked to imagine/recall music. Answer: **F**.

17. **H**

Drs Stewart and Halpern are now working together to recruit survey participants for a study looking at whether people at different stages of life experience earworms differently. Answer: **H**.

18. **wrist**

18: "a device attached to their **wrist** recorded their movements." **19:** "more gray matter in areas associated with processing **emotions**." **20:** Dr Stewart likens earworms to 'sonic screen savers' that **keep the mind entertained** while it is otherwise unoccupied — earworms *entertain* the idle brain, so the natural fit here is **entertained**. **21:** "the more **challenging** the activity, the less likely the volunteers were to hear the music." Answers: **wrist / emotions / entertained / challenging**.

19. emotions

18: "a device attached to their **wrist** recorded their movements." **19:** "more gray matter in areas associated with processing **emotions**." **20:** Dr Stewart likens earworms to 'sonic screen savers' that **keep the mind entertained** while it is otherwise unoccupied — earworms *entertain* the idle brain, so the natural fit here is **entertained**. **21:** "the more **challenging** the activity, the less likely the volunteers were to hear the music." Answers: **wrist / emotions / entertained / challenging**.

20. entertained

18: "a device attached to their **wrist** recorded their movements." **19:** "more gray matter in areas associated with processing **emotions**." **20:** Dr Stewart likens earworms to 'sonic screen savers' that **keep the mind entertained** while it is otherwise unoccupied — earworms *entertain* the idle brain, so the natural fit here is **entertained**. **21:** "the more **challenging** the activity, the less likely the volunteers were to hear the music." Answers: **wrist / emotions / entertained / challenging**.

21. challenging

18: "a device attached to their **wrist** recorded their movements." **19:** "more gray matter in areas associated with processing **emotions**." **20:** Dr Stewart likens earworms to 'sonic screen savers' that **keep the mind entertained** while it is otherwise unoccupied — earworms *entertain* the idle brain, so the natural fit here is **entertained**. **21:** "the more **challenging** the activity, the less likely the volunteers were to hear the music." Answers: **wrist / emotions / entertained / challenging**.

22. D

'Repetition leads to familiarity which leads to anticipation, which is satisfied by hearing the song,' says John Seabrook. Answer: **D**.

23. A

The idea that we have full control over our thought processes is an illusion, says psychologist Lauren Stewart. Answer: **A**.

24. B

Even tunes you may have heard but didn't pay attention to can worm their way into your subconscious, says Ira Hyman. Answer: **B**.

25. C

Imaging studies by Andrea Halpern show that **deliberately imagining music and actually listening to music activate many of the same neurological networks**. Answer: **C**.

26. B

So, when the mind hits a part of a song it can't remember, it loops back rather than moving on, Dr Hyman says. Answer: **B**.

Passage 24 — Nanotechnology: the science of the very small

14. bricks

in the same way that a child may use ____ ≈ para A: "we are now able to build new structures from atoms and molecules as if we were using the bricks of a children's construction toy".

Answer: bricks.

15. diameter

comparing the exact ____ of one atom with a human hair ≈ para A: "The diameter of an atom is typically one or three tenths of a nanometre while that of the human hair is around 1,000,000 nm". Answer: diameter.

16. sunscreen

a long-lasting ____, developed in the 1990s ≈ para B: "In the 1990s, there were safety issues with many of the sunscreen products available, so scientists at Oxford University set about finding a solution". Answer: sunscreen.

17. E

comparison between two of the world's vital natural resources ≈ para E: "many believe that water, which we take for granted, is going to become more valuable than oil". Answer: E.

18. F

how nanotechnology could be dangerous for humans ≈ para F: "There are concerns that potentially harmful nano-products might enter the body via the gut". Answer: F.

19. A

nanotechnology has influenced modern authors ≈ para A: "Their prominence has inspired science fiction". Answer: A.

20. G

achieved more than was anticipated a decade ago ≈ para G: "Healthcare is an area where the application of nanotechnology is already exceeding the expectations of ten years ago".

Answer: G.

21. H

financial as well as ecological benefits ≈ para H: "enabling us to make use of our resources more effectively, thereby protecting our environment, and creating new business opportunities". Answer: H.

22. D

more time is required to realise nanotechnology's potential ≈ para D: "it will be another decade or so before some of these ideas are deployed". Answer: D.

23. **C or D**

Correct pair is **C & D** ≈ para D: "**There are, however, many technical problems involved, and it will be another decade or so before some of these ideas are deployed**" (C), and "**We are already using nanoparticles and nanoporous materials to capture harmful greenhouse gases**" (D).

24. **C or D**

Same pair as Q23: **C & D** — see explanation above (either letter counts for either blank).

25. **B or D**

Correct pair is **B & D** ≈ para F: "**be adapted to help preserve and package food**" (B), and "**better delivery of pesticides in the field**" (D).

26. **B or D**

Same pair as Q25: **B & D** — see explanation above (either letter counts for either blank).

Passage 25 — Next to Nature, But What Is Art?

1. **C**

Para C: the moulder **was judged to have no rights to the image... he was specifically held not to be an artist**. Answer: **C**.

2. **E**

Para E: the cast of a giant's hand from 1889 is **an item that would now sit happily in any commercial or public gallery**. Answer: **E**.

3. **B**

Para B: **the casts were used for medical research**. Answer: **B**.

4. **F**

Para F: **does it interest the eye, excite the brain, move the mind to reflection and involve the heart?** Answer: **F**.

5. **D**

Para D: painters have always used **technical back-up such as studio assistants to do the boring bits**. Answer: **D**.

6. **NO**

life-casting was **considered inferior to sculpture**, and its speed and realism were seen as limitations, not virtues admired by sculptors. Answer: **NO**.

7. **NOT GIVEN**

Rodin is quoted saying life-casting **'happens fast but it doesn't make Art'**, but no 'shower process' is mentioned anywhere. Answer: **NOT GIVEN**.

8. **NO**

despite Gauguin's worries, **painting has proved robust**. Answer: **NO**.

9. **NOT GIVEN**

the passage never directly compares the amount of skill life-casting requires against sculpture. Answer: **NOT GIVEN**.

10. **YES**

each new art movement implies a reassessment of what has gone before. Answer: **YES**.

11. **YES**

works of art **float free of their creators' intentions; few of us can look at a medieval altarpiece as its painter intended**. Answer: **YES**.

12. **B**

the contradiction between unexpected size and verisimilitude is named as the piece's most significant impact. Answer: **B**.

13. **D**

what counts is the surviving object and our response to it. Answer: **D**.

Passage 26 — Orientation of Birds

14. **visual memory**

such birds possess a highly developed visual memory. Answer: **visual memory**.

15. **migration direction**

they appear to choose their normal migration direction even if they are released in a different place from their usual starting point. Answer: **migration direction**.

16. **destination**

they **maintain the correct direction, but fly west of their destination, and so fail to arrive**. Answer: **destination**.

17. **landmarks**

the albatross **clearly couldn't rely on any landmarks over the vast expanse of the pacific ocean**. Answer: **landmarks**.

18. laysan albatross

a laysan albatross returned to its nesting area on midway island in the middle of the pacific.
Answer: laysan albatross.

19. C

Para C: if the birds are transported 800 kilometres from their nest, it is only by good fortune that they find their way back... released within this area, the birds soon make their return; release them outside it and far fewer return. Answer: C.

20. B

Para B: birds are caught... they are taken to some spot, released, and the percentage of returns is recorded. the distance can be varied, and the direction, as well as the method of transporting them. Answer: B.

21. C

Para C: birds' eyes have a power of resolution two to three times greater than ours, enabling them to pick up very fine details. Answer: C.

22. G

Para G: schüz caught first-year european storks and released them later, after the departure of the adult storks at a time when they normally make their south-west autumn migration to africa. Answer: G.

23. C

domestic pigeons have been trained to peck at a certain point on an aerial photograph... four years later the birds were still able to respond to this training. Answer: C.

24. F

seven out of nine alpine swifts were recaptured at their nests after being displaced some 1400 kilometres. Answer: F.

25. A

in spite of the fact that there were no adults to guide them, the birds unanimously headed southwest. Answer: A.

26. D

this behaviour differs in the different age groups. the adults... found their wintering area... whereas the juveniles sought their winter quarters to the southeast of their real position.
Answer: D.

Passage 27 — Robotic Approach to Crop Breeding

1. heart disease

‘Recognised as the type of fatty acid most able to protect against **heart disease**.’ Answer: **heart disease**.

2. drought

‘aims to develop more **drought** tolerant sunflowers.’ Answer: **drought**.

3. transpiration

‘**Transpiration**...is thought to facilitate the passage of minerals through the plant and is vital for healthy plants.’ Answer: **transpiration**.

4. wind and humidity

‘**Wind** can affect leaf temperature...allow it to examine the effects of other variables such as **humidity**.’ Answer: **wind and humidity**.

5. H

Para H: **the sunflower canopy had to be complete, with no visible soil, so that the thermometers would only measure the temperature of the plants and not the surrounding environment.** Answer: H.

6. F

Para F: **they designed a robot with two infrared thermometers set at 180 degrees to each other. The robot runs on an oblong track around the experimental plot.** Answer: F.

7. C

Para C: **plants undergo photosynthesis to produce energy...transpiration is the mechanism by which plants lose water through their leaves.** Answer: C.

8. I

Para I: **the team and their robot have already made a major breakthrough in the Australian wheat industry with Drysdale Wheat.** Answer: I.

9. G

Para G: **if the skies darken due to rain, heavy cloud cover or sunset, the robot makes its way back to the garage to wait.** Answer: G.

10. B

Para B: **a robot developed by the Australian National University's Research School of Biological Sciences.** Answer: B.

11. **H**

Para H: **eight varieties of sunflowers were examined.** Answer: **H.**

12. **D**

Para D: **by measuring the temperature of the leaves, scientists can determine how much water is being lost through transpiration.** Answer: **D.**

Passage 28 — Roller Coaster: The Great Fairground Attraction

14. **chain**

Para B: **a long length of chain running up the hill under the track.** Answer: **chain.**

15. **gear**

Para B: **wound around a gear at the top of the hill and another one at the bottom of the hill.**
Answer: **gear.**

16. **motor**

Para B: **The gear at the bottom of the hill is turned by a motor.** Answer: **motor.**

17. **ice**

Para F: **a long, steep, wooden slide covered in ice.** Answer: **ice.**

18. **waxed**

Para G: **the French started building waxed slides instead.** Answer: **waxed.**

19. **wheels**

Para G: **To help the sleds move down these slides, they added wheels.** Answer: **wheels.**

20. **coal**

Para H: **originally to provide an easy way to send coal to the railway.** Answer: **coal.**

21. **steam engines**

Para H: **They were eventually replaced with steam engines.** Answer: **steam engines.**

22. **TRUE**

Para F describes the original, unmodified Russian ice slide; Para G: **It seems that the idea was then imported into France... the French started building waxed slides instead. To help the sleds move down these slides, they added wheels, and in 1817, for the first time, a train was attached to the track** — France is the first place any change to the basic slide is described. Answer: **TRUE.**

23. **FALSE**

Para K: **Following the Great Depression, a decline in roller coaster production began in the early 1930s** — directly contradicts "continued to increase" through the 1930s. Answer: **FALSE**.

24. **NOT GIVEN**

Para K: **a second roller-coaster boom in the 1970s... introduced a slew of innovative tubular steel coasters** — no reason (like public demand) is given for why this happened. Answer: **NOT GIVEN**.

25. **TRUE**

Para K–L: the boom covered "the 1970s and the beginning of the 1980s", then **a decline in interest for the rest of the decade**, before a new boom "since the early 1990s" — so most of the 1980s (after its early boom) was in decline compared to the 1970s/1990s booms. Answer: **TRUE**.

26. **TRUE**

Para L: **New launching techniques and other technological developments have opened up a world of options for designers... faster, taller and more twisted rides**. Answer: **TRUE**.

Passage 29 — Saturn Spectacular

1. **iii**

Para A: **scientists are unsure about the origin of the rings, their exact chemical composition, and why they behave in the way that they do**. Answer: **iii**.

2. **iv**

Para B: introduces **Saturn's giant moon Titan** and its unique atmosphere. Answer: **iv**.

3. **vii**

Para C: **living on Titan would be no walk in the park** — extreme cold, no liquid water or oxygen. Answer: **vii**.

4. **viii**

Para D: the cold **keeps things in a state of preservation**, unlike Earth where such reactions are long gone. Answer: **viii**.

5. **i**

Para E: **Saturn is only half as dense as Jupiter... releases less heat... storms are more powerful** — a direct comparison. Answer: **i**.

6. **vi**

Para F: scientists hope studying these differences will **reveal how giant planets form** and answer outstanding questions. Answer: **vi**.

7. **B**

Galileo used a **crude telescope**, which caused him to misinterpret the rings as separate bodies. Answer: **B**.

8. **D**

Titan's atmosphere consists mainly of **nitrogen (the primary component of our own air)**. Answer: **D**.

9. **C**

extreme cold, **no liquid water and no oxygen** would make life extremely difficult. Answer: **C**.

10. **C**

a thick haze of ammonia crystals **obscures the colourful banding seen so easily on Jupiter**. Answer: **C**.

11. **photographs**

the probe produced **photographs that finally show Saturn with crystalline clarity**. Answer: **photographs**.

12. **40 years**

interpreting all the data **might take 40 years**. Answer: **40 years**.

13. **weather systems**

studying these differences will reveal how giant planets form and how **weather systems work**. Answer: **weather systems**.

Passage 30 — Skyscraper Farming

14. **vii**

world's resources of fertile soil are rapidly deteriorating... **24 per cent of the world's 11.5 million hectares of cultivated land has already undergone human-induced soil degradation**. Answer: **vii**.

15. **ix**

the global population is expected to exceed nine billion by 2050... **food production will have to go up by 70 per cent**. Answer: **ix**.

16. **i**

dickson despommier claims that vertical farming could boost crop yields many times over. a single 20-storey vertical farm could theoretically feed 50,000 people. Answer: **i**.

17. **iii**

hokuto corporation is a model example of how a vertical farm can be profitable. with 28 vertical mushroom farms operating across the country, it produces some 68,000 tons of mushrooms annually. Answer: **iii.**

18. **iv**

despommier sees them as being capable of helping centers of displaced persons - such as refugee camps. Answer: **iv.**

19. **vi**

vertical farms could help to enrich biodiversity... vertical farming could dramatically cut the utilization of fossil fuels. Answer: **vi.**

20. **erosion**

soil degradation is happening particularly through erosion, according to a recent study by the uk government office for science. Answer: **erosion.**

21. **operating costs**

there are problems such as initial investment and operating costs are too great. Answer: **operating costs.**

22. **helicopters**

the mobile units could become mobile units that could be picked up by helicopters and dropped into the middle of a crisis zone. Answer: **helicopters.**

23. **B**

yamanoko goes on to highlight the relative cost-effectiveness of his organization's farming practices. Answer: **B.**

24. **A**

despommier is concerned about the harm which farming has done to the world's landscape... particularly the elimination of hardwood forests. Answer: **A.**

25. **A**

despommier claims that vertical farming could boost crop yields many times over. Answer: **A.**

26. **C**

this can, in turn, help to improve the productivity of conventional farms. Answer: **C.**

Passage 31 — Sports Science

1. C

Para C: in the long jump, **a radical jumping strategy change... resulted in a substantial increase in personal best.** Answer: C.

2. H

Para H: **reliving your best performances is one of the best ways to build confidence.** Answer: H.

3. B

Para B: biomechanists use **video, motion tracking, force transducers and instruments to measure electrical muscle activity.** Answer: B.

4. C

Carre: **we get it from all angles** – industry wants faster equipment, governing bodies want to preserve the game. Answer: C.

5. D

Maynard: athletes are trained to **refocus quickly and put themselves back on track in case something goes wrong mid-event.** Answer: D.

6. A

Fowler: **one of our elite jumpers made a radical jumping strategy change... substantial increase in their personal best.** Answer: A.

7. A

Fowler: **we need to know which muscles are working when, and how hard, to understand technique and co-ordination.** Answer: A.

8. B

Whyte: it's about **targeting each of these determinants of performance and training at the correct intensity, for the correct duration, at the correct frequency.** Answer: B.

9. visualisation

visualisation is discussed in **Para H: research also suggests visualisation is almost as good as practice.** Answer: **visualisation.**

10. positive distraction

Para G: anything that would be a **positive distraction.** Answer: **positive distraction.**

11. refocus quickly

Para G: they are trained to **refocus quickly** and put themselves back on track. Answer: **refocus quickly.**
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12. aerobically

Para D: muscles work **aerobically** for as long as possible. Answer: **aerobically**.

13. lactic acid

Para D: muscles produce **lactic acid**, which leads to muscle fatigue. Answer: **lactic acid**.

Passage 32 — Stress Less

1. C

Elsner says **doing well on a demanding workload is typical of senior executives**. Answer: **C**.

2. A

Plumridge: **I'll promise a client that I'll do something tomorrow... when I know it's not going to happen**. Answer: **A**.

3. A

he relies on the advice of colleagues: **just a fresh pair of eyes over an issue can help**. Answer: **A**.

4. D

his research shows **the major predictor of stress is the level of job control a person has**. Answer: **D**.

5. B

their study shows **people always believe that they will be less busy in the future**. This is a misapprehension. Answer: **B**.

6. C

Plumridge names **sleep, scheduling and family** as his three warning signs – not concentration. Answer: **C**.

7. D

the key is **not to focus on relief – a game of golf or a massage – but to reassess workloads**. Answer: **D**.

8. B

meditation can alter the biochemistry of the brain and the way that brains and bodies react to stress. Answer: **B**.

9. injury

stress causes the most lost time of any workplace **injury**. Answer: **injury**.

10. **16.6 weeks**

employees suffering stress are absent from work for an average of **16.6 weeks**. Answer: **16.6 weeks**.

11. **7%**

claims for stress-related psychological injury accounted for **7%** of insurance claims. Answer: **7%**.

12. **massage**

the key is not to focus on relief – a game of golf or a **massage**. Answer: **massage**.

13. **workloads**

the key to dealing with stress is to reassess **workloads**. Answer: **workloads**.

Passage 33 — Tackling Hunger in Msekeni

1. **iii**

Para A: a purpose-built classroom **has been emptied of pupils and turned into a storeroom for sacks of grain** – an unexpected use of a school building. Answer: **iii**.

2. **x**

Para B: **no war lays waste Malawi, nor is the land unusually crowded or infertile, but Malawians still have trouble finding enough to eat** – ruling out the usual explanations for hunger. Answer: **x**.

3. **viii**

Para C: describes how **donors such as the World Food Programme (WFP) provide the food** and local volunteers cook it – the mechanics of the programme. Answer: **viii**.

4. **ix**

Para D: **when the school's feeding programme was introduced, enrolment at Msekeni doubled**. Answer: **ix**.

5. **vi**

Para E: with no new teachers, **you would expect standards to fall even further. But they have not** – pass rates actually improved dramatically. Answer: **vi**.

6. **i**

Para F: **better nutrition makes for brighter children... more and better food helps brains grow and develop**. Answer: **i**.

7. **iv**

Para G: discusses **on a global scale, the good news is that people are eating better than ever before**, citing WHO statistics. Answer: **iv**.

8. **snacks**

girls are given **extra snacks to take home**. Answer: **snacks**.

9. **firewood**

families sent children out to **gather firewood** or help in the fields. Answer: **firewood**.

10. **85%**

pass rates at Msekeni improved dramatically, from 30% to **85%**. Answer: **85%**.

11. **50%**

Homo sapiens has grown **50% bigger** since the industrial revolution. Answer: **50%**.

12. **A / C**

A: half the lessons take place in the shade of yellow-blossomed acacia trees – some children are taught outdoors. **C: the influx of new pupils is not accompanied by an increase in the number of teachers** – no new staff recruited. Answers: **A, C**.

13. **A / C**

A: half the lessons take place in the shade of yellow-blossomed acacia trees – some children are taught outdoors. **C: the influx of new pupils is not accompanied by an increase in the number of teachers** – no new staff recruited. Answers: **A, C**.

Passage 34 — The City of Great Zimbabwe

1. **G**

Para G: deterioration due to **variations in temperature and soil moisture content, as well as pressure from tourism, encroaching invasive vegetation, and incorrect preservation methods**. Answer: **G**.

2. **B**

Para B: the west enclosure is thought to have been **the residence of successive chiefs**. Answer: **B**.

3. **F**

Para F: glass beads, porcelain and coins **testify to long-standing trade with the outer world**. Answer: **F**.

4. **A**

Para A: in 1986 it was designated both a national monument and a **World Heritage Site**. Answer: **A**.

5. **G**

Para G: today it is **managed by the Zimbabwean Ministry of Home Affairs**. Answer: **G**.

6. **B**

Para B: the city is **divided into three areas: the Hill Complex, the Great Enclosure and the Valley Complex**. Answer: **B**.

7. **A**

Para A: although there are examples of similar work elsewhere, **none are as distinguished and imposing as Great Zimbabwe**. Answer: **A**.

8. **E**

Para E: reasons for abandonment are debated, but suggestions include **a shortage of food for the overpopulated city, and deforestation**. Answer: **E**.

9. **B**

the Great Enclosure **is oval shaped**. Answer: **B**.

10. **A**

in the Hill Complex, the east enclosure **is considered to have served a ritual purpose**. Answer: **A**.

11. **C**

the Valley Complex features **an impressive display of V-shaped and checked wall decorations**. Answer: **C**.

12. **B**

the Great Enclosure's conical tower **may have functioned as a granary warehouse**. Answer: **B**.

13. **flag**

a design based on the birds became a national symbol, incorporated into the Zimbabwe **flag**. Answer: **flag**.

14. **tourism**

the site suffers from **pressure from tourism** among other factors. Answer: **tourism**.

15. **standards**

there is a need for conservation **standards that meet the international benchmark**. Answer: **standards**.

16. publications

limited revenue is generated by **the sale of publications** and charging an entrance fee. Answer: **publications**.

Passage 35 — The Dingo Debate

1. E

Para E: the fence **separates the main types of livestock found in Australia** – cattle to the northwest, sheep to the southwest. Answer: E.

2. D

Para D: **dingoes will kill them simply to lessen competition**. Answer: D.

3. C

Para C: **rabbits provide a huge prey base for two other introduced species: the feral cat and the red fox**. Answer: C.

4. B

Para B: **about 4,000 years ago, Asian sailors introduced dingoes to Australia**. Answer: B.

5. F

Para F: Allen's study found **some of the highest rates of calf predation occurred in areas that had been baited**. Answer: F.

6. A

Para A: dropping poisoned meat, **this practice is known as baiting**. Answer: A.

7. D

Para D: dingo packs **generally have only one fertile female, which limits their rate of increase**. Answer: D.

8. D

Clifford: he is **not convinced that they manage to catch cats that often, believing they are more likely to catch small native animals instead**. Answer: D.

9. A

McKechnie complains that **graziers' livelihoods are threatened when dingoes prey on cattle**. Answer: A.

10. B

Johnson says dingoes **have contributed to few, if any, extinctions**. Answer: B.

11. **Tasmanian tiger**

as **the Tasmanian tiger disappeared completely from Australia, dingoes became Australia's top predators**. Answer: **Tasmanian tiger**.

12. **rabbits**

when rabbit numbers are significantly reduced, fox and cat populations are left with little to eat besides native mammals. Answer: **rabbits**.

13. **overpopulation**

slow reproduction is **an adaptation to prevent overpopulation in the arid environment**. Answer: **overpopulation**.

Passage 36 — The Fascinating World of Attine Ants

14. **H**

Para H: the leaf-cutter ants are accomplishing feats beyond the power of humans — **growing a monocultural crop year after year without disaster** and using an antibiotic without causing resistance. Answer: **H**.

15. **C**

Para C: the lower attines' nests are smaller — **their nests are smaller and their techniques more primitive** — compared to leaf-cutter nests. Answer: **C**.

16. **B**

Para B: **15 percent of the leaf production of tropical forests disappears down the nests of leafcutter ants**. Answer: **B**.

17. **F**

Para F: **other ants won't go near it and the ants associated with the garden just starve to death**. Answer: **F**.

18. **G**

Para G: **a possible reason for the different varieties of fungus in the lower attine mushroom gardens** — to stay ahead of Escovopsis. Answer: **G**.

19. **D**

Para D: the leaf-cutters' fungus was **descended from a single pure strain, propagated for at least 23 million years**. Answer: **D**.

20. **B**

Para C: the lower attines **feed their gardens with similar leaves which have fallen on the ground**. Answer: **B**.

21. **A**

Para C: leaf-cutter workers come in made-to-fit sizes, including **miniature workers to seed them with fungus and clean off alien growths**. Answer: A.

22. **C**

Para B: 'the attine ants' achievement' (the whole tribe, not just leaf-cutters) allows them to **consume, courtesy of their mushroom's digestive powers, the harvest of tropical forests whose leaves are laden with poisonous chemicals**. Answer: C.

23. **A**

Para D: the leaf-cutters' fungus was **descended from a single pure strain, propagated for at least 23 million years**. Answer: A.

24. **A**

Para A: **a single leaf-cutter nest may contain a thousand such chambers**. Answer: A.

25. **B**

Para E: **a century of ant research did not provide any evidence for his idea** — textbooks even claimed the opposite. Answer: B.

26. **D**

Para H: the patch **harbors a particular kind of bacterium, one well known to the pharmaceutical industry and the source of half the antibiotics used in medicine**. Answer: D.

Passage 37 — The Gender Gap in New Zealand's High School Results

1. **pass rates**

the issue is the difference in pass rates between the sexes. Answer: pass rates.

2. **decade**

high school girls have been outperforming boys academically for more than a decade. Answer: decade.

3. **Australia**

it is an international phenomenon, and within australia was the subject of much debate and controversy. Answer: Australia.

4. **C**

Para C: **ncea... is useful to employers and to universities because it provides a fine-grained picture of pupils performance in every aspect of a subject**. Answer: C.

5. **B**

Para B: **the new zealand state ministry of education says it is taking the issue seriously. it is working with a reference group on boys' education.** Answer: B.

6. **D**

Para D: **the discrepancy in reading and writing skills between males and females shows up as early as preschool.** Answer: D.

7. **E**

Para E: **new zealand european girls most represented at the top and new zealand pacific island boys at the bottom... ethnic differences... may be more significant.** Answer: E.

8. **H**

believe there is still resistance within the education ministry towards doing anything about the problem. Answer: H.

9. **E**

in most parts of the curriculum, for example in maths, there isn't really a gender gap. but literacy is a different matter. Answer: E.

10. **G**

the written content of ncea papers is more demanding than the previous system of secondary school qualifications. Answer: G.

11. **C**

it became clear that the difference was, instead, to do with the quality of instruction. Answer: C.

12. **F**

boys' competitive instinct tends to come out later in their schooling years... by 17, boys are catching up academically with the girls. Answer: F.

13. **B**

it is easier in a boys' school, where activities cannot be "captured by girls". Answer: B.

Passage 38 — The Global Effect of Food Systems

1. **C**

Para C: **farming alone accounts for 10–12% of global greenhouse gas emissions.** Answer: C.

2. **F**

Para F: food systems should **cause less harm to the climate and be more resilient to the impacts** of global warming. Answer: **F**.

3. **B**

Para B: **wheat yields fell by 5.5% and maize yields by 3.8%** due to rising temperatures. Answer: **B**.

4. **A**

Para A: **few consider the impacts of the food they eat**. Answer: **A**.

5. **B**

Para B: food and agriculture **are massively impacted by climate change... and are also, simultaneously, major contributors to it**. Answer: **B**.

6. **F**

Para F: there's **no 'one size fits all' model food system to implement across the world**. Answer: **F**.

7. **carbon footprint**

people worry about their **individual carbon footprint**. Answer: **carbon footprint**.

8. **electricity**

people think about **how much electricity they use in their home**. Answer: **electricity**.

9. **greenhouse gases**

food systems account for **roughly one quarter of all manmade emission of greenhouse gases**. Answer: **greenhouse gases**.

10. **transportation**

that's more than the entire global **transportation** sector. Answer: **transportation**.

11. **A / C**

A: forests are massive carbon sinks... which remove carbon dioxide from the atmosphere and store it. C: 80% of deforestation is carried out to create farmland. Answers: **A, C**.

12. **A / C**

A: forests are massive carbon sinks... which remove carbon dioxide from the atmosphere and store it. C: 80% of deforestation is carried out to create farmland. Answers: **A, C**.

13. **C / D**

C: 30–40% of the food produced in the world is never eaten. D: if food waste were a country, it would be the world's third largest contributor to global warming. Answers: **C, D**.

14. **C / D**

C: 30–40% of the food produced in the world is never eaten. D: if food waste were a country, it would be the world's third largest contributor to global warming. Answers: **C, D.**

15. **A / E**

A: retailers can prevent waste by finding outlets for slightly flawed or blemished goods. E: in the developing world, improving infrastructure along the food chain – including cold storage – would prevent food loss. Answers: **A, E.**

16. **A / E**

A: retailers can prevent waste by finding outlets for slightly flawed or blemished goods. E: in the developing world, improving infrastructure along the food chain – including cold storage – would prevent food loss. Answers: **A, E.**

Passage 39 — The Impact of Invasive Species

14. **C**

Humans have always transferred a variety of species from one region to another. Answer: **C.**

15. **E**

the invading jellyfish underwent a population explosion in the space of just six years, the entire marine ecosystem was transformed. Answer: **E.**

16. **H**

It is also useful to learn to identify invasive species, and report any sightings to wildlife organisations. Answer: **H.**

17. **F**

The impact of the water hyacinth on native water plants is largely unstudied. Answer: **F.**

18. **G**

ordinary people outside the scientific community have a very limited understanding of the threat posed by invasive species. Answer: **G.**

19. **D**

Some scientists have even argued that in these cases the introduction enhances the diversity of animal and plant life in that specific area. Answer: **D.**

20. **D or E**

pets such as snakes or turtles which are intentionally set free into the wild (D); climate change has enabled some invasive plant species to colonise new areas (E). Answers: **D, E.**

21. **D or E**

pets such as snakes or turtles which are intentionally set free into the wild (D); climate change has enabled some invasive plant species to colonise new areas (E). Answers: **D, E.**

22. **B or D**

they imported it to a horticultural exhibition in New Orleans in 1884 (B, late 19th century); damaging boat engines and even blocking cooling pipes for power plants, occasionally leading to massive blackouts (D). Answers: **B, D.**

23. **B or D**

they imported it to a horticultural exhibition in New Orleans in 1884 (B, late 19th century); damaging boat engines and even blocking cooling pipes for power plants, occasionally leading to massive blackouts (D). Answers: **B, D.**

24. **biodiversity**

24: invasive species threaten biodiversity in many habitats. 25: without upsetting the ecological equilibrium of the area. 26: The jellyfish had a more serious impact on the ecosystem than the heavily polluted water (industrial waste). Answers: **biodiversity / equilibrium / waste.**

25. **equilibrium**

24: invasive species threaten biodiversity in many habitats. 25: without upsetting the ecological equilibrium of the area. 26: The jellyfish had a more serious impact on the ecosystem than the heavily polluted water (industrial waste). Answers: **biodiversity / equilibrium / waste.**

26. **waste**

24: invasive species threaten biodiversity in many habitats. 25: without upsetting the ecological equilibrium of the area. 26: The jellyfish had a more serious impact on the ecosystem than the heavily polluted water (industrial waste). Answers: **biodiversity / equilibrium / waste.**

Passage 40 — The Importance of Law

14. **iv**

The law influences all of us virtually all the time... It affects us from the embryo onwards.
Answer: **iv.**

15. **vii**

for the general population to help actually shape the law in many countries. Answer: **vii.**

16. **i**

IT expertise is compared with legal expertise: But legal understanding is just as vital and as universally needed. Answer: **i.**

17. **vi**

The number of laws has never been greater. Answer: **vi**.

18. **ii**

lawyers are not universally admired. Anti-lawyer jokes have a long history going back to the ancient Greeks. Answer: **ii**.

19. **viii**

it is no more just to say that lawyers are bad because they make a living from people's problems than it is to make the same accusation in respect of nurses or IT consultants.
Answer: **viii**.

20. **C or D**

(C) **legal understanding is just as vital and as universally needed** [as computer/IT skills]. (D) IT knowledge is such that **many parts of commercial and social life today would seize up in minutes** without it — the passage draws the same parallel for legal experts. Answers: **C, D**.

21. **C or D**

Same evidence as Q20 — this pair shares one answer set. Answers: **C, D**.

22. **truth**

22: lawyers' concerns **put personal reward above truth**. **23:** "just as there are some scientists, **journalists** and others in that category." **24:** the same accusation "in respect of **nurses** or IT consultants." **25:** public law work "relies on considerable professional **dedication**." **26:** legal work is "primarily concerned with drafting **documents**." Answers: **truth / journalists / nurses / dedication / documents**.

23. **journalists**

22: lawyers' concerns **put personal reward above truth**. **23:** "just as there are some scientists, **journalists** and others in that category." **24:** the same accusation "in respect of **nurses** or IT consultants." **25:** public law work "relies on considerable professional **dedication**." **26:** legal work is "primarily concerned with drafting **documents**." Answers: **truth / journalists / nurses / dedication / documents**.

24. **nurses**

22: lawyers' concerns **put personal reward above truth**. **23:** "just as there are some scientists, **journalists** and others in that category." **24:** the same accusation "in respect of **nurses** or IT consultants." **25:** public law work "relies on considerable professional **dedication**." **26:** legal work is "primarily concerned with drafting **documents**." Answers: **truth / journalists / nurses / dedication / documents**.

25. dedication

22: lawyers' concerns **put personal reward above truth.** **23:** "just as there are some scientists, **journalists** and others in that category." **24:** the same accusation "in respect of **nurses** or IT consultants." **25:** public law work "relies on considerable professional **dedication.**" **26:** legal work is "primarily concerned with drafting **documents.**" Answers: **truth / journalists / nurses / dedication / documents.**

26. documents

22: lawyers' concerns **put personal reward above truth.** **23:** "just as there are some scientists, **journalists** and others in that category." **24:** the same accusation "in respect of **nurses** or IT consultants." **25:** public law work "relies on considerable professional **dedication.**" **26:** legal work is "primarily concerned with drafting **documents.**" Answers: **truth / journalists / nurses / dedication / documents.**

Passage 41 — The International Style

1. C

Para C: **interiors showcased open spaces and fluid movements between separate areas of the building.** Answer: C.

2. E

Para E: **arts organisations rarely opted for its austerity.** Answer: E.

3. F

Para F: buildings in this style **demanding very little of architects in the way of imagination,** frustrating a younger generation. Answer: F.

4. B

Para B: the Empire State Building's top **is decidedly art deco, yet the uniform shaft of the lower two thirds represents a pronounced step in a new direction.** Answer: B.

5. C

Para C: architects created blockish, flat-roofed skyscrapers **using steel, stone and glass.** Answer: C.

6. D

Para D: rather than lauding institutions through features such as **Romanesque columns.** Answer: D.

7. commercial and civic

a steep rise in demand for commercial and civic premises. Answer: **commercial and civic.**

8. ornamentation

architects broke with the past by **rejecting virtually all nonessential ornamentation**. Answer: **ornamentation**.

9. ribbon design

a typical facade has an instantly recognisable **ribbon design**. Answer: **ribbon design**.

10. wealth and prestige

pre-modern architecture was typically designed to display **the wealth and prestige** of its landlords or occupants. Answer: **wealth and prestige**.

11. function and practicality

this focus on **function and practicality** reflected a desire to 'get on with business'. Answer: **function and practicality**.

12. D

detractors felt it represented **an obsession with profit and 'the bottom line'**. Answer: **D**.

13. B

the international style was **ubiquitous across key urban centres, dominating skylines**. Answer: **B**.

Passage 42 — The Lost City

1. B

Para B: it would **take far too long to excavate the site using traditional techniques alone**, so modern technology was used as well. Answer: **B**.

2. G

Para G: they were **staggered to discover that the walls were made entirely from stone and that the gate would have stood at least ten metres high**. Answer: **G**.

3. C

Para C: Dr Summers used a hand-held balloon and later a **hot-air balloon** to photograph the site from above. Answer: **C**.

4. A

Para A: many archaeologists believe these are the remains of **the fabled city of Pteria**, described by Herodotus. Answer: **A**.

5. **hot-air balloon**

Dr Summers rented a **hot-air balloon** and floated over the site, taking pictures. Answer: **hot-air balloon**.

6. **iron particles**

the **iron particles** in buried structures can be permanently reoriented by heat. Answer: **iron particles**.

7. **compass needle**

reoriented **like a compass needle**, to align with the Earth's magnetic field. Answer: **compass needle**.

8. **metal probe**

shooting pulses into the ground through a thin **metal probe**. Answer: **metal probe**.

9. **mudbrick**

stone and mudbrick are poor conductors. Answer: **mudbrick**.

10. **damp soil**

looser, **damp soil** conducts very well. Answer: **damp soil**.

11. **spring season**

the project has a **spring season**, when most of the resistivity work is done. Answer: **spring season**.

12. **clarify**

the team wants to **clarify** pictures from the magnetometry. Answer: **clarify**.

13. **B**

remote sensing does not reveal everything... but it shows the most interesting sub-surface areas of the site. The archaeologists can then excavate these using traditional techniques.
Answer: **B**.

Passage 43 — The Mbuti of the Ituri Forest

1. **iii**

Para A: the whole paragraph describes **both Mbuti hunter-gatherers and Bantu village-dwelling cultivators** sharing the forest and how population is spread across it. Answer: **iii**.

2. **iv**

Para B: describes **starch foods, honey, and forest supplies** exchanged between the two groups, plus trade with no fixed terms. Answer: **iv**.

3. **vii**

Para C: villagers **organize the resources needed for a wedding feast and delegate responsibilities at a funeral**, and mediate conflicts – support at key life events. Answer: **vii**.

4. **i**

Para D: **Mbuti even take precedence over village ritual authorities** in the harvest festival, and participate in coronation ceremonies – an elevated, symbolic role. Answer: **i**.

5. **vi**

Para E: the relationship **has been flexible enough to survive a number of migrations and political upheavals**, extending even to the newer Baungwana group. Answer: **vi**.

6. **C**

the whole passage describes mutual reciprocity: Mbuti need starch foods from villagers, villagers need **forest supplies and spiritual mediation** from Mbuti. Answer: **C**.

7. **B**

Villagers have **a fairly predictable yearly cycle of activities**, while Mbuti, as nomads, are highly unpredictable. Answer: **B**.

8. **C**

for perhaps two millennia, the Ituri Forest has been home for both Mbuti hunter-gatherers and Bantu cultivators. Answer: **C**.

9. **A**

Mbuti are considered by all as **the guardians of the forest, being the most familiar with the forest and the most efficient at extracting its resources**. Answer: **A**.

10. **three**

averaging less than three persons per km². Answer: **three**.

11. **honey**

Mbuti provide villagers with 'prestige foods', for example **honey** from the forest. Answer: **honey**.

12. **cash**

in most cases no cash is involved, and donations may even take the form of gifts. Answer: **cash**.

13. **harvest**

in the **harvest festivals** of cultivators, Mbuti even take precedence over village ritual authorities; only the third crop is for the mwami. Answer: **harvest**.

Passage 44 — The Museum Dilemma

1. **F**

the Leonardo da Vinci cartoon was **damaged by gunshot... despite the painting being protected behind a glass screen** – deliberate damage cannot be controlled for. Answer: **F**.

2. **B**

the platform was meant to prevent damage, but **inadvertently contributed to extensive damage when a young boy tripped on it**. Answer: **B**.

3. **I**

laser alarms were used here, but **this idea has not been taken up universally** due to the aesthetics/noise trade-off. Answer: **I**.

4. **D**

the augmented reality project was **initially trialled at White Sands National Monument... with great success**. Answer: **D**.

5. **G**

the same AR idea deployed here failed because **no one engaged with it**, due to reluctance to download the app. Answer: **G**.

6. **E**

visitors like to **handle objects at the same time for extra realism, such as at Culloden Battlefield**. Answer: **E**.

7. **NO**

famous museums like the Louvre attract crowds willing to pay steep fees; it is **the 55,000 local museums** that struggle for funds, not the famous ones. Answer: **NO**.

8. **NOT GIVEN**

no statistic or survey about how many people have visited (or not visited) their local museum is given anywhere in the passage. Answer: **NOT GIVEN**.

9. **NO**

written warnings tend to be ignored; gallery attendants are the preferred (though more expensive) option. Answer: **NO**.

10. **NOT GIVEN**

it was **the platform** – not the rope barrier itself – that a boy tripped on; the passage never states rope barriers themselves cause tripping. Answer: **NOT GIVEN**.

11. **YES**

exhibits that encourage engagement can mean **visitors then assume it is acceptable to get up close and personal with all works of art**. Answer: YES.

12. **NO**

although paintings lack the same tactile allure, **they too can suffer similar consequences** to sculptures. Answer: NO.

13. **YES**

the Scotland study found **visitors still see information about the place as important, whether the experience is virtual or not**. Answer: YES.

Passage 45 — The Plan to Bring an Asteroid to Earth

14. **v**

it actually starts to seem like a reasonable idea — capturing an asteroid is achievable with today's technology, not science fiction. Answer: v.

15. **viii**

materials mined there could be more easily taken off the asteroid and shuttled around the solar system — practical (base), economic (cheaper materials) and research (scientific data) justifications together. Answer: viii.

16. **i**

Engineers would need to be absolutely certain they could control such a potentially dangerous object. Answer: i.

17. **vii**

researchers could always start with a smaller asteroid perhaps 2 to 10 meters across. Answer: vii.

18. **iv**

Convincing taxpayers to foot such a bill could be difficult. Answer: iv.

19. **E**

useful practice for planetary defense missions in the event of a threat from a meteoroid or comet from space approaching our planet. Answer: E.

20. **D**

You're moving the largest mother lode imaginable. Answer: D.

21. **B**

NASA isn't going to want to go to something that is smaller than our spaceships. Answer: B.

22. **G**

Investigations of asteroids can be done much more cheaply with an unmanned spacecraft. Answer: G.

23. **base**

23: provide a stationary base from which to launch missions farther into space. 24: because of our planet's strong force of gravity. 25: some are full of metals, which can be mined. 26: up to one-quarter water, used for life-support or fuel. Answers: base / gravity / metals / water.

24. **gravity**

23: provide a stationary base from which to launch missions farther into space. 24: because of our planet's strong force of gravity. 25: some are full of metals, which can be mined. 26: up to one-quarter water, used for life-support or fuel. Answers: base / gravity / metals / water.

25. **metals**

23: provide a stationary base from which to launch missions farther into space. 24: because of our planet's strong force of gravity. 25: some are full of metals, which can be mined. 26: up to one-quarter water, used for life-support or fuel. Answers: base / gravity / metals / water.

26. **water**

23: provide a stationary base from which to launch missions farther into space. 24: because of our planet's strong force of gravity. 25: some are full of metals, which can be mined. 26: up to one-quarter water, used for life-support or fuel. Answers: base / gravity / metals / water.

Passage 46 — The Power of Music

14. **D**

even three-year-olds have been shown to link music in a major mode to happy faces. Answer: D.

15. **B**

archaeologists have found musical instruments played by neanderthals at least 50,000 years ago. Answer: B.

16. **C**

the long-standing principle among psychologists that our response to music is an acquired one. Answer: C.

17. **G**

the rhythms also triggered activity in the motor regions of the brain — alongside the hearing areas. Answer: G.

18. **E**

the same change in pitch works as an emotional telltale in communication between some mammals, and is also present in human speech. Answer: E.

19. **physical health**

19: they should be in first-rate physical health but musically untrained. 20: carried out brain scans on volunteers who were listening to rhythmic sounds. 21: why the simple act of listening to music might help disabled patients. 22: they were able to improve their walking ability. Answers: **physical health / brain scans / disabled / walking.**

20. **brain scans**

19: they should be in first-rate physical health but musically untrained. 20: carried out brain scans on volunteers who were listening to rhythmic sounds. 21: why the simple act of listening to music might help disabled patients. 22: they were able to improve their walking ability. Answers: **physical health / brain scans / disabled / walking.**

21. **disabled**

19: they should be in first-rate physical health but musically untrained. 20: carried out brain scans on volunteers who were listening to rhythmic sounds. 21: why the simple act of listening to music might help disabled patients. 22: they were able to improve their walking ability. Answers: **physical health / brain scans / disabled / walking.**

22. **walking**

19: they should be in first-rate physical health but musically untrained. 20: carried out brain scans on volunteers who were listening to rhythmic sounds. 21: why the simple act of listening to music might help disabled patients. 22: they were able to improve their walking ability. Answers: **physical health / brain scans / disabled / walking.**

23. **C**

Chen: could also prove useful with other impairments such as those involved in sound production — e.g. a stutter. Answer: C.

24. **B**

Zatorre: this ability could be enhanced by training. Answer: B.

25. **A**

Cook: analysis of how people respond to notes suggests a link with how our brains interpret certain sounds. Answer: A.

26. **B**

Zatorre: **pleasurable music activates brain circuitry which has been in existence in the human brain** — shared with rats, linked to biological rewards. Answer: **B**.

Passage 47 — The Purpose of Facial Expressions

14. **E**

it is not reliable one hundred percent of the time. Answer: **E**.

15. **B**

a smile may prompt people to approach, a scowl may warn them to stay away. Answer: **B**.

16. **F**

it may be uncomfortable or inconvenient for us to let others know our emotions. Answer: **F**.

17. **C**

expressions occur most often during pivotal points, the turning points in social interactions. Answer: **C**.

18. **A**

a large number of electrical impulses also occur, creating specific facial expressions and other physiological changes. Answer: **A**.

19. **E**

Campos: **the only thing you need to study in a car is the transmission** — the face is just one component. Answer: **E**.

20. **B**

Fridlund: **expressions evolved to set off certain behaviors in other people**. Answer: **B**.

21. **E**

Campos: **there is profound agreement that the face, along with the voice, body posture and hand gestures, forecasts to outside observers what people will do next**. Answer: **E**.

22. **D**

Camras: **there are many situations where emotion is experienced, yet no basic facial expression is displayed**. Answer: **D**.

23. **C**

Russell: **there is very little evidence supporting such a connection**. Answer: **C**.

24. **smile**

24: a smile may prompt people to approach. 25: they are holding a dialog with another person internally. 26: Fridlund thinks of facial expressions as tools for influencing social interactions. Answers: **smile / dialog / tools.**

25. **dialog**

24: a smile may prompt people to approach. 25: they are holding a dialog with another person internally. 26: Fridlund thinks of facial expressions as tools for influencing social interactions. Answers: **smile / dialog / tools.**

26. **tools**

24: a smile may prompt people to approach. 25: they are holding a dialog with another person internally. 26: Fridlund thinks of facial expressions as tools for influencing social interactions. Answers: **smile / dialog / tools.**

Passage 48 — The Return of Monkey Life

14. **C**

they produce much lower levels of toxin than do well-established, old-growth trees. Answer: **C.**

15. **G**

We're moving away from that vision and beginning to consider ways in which commercial activities may become a tool for the conservation of primates. Answer: **G.**

16. **C**

keep their systems primed by sampling a variety of plants and then focusing on a small number of the most nutritious food items. Answer: **C.**

17. **B**

by the 1980s less than two percent remained undisturbed, and less than one percent was protected. Answer: **B.**

18. **reproduction**

18: Howler reproduction is faster than that of other native monkey species. 19: do not depend on large areas of fruiting trees. 20: ability to thrive on leaves loaded with toxins. 21: have suffered during the long, ongoing drought in the area. Answers: **reproduction / fruit(s) / toxins / drought.**

19. **fruit**

18: Howler reproduction is faster than that of other native monkey species. 19: do not depend on large areas of fruiting trees. 20: ability to thrive on leaves loaded with toxins. 21: have suffered during the long, ongoing drought in the area. Answers: **reproduction / fruit(s) / toxins / drought.**

20. **toxins**

18: Howler reproduction is faster than that of other native monkey species. 19: do not depend on large areas of fruiting trees. 20: ability to thrive on leaves loaded with toxins. 21: have suffered during the long, ongoing drought in the area. Answers: reproduction / fruit(s) / toxins / drought.

21. **drought**

18: Howler reproduction is faster than that of other native monkey species. 19: do not depend on large areas of fruiting trees. 20: ability to thrive on leaves loaded with toxins. 21: have suffered during the long, ongoing drought in the area. Answers: reproduction / fruit(s) / toxins / drought.

22. **B**

species of Indigenous trees which had been absent for decades began to invade the abandoned pastures. Answer: B.

23. **A**

the other primates were once common in this area, too, but vanished, leaving only howlers. Answer: A.

24. **C**

monkeys bring underappreciated benefits to such plantations, dispersing the seeds of fruits. Answer: C.

25. **B**

Capuchins were the first to begin using the reborn forests, followed by howlers. Eventually, even spider monkeys returned. Answer: B.

26. **A**

vanished after the Pan-American Highway was built nearby in the 1950s. Answer: A.

Passage 49 — The Role of Mothers in the Origins of Music

14. **G**

vocal music, like speech, leaves no physical traces that could help us determine its ancient origins. Answer: G.

15. **C**

a fetus begins hearing nearly four months before birth, as it regularly hears its mother's voice, heartbeat, and digestive sounds. Answer: C.

16. **D**

Some key aspects are that music is meaningful and can be used to communicate information — part of Parncutt's list of 'universal aspects of music'. Answer: D.

17. **B**

although it might appear amusing or even nonsensical on the surface, 'motherese' is actually a sophisticated form of communication. Answer: B.

18. **A**

'motherese' arose as a way to strengthen this maternal bond and to help infants survive. Answer: A.

19. **C**

Trehub warns: many ideas about music's beginnings are speculative. Answer: C (Sandra Trehub).

20. **D**

McLachlan: in work, music could help distract from monotonous tasks. Answer: D (Neil McLachlan).

21. **B**

McPherson: by evaluating all these theories, he constructs a compelling argument. Answer: B (Gary McPherson).

22. **A**

'motherese' arose as a way to strengthen this maternal bond and to help infants survive. Answer: A (Richard Parncutt).

23. **D**

the 'motherese' theory does not explain the origins of non-vocal music. Answer: D (Neil McLachlan).

24. **courtship**

birds singing during courtship rituals. Answer: courtship.

25. **pleasure**

music exists to bring us pleasure, similarly to the effect of a drug. Answer: pleasure.

26. **tool**

music developed as a tool for play, helping children learn and gain other non-musical skills. Answer: tool.

Passage 50 — The Science of Sleeping on It

1. **E**

Para E: **this can only be applied to situations in which one already has some background knowledge or information.** Answer: **E.**

2. **A**

Para A: **it is often said that studying before sleep is a good way to retain information.** Answer: **A.**

3. **G**

Para G: **the brain wants to acknowledge a thought and move on to the next one, but in reality, we tend to latch onto one thought and ruminate on it.** Answer: **G.**

4. **F**

Para F: **all functions of the brain must have some practical usage, and are rooted in evolution and natural selection, which may explain why the brain utilizes dreams to help break down, analyze, and solve problems.** Answer: **F.**

5. **sound clip**

a unique sound clip would play in the background. Answer: **sound clip.**

6. **success rate**

their success rate improved across the board. Answer: **success rate.**

7. **problem solving**

rapid eye movement (rem) sleep enhances creative problem solving. Answer: **problem solving.**

8. **unrelated**

it stimulates associative networks, which allow the brain to make connections between unrelated concepts or ideas. Answer: **unrelated.**

9. **emotions**

the brain picks out emotions that occurred during or shortly after something happened. Answer: **emotions.**

10. **C**

this rehearsal appears to distinguish meaningful patterns from unhelpful "background noise". Answer: **C.**

11. **B**

getting up in the middle of the night to work on a problem, however, is not particularly helpful, as it will just leave you exhausted in the morning. Answer: **B**.

12. **D**

this research adds to a growing list of literature suggesting that sleep can reorganise information to facilitate problem-solving. Answer: **D**.

13. **E**

all functions of the brain must have some practical usage, and are rooted in evolution and natural selection. Answer: **E**.

Passage 51 — The Shape of Bird Eggs

1. **C**

Para C: researchers concentrated on the egg's **soft thin inner membrane**. Answer: **C**.

2. **E**

Para E: a bird which spends lots of time in the air requires a compact, long, streamlined body, and this **best accommodates an elongated egg**. Answer: **E**.

3. **F**

Para F: **penguins lay pointed, asymmetrical eggs too, even though they are flightless**. Answer: **F**.

4. **G**

Para G: the results **could be of value to the egg industry**, perhaps by helping in the production of more durable eggs. Answer: **G**.

5. **B**

Para B: the program **can select the image of any egg and calculate its length, width and shape**. Answer: **B**.

6. **D**

Para D: Stoddard and her researchers **found neither of these hypotheses to be persuasive**. Answer: **D**.

7. **A**

Para A: eggs all basically serve one function: **nourish and protect the developing chick inside**. Answer: **A**.

8. **C / E**

C: Stoddard headed an **interdisciplinary team of evolutionary biologists, computer scientists, mathematicians and physicists**. **E:** they wrote **Eggxtractor**, a tool to calculate egg length, width and shape from photos. Answers: **C, E**.

9. **C / E**

C: Stoddard headed an **interdisciplinary team of evolutionary biologists, computer scientists, mathematicians and physicists**. **E:** they wrote **Eggxtractor**, a tool to calculate egg length, width and shape from photos. Answers: **C, E**.

10. **A / C**

A: the inner membrane is **essential in fixing the egg's shape**. **C:** **members of each closely associated species tended to lay eggs with the same characteristic shape**. Answers: **A, C**.

11. **A / C**

A: the inner membrane is **essential in fixing the egg's shape**. **C:** **members of each closely associated species tended to lay eggs with the same characteristic shape**. Answers: **A, C**.

12. **frequency**

egg shape reflects how well the birds can fly and therefore their **frequency** of flight. Answer: **frequency**.

13. **skeletons**

birds needing aerodynamic bodies have smaller, less heavy **skeletons**. Answer: **skeletons**.

14. **adaptation**

penguins' bodies may be an **adaptation** to allow them to swim. Answer: **adaptation**.

15. **production**

the results could help in the **production** of more durable eggs. Answer: **production**.

16. **dinosaurs**

she wants to study egg shape changes when birds evolved from **dinosaurs**. Answer: **dinosaurs**.

Passage 52 — The Study of Laughter

1. **B / C**

B: laughter is **undoubtedly the best medicine** for adults (cardiovascular/immune effects), and children need laughter to **form positive emotional bonds, gain new social skills**. **C:** **laughter is clearly not unique to us** (apes, dogs). Answers: **B, C**.

2. **B / C**

B: laughter is **undoubtedly the best medicine** for adults (cardiovascular/immune effects), and children need laughter to **form positive emotional bonds, gain new social skills**. **C:** laughter is clearly **not unique to us** (apes, dogs). Answers: **B, C**.

3. **YES**

laughter is a universal phenomenon, and one of the most common things we do. Answer: **YES**.

4. **YES**

we have **powerful systems in our brain which respond to pleasure, happiness and joy**. It's also involved in **release of fear**. Answer: **YES**.

5. **NOT GIVEN**

the passage says **its function is surely communication**, but never explicitly states this is the ONLY purpose (other effects like health benefits are also discussed). Answer: **NOT GIVEN**.

6. **NO**

the causality is reversed: **it activates the cardiovascular system, so heart rate and blood pressure increase... causing blood pressure to fall again**. Answer: **NO**.

7. **NOT GIVEN**

the passage says some studies **suggest** this, but **no rigorously controlled studies have confirmed these effects**. Answer: **NOT GIVEN**.

8. **fear**

Cardoso spent years investigating the neural basis of **fear** in rats. Answer: **fear**.

9. **scared**

when alone, rats were **scared** and uncomfortable. Answer: **scared**.

10. **vocalisations**

juvenile rats make short **vocalisations**, pitched too high for humans to hear. Answer: **vocalisations**.

11. **primates**

laughter may have evolved long before **primates**. Answer: **primates**.

12. **purposes**

panting could be something similar but with entirely different **purposes**. Answer: **purposes**.

13. **evolutionary advantages**

and different **evolutionary advantages**. Answer: **evolutionary advantages**.

Passage 53 — The Tasmanian Tiger

14. striped

it was given the name 'tiger' because it had striped fur. Answer: striped.

15. Australia

many types of thylacine roamed across australia. Answer: Australia.

16. 12 million years

when a period of climate change cooled australia about 12 million years ago, the numbers of these ancient thylacines began to decline. Answer: 12 million years.

17. Tasmania

so that tasmania, a large island to the south of australia, was then the last remaining place where thylacines existed. Answer: Tasmania.

18. Europeans

europeans with sheep, dogs and a great indifference to native flora and fauna, seem to have brought about their extinction. Answer: Europeans.

19. B

Randolph Rose: take it from me, the tiger is gone. Answer: B.

20. A

Hans Naarding: if there are thylacines out there, they are better off right where they are. Answer: A.

21. D

Nick Mooney: many sincere seekers are victims of obsession. Answer: D.

22. C

David Pemberton: despite scientific thinking that a relatively large number of animals is required to sustain a population, the florida panther is down to a dozen or so animals... and is still ticking along. Answer: C.

23. A

Hans Naarding: i was besieged by television crews, including four or five from japan, and others from the united kingdom, germany, new zealand and south america. Answer: A.

24. **D**

Nick Mooney: **this particular account of a sighting and the photographs submitted as proof amount to one of the most convincing cases for the species' survival that he has seen.** Answer: D.

25. **B**

government and private search parties combed the region, but no further sightings were made. A search followed, but it found nothing further. Answer: B.

26. **D**

the coelacanth **was thought to have died out with the dinosaurs... until a specimen was dragged to the surface** in 1938 — proving experts wrong about its extinction. Answer: D.

Passage 54 — Undoing Our Emotions

14. **C**

Para C: **even those who normally were slow to bounce back could be coached to recover more quickly by being told to view the experiment as a challenge, rather than a threat.** Answer: C.

15. **A**

Para A: **the emotions expressed in these writings were predictive of how long they would live: those with upbeat autobiographies lived more than ten years longer.** Answer: A.

16. **D**

Para D: negative emotions **give a heightened sense of detail that makes us hypersensitive to minute clues related to the source of a threat, but that also produces tunnel vision in which we ignore anything unrelated to the danger.** Answer: D.

17. **C**

Para C: **those who viewed the experiment with good-humoured excitement saw their heart rates return to normal much more quickly than those who were angry about being fooled.** Answer: C.

18. **A**

Para A: **the researchers were seeking material to confirm earlier studies hinting at a link between having a good vocabulary in youth and a low risk of alzheimer's disease... what they found was even more amazing.** Answer: A.

19. **E**

Para E: Fredrickson showed students **some film clips... students who had seen the humorous or peaceful clips were more likely to match objects according to broad impressions.** Answer: E.

20. **D**

Susan Folkman: **these people frequently experience positive emotions.** Answer: **D.**

21. **C**

Brooks Gump: **high cardiovascular reactivity could be linked to increased chances of a heart attack.** Answer: **C.**

22. **A**

Deborah Danner: **the results were particularly striking because all members of the convent lived similar lifestyles, eliminating many variables.** Answer: **A.**

23. **A**

Danner points out that **no one has been able to determine why positive emotions might have such life-extending effects.** Answer: **A.**

24. **focus**

negative emotions provided focus, which was important for surviving in life-or-death situations. Answer: **focus.**

25. **friendship**

if you are building a friendship through being playful, that friendship is a lasting resource. Answer: **friendship.**

26. **society**

not only are these emotions good for the individual, but they are also good for society. Answer: **society.**

Passage 55 — Urban Wildlife and Citizen Science

1. **G**

Para G: **there are downsides too... Still, the benefits appear to outweigh the risks.** Answer: **G.**

2. **C**

Para C: using the general public gives **the ability to have far more cameras out in the field than any single researcher could manage.** Answer: **C.**

3. **E**

Para E: **the introduction of the grey squirrel is the main factor in the red squirrel's decline, due to competition and disease.** Answer: **E.**

4. **A**

Para A: **cities accommodate only 8% of the bird species and 25% of the plants** that would have lived there before urban development. Answer: **A**.

5. **F**

Para F: the local conservation group in Formby **supplies additional food, and employs dedicated squirrel officers** to protect red squirrels. Answer: **F**.

6. **B**

Para B: **when data is limited, it is difficult to understand the bigger picture**: we can't know if animal populations are becoming more or less abundant. Answer: **B**.

7. **G**

Para G: **managing urban sites like the one in Formby may be a better alternative** than rural conservation, given rising urbanisation. Answer: **G**.

8. **biodiversity**

loyiha **biodiversity**ni baholashga yordam beradi. Answer: **biodiversity**.

9. **spotter**

ular ro'yxatdan o'tib **spotter** bo'lishlari mumkin. Answer: **spotter**.

10. **34**

ma'lumotlar bazasida **34 species** qayd etilgan. Answer: **34**.

11. **squirrel**

eng ko'p qayd etilgan hayvon – **the American grey squirrel**. Answer: **squirrel**.

12. **raccoon**

a **North American raccoon**, later moved to a wildlife park. Answer: **raccoon**.

13. **C**

the passage centres on citizen-science efforts like **MammalWeb** and the **Formby** volunteers – local residents actively aiding conservation, not a purely negative or narrow focus. Answer: **C**.

Passage 56 — Viking Ireland

1. **v**

Para B questions whether the monks' written texts are reliable: **the raids were not the first contact, as the old texts suggest**. Answer: **v**.

2. **viii**

Para C: after summer raids, the Vikings **decided to stay, and built settlements for the winter**.
Answer: **viii**.

3. **vi**

Para D: carbon dating (a research technique) gave Simpson results she did not expect – **these dates seem impossibly early**. Answer: **vi**.

4. **iv**

Para E focuses on one skeleton and **revealed the most about the lives and hardships of Vikings at this time**. Answer: **iv**.

5. **vii**

Para F: isotopic tests show **the two men with the spinal deformity had spent their childhood in Scandinavia**, others in Ireland/Scotland. Answer: **vii**.

6. **iii**

Para G: Simpson and Johnson stress **more excavations and tests will be needed**, and both possibilities need further research. Answer: **iii**.

7. **C**

Ruth Johnson describes how, **after some years of profitable plunder, they decided to stay**.
Answer: **C**.

8. **B**

Gareth Williams says **the raids were not the first contact, as the old texts suggest**. Answer: **B**.

9. **A**

The dates were not what Simpson had thought they would be. Answer: **A**.

10. **B**

Williams suggests **something in the water or soil in Dublin skewing the data**. Answer: **B**.

11. **standards**

he was approximately 1.70m **tall by the day's standards**. Answer: **standards**.

12. **torso**

he had **the muscular torso and upper limbs** from hard rowing. Answer: **torso**.

13. **childhood**

stresses associated with heavy lifting **beginning in childhood**. Answer: **childhood**.

14. **spine**

a congenital **deformity at the base of his spine**. Answer: **spine**.

15. **weapons**

unlike the others, **he was not buried with weapons**. Answer: **weapons**.

Passage 57 — Walking and shoes in eighteenth-century London

14. **A**

employment of people to keep streets clean ≈ para A: "**led to the employment of cleaners and street sweepers**". Answer: **A**.

15. **F**

specific locations where fashionable people gathered to walk ≈ para F: "**Parks such as St James's and Kensington Gardens attracted crowds of strollers during the season**". Answer: **F**.

16. **D**

awkward devices women used to protect their shoes ≈ para D: "**Pattens—overshoes with raised iron rings—were commonly worn by women to lift their delicate footwear above the filth**. These contraptions were noisy and awkward". Answer: **D**.

17. **C**

first laws introduced to improve the condition of streets ≈ para C: "**The first legislation concerning paving in London was passed in 1762**". Answer: **C**.

18. **G**

criticism from those concerned about morality ≈ para G: "**The new culture of walking also had its critics. Moralists complained that the freedom to stroll encouraged idleness and frivolous display**". Answer: **G**.

19. **B**

walking in the city vs walking in the countryside ≈ para B: "References to walking are not abundant in early eighteenth-century literature, **except for writing about the countryside, where conditions were far better**. Walking in London was regarded as inconvenient and even dangerous". Answer: **B**.

20. **poor**

associated mainly with people who were ____ ≈ para B: "**Walking in urban areas was generally associated with the poor**". Answer: **poor**.

21. **dirt**

carriages, horses, and ____ ≈ para C: "**safe from the carriages, horses and dirt**". Answer: **dirt**.

22. leather

made from thick ____ ≈ para D: shoes "were typically made of stout **leather** with thick soles" — "made from" points to the material (leather), not the component (soles). Answer: **leather**.

23. fit

improved the ____ **of shoes** ≈ para F: specialist lasts for left and right feet "**This improved fit** and comfort". Answer: **fit**.

24. B

Para E: "As **paving spread and streets became cleaner**, shoe design evolved to reflect new **possibilities. Lighter materials... became fashionable**". Answer: **B**.

25. C

Para C: "**Walking was no longer solely a necessity for the poor but had become a respectable activity for the upper classes**" — combined with walking already being common among the poor, this spans all classes. Answer: **C**.

26. C

Para G opens with "**The new culture of walking also had its critics**" and details moralist objections — the whole paragraph's purpose is showing disapproval existed. Answer: **C**.

Passage 58 — Waste Production

1. Japan

Japan takes their twisted metal and sells it back to them as cars. Answer: **Japan**.

2. Western states

Western states have room for more, although dumps and their toxic by-products are seldom welcome neighbors. Answer: **Western states**.

3. Air pollution

burning waste raises concerns about **air pollution** from heavy metals such as lead, cadmium, copper and mercury, which vaporize in intense heat. Answer: **Air pollution**.

4. Steel

Recycling steel is less expensive than mining ore. Answer: **Steel**.

5. Paper

yard trimmings and wooden construction debris are **two of the largest components of landfills after paper**. Answer: **Paper**.

6. **17%**

Seattle recovers 45 per cent of its waste compared with a **17 per cent average for the rest of the country**. Answer: **17%**.

7. **NOT GIVEN**

the passage says **we burn 16 per cent now and could burn more** – a possibility, not a recommendation that the USA should do so. Answer: **NOT GIVEN**.

8. **YES**

Success is pinned to a **‘pay as you throw’ program: residents who recycle best pay the least for removal of their trash**. Answer: **YES**.

9. **NOT GIVEN**

Lynn Scarlett is from a Los Angeles think tank, but the passage never discusses enforcement of recycling laws specifically in Los Angeles – the enforcement gap it describes is **in Germany**. Answer: **NOT GIVEN**.

10. **NOT GIVEN**

the passage describes **Germany’s 1991 packaging-recycling law** but never claims it is the most advanced in the world. Answer: **NOT GIVEN**.

11. **YES**

some dots appeared that were not backed by recycling contracts. Answer: **YES**.

12. **YES**

supporters say **it just takes time for regional markets to evolve and close the loop from manufacturer to consumer and back to manufacturer again**. Answer: **YES**.

13. **YES**

‘We underestimated people’s response to recycling,’ said David Dougherty – they recycled so well it created an over-supply. Answer: **YES**.

Passage 59 — What's Happening to Our Food?

14. **C**

Para C: Wagstaff agrees this may **cause some loss of flavor**, compared with a home-grown tomato left to ripen. Answer: **C**.

15. **B**

Para B: Decker and Davis both agree **any differences in nutrient levels are relatively small**. Answer: **B**.

16. **A**

Para A: **it's like taking a glass of orange juice and adding water to it** — the concentration of nutrients drops. Answer: **A**.

17. **C**

Para C: **fruit and vegetables in supermarkets might look shiny and fresh, but often they were picked several days earlier**. Answer: **C**.

18. **D**

Para D: **one in five adults in the UK regularly buys fruit and vegetables in this form every week** (fresh-cut). Answer: **D**.

19. **A**

Para B (Davis): people who lack iron, magnesium and calcium **increase susceptibility to lots of different problems**. Answer: **A – Donald Davis**.

20. **F**

Para D (Buttriss): **if their availability means that such vegetables are consumed in greater quantities, then the net effect is beneficial**. Answer: **F – Judy Buttriss**. *Note: the answer key you supplied showed E (Catherine Collins) here, but the passage attributes this exact quote to Judy Buttriss — I've used F to match the passage text; let me know if you'd rather keep E.*

21. **C**

Para B (Decker): **lifespans have got longer, people are bigger and stronger, and a lot of that has to do with the food supply being better**. Answer: **C – Eric Decker**.

22. **B**

Para B (Finglas): **we eat a wider range of foods today than we did 10 years ago, let alone 40 years ago**. Answer: **B – Paul Finglas**.

23. **enzymes**

Para C: **lots of these reactions are driven by enzymes**. Answer: **enzymes**.

24. **bruising**

Para C: **the bruising and other types of damage that would occur if ripe fruits were transported**. Answer: **bruising**.

25. **light**

Para C: glucosinolates in cabbage **deplete in the light**. Answer: **light**.

26. **peas**

Para D: **frozen peas are much more nutritious than those you buy ready to shell**. Answer: **peas**.

Compiled by @bekhruzposts

Passage 60 — Why Do We Need Sleep?

1. **D**

sleep-deprived suspects held by the police will often confess to anything in exchange for rest. Answer: D.

2. **E**

each NREM stage is characterized by a distinctive... pattern of brain activity, as shown on an EEG. Answer: E.

3. **C**

the industrial revolution brought us alarm clocks and fixed work schedules; we can no longer sleep in. Answer: C.

4. **D**

lack of sleep has been directly tied to obesity: the body overproduces ghrelin, the hunger hormone. Answer: D.

5. **G**

Dr Ellenbogen and others argue against shortcuts, warning against artificially manipulating sleep with drugs. Answer: G.

6. **F**

REM sleep was at first believed to be only a variation of a stage in NREM sleep... [but] was recognized as the second major phase of sleep. Answer: F.

7. **A**

a 24 percent increase in heart attacks and a jump in fatal car crashes. Answer: A.

8. **A / D**

A: deep NREM sleep is as essential to our brain as food is to our body, and produces growth hormones. D: each of the four stages is characterized by a distinctive... pattern of brain activity, as shown on an EEG. Answers: A, D.

9. **A / D**

A: deep NREM sleep is as essential to our brain as food is to our body, and produces growth hormones. D: each of the four stages is characterized by a distinctive... pattern of brain activity, as shown on an EEG. Answers: A, D.

10. **B / E**

B: REM is identified by distinctive eye movements. E: some theorists argue REM is when we are most intelligent, insightful, creative, and free. Answers: B, E.

11. **B / E**

B: REM is identified by **distinctive eye movements**. **E:** some theorists argue REM is when we are **most intelligent, insightful, creative, and free**. Answers: **B, E**.

12. **fatigue**

the resulting **fatigue** contributes to more than a million auto accidents each year. Answer: **fatigue**.

13. **insomnia**

insomnia is by far the most prevalent of these problems. Answer: **insomnia**.

14. **emergency**

the brain can wake us **when it perceives an emergency like the cry of a child**. Answer: **emergency**.

15. **depression**

people who regularly sleep less than six hours have **an elevated risk of developing depression**.
Answer: **depression**.

16. **dreams**

the content of our **dreams** each night in REM sleep is important in the processing of new memories.
Answer: **dreams**.